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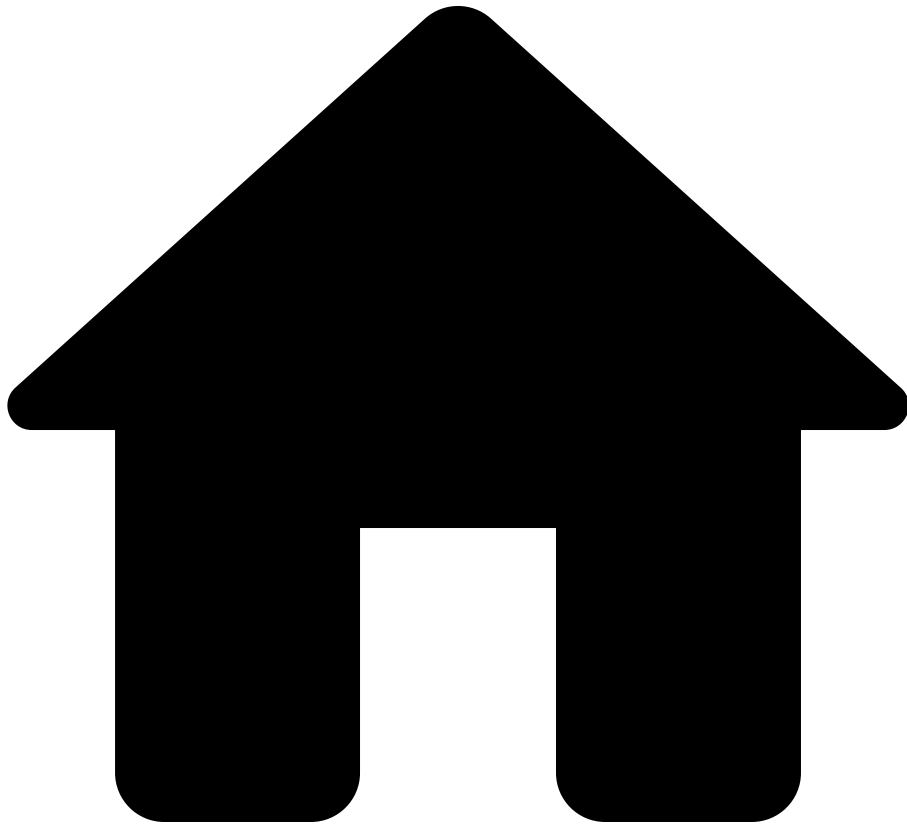
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Expressions and Types

You can think of NQE Checks as queries. Each query is actually an expression. There are many kinds of expressions, which weâll detail in the following sections. You have already seen some examples in the sections above, for example `foreach ...` `select ...` expressions.

Each expression has a value and a type. The value of the expression is the result produced by the expression. For example, the expression `12 + 13` has value `25`. The type of an expression describes the kind of value produced by an expression. For example, the type of the expression `12 + 13` is number. As we describe the forms of expressions below, weâll describe the type of each expression form.

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Numbers

Numbers can be written as normal:

- *Integers*: For example `16` to denote the integer (base 10) number 16.
- *Floats*: For example, `3.14` to denote the decimal number 3.14.

- In a local declaration such as `x = 2.3;`, hovering over `x` shows a tooltip indicating its type is `Float`, whereas hovering over `x` in `x = 2;` shows `Integer`.

Range of Numeric Values

Numbers (both integer and floating point) range from -9,223,372,036,854,775,807 to 9,223,372,036,854,775,807.

The closer to the boundaries, the greater the likelihood for rounding errors; the small floating-point number `-9223372036854771.5` rounds to `-9223372036854772`. Likewise `9223372036854775` results in the approximate number `9223372036854776`.

Arithmetic Operators

The following operators work on both integers and floats.

- `-x` : Negation
- `x + y` : Addition
- `x - y` : Subtraction
- `x * y` : Multiplication
- `x / y` : Division
- `x % y` : Modulo
- `x ^ y` : Exponentiation
- `sum(collection)` : The sum of a *collection of integer numbers* or a *collection of floating-point numbers*.
 - For example, on integers, `sum([1, 2, 3])` equals `6`.
 - Likewise, on floats, `sum([1.1, 2.1, 3.1])` equals `6.300000000000001`.
 - Keep in mind that *floating-point arithmetic is generally not precise!*
 - You can define a collection in various ways, not just as a collection of literal numbers. See [Collections](#) for more detail on working with lists and bags.
 - Importantly, a collection's elements must all be of the same type. As such, `sum([1.1, 2, 3.1])` has a type-error since the collection is not well-formed. The fix is to replace *integer* `2` with `2.0` or `2.` to have it be treated as a *float*.

Coercion Operators

To turn a literal *integer* such as `2` into a *float* we simply write `2.` or `2.0`.

- However, if we have any integer-valued expression, such as `x` or `x + 10`, where `x` is assigned an integer, then we can convert it to a *float* by writing `float(x)` or `float(x + 10)`, respectively.
- Conversely, if we have a float-valued expression `E`, then we can convert it to an *integer* by writing `int(E)`.
 - Note that `int(E)` just truncates any decimal points; e.g., `int(2.9)` results in `2`.

The following table lists the functions that convert an integer to a float:

Method	Description
<code>int(x)</code>	Truncates decimal points
<code>floor(x)</code>	Rounds down to the nearest smaller integer
<code>ceil(x)</code>	Rounds up to the nearest larger integer
<code>round(x)</code>	Rounds to the nearest integer

Note that `int(x)` and `floor(x)` have the same behavior.

See also

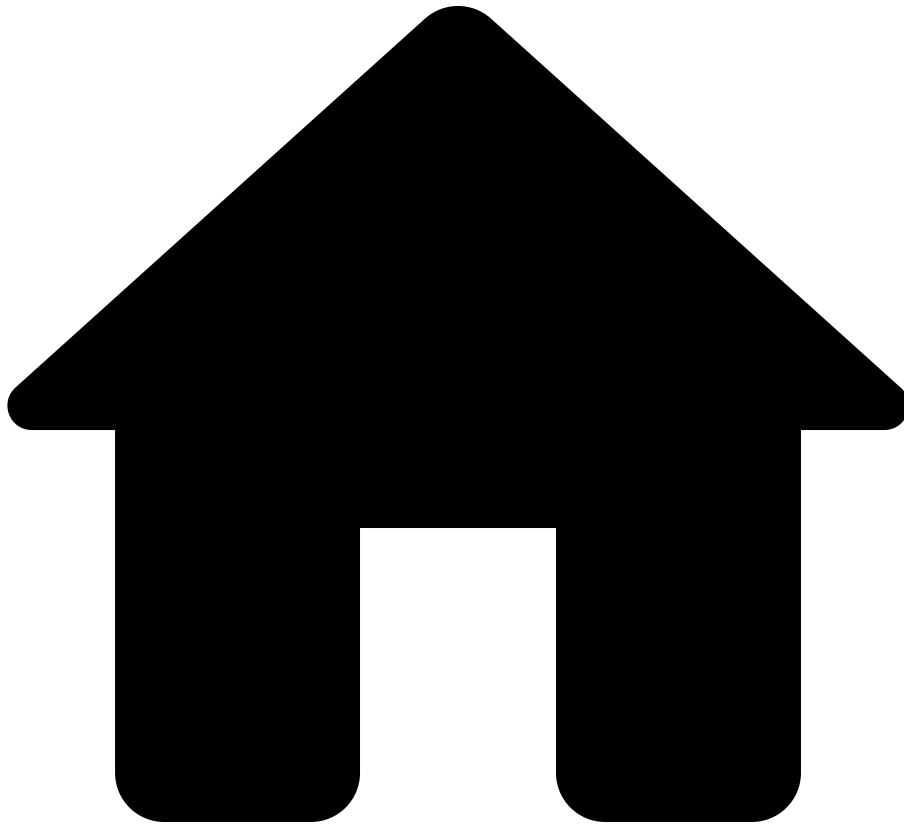
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Functions

- [float](#)
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Strings

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A string can be written inline between a single pair of double quotes or via a multi-line block between three pairs of double-quotes.

Inline

An example of an inline string is `"abc"`. A double quote can appear in an inline string by escaping it with `\` as in `"ab\"cd"`. Here is a list of some characters that can be expressed via escaping:

- `"` via `\"`,
- a tab via `\t`,
- a new line via `\n`, and
- `\` via `\\`.

Block

Here is an example of a block string, which is defined using three pairs of double quotes.

```
myMultiLineString =  
"""  
This is the first line of my "string"  
and this is the last line.  
""";
```

where the text is interpreted literally. In particular, all white space is included and all escape sequences are ignored.

Operations

The following operations are available on strings.

- `length(x)`: the number of characters in a string. For example, `length("abc")` is `3`.
- `x + y`: the result of concatenating `x` and `y`. For example, `"abc" + "def"` evaluates to `"abcdef"`.
- `matches(string, pattern)`: returns `true` if and only if `pattern` occurs in `string`, case-sensitive. For example, `matches("def", "def")` and `matches("abcdefefg", "*def*")` both return `true`.
- `substring(s, b, e)`: the substring of string `s` starting from index `b` (inclusive) and ending at index `e` (exclusive). For example, `substring("abcdef", 0, 6)` is `"abcdef"`,

`substring("abcdef", 1, 3)` is `"bc"` (empty string), and
`substring("abcdef", 1, 1)` is `" "`.

- `prefix(s, n)` : the first `n` characters of string `s` . For example,
`prefix("abcdef", 3)` is `"abc"` , `prefix("abcdef", 0)` is `" "`
(empty string), and `prefix("abcdef", 10)` is `"abcdef"` .
- `suffix(s, n)` : the last `n` characters of string `s` . For example,
`suffix("abcdef", 3)` is `"def"` , `suffix("abcdef", 0)` is `" "`
(empty string), and `suffix("abcdef", 10)` is `"abcdef"` .
- `toUpperCase(x)` and `toLowerCase(x)` : respectively change all alphabetic characters in the string to upper- and lower-case. For example,
`toUpperCase("UpperCase")` is `"UPPERCASE"`
and `toLowerCase("LowerCase")` is `"lowercase"` .
- `replace(text, target, replacement)` : replaces every occurrence of `target` in `text` with `replacement` . All three arguments must be strings. For example, `replace("abc-def", "-", "_")` is `"abc_def"` ,
`replace("abc-def", "-", "")` is `"abcdef"` and `replace("abab", "a", "cd")` is `"cdbcd b"` .

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Booleans

Boolean values are written as: `true` and `false`. A variety of logical operators are available:

- `x && y` : AND
- `x || y` : OR
- `!x` : NOT

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If Expressions

If expressions allow us to select a value based on a boolean (true or false) value. For example, the expression:

```
if x > 10 then "big" else "small"
```

will evaluate to the string "big" if `x` is greater than 10 and will evaluate to the string "small" otherwise. Let's look at another example of an if expression:

```
foreach device in network.devices
let dcName = if prefix(device.name, 3) == "atl"
              then "ATLANTA"
              else "PHOENIX"
select { device: device.name, dataCenterName: dcName }
```

In this example, if the field `name` in `device`'s first 3 letters are "atl", then `dcName` is "ATLANTA", otherwise `dcName` will be "PHOENIX". Either way, `dcName` will always be set to a string. **Note:** the two branches of an if expression must contain values of the same type, in both of these cases, string.

Here are some more examples:

- `daysInYear(year) = if isLeapYear(year) then 366 else 365;`
- `maximum(x, y) = if x > y then x else y;`
- `makeEven(x) = if x % 2 == 0 then x else x * 2;` . A function that returns the argument if it was even, or returns the argument multiplied by two if it was odd.

A note for users familiar with other programming languages. Many programming languages, such as Python and Java, have if statements that differ from NQE's if expressions. In those languages, if statements control execution of two different blocks of statements. In contrast, NQE's if expressions chooses between two different expressions. However, languages (e.g. Python and Java) also typically have "ternary conditional" expressions that are analogous to NQE's if expression. The following examples are equivalent to NQE's `if boolean then thenValue else elseValue` :

- In Python: `thenValue if boolean else elseValue` .
- In Java: `boolean ? thenValue : elseValue` .

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IP Addresses

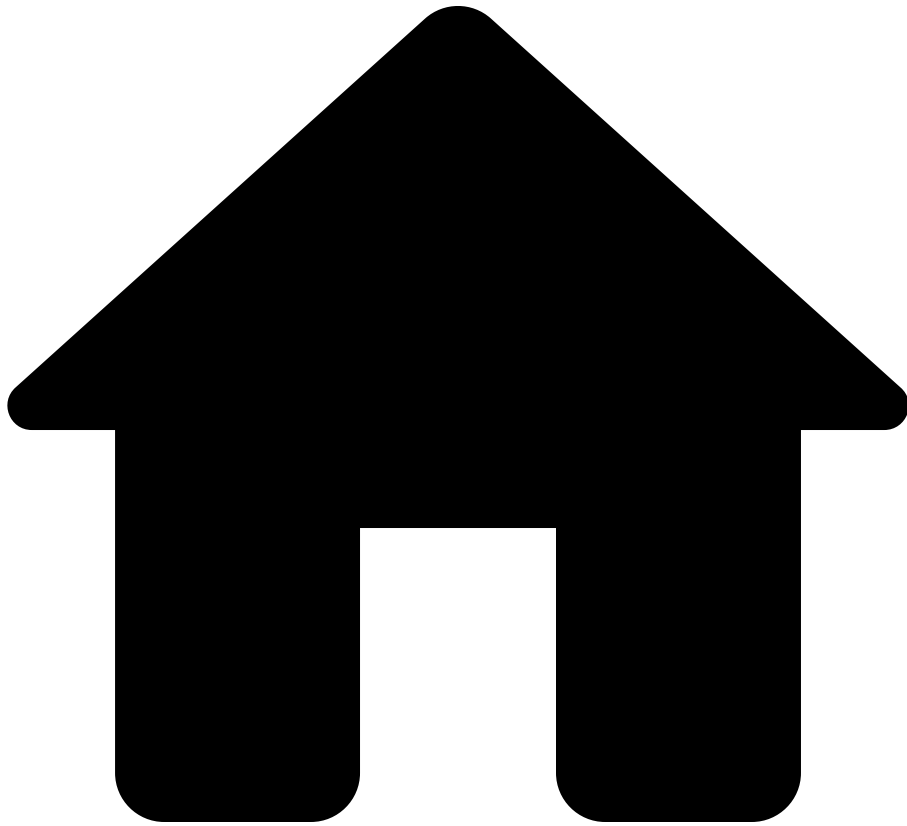
IP addresses can be written like this: `ipAddress\("1.2.3.4"\)` for IPv4 or `ipAddress\("2001:0db8:85a3:0000:0000:8a2e:0370:7334"\)` for IPv6. Such values have type `IpAddress`.

If `ip` has type `IpAddress`, then it can be inverted via `-` like this: `-ip`. For example, `-ipAddress("0.0.0.255")` is an expression of the IP address `255.255.255.0`.

In addition, the following functions work on IP addresses:

- [`toNumber\(ipv4Address\)`](#): The numeric value corresponding to an IPv4 address. For example, `toNumber(ipAddress("0.0.1.2"))` is `258`. Note that this can only be applied to IPv4 addresses.
- [`ipv4Address\(integer\)`](#): The IPv4 address corresponding to a given integer in the range `0` to `2 ^ 32 - 1`. For example, `ipv4Address(0)` denotes `0.0.0.0` and `ipv4Address(1234)` denotes `0.0.4.210`.
- [`isIPv4\(ipAddress\)`](#): Tests whether `ipAddress` is an IPv4 address. For example, `isIPv4(ipAddress("1.2.3.4"))` returns `true` and `isIPv4(ipAddress("2001:0db8:85a3:0000:0000:8a2e:0370:7334"))` returns `false`.
- [`isIPv6\(ipAddress\)`](#): Tests whether `ipAddress` is an IPv6 address. For example, `isIPv6(ipAddress("2001:0db8:85a3:0000:0000:8a2e:0370:7334"))` returns `true` and `isIPv6(ipAddress("1.2.3.4"))` returns `false`.
- [`fromTo\(ipAddress, ipAddress\)`](#): List of IP addresses between two `ipAddress`.

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IP Subnets

IP subnets can be obtained from the data model or via pattern matching against configuration text (see Section [Details on Patterns](#) for how to match subnets). In addition, there are three ways to construct IP subnets from other values:

- `ipSubnet(string)` : Construct a subnet from a string, such as `ipSubnet("1.2.3.4/24")` .
- `ipSubnet(ipAddress, number)` : Construct a subnet from an IP address and length, such as `ipSubnet(ipAddress("1.2.3.4"), 24)` .
- `ipSubnet(ipAddress, netmask)` : Construct a subnet from an IP address and a net mask, such as `ipSubnet(ipAddress("1.2.3.4"), ipAddress("255.255.128.0"))` which is equivalent to `ipSubnet("1.2.3.4/17")` .

In addition, the following functions work on subnets:

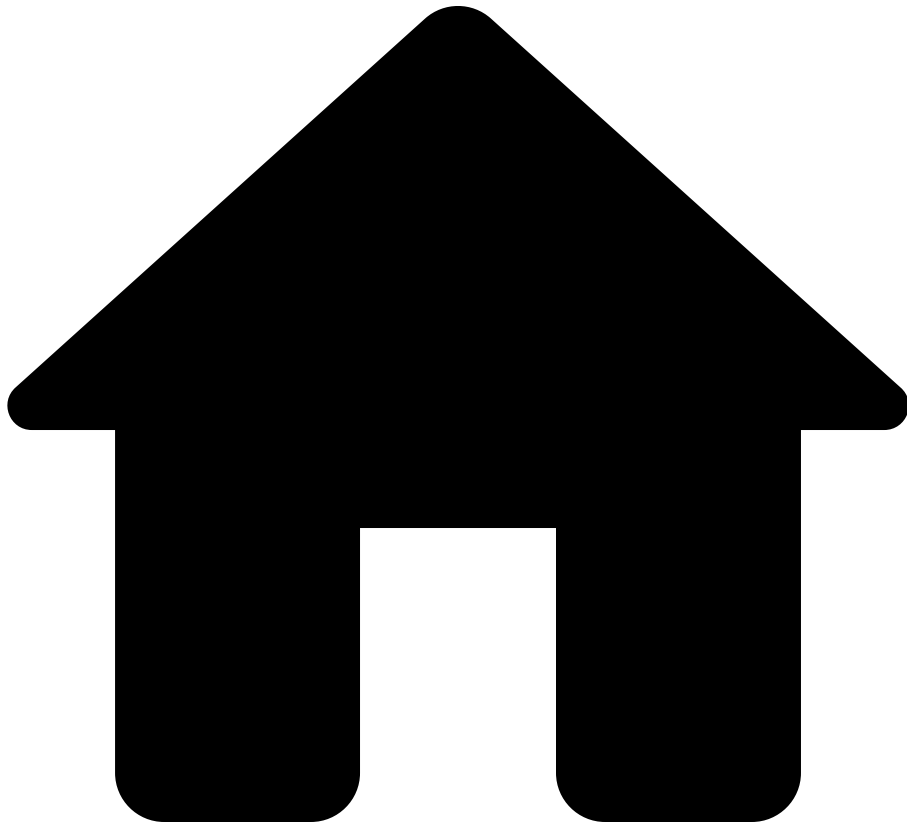
- `length(subnet)` : the length of the subnet. For example, `length(ipSubnet("1.2.3.4/24"))` is `24` .
- `address(subnet)` : returns the IP address for the subnet. For example, `address(ipSubnet("1.2.3.4/24"))` is `ipAddress("1.2.3.4")` .
- `networkAddress(subnet)` : returns the network IP address for the subnet, that is the IP address for the subnet, with all host bits set to 0. For example, `networkAddress(ipSubnet("1.2.3.4/24"))` is `ipAddress("1.2.3.0")` .
- `broadcastAddress(subnet)` : returns the broadcast IP address for the subnet, that is, the IP address for the subnet, with all host bits set to 1. For example, `broadcastAddress(ipSubnet("1.2.3.4/24"))` is `ipAddress("1.2.3.255")` .
- `ip in subnet` and `ip not in subnet` : returns a boolean value indicating whether the given IP address (the first argument) is a member of the given IP subnet (the second argument). For example, `ipAddress("1.2.3.4") in ipSubnet("1.2.3.4/24")` would be `true` and `ipAddress("1.2.3.4") not in ipSubnet("1.2.3.4/24")` would be `false` .

- [isIPv4\(ipSubnet\)](#) : Tests whether `ipSubnet` is an IPv4 subnet. For example, `isIPv4(ipSubnet("192.168.1.0/24"))` returns `true` and `isIPv4(ipSubnet("2001:0db8:abcd:0000::/56"))` returns `false`.
- [isIPv6\(ipSubnet\)](#) : Tests whether `ipSubnet` is an IPv6 subnet. For example, `isIPv6(ipSubnet("2001:0db8:abcd:0000::/56"))` returns `true` and `isIPv6(ipSubnet("192.168.1.0/24"))` returns `false`.

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- [length](#)
- [address](#)
- [networkAddress](#)
- [broadcastAddress](#)

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MAC Addresses

MAC addresses can be written like this: `macAddress("11:22:33:44:55:66")`.

The `ouiVendors(mac)` function returns a list of `Vendor` s associated with a MAC address. For example, `ouiVendors(macAddress("00:05:00:00:00:0D"))` returns a list equivalent to `[Vendor.CISCO]` and

`ouiVendors(macAddress("F8:0D:AC:FF:FF:FF"))` returns a list equivalent to `[Vendor.HP, Vendor.ARUBA]` .

The `ouiAssignee(mac)` function returns the name of the OUI assignee (as a `String`) associated with a MAC address. For example, `ouiAssignee(macAddress("00:55:DA:50:00:00"))` returns the string `"Nanoleaf"` . The function can return a `null` value if the assignee for the MAC address is not known.

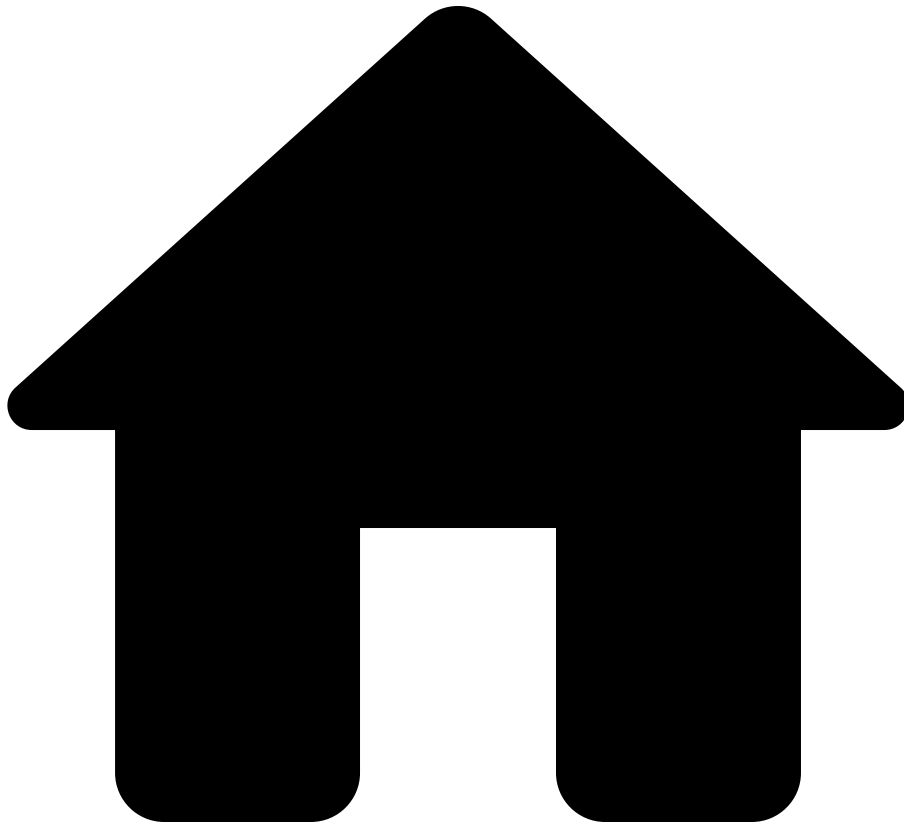
Source of MAC Vendor Information

Forward derives vendor information for MAC addresses using the publicly available IEEE Organizationally Unique Identifier (OUI) databases. These include the official lists maintained by the IEEE Standards Authority:

- [OUI](#)
- [MAM](#)
- [OUI-36](#)

The Forward platform periodically refreshes this data to ensure MAC vendor lookups reflect the latest assignments published by IEEE.

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Time

There are three types to represent three aspects of time and various ways to combine with with operators.

Types

- *Date*: A calendar day such as October 29, 1929.
- *Timestamp*: A date and time in UTC with second-level precision such as midnight UTC on January 1, 2000.
- *Duration*: An interval of time such as 23 hours, 56 minutes, and 4 seconds.

Date

A `Date` represents a calendar day. An example is October 29, 1929. A `Date` can be created with the `date` function.

The range of a `Date` is `-999999999-01-01` to `999999999-12-31`.

Operators

A `Date` can appear as the first operand in the following operations.

- `Date - Date = Duration`
- `Date - Duration = Date`
- `Date + Duration = Date`

These operations will raise an error if the result exceeds the supported range of the return type.

Timestamp

A `Timestamp` represents a date and time in UTC with precision to the second. An example is midnight UTC on January 1, 2000. A `Timestamp` can be created with the `timestamp` function.

The range of a `Timestamp` is `-1000000000-01-01T00:00:00Z` to `1000000000-12-31T23:59:59Z`.

Operators

A `Timestamp` can appear as the first operand in the following operations.

- `Timestamp - Timestamp = Duration`
- `Timestamp - Duration = Timestamp`

- `Timestamp + Duration = Timestamp`

These operations will raise an error if the result exceeds the supported range of the return type.

Duration

A `Duration` is an interval of time. An example is 23 hours, 56 minutes, and 4 seconds. A `Duration` can be created with the `duration` function as well as the `days`, `hours`, `minutes`, and `seconds` functions.

The uptime of a device is an example of a `Duration`. See the `uptime` field of [System](#).

The range of `Duration` is `-9223372036854775807` to `9223372036854775807` seconds, which is over 291,672,107,014 years.

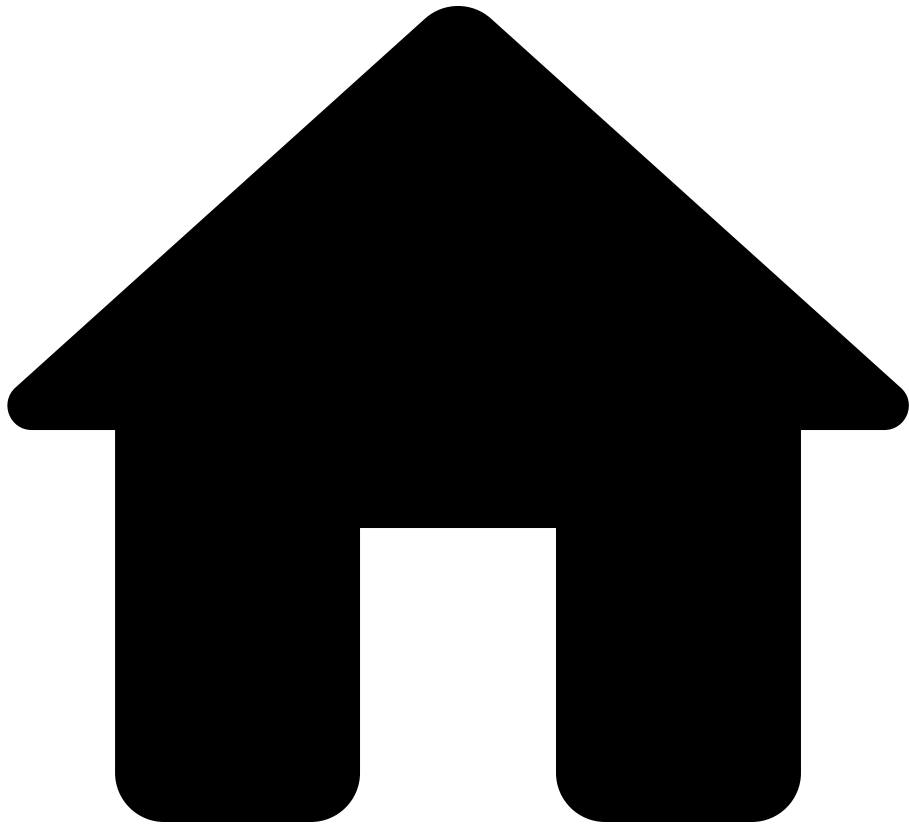
Operators

New `Duration`s can be created from existing ones with these operators.

- `-x` : Negation
- `x + y` : Addition
- `x - y` : Subtraction

Addition and subtraction will raise an error the resulting interval of time exceeds the range supported by `Duration`.

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Collections

NQE provides three types for working with collections of values.

Type	Description
<i>Bag</i>	

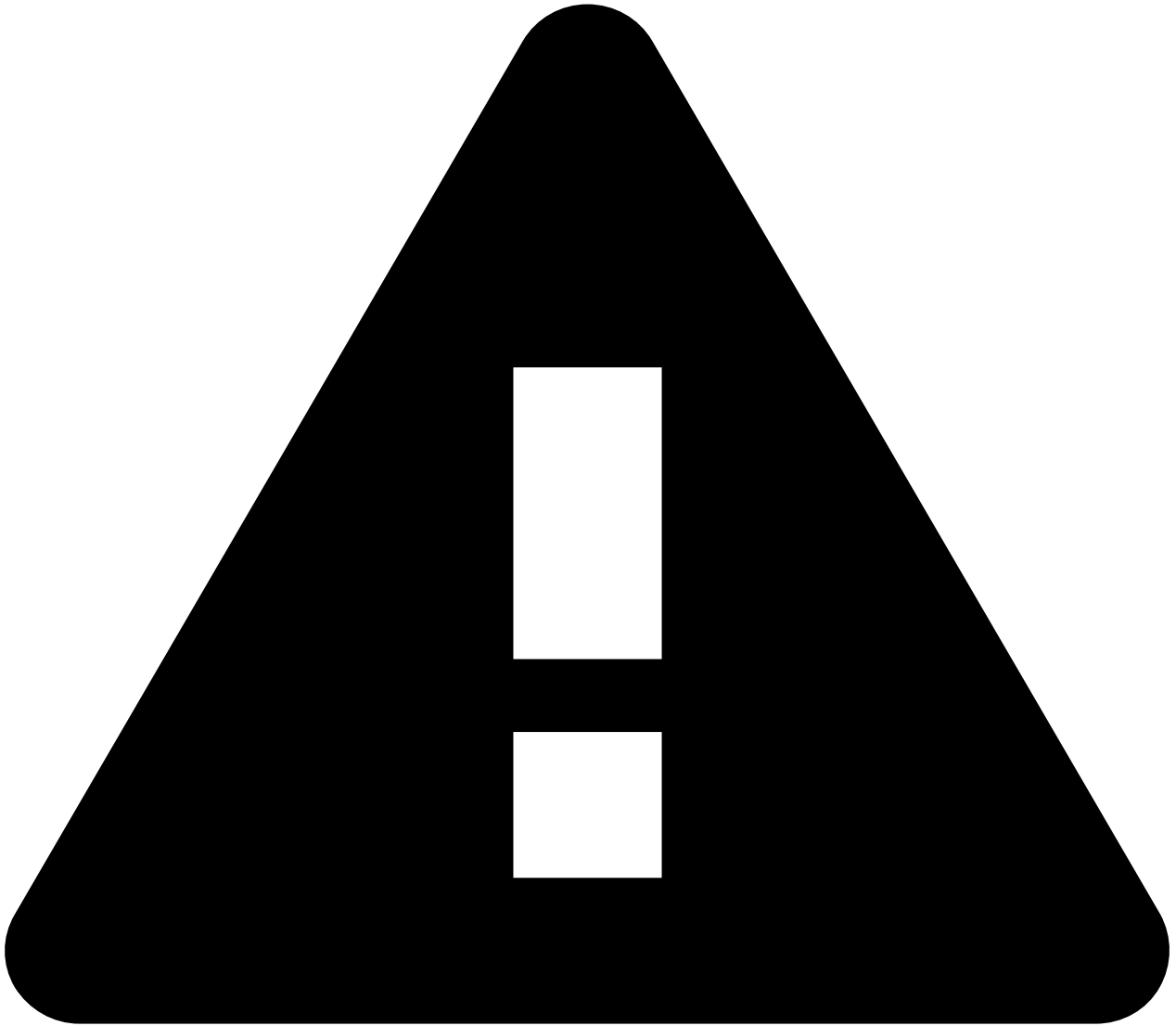
Type	Description
	An unordered collection of elements where elements may appear any number of times.
<i>List</i>	Like Bag, but the elements are arranged in a specific order.
Set	Like Bag, but each element can appear only once. In other words, an unordered collections of elements where multiplicity does not matter.

While Bag and List are generic datatypes that support any type of element, the Set datatype is currently focused on representing sets of IP addresses. Bags and Lists support iteration, but Sets do not. This remainder of this guide focuses only on Bags and Lists. See [the guide for Sets](#) for more information on Sets.

The primary difference between Bags and Lists is that Lists are ordered collections, whereas bags are unordered. Ordered collections have special operations that make use of the fact that the collection is a sequence of values. Examples include [list indexing](#) and [limit](#).

The following query retrieves all devices, sorts them by name, and limits the results to the first 10:

```
foreach device in network.devices
select { Name: device.name, Platform: device.platform }
order by Name ascending
limit 10
```



Important Language Changes

NQE 25.11 introduced stricter typing for Bags and Lists that may affect existing queries. If you have queries written before 25.11, please review the [Breaking Changes and Migration](#) section.

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Breaking Changes and Migration

Changes in NQE 25.11

NQE 25.11 introduced stricter typing rules for Bags and Lists that may affect existing queries. In particular, some collection-valued expressions are now typed as **Bags** rather than Lists. This section outlines the key changes and how to migrate your code.

Bag/List Comparisons Generate Warnings

Comparing a Bag with a List using equality (`==` , `!=`), relational operators (`<` , `<=` , `>` , `>=`), or membership operators (`in` , `not in`) is **allowed but discouraged** since List comparisons care about the positional order of elements, while Bag comparisons are order-insensitive. Such comparisons will generate warnings, and you should fix them with one of the suggested actions.

When you compare a Bag with a List, or check membership of a List in a Bag collection (or vice versa), the List is implicitly converted to a Bag (losing its order), and then Bag comparison semantics are used. This can lead to unexpected results if you intended to compare order-sensitive Lists.

Example of changed behavior:

Before 25.11: This worked without any issues.

```
deviceNames = foreach device in network.devices select
device.name;
{[
  isEqual: deviceNames == ["device1", "device2"]
]}
```

Here, `isEqual` is true precisely when *there are two devices, the very first is named "device1" and the very last is named "device2"*.

After 25.11: Square brackets `[]` are typed as Lists. As such, comparing a List with a Bag uses Bag comparison semantics (order-insensitive), which could give you unexpected results. As such, you will get a warning.

```
deviceNames = foreach device in network.devices select
device.name; // This is now a Bag of strings
{[
  isEqual: deviceNames == ["device1", "device2"]
]}
```

Here, `isEqual` is true precisely when *there are two devices, one is named "device1" and the other is named "device2", in no specified order.*

Migration strategy: *Make both operands the same type!*

Option 1: If order matters then explicitly use [order by](#) to get a List.

```
// Use "order by" clause to sort the devices
alphabetically by name
deviceNames = foreach device in network.devices
  let name = device.name
  select name
  order by name;
{[
  isEqual: deviceNames == ["device1", "device2"]
]}
```

Option 2: If order does not matter then use [bag](#) to get a Bag.

```
deviceNames = foreach device in network.devices select
device.name; // This is a Bag of strings

// Ensure the comparison is done against two bags.
{[
  isEqual: deviceNames == bag(["device1", "device2"])
]}
```

Creating Collections

There are several ways to construct Bags and Lists in NQE.

Literals

The simplest way to create a list collection is by writing a list literal, a comma-separated sequence of expressions within brackets:

```
numbers = [1, 2, 3, 4, 5];
names   = ["router1", "router2", "router3"];
mixed   = [[10, 20], [30, 40]]; // list of lists
```

The Bag function

We can wrap a list literal with the [bag](#) method to create a bag collection.

```
integers = bag([1, 2, 3, 4, 5]); // a bag of integers
```

Since bags ignore order, the following expression is true:

```
bag([1, 2, 1]) == bag([2, 1, 1])
```

Comprehensions

Collections can be created using `foreach ... select ...` expressions, also known as [comprehensions](#).

A comprehension evaluates to a **Bag** if the source collection is a Bag (and there is no [order by](#) clause).

```
// When iterating over a Bag (like network.devices), the
result is a Bag
deviceNames = foreach device in network.devices select
device.name; // Bag<String>
```

A comprehension evaluates to a **List** if the source collection is a List (or if there is an [order by](#) clause).

```
// When iterating over a List, the result is a List
numbers = [1, 2, 3, 4]; //
List<Integer>
doubled = foreach n in numbers select n * 2; //
List<Integer>: [2, 4, 6, 8]
```

Concatenation

Collections can be combined using the `+` operator, which concatenates two collections: It preserves order if both collections are lists, and otherwise produces a bag.

```
list1 = [1, 2, 3];
list2 = [4, 5, 6];
combined = list1 + list2; // == [1, 2, 3, 4, 5, 6]
```

Combining Bags and Lists results in a bag:

```
myList = [1, 2, 3];
myBag = bag([4, 5, 6]);
combined = myList + myBag; // == bag([1, 2, 3, 4, 5, 6])
// == bag([2, 4, 6, 5, 3, 1])
```

Difference

The `-` operator removes all elements from the first collection that appear in the second one. The resulting difference is an ordered collection exactly when the first argument is ordered.

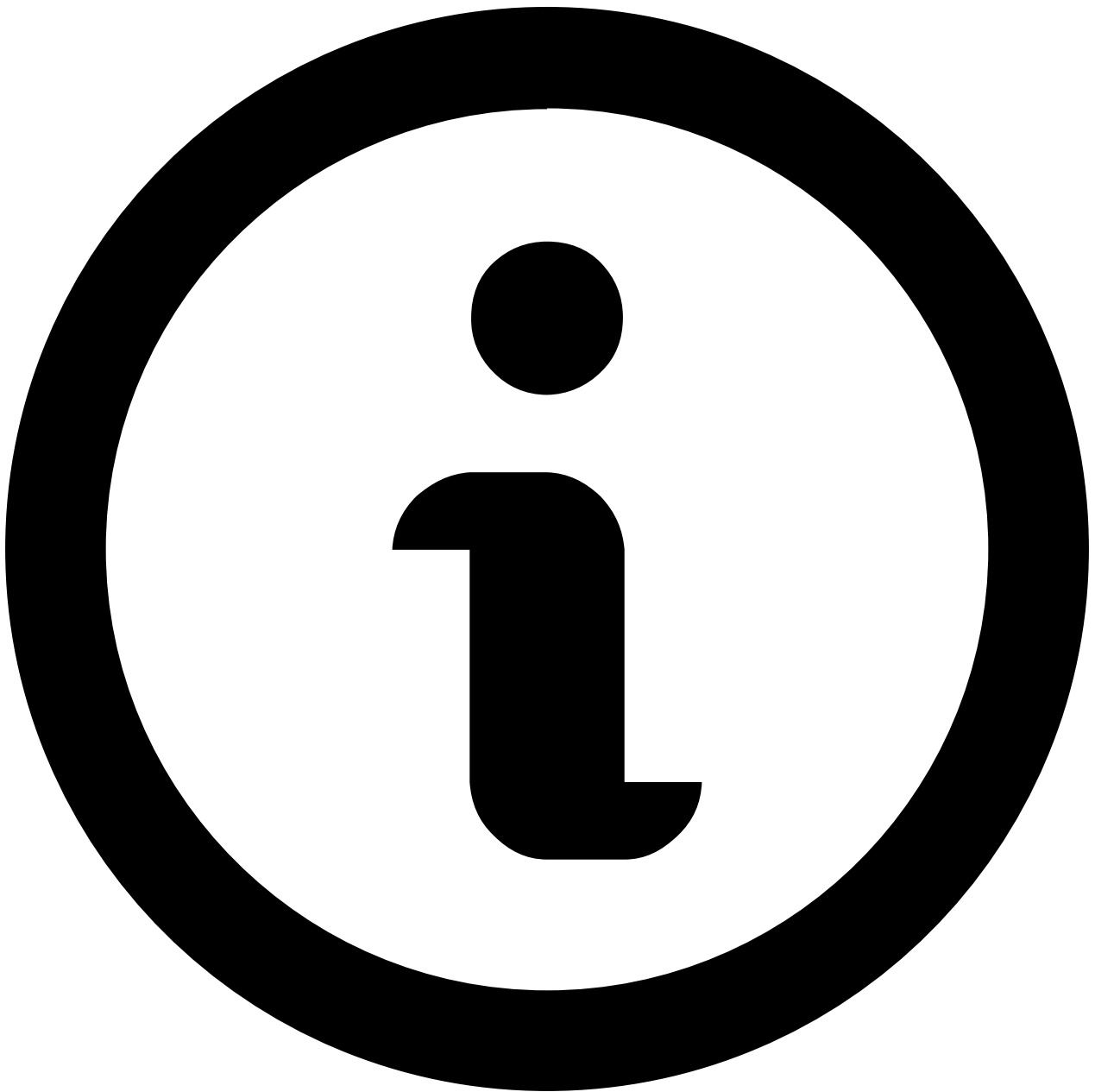
Removing from a bag results in a bag:

```
all = bag([1, 2, 3, 4, 5, 4]);
exclude = [2, 4];
```

```
difference = all - exclude; // == bag([1, 3, 5])  
// == bag([5, 1, 3])
```

Removing from a list results in a list:

```
all = [1, 2, 3, 4, 5, 4];  
exclude = bag([2, 4]);  
difference = all - exclude; // == [1, 3, 5]
```

note

Notice that the item `4` is duplicated in `all`, however the resulting `difference` removes all occurrences.

Builtins That Return Collections[ââ](#)

Many builtin functions return collections as their result. For example:

- [fromTo](#)(start, end): Returns a list of numbers from `start` to `end` (inclusive).

- [regexMatches\(text, regex\)](#): Returns a list of all regex matches found in the text
- [splitJsonObjects\(json\)](#): Returns a list of JSON object strings
- [patternMatches\(config, pattern\)](#): Returns a bag of pattern matches from configuration
- [blockMatches\(config, pattern\)](#): Returns a bag of block matches from configuration

See the [standard library reference](#) for more functions that return collections.

Converting Between Bags and Lists

bag Function

Convert a List to a Bag using the `bag(list)` function:

```
orderedDevices = [/* ... */];    // List
deviceSet = bag(orderedDevices); // Bag
```

This is useful to make explicit the implicit downcasts when using comparison operators (`==`, `!=`, `<`, `<=`, `=>`, `>`), or membership operators (`in`, `not in`) with a Bag and a List, which produce a type check warning and will eventually produce a type check error. See [Comparing Bags and Lists](#) and [Membership Testing](#) for more details.

Lists are subtypes of Bags

Since a list is just a bag with a specified order, a list can be used anywhere a bag is expected --we just "forget" the ordering of the list to obtain a bag.

If-Then-Else: When branches return different types:

```
aList      = [1, 2, 3];           //
List<Integer>
aBag       = bag([3, 4, 5]);      //
Bag<Integer>
condition  = false;
combined   = if condition then aList else aBag; //
Bag<Integer>
```

User-Defined Functions: When a function expects a Bag parameter but receives a List:

```
processDevices(devices: Bag<Device>) : Integer =  
length(devices);  
  
[ { useCase1: processDevices(["Device 1", "Device 2"])  
  , useCase2: processDevices(bag(["D 10", "D 20"]))  
}]
```

Notice that `useCase1` works since we automatically get to use `processDevices` as if it were typed `processDevices(devices: List<Device>) : Integer` as well —since List is a subtype of Bag.

List Indexing

Lists support zero-based indexing to access individual elements. Use square brackets with an integer index to retrieve an element at a specific position.

```
numbers = [10, 20, 30, 40, 50];  
first   = numbers[0];    // 10  
second  = numbers[1];    // 20  
last    = numbers[4];    // 50
```

When an index is out of bounds (negative or greater than or equal to the list length), a runtime error is emitted.

```
numbers = [10, 20, 30];  
tooLarge = numbers[10]; // ðŸ’¥ Error: Index too large  
negative = numbers[-1]; // ðŸ’¥ Error: Index must be  
non-negative
```

Notes:

- List indexing only works with Lists, not Bags (bags are unordered and don't support indexing)
- Indices must be integer values

- Attempting to index a bag results in a type error
- Out-of-bounds access results in a runtime error

Ordering

One of the most powerful features of Lists is the ability to specify and maintain order.

The `order by` Clause

The `order by` clause sorts a comprehension by one or more expressions, resulting in an ordered collection.

```
1. foreach device in network.devices
2. select { Name: device.name, Platform: device.platform }
3. order by Name asc
```

Notice that line (3) orders all records specified in line (2) by the `Name` field in ascending order.

To order by names in descending order, we can use the `desc` keyword:

```
1. foreach device in network.devices
2. select { Name: device.name, Platform: device.platform }
3. order by Name desc
```

Instead of `asc` and `desc`, for readability you can use `ascending` and `descending` to specify sorting direction.

```
1. foreach device in network.devices
2. select { Name: device.name, Platform: device.platform }
3. order by Name ascending
```

You can sort by multiple columns: Line (3) below orders by the `Platform` field of each record, specified in line (2), in descending order and then orders by the `Name` field of each record in ascending order.

```
1. foreach device in network.devices
2. select { Name: device.name, Platform: device.platform }
3. order by Platform desc, Name asc
```

Instead of `asc` and `desc`, for readability you can use `ascending` and `descending` to specify sorting direction.

```
1. foreach device in network.devices
2. select { Name: device.name, Platform: device.platform }
3. order by Platform descending, Name ascending
```

Finally, you can even sort by local `let` variables:

```
/** ðŸŒˆ2 Results are ordered by platform ascending,
 *     then by model descending, then by name;
 *     however, the results only show the name and platform.
 */
foreach device in network.devices
let Platform = device.platform
// ðŸŒˆ%ï, This is used for sorting only & not shown in the
rows.
let Model = Platform.model
select {
  Name: device.name,
  Platform
}
order by Platform asc, Model desc, Name ascending
```

Important: The `order by` clause requires each ordering expression to be a **name** that refers to either:

- A field of the record returned by the `select` clause, or
- A variable defined in a `let` clause

You cannot use arbitrary expressions directly in `order by`. If you need to sort by a computed value, bind it to a name using `let` first.

```

bad    = foreach device in network.devices
        select device
        order by device.name; // ðŸ’¥«ERROR: â€œorder byâ€œ
clause expects an identifier, not an expression

works  = foreach device in network.devices
        let name = device.name
        select device
        order by name; // ðŸ’¥ Works

```

Natural String SortingðŸ’¥

By default, strings are sorted letter-by-letter, treating numbers as single characters. This means "device10" comes before "device2" (because the character "1" comes before "2"). For more intuitive sorting of strings containing numbers, use the `natural` modifier (see [natural sort order](#)):

```

foreach device in network.devices
let name = device.name
select { name }
order by name ascending natural

```

With `natural`, strings are sorted with natural number ordering: "device1", "device2", "device10" (rather than "device1", "device10", "device2").

```

// Sorting interface names naturally
foreach iface in device.interfaces
let name = iface.name
select { name }
order by name ascending natural // GigabitEthernet0/1,
GigabitEthernet0/2, GigabitEthernet0/10

```

ðŸ’¥ Note: The `natural` modifier can only be used with String-typed columns. Using it with numbers, dates, or other types will produce a type-checking error.

The `orderBy` Function

To sort collections directly, use the `orderBy(collection, keySelector)` function:

```
getDeviceName(device) = device.name;  
  
devicesSortedByName = orderBy(network.devices, getDeviceName);
```

For descending order, negate numeric keys or use the comprehension syntax.

The `order` Function

To sort a collection by its natural order without specifying a key selector, use the `order(collection)` function:

```
devices = order(network.devices);
```

The `order` function sorts elements using their natural ordering - the same ordering used by comparison operators like `<` and `>`. See [Comparisons](#) for details on how different types are ordered.

This is equivalent to the following comprehension:

```
devices = foreach device in network.devices  
           select device  
           order by device ascending;
```

The `order` function always returns a List in ascending order. For descending order, use the comprehension syntax with `order by ... descending`.

Limiting Results

Basic Use

The `limit` clause restricts the number of elements returned from a list:

```
foreach device in network.devices
select { Name: device.name }
order by Name ascending
limit 10 // Only first 10 devices
```

Without An `order by` Clause

While `limit` is commonly used with `order by` to get the "top N" results, it can be used without `order by` as long as the comprehension produces a list. For example, iterating over an already-ordered list will allow `limit` to work without requiring `order by`. For instance, if you already have your devices ordered, then you can do:

```
devicesList = ["device1", "device2", "device3"];

foreach device in devicesList
select { device }
limit 2 // No "order by" clause needed since
        "devicesList" is a list
```

Limiting by an expression

The limit value can be any expression that evaluates to a number, not just a numeric literal. For example, the following query gets half of the possible devices.

```
foreach device in network.devices
select { Name: device.name }
order by Name ascending
limit length(network.devices) / 2
```

Note: The `order by` and `limit` clauses can appear in nested queries, not just at the outermost level. This allows you to sort and limit intermediate results within a larger query.

The limit function

The [limit](#) function can also be applied directly to collections:


```

getDeviceName(device)    = device.name;
devicesSortedByName      = orderBy(network.devices,
getDeviceName);
firstFiveDevicesByName   = limit(devicesSortedByName, 5);

```

Operations [↗](#)

Membership Testing [↗](#)

Test whether an element is in a collection using `in` or `not in`:

```

[ { present: "Jasim" in ["Jane", "Jasim", "Jerry"] // = true
  , notPresent: 46 not in bag([3, 7, 365])          // = true
} ]

```

This can be used in comprehensions to filter out results:

```

invalidDevices = ["Device1", "Device5", "Device9"];

foreach device in network.devices
let Name = device.name
where Name not in invalidDevices
select { Name }

```

Note: Checking membership of a List in a Bag collection (or vice versa) generates a warning, similar to comparing Bags and Lists directly. For example, `[1, 2] in [bag([1, 2])]` will emit a warning. To avoid warnings, ensure the item type matches the collection's item type (e.g., use `bag([1, 2]) in [bag([1, 2])]` instead). See [Bag/List Comparisons Generate Warnings](#) for migration strategies.

Comparison Operators [↗](#)

Collections can be compared using `<`, `<=`, `>`, `>=` operators.

- Lists are compared lexicographically (element by element).

- Bags are compared by sorting them (using the natural order) then doing list comparisons.

```
[{ true1: [1, 2, 3] < [1, 2, 4]           // = true since 3 <
4 at first differing position
, true2: [1, 2]    < [1, 2, 4]           // = true since
first list is a prefix of the second list
, true3: bag([2, 1, 1]) < bag([1, 3, 1]) // = true since [1,
1, 2] < [1, 1, 3] is true
}]
```

Equality

Lists are equal if and only if they have the same elements in the same order:

```
[1, 2, 3]  == [1, 2, 3]    // = true
[1, 2, 3]  == [3, 2, 1]    // = false; different order
[1, 2, 3]  == [1, 2, 3, 4] // = false; different length
```

Bag equality is order-independent but cares about multiplicity:

```
bag([1, 2, 3]) == bag([3, 2, 1]) // = true; same elements,
order ignored
bag([1, 2])    == bag([2, 1])    // = true; same elements,
order ignored
bag([1, 2])    == bag([2, 1, 1]) // = false; 1
occurs more times in the right than in the left
```

Comparing Bags and Lists

Comparing a Bag with a List using equality or relational operators is allowed but generates a warning. When comparing a Bag with a List, the List is implicitly converted to a Bag (losing order information), and Bag comparison semantics are applied.

```
myList  = [1, 2, 3];
myBag   = bag([3, 2, 1]);
```

```
myList == myBag // Warning: comparing List with Bag
                // Result: true (order is ignored, both
                // become bags for comparison)
```

To avoid warnings and make your intent clear, convert both operands to the same type:

```
// If order matters
myList == order(myBag) // No warning; false since [1, 2, 3] != [3, 2, 1]

// If order doesn't matter
bag(myList) == myBag    // No warning; true
```

For more details about language changes that may affect existing queries, see [Breaking Changes and Migration](#).

Extracting Single Elements

Use `the(collection)` to extract a single element from a `collection`. This function returns:

- The single element if the collection contains exactly one element
- `null` if the collection is empty or contains more than one element

```
// The device named "core-router-1", if present; else null
coreRouter1Device = the(
  foreach device in network.devices
  where device.name == "core-router-1"
  select device
);
```

This is particularly useful when you expect a unique result but want to handle the case gracefully when the assumption doesn't hold.

Deduplication

Remove duplicate elements from a collection using [distinct](#). For lists, this preserves order: The first occurrence of an element is kept and all other subsequent occurrences are dropped.

```
list      = [1, 2, 3, 1, 4, 2];
uniqueList = distinct(list); // [1, 2, 3, 4]

bag       = bag([1, 2, 3, 1, 4, 2]);
uniqueBag = distinct(bag); // == bag([1, 2, 3, 4])
           // == bag([2, 4, 3, 1])
```

List-Specific Data Model Fields

Several data model fields return Lists instead of Bags to reflect their inherent ordering. Examples include:

- `Device.aclEntries` and `Device.natEntries`
- `IfaceAcls.inboundAclNames` and `IfaceAcls.outboundAclNames`
- `NetworkAcl.ingressRules` and `NetworkAcl.egressRules`
- `SecurityGroup.ingressRules` and `SecurityGroup.egressRules`
- `LoadBalancer.loadBalancerRules` and `LoadBalancer.outboundRules`

See Also

Guides

- [Comprehensions](#) - In-depth coverage of foreach expressions
- [Comparisons](#) - How values are ordered and compared

Functions

The following operations are available on collections.

- **fromTo(start, end)** Creates a range of numbers
`[start, start + 1, start + 2, ..., end]`.
- **bag(list)**: Converts a list to a bag
- **length(collection)**: Gets the number of elements in a collection
- **limit(list, n)**: Gets the first `n` elements from a list
- **item in collection**: Checks that `item` is an element of `collection`. For example, `"b" in ["a", "b"]` returns true.
- **item not in collection**: Checks that `item` is *not* an element of `collection`.
- **distinct(collection)**: Removes duplicates from a collection.
- **orderBy(collection, keySelector)**: Sorts a collection by applying the key selector function to each element. The result is a list.
 - This can be written as a comprehension

```
orderBy(collection, selector)
=
foreach item in collection
  let key = selector(item)
  select item order by key ascending
```

- Often, we want to order a collection by the natural order on elements; that is, the same order used for comparing items with relational expressions such as less-than (`<`) and greater-than (`>`). See Section [comparisons](#) for more detail on the order of values. This is done by **order(collection)** ---no selector function required.
- **max(collection)**: Gets the maximum element in a collection, or `null` if the collection is empty.
 - Note: `max(collection)` is the last element of `order(collection)`.

- **`min(collection)`**: Gets the minimum element, or `null` if the collection is empty.

- Note: `min(collection) = order(collection)[0]`

- **`maxBy(collection, keySelector)`**: Gets the element with the maximum key value, or `null` if the collection is empty.

- This gets the greatest item in the `collection`, where items are ordered by the value returned by applying `keySelector` to each item.

- Note: `maxBy(collection, keySelector)` is the last element of `orderBy(collection, keySelector)`.

- **`minBy(collection, keySelector)`**: Gets the element with minimum key value, or `null` if the collection is empty.

- This gets the least item in the `collection`, where items are ordered by the value returned by applying `keySelector` to each item.

- Note:

- `minBy(collection, keySelector) = orderBy(collection, keySelector)[0]`.

- **`the(collection)`**: Extracts the single element from a collection, if it has a unique element; else `null`.

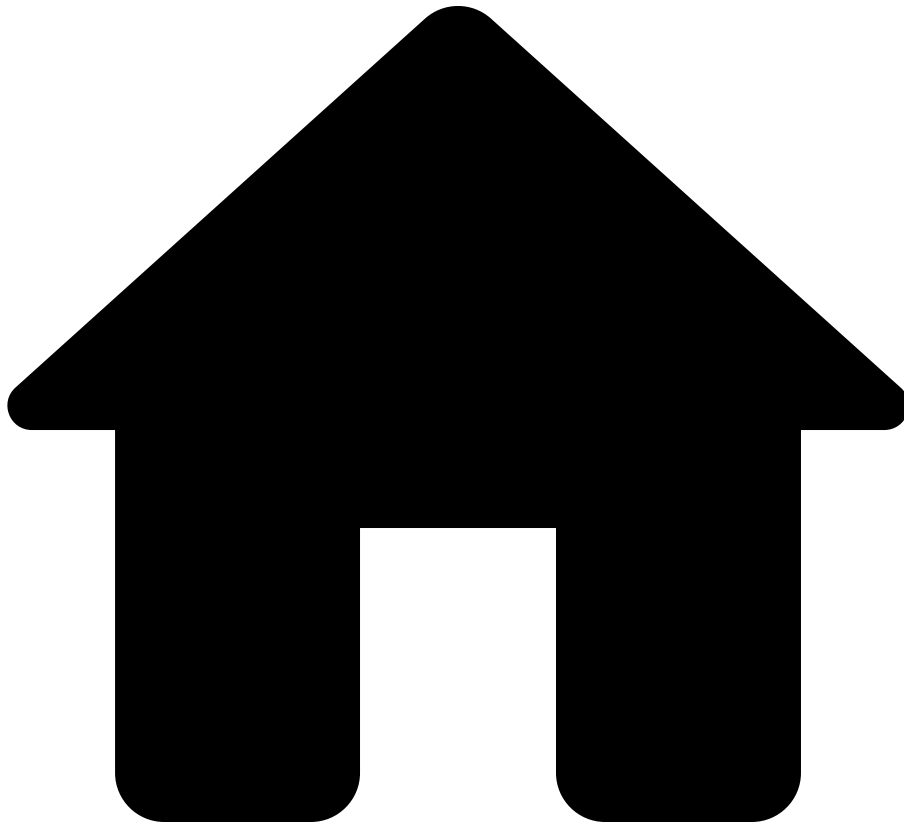
- **`sum(collection)`**: Sums the numeric elements in a given collection.

- **`join(delimiter, collection)`**: Concatenates together the string elements in `collection` using `delimiter` between each element.

- **`any(collection)`**: Checks if any boolean element is true

- **`all(collection)`**: Checks if all boolean elements are true

.



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- Sets

Sets

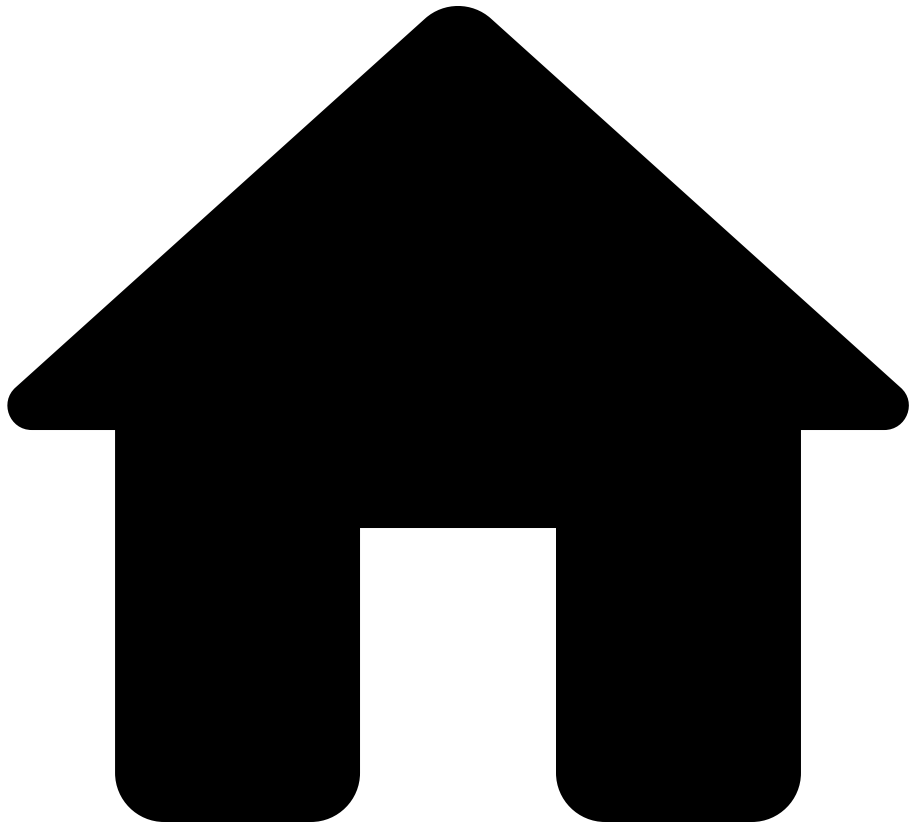
A `Set` is a collection of elements that are unique (i.e. that does not contain duplicates). The only way to create a `Set` from other types is to use the [ipAddressSet](#) function.

The following operations are available on sets.

- [isEmpty\(set\)](#) : Returns a `Bool` indicating if `set` contains no elements.

- `A + B` : Returns the union of sets `A` and `B` .
- `A - B` : Returns the differences of the sets `A` and `B` , which contains the elements in `A` that are not in `B` .
- `intersect(set1, set2)` : Returns the intersection of `set1` and `set2` .
- `ip in set` : Returns a `Bool` indicating if the [IpAddress](#) `ip` is contained in `set` .
- `subnet in set` : Returns a `Bool` indicating if the set of [IpAddress](#) es represented by the [IpSubnet](#) `subnet` is a subset of `set` .
- `subset in superset` : Returns a `Bool` indicating if `subset` is a subset of `superset` .

.



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- Records

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Records

Construction [↗](#)

A record is an object that has zero or more name-value pairs. A record can be denoted by record literal. For example, `{name: "Bob", age: 10 + 20}` is a record with two

fields, `name` and `age`. The `name` field is a string with value `"Bob"`, while the `age` field is a number with value `30`.

Records can be arbitrarily nested and have values of any type. For example, these are all legal records:

- `{a: 1, b: {c: 2, d: 3}}` is a record with two fields, `a` and `b`, where the value of `b` is a record with two further fields, `c` and `d`.
- `{name: "Bob", age: 30, friends: ["John", "Jane"]}` is a record where the `friends` field is a list of strings.

Additionally, a field name can be defined using double quotes and can be any [inline string](#). Here are two examples.

- `{ "Device Count": 35971 }` is a record where the only field name includes spaces. A space is a special character.
- `{ "Management IP(s)": [ipAddress("1.1.1.1"), ipAddress("2.2.2.2")] }` is a record where the only field name includes the special characters `(` and `)`.

Selection

. Operator

The primary operation to consume a record is the dot-operator `.`, which selects a field value from a record. For example:

- `{name: "Bob", age: 10 + 20}.name` has value `"Bob"`.
- `r.age` is `30` if `r` is bound to `{name: "Bob", age: 10 + 20}`.
- `r.age` terminates the execution of the query with an error if `r` is bound to `null`.

?. Operator

Another operation to consume a record is the null-conditional dot-operator, which behaves the same as the dot-operator, except that it propagates `null` when applied to a `null` record instead of raising an error. In other words, the `?.` operator selects a

field value from a record if the record is not `null` but returns `null` if the record is `null`. For example:

- `{name: "Bob", age: 10 + 20}?.name` has value `"Bob"`.
- `r?.age` is `null` if `r` is bound to `null`.

If `r` is bound to `null`, then the execution of `r?.b.c` will terminate with an error because `r?.b` returns `null` and then the attempt to access field `c` on that `null` value will fail.

`[]` and `?[]` Operators

A record field containing a special character cannot be selected using either of the previous two operators: `.` and `?.`. Instead, such a field is selected using the square-brackets operator `[]`, which looks like `r["Device Count"]` for some record `r`. Just like with the `.` operator, the `[]` operator can be preceded by a `?` to form the null-conditional square-brackets operator `?[]`, which looks like `r?["Device Count"]` for some record `r`.

Here are self-contained examples using the `[]` and `?[]` operators.

- `r = { "Device Count": 35971 }; deviceCount = r["Device Count"];` assigns the values `35971` to `deviceCount`.
- `r = null : { "Device Count": Integer }; deviceCount = r?["Device Count"];` assigns the `null` to `deviceCount`.

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- [NQE Language](#)
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- Comparisons

Comparisons

You can compare two values for equality, and order via the typical operators, like `==`, `!=`, `<`, `<=`, `>` and `>=`. For example, `interface.mtu == 1000` and `1000 < interface.mtu` are valid expressions.

The order of items that have the same type (used by `<` and other relational operators) is as follows:

- **Numbers:** numerical order. For example, `1` is less than `2`, and `1.2` is less than `3.4`.
- **Strings:** alphabetical order. For example, `"ac"` is less than `"bb"`.
- **Booleans:** `false` is less than `true`.
- **IP addresses:** numerically. For example, `ipAddress("1.2.3.4")` is less than `ipAddress("1.2.4.5")`.
- **IP subnets:** ordered by prefix length first. For subnets with equal prefix lengths, the subnets are ordered numerically by host part. For example, `ipSubnet("1.2.3.4/24")` is less than `ipSubnet("1.2.3.4/25")`, but greater than `ipSubnet("1.2.3.3/24")`.
- **MAC addresses:** ordered numerically. For example, `macAddress("11:22:33:44:88:99")` is less than `macAddress("11:22:33:44:66:77")`.
- **Date:** chronological order. For example, `2000-01-01` is before `2000-01-02`.
- **Timestamp:** chronological order. For example, `2000-01-01T00:00:00Z` is before `2000-01-01T00:00:01Z`.
- **Duration:** numerically. For example, `1` second is less than `2` seconds.
- **Lists:** compared lexicographically; for example `[1, 4]` is less than `[2, 3]`.
- **Bags:** compared by first converting to lists using the natural order, then compared as lists. For example, `bag([2, 1, 1]) < bag([1, 3, 1])` is true since `[1, 1, 2] < [1, 1, 3]` is true.
- **Records:** two records are ordered lexicographically based on their field values, with fields compared in order of alphabetically increasing field name. For example `{x: 1, z: "b"}` is less than `{x: 2, z: "a"}`, but is greater than `{x: 1, z: "a"}`.
- **OneOfs:** two `OneOf` values are compared based on alphabetical order of their alternative names. For example, `AdminStatus.DOWN` is less than `AdminStatus.UP`. If the two values have the same alternative, then they are ordered based on the ordering of any values they contain.
- A null value is less than all other values (except itself, to which it is equal).

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- [NQE Language](#)
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- OneOfs and Enumerations

OneOfs and Enumerations

A variety of enumerated constants are provided within the data model. For example, the `OperStatus` enumeration has named constants: `UP`, `DOWN`, `TESTING`, `UNKNOWN`, `DORMANT`, `NOT_PRESENT`, `LOWER_LAYER_DOWN`. These can be referenced using the enumeration name followed by the constant name. For example: `OperStatus.UP` or `OperStatus.NOT_PRESENT`.

Likewise, the data model contains several OneOf (also known as disjoint sum or tagged union) data types. A OneOf data type is essentially a tagged union of different types. It consists of several different possible types of values, distinguished by tags. For example, the data model defines a `SubInterfaceVlan` type with two different possible types of instances: `VLAN_ID(v)` (where `v` is a `VlanID`) and `QINQ_ID(s)` (where `s` is a string in a particular format).

Writing OneOf values is similar to writing enumerated values. The following are legal examples:

- `SubInterfaceVlan.VLAN_ID(123)`
- `SubInterfaceVlan.QINQ_ID("1.2")`

This is similar to enumerated constants, except that for enumerated constants there is no value in parentheses.

Consuming OneOf values, including enumerations is done via case-analysis using a `when` expression. The general form of a `when` expression is:

```
when expression is pattern1 -> expression1; pattern2 ->
expression2; ...; patternN -> expressionN
```

Here, `expression` is the expression to be analyzed and must have a OneOf type (e.g. a variable holding an enum value). After the `is` keyword, there are any number of cases listed, where each case consists of a pattern, such as `pattern1` and a case expression, such as `expression1`. The `->` characters separate the pattern and the case expression, and must be present. The `;` character separates consecutive cases, and must be present. The patterns `pattern1`, `pattern2`, ... `patternN` are matched against the expression in sequence. When one matches, the case expression for that pattern is used as the overall value of the `when` expression. There are three forms of patterns:

- `Con` : matches values of the specific OneOf constructor `Con`. For example, `VLAN_ID` (one of the two possible `SubInterfaceVlan` constructors as shown above).
- `Con(var)` : matches values of the specific OneOf constructor `Con`. For example, `VLAN_ID(vlanId)`. Also binds the value contained in the OneOf constructor to the variable `var` specified in the pattern. The case expression for this pattern is then evaluated with `var` in scope. This pattern form is only

permitted when the OneOf constructor `Con` has a data value associated with it (e.g. this cannot be used to match against an enum value).

- `otherwise` : matches any value. Can only appear as the final item in the list of cases. Useful when many of the cases compute the same result.

Here are some examples to illustrate `when` expressions, assuming that `v` is a variable in scope that has type `SubInterfaceVlan` :

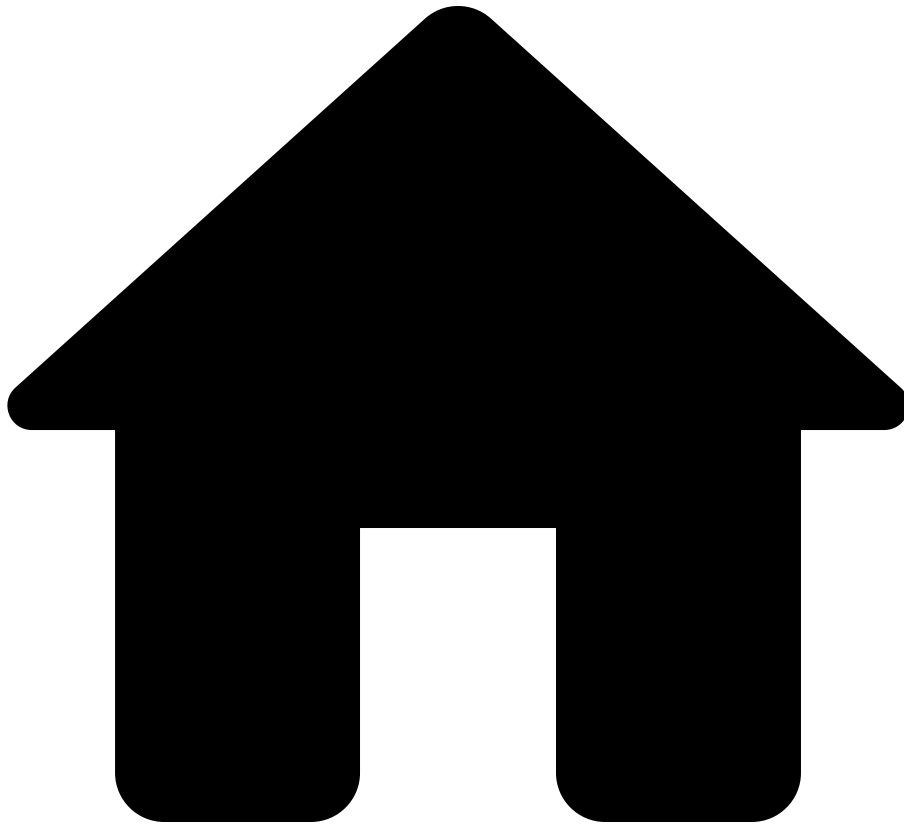
- `when v is QINQ_ID(s) -> s + "!"; VLAN_ID -> "def"`

Performs case analysis on `v` , splitting into two different cases. The first case tries to match `QINQ_ID` values and if successful binds the value contained inside the `QINQ_ID` value as `s` and then returns `s` appended with an exclamation mark. The second case handles the `VLAN_ID` case, ignoring the data inside the `VLAN_ID` constructor (unlike the first case).

- `when v is QINQ_ID -> "abc"; otherwise -> "def"`

Matches `QINQ_ID` values, ignoring the value inside the `QINQ_ID` constructor. The second case uses an `otherwise` pattern, to match the remaining values.

.



- Reference
- [NQE Language](#)
- [Guides](#)
- Missing Values and Null Expressions

Missing Values and Null Expressions

Data are sometimes missing in the data model (marked as **nullable** in the *Data Model* tab). For example, the mtu of an interface may be missing. To determine if an expression produces a value or is missing, the `isPresent(value)` operator can be used.

A missing value can also be introduced by the query using the syntax `null : T` for any type `T`. For example, `null : Integer` denotes a null numeric value, while `null : List<Device>` denotes a null value whose type is a list of `Device` records.

This expression can be useful when performing a case analysis, if you prefer to ignore some possible cases. For example, a `SubInterface` record has a `vlan` field whose type is `SubInterfaceVlan`, which is a `OneOf` type, with two subtypes:

`VLAN_ID(vlan)` where `vlan` is a `Integer` and `QINQ_ID(outerInner)` where `outerInner` is a `String`. To get just the vlan number for a subinterface record held in a variable called `subiface`, you could use the following expression:

```
when subiface.vlan is VLAN_ID(vlan) -> vlan; otherwise ->
null : Integer
```

The expression can be used in a larger context, for example within a `where` clause of a query to filter the results to just subinterfaces that have a vlan number, as in the following fragment:

```
...
let vlan = when subiface.vlan is VLAN_ID(vlan) -> vlan;
otherwise -> null : Integer
where isPresent(vlan)
...
```

See [OneOfs and Enumerations](#) for information on `OneOf` types and `when` expressions.

.



- Reference
- [NQE Language](#)
- [Guides](#)
- Snapshot Data

Snapshot Data

Snapshot data is made available via the `network` variable, which is always in scope. For example, `network.devices` denotes the [collections](#) of devices in the network (as a Bag, which is an unordered collection).

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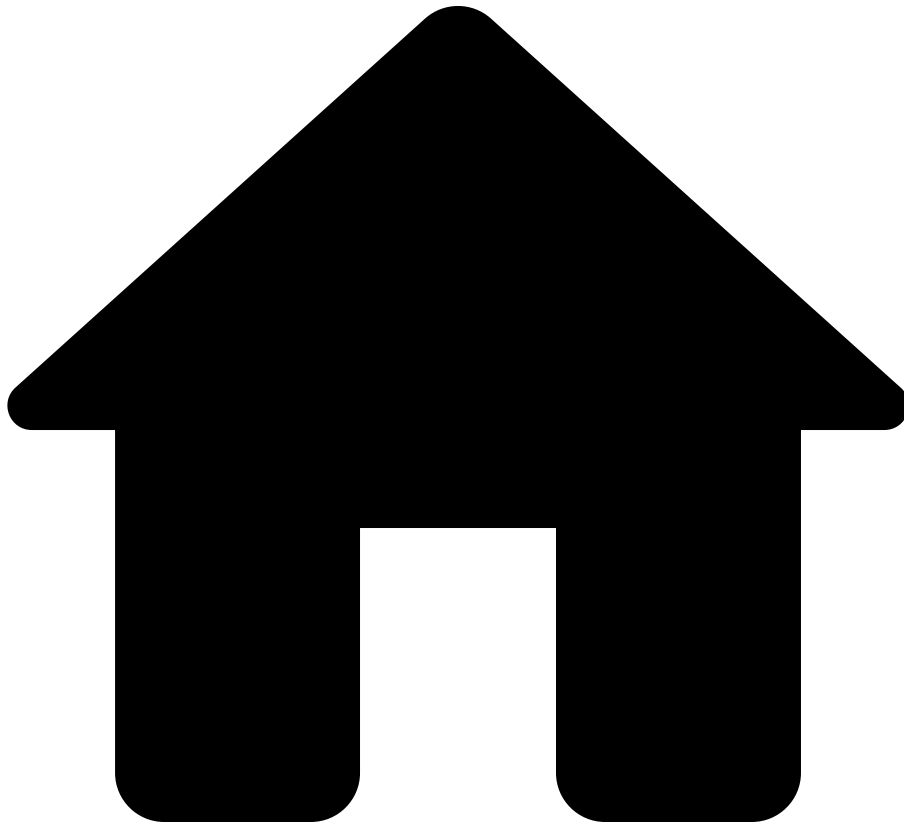


- Reference
- [NQE Language](#)
- [Guides](#)
- Device Groups

Device Groups

Each device has a field named `groupNames` that contains the names of the device groups that contain this device. With this, it is easy to write a query that only considers the devices in a specific group as well as the union or intersection of two groups. See the Examples tab for these queries.

.



- Reference
- [NQE Language](#)
- [Guides](#)
- User-defined constants and functions

On this page

User-defined constants and functions

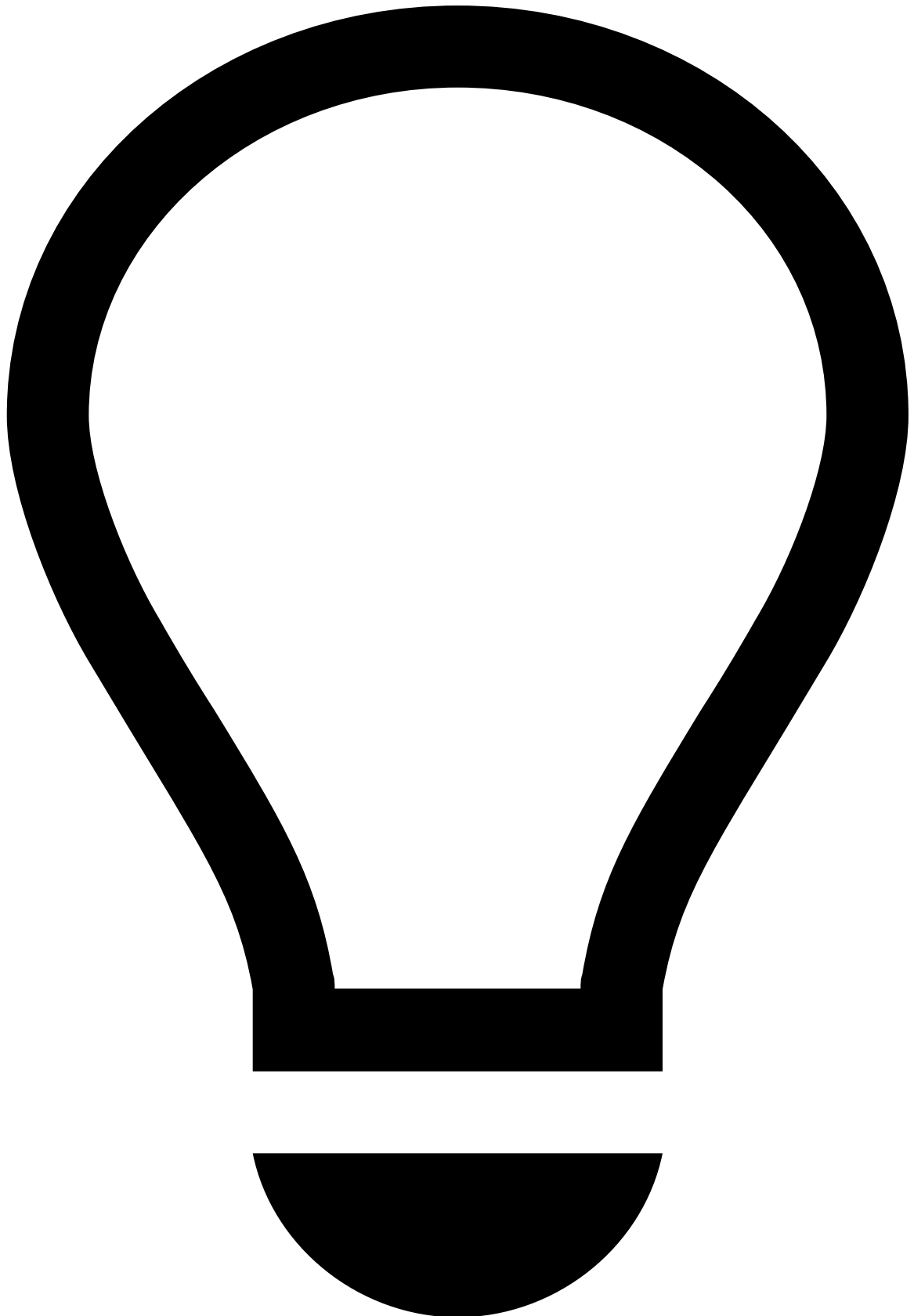
Overview [↗](#)

To help in writing clear queries, NQE supports user-defined constants and functions, known as *declarations*. Declarations are defined before the main expression of the query.

For example, here is a query that uses one user-defined constant and one user-defined function:

```
threshold = 10;
hasManySubInterfaces(iface) = length(iface.subinterfaces) >
threshold;

foreach device in network.devices
foreach iface in device.interfaces
where hasManySubInterfaces(iface)
select {
  device: device.name,
  iface: iface.name
}
```



Note

- Each declaration is terminated by a semicolon.
- Each declaration can have zero or more parameters, enclosed in parentheses, and comma-separated. If there are zero parameters, the parentheses can be omitted.

- Values of any type—not just primitive types—can be passed as arguments to user-defined functions.

Type Annotations

A declaration can be annotated with an [NQE type](#). Type annotations serve to document your intentions about the parameter or return type and can be used by the NQE type checker to provide helpful messages about problems in logic that uses the declaration or in the declaration definition itself.

Return Type Annotations

A declaration can be annotated with the expected type it should return. For example, we can extend the definition of the user-defined declarations above to include the types:

```
threshold : Integer = 100;  
hasManySubInterfaces(iface) : Bool =  
  length(iface.subinterfaces) > threshold;
```

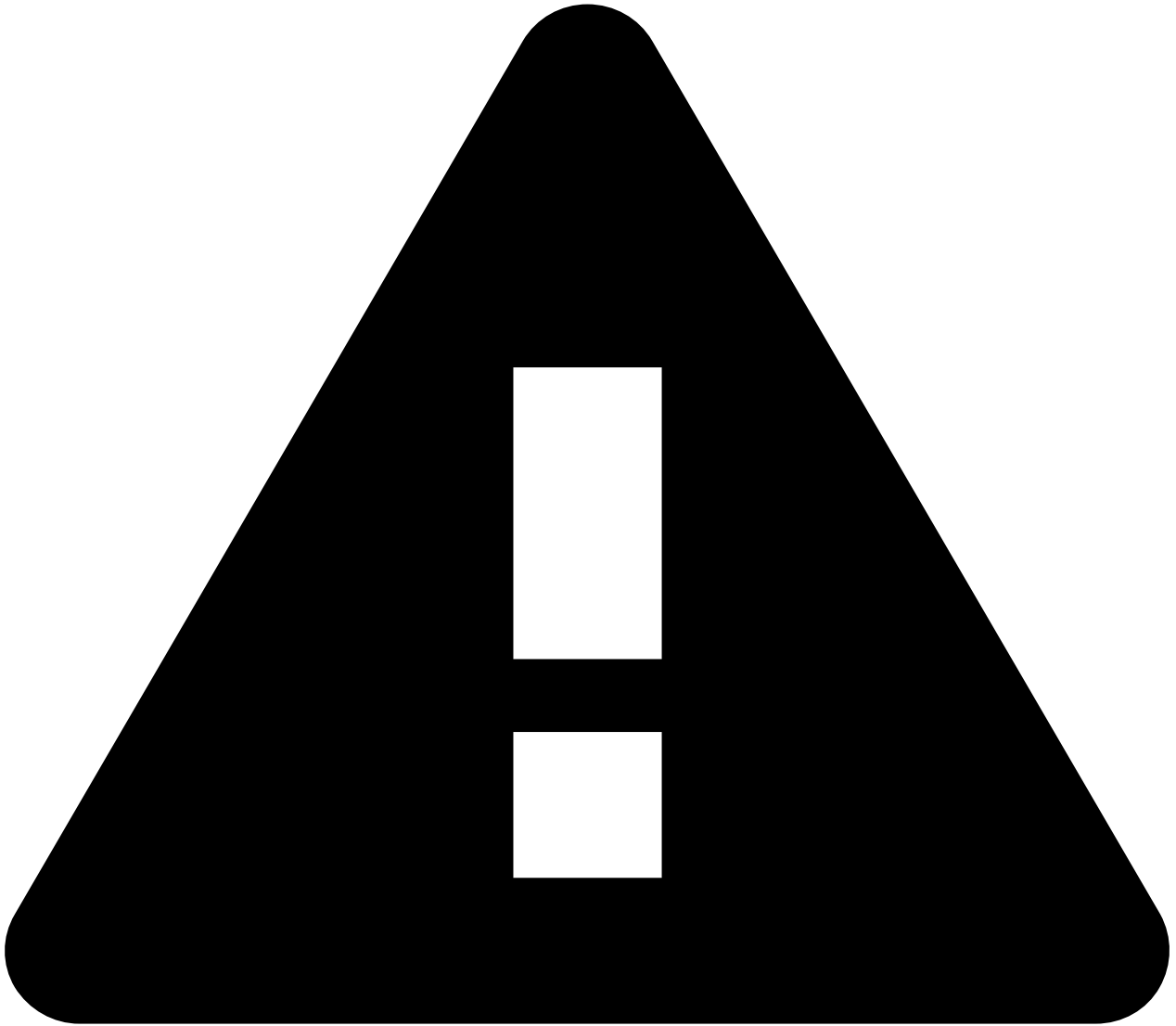
This says that the value of `threshold` has type `Integer` (i.e. is a number), and that the function `hasManySubInterfaces` returns a value of type `Bool` (true or false).

Parameter Type Annotations

Each parameter can also be annotated with the expected type it should receive. For example, we can again extend the definition of `hasManySubInterfaces` by annotating the `iface` parameter:

```
hasManySubInterfaces(iface : Iface) : Bool =  
  length(iface.subinterfaces) > threshold;
```

This says that the function `hasManySubInterfaces` takes an `iface` of type `Iface` (i.e. is an interface) and returns a value of type `Bool` (true or false).



Note

- You must either annotate all parameters with types or none of them.

Type References [↗](#)

The `i face` parameter above was referred to by its `I face` type name. NQE also allows us to write the type annotation with a type that contains just the fields we need to use. For example, instead of using `I face`, we can write:

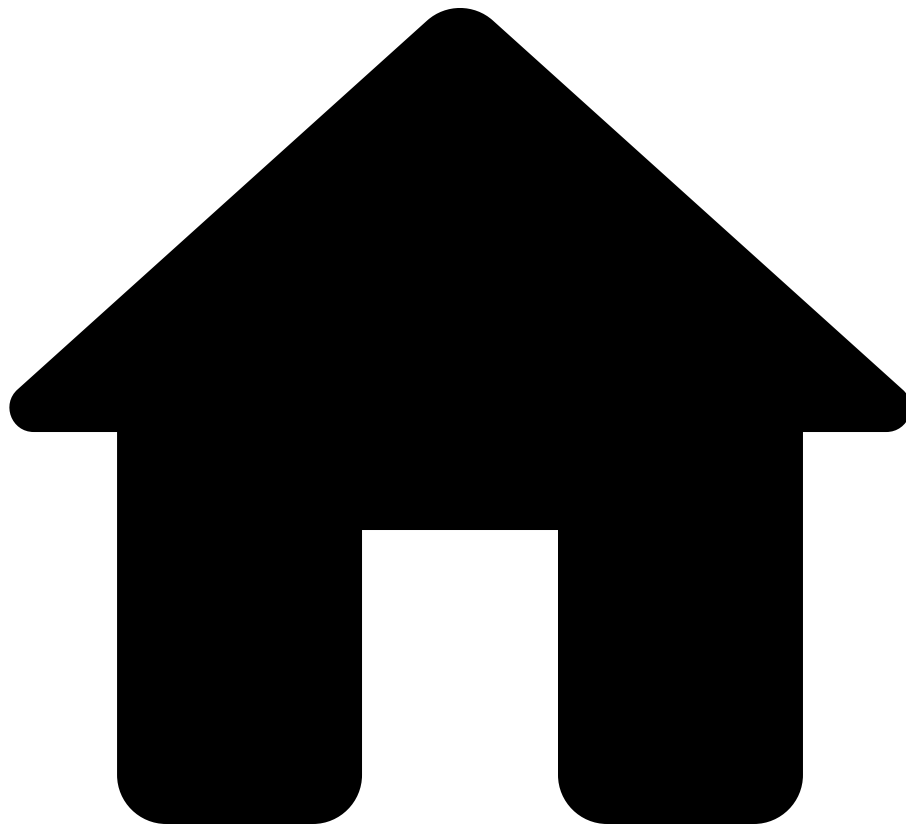
```
hasManySubInterfaces(iface : { subinterfaces:
List<SubInterface> }) : Bool =
    length(iface.subinterfaces) > threshold;
```

This enables the function `hasManySubInterfaces` to not only accept a parameter of type `Iface` but also accept a parameter of any kind of record that *at least* has a field named `subinterfaces` of type `List<SubInterface>`. This is useful in scenarios where you want to write a function that is more flexible and can be reused in more than one call.

Possible Types in NQE [↗](#)

The above example shows a few different types, namely `Integer`, `Bool`, named types (`Iface`), records, and lists, but many more types are available in NQE. To see all available types in NQE, see [NQE Types](#).

.



- Reference
- [NQE Language](#)
- [Guides](#)
- Comprehensions

On this page

Comprehensions

Comprehensions provide a powerful way of defining new collections in terms of some other data, and are the primary way of writing checks. Comprehensions were briefly introduced above. Here we provide further details.

Before diving into the notation, let's look at the essential ideas of comprehensions as they occur in everyday, informal language. Sometimes, we need to talk about collections of things without actually listing them out explicitly. For example, if you are talking about a particular car, you could hear someone talk about:

- The prices of the parts in this car that are made of steel and weigh over 500 grams

This phrase is referring to a specific collection of things (some specific prices) without actually listing them out. Of course, you could actually list these prices out, and get a list like \$100, \$2.30, etc, but this would be a very inefficient way of communicating. Instead, in this phrase, the collection of prices is expressed using the following ingredients:

- A **condition** that defines a set of things, namely the set of things that satisfies the condition. In this example, the condition is that the thing must be a part of this car and it must be made of steel and it must weigh over 500 grams.
- The **attribute** of the things of interest. In the example above the attribute is price.

A comprehension is just a precise notation for expressing the above sorts of ideas. For example, we could write the above collection of prices in comprehension notation as follows, where we assume that we can refer to a variable `thisCar` to stand for "this car" in the statements above, and that the car and parts have attributes as needed for the example:

```
foreach part in thisCar.parts where part.material == "steel" && part.weightInGrams > 500
select part.price // |----- Qualifier 1 -----| |----- Qualifier 2 -----| //
|----- Condition -----| |-- Attribute --|
```

Let's dissect this example. The comprehension has a **condition** and an **attribute**. The condition is itself built out of any number of **qualifiers**. In this example, there are two qualifiers. The first qualifier defines a condition, namely membership in the `thisCar.parts` collection, and it also gives a name to stand for any item satisfying this condition, namely `part`. Giving each item satisfying the condition a name is useful in order to specify further conditions on the items. In fact, the second qualifier, which starts with the `where` keyword, does exactly this: it requires that the matching parts also satisfy some conditions on materials and weight, using the name for the thing, `part`, just introduced in the first qualifier in order to express this additional condition. Finally, the attribute indicates that the things in the collection are the prices of the satisfying parts.

Let's use some sample data to illustrate the above ideas and see how this comprehension works in detail.

Suppose we have the following parts, which we've named `P1`, `P2` and `P3` in order to easily refer to them. We define these parts as NQE declarations, so that we can use them in writing the above comprehension. We also define `thisCar` that has exactly these parts:

```
P1 = {name: "muffler", material: "steel", weightInGrams: 5000,
price: 17};
P2 = {name: "door handle", material: "plastic", weightInGrams:
300, price: 1};
P3 = {name: "front seat", material: "fabric", weightInGrams:
5000, price: 86};
thisCar = {parts: [P1, P2, P3]};
```

With these declarations, we can try out the query above. For example, you can copy and run the following to try out our example:

```
P1 = {name: "muffler", material: "steel", weightInGrams: 5000,
price: 17};
P2 = {name: "door handle", material: "plastic", weightInGrams:
300, price: 1};
P3 = {name: "front seat", material: "fabric", weightInGrams:
5000, price: 86};
thisCar = {parts: [P1, P2, P3]};

foreach part in thisCar.parts
where part.material == "steel" && part.weightInGrams > 500
select {price: part.price}
```

The main query here is a slightly-modified form of the original example. Specifically, the only change is that the select expression is a record with the price as its single field. We make this change to satisfy a technical requirement on top-level queries. See [Section "Top-level Expression"](#) for details on this.

After the first qualifier, namely `foreach part in thisCar.parts` we have a set of satisfying assignments for `part`. They are just all the parts. We can list the satisfying assignments in a table, with one column for each variable introduced in the qualifiers so far. In this case, we just have a single variable, `part` so we will have just a single column for this variable:

part
<code>{name: "muffler", material: "steel", weightInGrams: 5000, price: 17}</code>
<code>{name: "door handle", material: "plastic", weightInGrams: 300, price: 1}</code>
<code>{name: "front seat", material: "fabric", weightInGrams: 5000, price: 86}</code>

The second qualifier imposes the condition `part.material == "steel" && part.weightInGrams > 500`, which limits the set of satisfying assignments to:

part
<code>{name: "muffler", material: "steel", weightInGrams: 5000, price: 17}</code>

Finally, the select computes item values for each of satisfying part, of which there is only one:

part	attribute
<code>{name: "muffler", material: "steel", weightInGrams: 5000, price: 17}</code>	<code>{price: 17}</code>

A Second Example: Two Variables

By using more qualifiers, comprehensions can define more complex collections. For example, suppose we wanted to refer to the total price of pairs of parts that have a part-subpart relationship where the parent part and child parts are made of different materials. If we assume that each part has a `parent` field that specifies the name of

its parent (if any), then we could express that precisely with the following comprehension:

```
foreach parent in thisCar.parts
foreach child in thisCar.parts
where parent.name == child.parent
where parent.material != child.material
select {totalPrice: parent.price + child.price}
```

Here the condition is about two parts that satisfy some specific relationship to each other. Therefore, the set of things that satisfy the condition is a set of pairs of parts. The names introduced for each part (`parent` and `child`), are useful when expressing the conditions. Again, the attribute expression indicates that the set consists of the total price for each pair of parts satisfying the given condition.

We can illustrate this more complex query by seeing how each qualifier changes the set of satisfying assignments to variables. For this example, let's suppose that we extend the parts with the `parent` field:

```
P1 = {name: "muffler", material: "steel", price: 17, parent: ""};
P2 = {name: "door handle", material: "plastic", price: 1, parent: "door"};
P3 = {name: "door", material: "steel", price: 10, parent: ""};
P4 = {name: "muffler widget", material: "steel", price: 2, parent: "muffler"};
thisCar = {parts: [P1, P2, P3, P4]};
```

After the first qualifier introduces the `parent` variable, the set of satisfying assignments are all `parent` values:

parent
{name: "muffler", material: "steel", price: 17, parent: ""}
{name: "door handle", material: "plastic", price: 1, parent: "door"}

parent
<code>{name: "door", material: "steel", price: 10, parent: ""}</code>
<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>

After the second qualifier introduces the `child` variable, the set of satisfying assignments are all combinations of parts, since no filtering has been applied to `parent` or `child` yet:

parent	child
<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>	<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>
<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>	<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>
<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>	<code>{name: "door", material: "steel", price: 10, parent: ""}</code>
<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>	<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>
<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>	<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>
<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>	<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>
<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>	<code>{name: "door", material: "steel", price: 10, parent: ""}</code>

parent	child
<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>	<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>
<code>{name: "door", material: "steel", price: 10, parent: ""}</code>	<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>
<code>{name: "door", material: "steel", price: 10, parent: ""}</code>	<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>
<code>{name: "door", material: "steel", price: 10, parent: ""}</code>	<code>{name: "door", material: "steel", price: 10, parent: ""}</code>
<code>{name: "door", material: "steel", price: 10, parent: ""}</code>	<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>
<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>	<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>
<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>	<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>
<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>	<code>{name: "door", material: "steel", price: 10, parent: ""}</code>
<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>	<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>

The third qualifier limits this set of assignments to just those where the `child.parent` matches `parent.name` :

parent	child
<code>{name: "muffler", material: "steel", price: 17, parent: ""}</code>	<code>{name: "muffler widget", material: "plastic", price: 2, parent: "muffler"}</code>
<code>{name: "door", material: "steel", price: 10, parent: ""}</code>	<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>

Finally the fourth qualifier limits to just the one case where the parent material is different than the child material:

parent	child
<code>{name: "door", material: "steel", price: 10, parent: ""}</code>	
<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>	

Finally, the `select` clause computes the attribute for each satisfying assignment

parent	child	attribute
<code>{name: "door", material: "steel", price: 10, parent: ""}</code>	<code>{name: "door handle", material: "plastic", price: 1, parent: "door"}</code>	<code>{totalPrice: 11}</code>

You can continue this way, by using more qualifiers to express conditions on any number of items. In every case, the condition specifies a set of assignments to variables defined in the condition. If there are 3 variables introduced in the condition, then each satisfying assignment provides a value for the 3 variables, and so on.

How Qualifiers Find Satisfying Assignments to Variables

In some cases, qualifiers can be reordered, but in general the order of qualifiers is important. Specifically, qualifiers are read from left-to-right. This matters because:

- Every qualifier can only make use of variables in scope at the qualifier.
- A `foreach` qualifier introduces a new variable which is in scope in the subsequent qualifier. Similarly, `let` and `group-by` qualifiers, introduced in the next section, also introduce variables that are in scope at the subsequent qualifier.

That means that this is legal:

```
foreach part in thisCar.parts
where part.material == "steel" && part.weightInGrams > 500
select part.price
```

but this is not:

```
where part.material == "steel" && part.weightInGrams > 500
foreach part in thisCar.parts
select part.price
```

because the first qualifier refers to a variable `part` which is not in scope at the first qualifier.

On the other hand, in some cases, the order does not matter. For example:

```
foreach part1 in thisCar.parts
foreach part2 in thisCar.parts
where part1 != part2 && part1.weightInGrams ==
part2.weightInGrams
select part1.price + part2.price
```

could be rewritten as:

```
foreach part2 in thisCar.parts
foreach part1 in thisCar.parts
```

```
where part1 != part2 && part1.weightInGrams ==  
part2.weightInGrams  
select part1.price + part2.price
```

(first two qualifiers swapped) without affecting the meaning.

Additional Forms of Qualifiers

So far, we have introduced two forms of qualifiers: `foreach` and `where` qualifiers. There are two further forms of qualifiers that you can use.

Let Qualifier

A `let` qualifier introduces a new variable in terms of existing variables. This allows you to write more readable comprehensions by naming intermediate values. For example, we can avoid the duplicated `x + y` expression in the following query:

```
foreach x in [1, 2, 3]  
foreach y in [3, 4, 1]  
where x + y > 4  
select x + y
```

by defining a variable `z` to stand for that value:

```
foreach x in [1, 2, 3]  
foreach y in [3, 4, 1]  
let z = x + y  
where z > 4  
select z
```

As another example, you could write the following to give a short name to the device's platform OS:

```
foreach device in network.devices  
let os = device.platform.os  
select {os: os}
```

Note: `let` is only allowed after the first `foreach` clause.

Group-By Qualifier

The final additional form of qualifier is the **group-by** qualifier. This qualifier allows you to refer to groups of items that share some common property.

Let's look at an example to illustrate the idea. Suppose we want to determine the set of part weights (and the count of parts for that weight) for which there is more than 1 part price. This set can be expressed with the following comprehension:

```
foreach part in thisCar.parts
group part.price as pricesForWeight by part.weightInGrams as
weight
where length(pricesForWeight) > 1
select {weight: weight, count: length(pricesForWeight)}
```

The group-by qualifier in the second qualifier introduces two new variables, `pricesForWeight` and `weight`. Each assignment to these two variables will correspond to a group of parts that all have the same value for `part.weightInGrams`. For each group, the assignment to the `weight` variable will be the weight of the parts in the group, given by the `part.weightInGrams` expression applied to any of the parts in the group, and the `pricesForWeight` variable will have the collection of prices (given by applying `part.price` expression to all the parts in the group) for the group.

Let's look at an example evaluation to see how this works. Consider our initial collection of parts above, which were:

```
P1 = {name: "muffler", material: "steel", weightInGrams: 5000,
price: 17};
P2 = {name: "door handle", material: "plastic", weightInGrams:
300, price: 1};
P3 = {name: "front seat", material: "fabric", weightInGrams:
5000, price: 86};
thisCar = {parts: [P1, P2, P3]};
```

Then after the first qualifier, the satisfying assignments to variables are:

part
{name: "muffler", material: "steel", weightInGrams: 5000, price: 17}
{name: "door handle", material: "plastic", weightInGrams: 300, price: 1}
{name: "front seat", material: "fabric", weightInGrams: 5000, price: 86}

Next, after the group-by qualifier, the satisfying assignments consist of the following assignments to the `weight` and `pricesForWeight` variables:

weight	pricesForWeight
300	[1]
5000	[17, 86]

We see that for each weight value in the satisfying assignments occurring after the first qualifier, we have one assignment with that `weight` value. For each assignment, the `pricesForWeight` variable is assigned the list of prices of parts with that weight value. Also, note that the satisfying assignments after the group-by qualifier no longer include assignments to variables occurring prior to the qualifier (in this case, just the `part` variable).

Then, the third qualifier places an additional condition on the assignments, namely that the `length(pricesForWeight) > 1`. The assignments are then filtered to:

weight	pricesForWeight
5000	[17, 86]

Finally, the select clause includes the `weight` and total length of `pricesForWeight` for each group:

weight	pricesForWeight	attribute
5000	[17, 86]	{weight: 5000, count: 2}

In general, the group-by qualifier has the form `group valueExpression as valuesVariable by keyExpression as keyVariable`. Each variable assignment to this group-by will correspond to a possible value, `v`, of `keyExpression` as applied to the variable assignments satisfying the previous qualifiers. For each such value `v`, the assignment will put `v` into `keyVariable`, and will put into `valueVariable`, the collection of `valueExpression` values for assignments where `keyExpression` equals `v`.

You can find further usages of the `group` statement under the *Examples* tab.

Select Distinct

The select clause of a comprehension can use the `distinct` keyword. When this is used, the resulting list contains the unique values from the list. In other words, the overall list is de-duplicated. For example:

```
foreach x in [1, 2, 1, 1]
select distinct {a: x}
```

will evaluate to the list `[{a: 1}, {a: 2} ]`, despite the repeated appearance of `1` in the list which `x` is ranging over.

order by Clause

The `order by` clause sorts the results of a comprehension. When an `order by` clause is used, the comprehension always returns a **List**. See [collections](#) for more detail on collections types.

Basic Ordering

The simplest form specifies one or more expressions to sort by:

```
foreach device in network.devices
select { Name: device.name, Platform: device.platform }
order by Name ascending
```

This sorts devices alphabetically by name in ascending order.

Multiple Sort Keys

You can sort by multiple expressions. When elements have equal values for the first expression, they are ordered by the second expression, and so on:

```
foreach device in network.devices
select {
  Platform: device.platform,
  Model: device.platform.model,
  Name: device.name
}
order by Platform ascending, Model descending, Name ascending
```

You can control sort direction using `asc` or `ascending`, and `desc` or `descending`.

Ordering and Let Qualifiers

You can use variables introduced by `let` qualifiers in `order by` clauses:

```
foreach device in network.devices
let Platform = device.platform
let Model = Platform.model
select {
  Platform,
  Name: device.name
}
order by Platform ascending, Model descending, Name ascending
```

Limit Clause

The `limit` clause restricts the number of results returned from a comprehension:

```
foreach device in network.devices
select { Name: device.name }
```



```
order by Name ascending
limit 10
```

This returns only the first 10 devices (alphabetically by name).

Limit Without Order By

If your `foreach` is over a list expression, you can use `limit` without an `order by` clause:

```
deviceList = ["device1", "device2", "device3"];

foreach device in deviceList
select device
limit 2
```

Nested Comprehensions

Comprehensions can appear at the top-level, or they can appear nested within another expression. However, when a comprehension is nested, it must be enclosed in parentheses. For example, the following expression is legal:

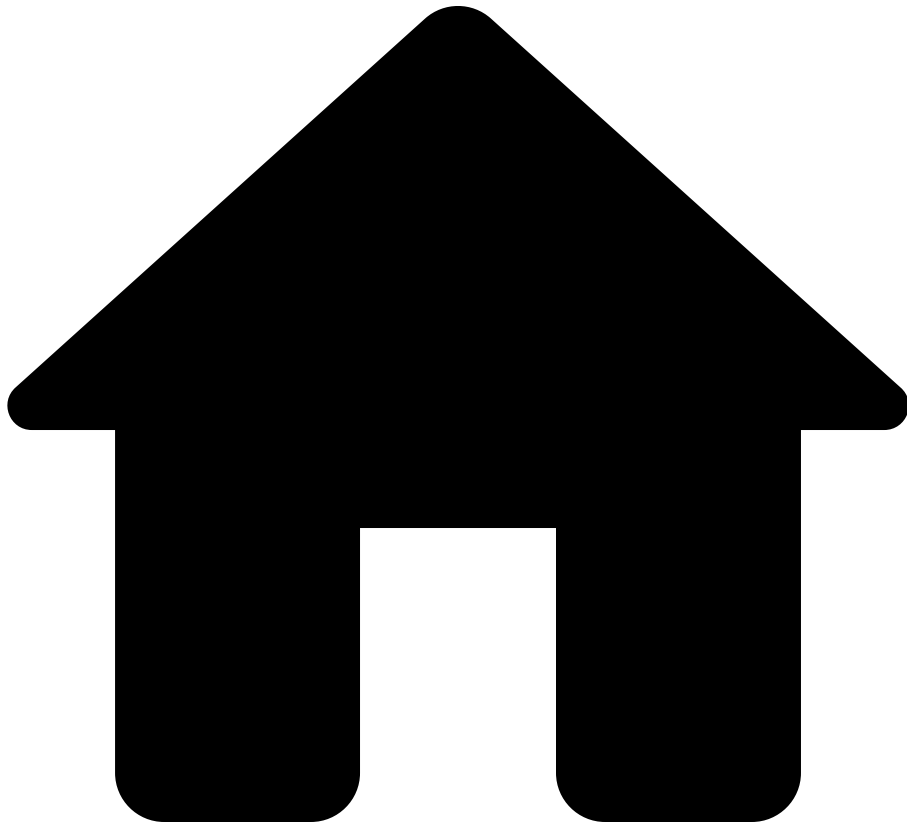
```
foreach d in network.devices
select {
  device: d.name,
  ifaceNames: (foreach iface in d.interfaces select
iface.name)
}
```

The above returns a list of records with `device` and `ifaceNames` fields. The `device` field has the device name, while the `ifaceNames` field has the names of all of the device's interfaces. The value of this latter field is computed via a nested comprehension, which is enclosed in parentheses.

Technical Restrictions^{ââ}

A comprehension always begins with `foreach ...` and ends with a single `select ...` expression. Optionally, the `select` clause can be followed by an `order by` clause and/or a `limit` clause.

.



- Reference
- [NQE Language](#)
- [Guides](#)
- Extracting Data from Configurations via Pattern Matching

On this page

Extracting Data from Configurations via Pattern Matching

The data model exposes device configurations via the `config` field of the `files` field of `Device` objects. Specifically, the `config` field contains a [collection](#) of config

line records. To reflect the nesting structure of config lines in device configurations, config line records have a `children` field which contains the immediate nested config lines of the parent config line. See the data model for more details.

Data can be extracted from configurations by pattern-matching against the lines, using either the [patternMatch](#) or the [patternMatches](#) functions. We explain both here, though typically, using `patternMatches` will lead to simpler query logic.

The `patternMatches` function takes two arguments: the first is a collection of config line records (which has type `Bag<ConfigLine>`), and the second is a pattern (which has type `Pattern<T>` for some type `T` that depends on the pattern). The function returns a bag of all successful matches of the given config lines against the given pattern. Each successful match is a record that has two fields: `line` holds the matched config line, while `data` holds the matched data. For example, the following query lists all TACACS servers, by device, and includes the text of the matched config line as well:

```
foreach device in network.devices
foreach match in patternMatches(device.files.config, `tacacs-
server host {server:string}`)
select {device: device.name, tacacsServer: match.data.server,
line: match.line.text}
```

In the above query, the `patternMatches` function is applied to all top-level lines of config in each device. The second argument to `patternMatches` is the pattern, which is enclosed in back-ticks (```). This pattern matches lines that start with the specific keywords `tacacs-server` and `host`, and then continue with a string of arbitrary non-whitespace characters. The function returns each successful match against the top-level lines of config. Because the pattern `tacacs-server host {server:string}` has a pattern `{server:string}`, the type of the pattern is `Pattern<{server:string}>`. Then for each match, `match.data` holds a record that has a single field named `server` whose type is a `string`. Note that the name of the field is determined by the name given to the component within the pattern.

The subsequent section has more details on the syntax and semantics of patterns. Also, see the [Examples](#) for more example checks using this capability.

This `patternMatch` function works similar to `patternMatches`, but takes a single string as its first argument. Instead of returning a bag of all successful matches, it returns a single match, or `null` if the match fails. For example, the following query lists all TACACS servers, by device:

```
foreach d in network.devices
foreach line in d.files.config
let match = patternMatch(line.text, `tacacs-server host
{server:string}`)
where isPresent(match)
select {device: d.name, tacacsServer: match.server}
```

Note that the result of `patternMatch(line.text, `tacacs-server host {server:string}`)` is bound to variable `match` using a `let` qualifier, rather than a `foreach` qualifier as was used in the above example using `patternMatches`. A `let` qualifier is used because `patternMatch` returns a single match, rather than a bag of matches. Also note the `isPresent` condition on the fourth line, which limits the query to just the successful matches.

In most cases, using `patternMatches` is recommended, since it leads to shorter, simpler query logic.

Details on Patterns [🔗](#)

As mentioned above, a pattern is written enclosed in backticks (``). A pattern can be matched against text and also extracts data for named components within the pattern.

A pattern consists of a space-separated **sequence of pattern items**, where each item is either

- a literal: matches a specific keyword, such as `tacacs-server` or `interface`,
- a pattern expression (described below), enclosed in curly braces, such as named components as in the query above (`{server:string}`).

The **sequence of pattern items** matches some text if each item matches, in sequence, against the text.

A pattern expression (enclosed within curly braces in a pattern) can be any of the following:

- A string literal, which matches the exact text. For example, `{"interface Vlan1"}`.
- `string`: matches any sequence of characters up to the next whitespace.
- `number`: matches any integer number (e.g. `0`, `2`, `-22`, or `98231`, but not `3.1415`).
- `float`: matches any decimal number (e.g. `0.1`, `.5`, `3.`, `-2.2` or `98.231`, but not `3`).
- `ipv4Address`: matches any IPv4 address, such as `10.22.31.99`.
- `ipv6Address`: matches any IPv6 address, such as `2001:0db8:85a3:0000:0000:8a2e:0370:7334`.
- `ipv4Subnet`: matches any IPv4 subnet, such as `20.30.0.0/16`.
- `ipv6Subnet`: matches any IPv6 subnet, such as `2001:0db8:85a3:0000:0000:8a2E:0370:7334/24`.
- `macAddress`: matches any MAC address, such as `AA:bb:33:44:55:66`, `00aa.11bb.22cc` or `00aa11.bb22cc`.
- A literal, enclosed in double-quotes: matches any specific sequence of characters. For example, `"tacacs-server"`, matches the literal string `tacacs-server`. Note that literals can already be written at the top-level of a pattern as described above. However, it is also possible to write literals within pattern expressions (enclosed by curly braces).
- `eof` matches when there is no input left. For example, `{string} {eof}` matches `"foo"` but not `"foo bar"`.
- `empty` always matches successfully and returns an empty record. This pattern is often useful in combination with the `|` pattern builder, because it allows you to indicate an optional argument in a pattern. For example `logging {"host" | empty} {ipv4Address}` matches commands like `"logging host 1.2.3.4"` and `"logging 1.2.3.4"` where the second argument can be `"host"` or can be omitted.

- `var:pattern`, where `var` is a variable name and `pattern` is a pattern expression: matches the same text that `pattern` matches. In addition, it extracts data: the result of matching `var:pattern` will be a record (aka object) with a field named `var` whose value is the value extracted from matching `pattern`. For example, if `name:string` is matched against the string `"abc.def"`, the match will succeed and return a record `{name: "abc.def"}` having a single `name` field whose value is the value extracted by the `string` pattern.
- `pattern1 pattern2 ... patternN`, is the sequence of several patterns: matches if each of the patterns can be matched in sequence. The return value is an object with fields for all the named components contained in the patterns.
- `pattern1 | pattern2`, is the "or" of the given patterns: matches if either pattern matches. The returned value is a record with two fields, named `left` and `right`. If `pattern1` matches then its value will be in the `left` field and the `right` field will be `null`. Likewise, if `pattern2` matches then its value will be in the `right` field and the `left` field will be `null`. For example, if `number | ipv4Address` were matched against text `"123"`, then the result would be `{left: 123, right: null}`, whereas if this pattern were matched against text `"1.2.3.4"`, the result would be `{left: null, right: 1.2.3.4}`. There is no backtracking, which means that if `pattern1` matches the next word then `pattern2` is never considered even if the rest of the pattern does not match the rest of the input.
- `pattern*`: matches the given pattern zero or more times. It always succeeds, and returns a list of the values matched by the pattern.
- `pattern+`: matches the given pattern one or more times. When it succeeds it returns a list of the values matched by the pattern.
- `!pattern`: matches whenever the given pattern does not match. It consumes no input and returns an empty record. To consume input and return a record containing the matched input, follow the negated pattern with the `string` pattern. For example, `{!"tag" string}` matches (and consumes) any word that is not "tag". This can be used to ignore elements up to an element of interest. For example, extracting the IP address from

```
Name Vendor Model OS Version IP Address
alt-app-lb01 F5 BIG-IP Virtual Edition 11.6.1
10.100.0.117
atl-ce01 Juniper vmx 14.1R5.4 147.75.201.115
```

is a bit difficult because the length of the model varies. However, this is achieved via the pattern

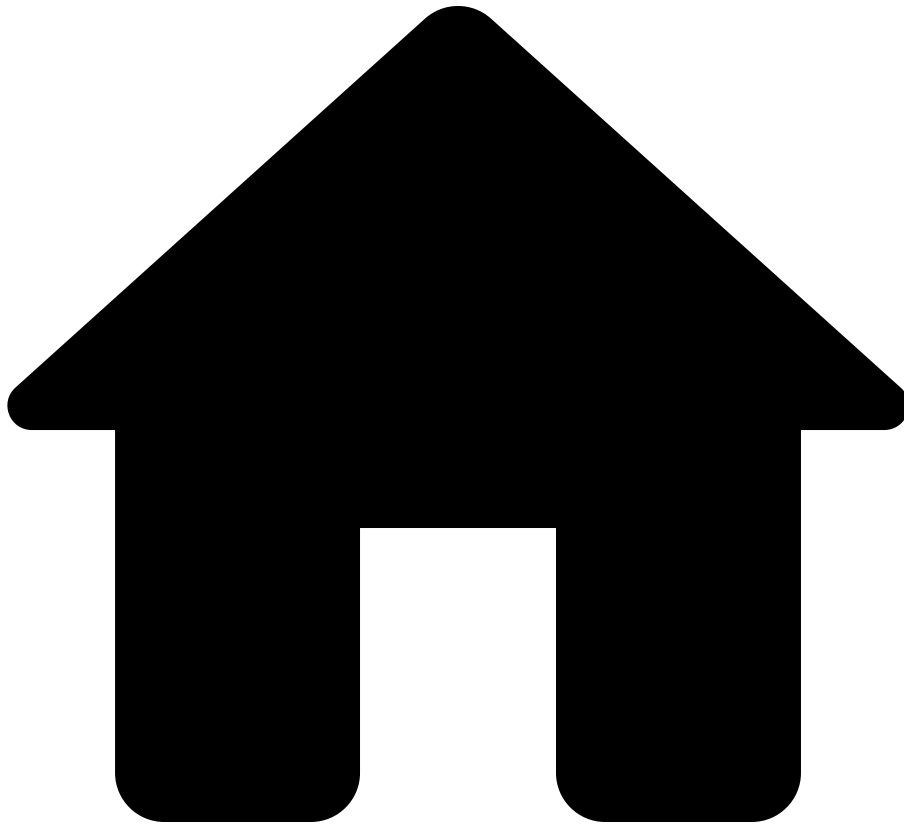
```
{(!ipv4Address string)*} {ipAddress:ipv4Address}
```

Extracting Configuration Blocks from Arbitrary Strings

The `parseConfigBlocks(os, text)` function can be used to parse a collection of configuration blocks from a string that holds a device configuration. In other words, an expression like `parseConfigBlocks(OS.NXOS, "interface eth1\n no shut\n")` could be used to parse some NXOS configuration in the given string. The resulting value can then be used with various pattern-matching functions.

This function may be useful to process the response to custom commands, for example the response to command `show running-config all`.

.



- Reference
- [NQE Language](#)
- [Guides](#)
- Primary Keys

On this page

Primary Keys

A **primary key** defines a uniqueness constraint on a query's output. The key specifies one or more columns whose combined value must uniquely identify each row in any given result. Declare a primary key by adding a `@primaryKey` annotation just before the main part of your query.

Examples

The following query shows routes and defines the primary key to be the combination of device name, VRF and route prefix:

```
@primaryKey(DeviceName, VRF, Prefix)
foreach device in network.devices
foreach networkInstance in device.networkInstances
let afts = networkInstance.afts
where isPresent(afts.ipv4Unicast)
let ipv4Unicast = afts.ipv4Unicast
foreach ipEntry in ipv4Unicast.ipEntries
where length(ipEntry.prefix) == 32
select {
  DeviceName: device.name,
  VRF: networkInstance.name,
  Prefix: ipEntry.prefix,
  "Next Hops": (foreach nextHop in ipEntry.nextHops
    select nextHop.ipAddress)
}
```

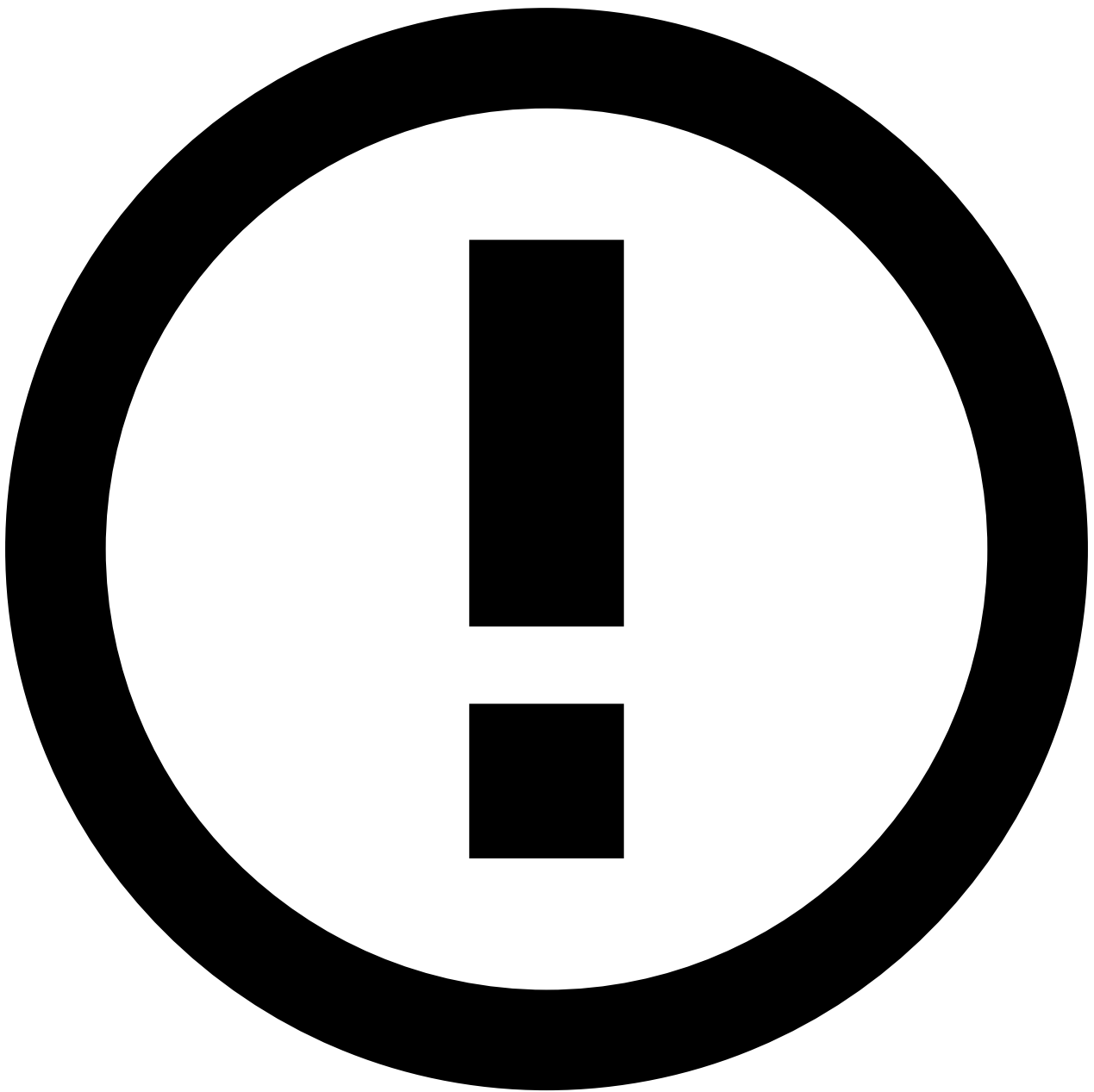
The following query defines a primary key on the union of classic and SNMP devices:

```
classicDevices =
  foreach device in network.devices
  select {
    name: device.name,
    locationName: device.locationName,
    profileName: ""
  };

snmpDevices =
  foreach snmpSource in network.endpoints
  select {
    name: snmpSource.name,
    locationName: snmpSource.locationName,
    profileName: snmpSource.profileName
  }
```

```
};  
  
@primaryKey(name)  
classicDevices + snmpDevices
```

This example shows that a primary key can be defined on any main expression, not just a comprehension.



Note

NQE History can only be enabled on a query that has a primary key.

Syntax

The attribute consists of the string `@primaryKey` followed by a parenthesized set of column names. The order of the columns does not matter.

```
@primaryKey(DeviceName, VRF, Prefix)
```

If a column name contains spaces, commas, or other special characters, it must be quoted with double quotes.

```
@primaryKey("Device Name", VRF, "Pre-fix")
```

Validation

A query has a valid primary key if the following are all satisfied:

- The query's main expression is annotated by `@primaryKey`.
- The `@primaryKey` annotation contains at least one column name.
- The `@primaryKey` annotation does not repeat any column names.
- Each column name in the `@primaryKey` annotation is present in the query result.

A query with an invalid primary key is itself invalid (and cannot be run).

Enforcement

When running a query with a primary key, if a duplicate key value is found, then the result is an error that includes an example of a duplicate key value.

Best Practices

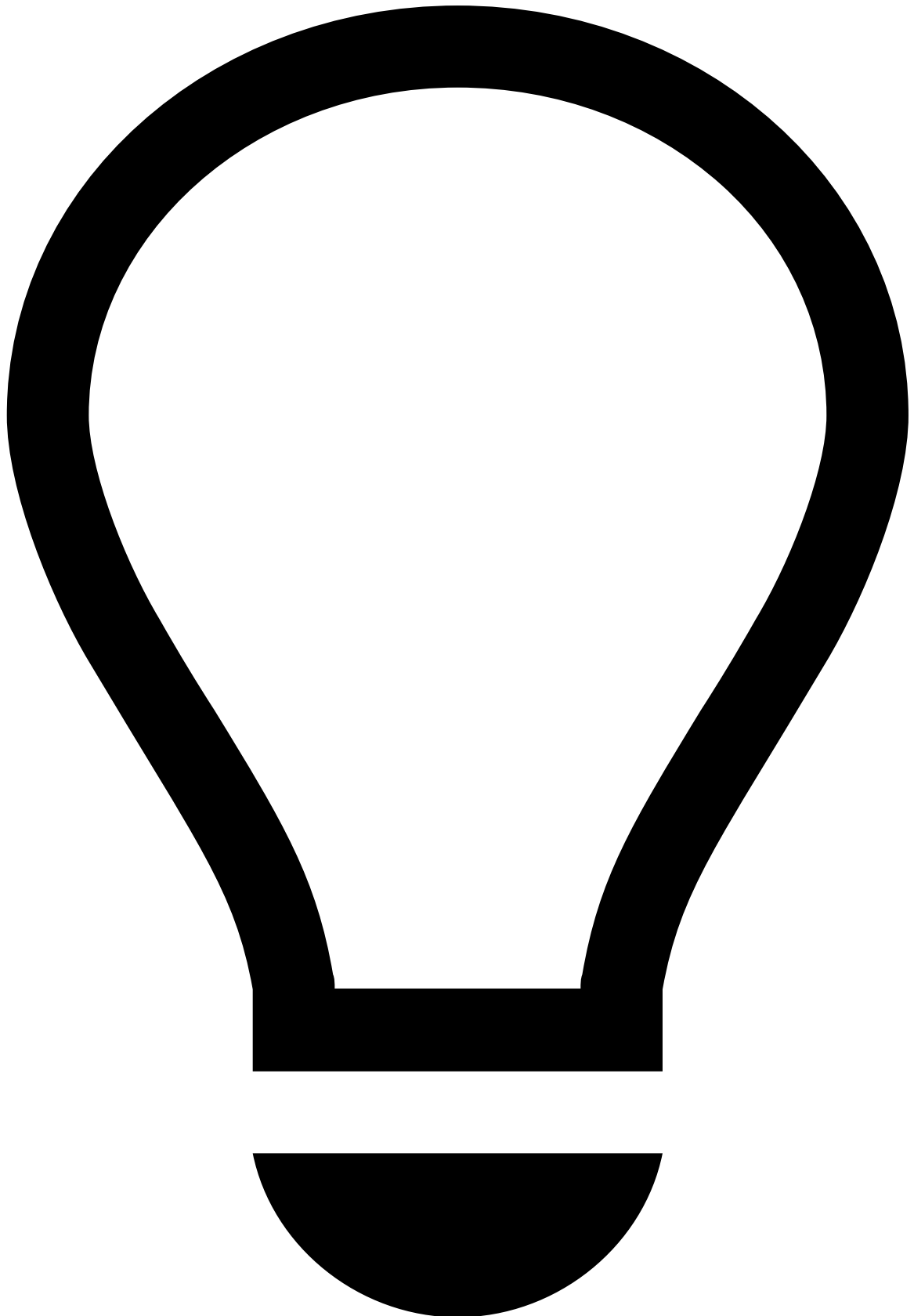
When selecting columns for the primary key, it is best to achieve the following goals:

- **Uniqueness:** Include enough columns so that the primary key value for each row is unique.

- **Stability:** Include columns such that the primary key value for each row is a stable. (Stable values do not change over time unless the identity of the value they represent changes.)
- **Minimal:** Exclude any column from the primary key if the resulting primary key still satisfies the uniqueness condition.

For example:

- For a query listing interfaces, use device name and interface name.
 - Don't include fields like interface description, since the interface description could change for the same interface and that would make the interface with the changed description appear as a different entry.
- For a query listing routes, use device name, VRF, and prefix.
- For a query listing BGP peers, use device name, VRF, and peer IP address.



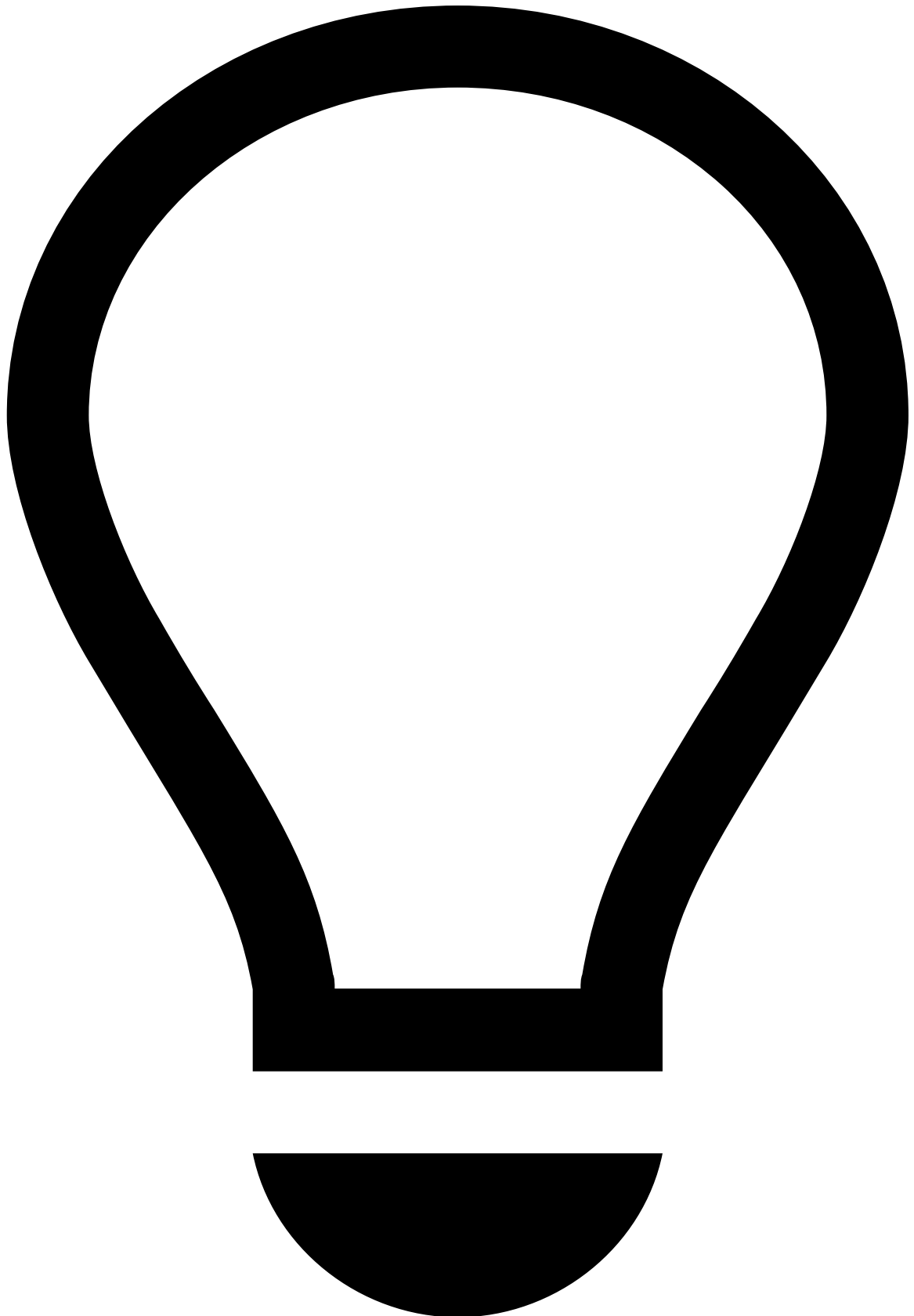
Note

There may be multiple possible choices for a primary key. Choose the primary key that best represents the natural identity of the entities in the query's results.

Diff Performance

When computing [NQE diffs](#), the diff engine needs to determine which columns to use to correlate rows between the before and after results. Without a declared primary key, the system infers one automatically — this works well for most queries. However, declaring a key explicitly is recommended as it gives you control over which columns are used and can significantly improve performance on large, wide tables (e.g. reducing diff times from several minutes to under a minute on million-row tables).

The diff results display a **key icon** on each column that is part of the key used for row correlation, making it easy to see which columns are driving the comparison.



Recommendation

Define a `@primaryKey` on any query whose diffs you rely on especially queries added to Inventory+ or verifications.

.



- Reference
- [NQE Language](#)
- [Guides](#)
- Top-level Expression

Top-level Expression

The top-level expression is the expression that defines the results of the check. The top-level expression must evaluate to a bag or list of records. See [the guide on collections](#) for more information about bags and lists.

In order to prevent overly large query results, NQE automatically rewrites checks whose resulting record's fields' values are not simple. A simple value is defined as a value that is either a basic value or a bag or list of basic values, where a basic value is defined as:

- A value of a primitive type, like `string`, `number`, `boolean`, or `ipAddress`, `ipSubnet`, `macAddress`, or
- An enumerated value.

For example, a list of `string` is a simple value, whereas a list of lists of `string` is not simple.

Whenever a returned record has a field whose value is not simple, that value is converted to a `string`. For example, this query:

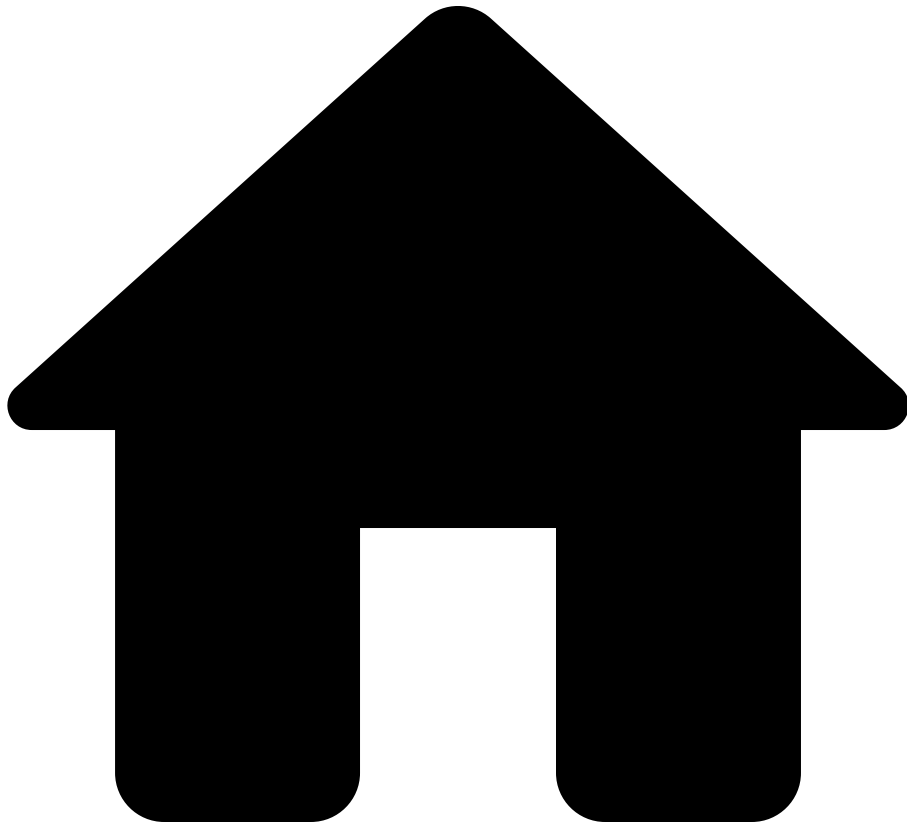
```
foreach device in network.devices
select { name: device.name, device: device }
```

will be rewritten to:

```
foreach device in network.devices
select { name: device.name, device: toStringTruncate(device) }
```

since `device.name` is a simple value (a `string`), but `device` is not a simple value. For more information on the behaviour of `toStringTruncate`, please see [toString](#).

•



- Reference
- [NQE Language](#)
- [Guides](#)
- Parameterized Queries

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Parameterized Queries

Instead of a main query, you can instead mark a function or variable declaration with an annotation that declares that function as the file's main query.

Variable Query Declarations

For example, before your main query may have been:

```
foreach device in network.devices
select { name : device.name }
```

Now you can instead write:

```
@query
deviceNames =
  foreach device in network.devices
  select { name: device.name };
```

This query can now be run in the same way as a main expression.

Parameterized Queries

Query declarations may also be given parameters. For example, we might have a function that finds all device interfaces that have too many subinterfaces:

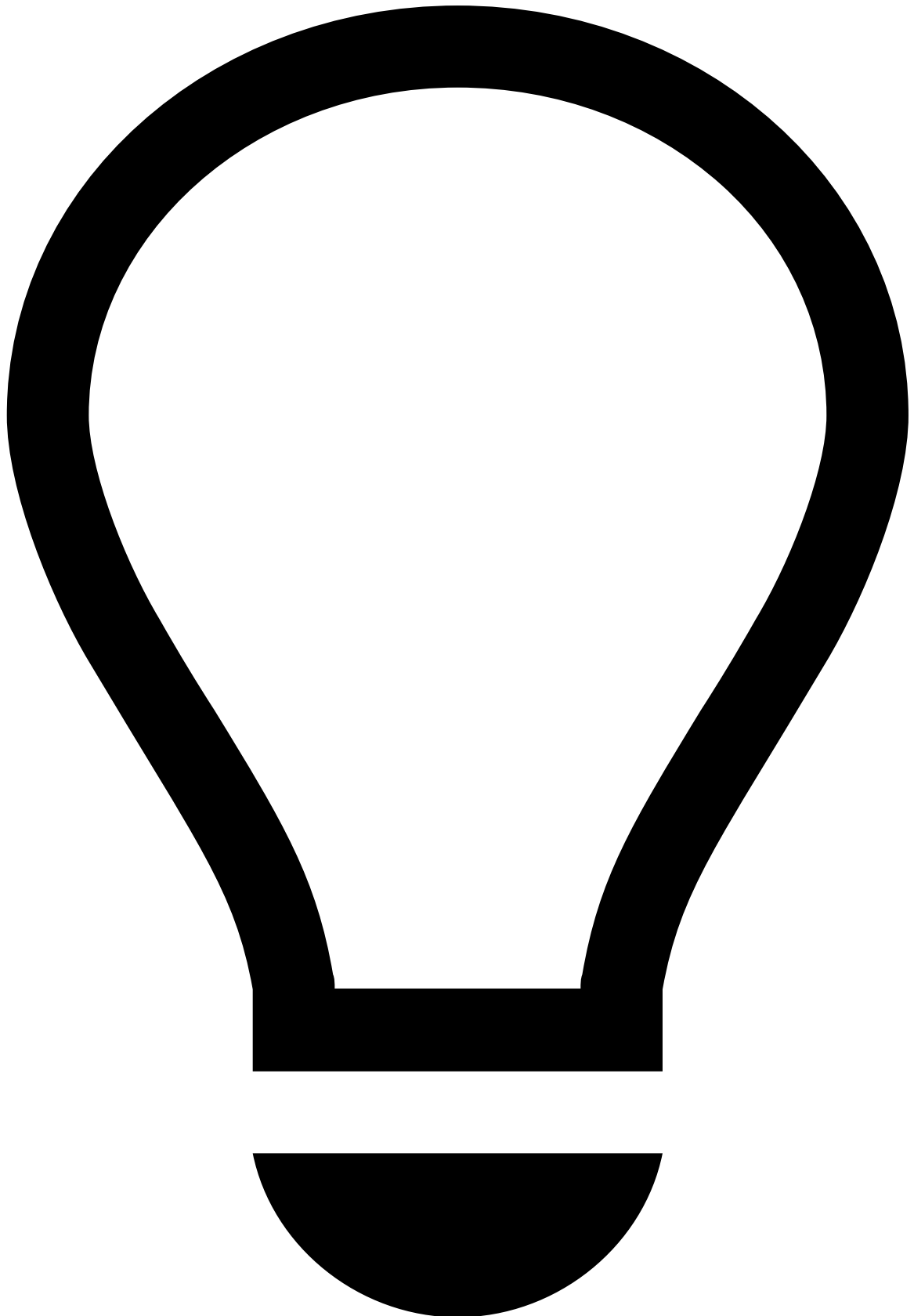
```
findBadInterfaces(threshold) =
  foreach device in network.devices
  foreach iface in device.interfaces
  where length(iface.subinterfaces) > threshold
  select { name : device.name, ifaceName : iface.name };
```

We can expose this function as a query that can be run with two changes to the declaration:

```
@query
findBadInterfaces(threshold : Integer) =
  foreach device in network.devices
  foreach iface in device.interfaces
```

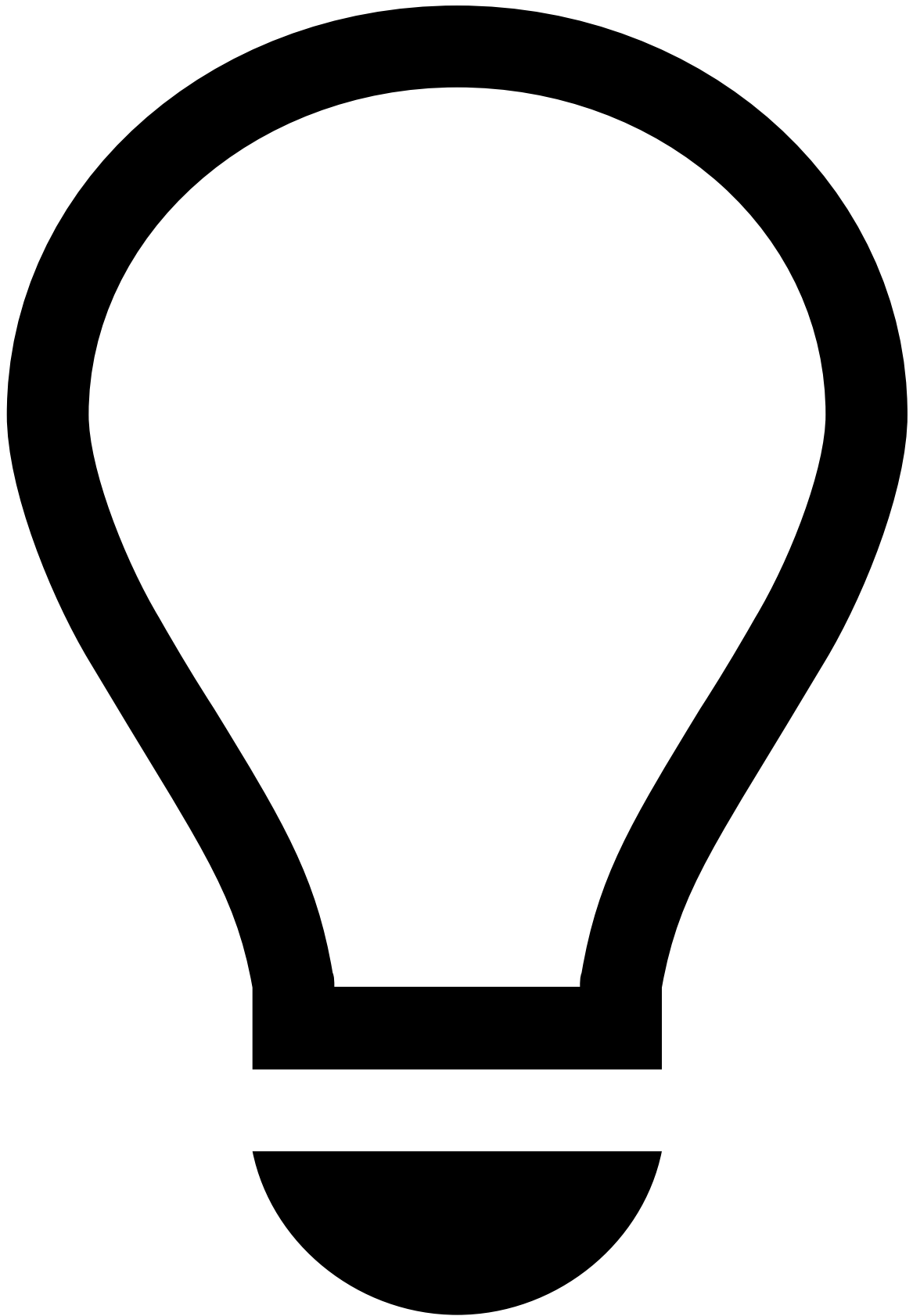
```
where length(iface.subinterfaces) > threshold  
select { name : device.name, ifaceName : iface.name };
```

- The first change is adding an `@query` annotation before the function's declaration.
- The second change is providing the type of the `threshold` parameter. Since `threshold` is a number, we declare `threshold` to have the type `Integer`.



Note

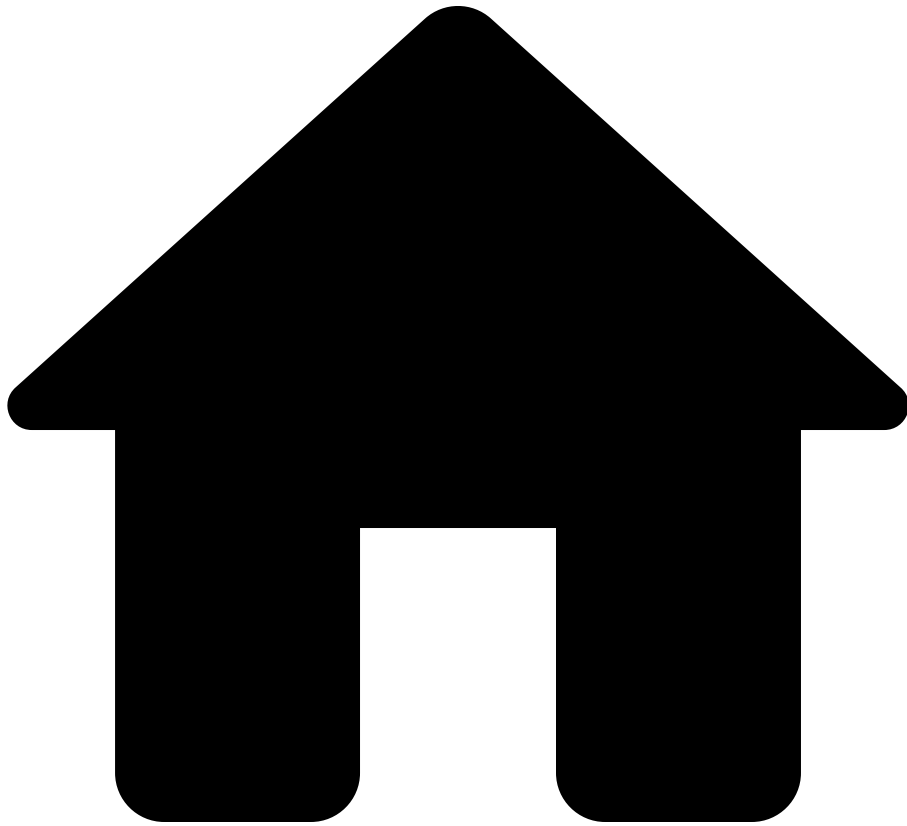
- All query parameters must have a type annotation. For more information, see [Parameter Type Annotations](#).



Note

- Return type annotations are also allowed on `@query` declarations, but are optional. For more information, see [Return Type Annotations](#).

-



- Reference
- [NQE Language](#)
- [Guides](#)
- NQE Types

NQE Types

NQE type literals are used in NQE queries in several places:

- [Missing Values](#)
- [User-Defined Variables and Functions](#)
- [Parameterized Queries](#)
- [extractJson built-in function](#)

NQE types also appear in diagnostic messages if there is a problem with a query definition.

Here is a list of all the types available in NQE and the syntax for each of these types:

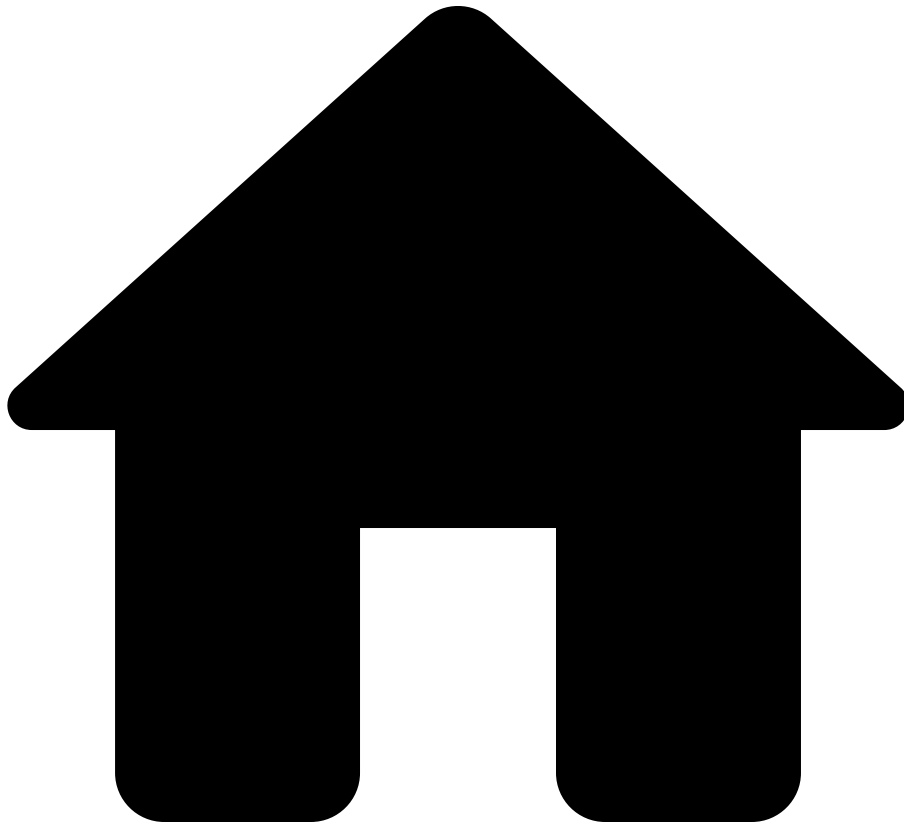
- `String` : a string such as `"foo"` (see [Strings](#)).
- `Integer` : a whole number such as `42` (see [Numbers](#)).
- `Float` : a decimal number such as `4.2` (see [Numbers](#)).
- `Bool` : a boolean such as `true` (see [Booleans](#)).
- `IpAddress` : an IP address such as `127.0.0.1` (see [IP Addresses](#)).
- `IpSubnet` : an IP subnet such as `127.0.0.1/4` (see [IP Subnets](#)).
- `MacAddress` : a MAC address such as `12:34:56:AB:CD:EF` (see [MAC Addresses](#)).
- `Date` : a date such as `October 29, 1929` (see [Time](#)).
- `Timestamp` : a date and time in UTC such as `midnight UTC on January 1, 2000` (see [Time](#)).
- `Duration` : an interval of time such as `23 hours, 56 minutes, and 4 seconds` (see [Time](#)).
- `List<T>` : an ordered collection whose elements are of type `T`. The type `T` can be replaced with any NQE type; for example, `List<IpAddress>` is the type of a list of IP addresses (see [Collections](#)).
- `Bag<T>` : an unordered collection whose elements are of type `T`. The type `T` can be replaced with any NQE type; for example, `Bag<Device>` is the type of a bag of devices (see [Collections](#)).
- `Regex<T>` : a regular expression whose match result is of type `T`. For example, a regular expression of type `Regex<{Location: String, ID: Integer}>` can be used to extract a value of type `Integer` when matched successfully against a string; one such regular expression is `re`(?<Location>\w{1,4})(?<ID: Integer>\d{3})``. For more information see the [Regular Expression Guide on Data Extraction](#).
- `Pattern<T>` : a pattern whose match result is of type `T`. For example, a pattern of type `Pattern<Integer>` will return a value of type `Integer` when matched successfully against a string; one such pattern is `{number}`. For more information see [Extracting Data from Configurations via Pattern Matching](#).
- `PatternBlocks<T>` : a block of patterns whose match result is of type `T`. For example, a pattern of type `PatternBlocks<Integer>` will return a value of type `Integer` when matched successfully against a string; one such pattern is

```
```\n{number}\n```\n
```

A function with a parameter of this type is [blockMatches](#) .

- `TypeName` : any type can be referred to by its name, which is any word in leading uppercase that contains only numbers and letters. Some examples include `AdminStatus` and `Device` (see the *Data Model* tab).
- `{ field1: type1, field2: type2, ..., fieldN: typeN }` : a record type (see [Records](#)), denoted as a list of fields and their associated types. For example, `{ device: Device, ipAddr: List<IpAddress> }` is the type of a record containing a "device" field of type `Device` and an "ipAddr" field that is a list of IP addresses. To declare an optional field, follow the field name with a question mark; for example `{name: String, mtu?: Integer}` is a record with a `name` field and an optional `mtu` field.

.



- Reference
- [NQE Language](#)
- [Guides](#)
- Imports

On this page

## Imports

While using NQE, you may find yourself needing the same functionality in multiple queries. Imports allow you to share values and functions without copying and pasting code. In the following example, we will write logic for a car shop, utilizing imports to make our code easy to read and update.

For our car shop, we will need to keep a record of the cars in our inventory. Our car inventory will be represented as a list of records. For simplicity, our car shop will only hold one car for each car model. Let's create a directory called `Shop` and within that directory a file called `Inventory`. These are the contents of the file `Shop/Inventory`:

```
export inventory =
 [{ make: "Tesla", model: "3", isOnSale: true, isClean:
false },
 { make: "McLaren", model: "F1", isOnSale: false, isClean:
false },
 { make: "Toyota", model: "Prius", isOnSale: true, isClean:
true }
];
```

Note that we marked the value `inventory` as an `export`, which means that we can now access the value `cars` in files other than `Shop/Inventory`, so as long as we import `Shop/Inventory`. Now let's write a check that ensures that all cars that are on sale are clean, which we will write in the file `Checks/On Sale Cars Are Clean`:

```
import "Shop/Inventory";

foreach car in inventory
select {
 make: car.make,
 model: car.model,
 // A car is a check violation if the car is on sale and not
clean
 violation: car.isOnSale && !car.isClean
}
```

We import from a file with a statement of the form `import "path/to/module";` where `path/to/module` is the location of the file we would like to import. The path is absolute (not relative) and slashes separate directories and the file name. Note also that when we import a file, we only have access to those functions and variables that are marked `export`.

Now that we have defined our car inventory as an export, we can use our car inventory in multiple different checks. Let's now write a check that we are only selling car makes that we like in a file called `Checks/Sell Only Liked Makes` :

```
import "Shop/Inventory";

// We only sell good makes at our shop
makesWeLike = ["Tesla", "McLaren", "BMW"];

doWeLikeMake(make) = make in makesWeLike;

foreach car in inventory
where car.isOnSale
select { make: car.make, model: car.model, violation: !
doWeLikeMake(car.make) }
```

But there's a new issue at our shop: we are cleaning cars with makes that we don't like... That simply won't do! Let's write a check that ensures that any car that we don't like is dirty, in a file called `Checks/Unliked Car Makes Are Dirty` :

```
import "Shop/Inventory";

// We only sell good makes at our shop
makesWeLike = ["Tesla", "McLaren", "BMW"];

doWeLikeMake(make) = make in makesWeLike;

foreach car in inventory
where !doWeLikeMake(car.make)
select { make: car.make, model: car.model, violation:
car.isClean }
```

See how we copy-and-pasted the `doWeLikeMake` function in order to find out whether we like a car make? As well as values, we can also export and import functions, so long as we tell NQE what type the function's arguments are. This will help us to avoid

duplicating the function `doWeLikeMake` between these two new checks. Let's export our `doWeLikeMake` from a file called `Shop/Makes We Like`:

```
import "Shop/Inventory";

// We only sell good makes at our shop
makesWeLike = ["Tesla", "McLaren", "BMW"];

export doWeLikeMake(make: String) = make in makesWeLike;
```

Now we can rewrite the check in `Checks/Sell Only Liked Makes` to:

```
import "Shop/Inventory";
import "Shop/Makes We Like";

foreach car in inventory
where car.isOnSale
select { make: car.make, model: car.model, violation: !
doWeLikeMake(car.make) }
```

And we can also rewrite the check in `Checks/Unliked Car Makes Are Dirty`:

```
import "Shop/Inventory";
import "Shop/Makes We Like";

foreach car in inventory
where !doWeLikeMake(car.make)
select { make: car.make, model: car.model, violation: !
car.isClean }
```

The rewritten checks are a significant improvement! Imports and exports allowed us to reuse code with ease. Also, notice that if we ever want to add another brand we like, we can just add a new make, say Ferrari, to `makesWeLike`:

```
// We only sell fancy makes at our shop
makesWeLike = ["Tesla", "McLaren", "BMW", "Ferrari"];
```

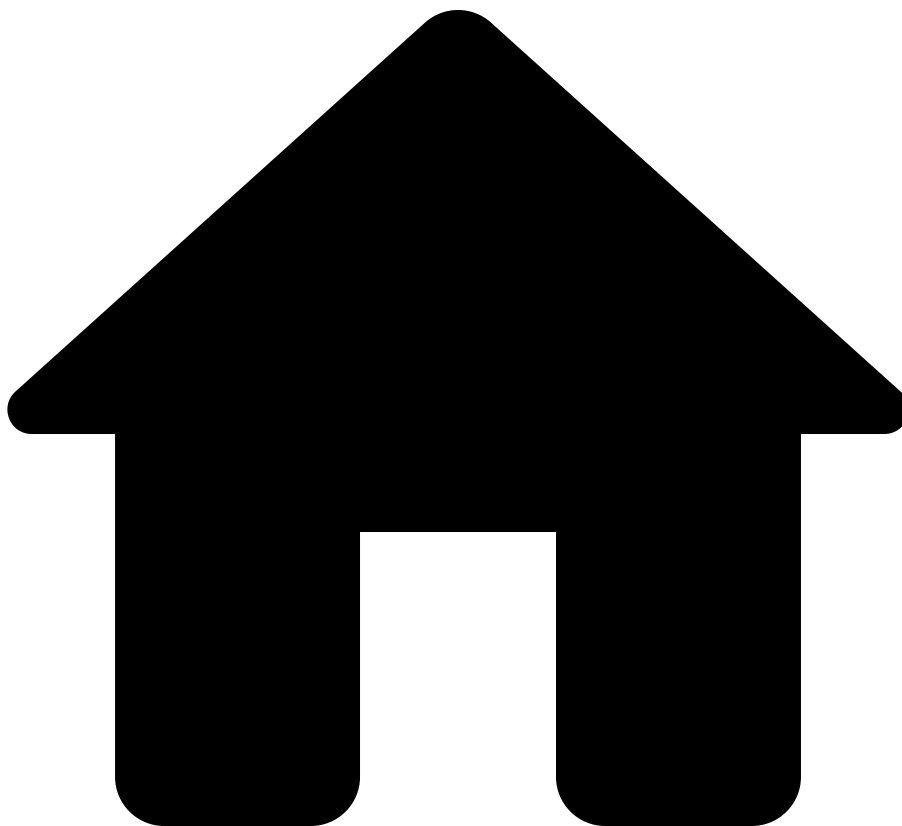
## Importing from Forward Library [↗](#)

To import from the Forward Library, start the path with `@fwd` . Here is an example of that.

```
import "@fwd/Interfaces/Interface IPs";

foreach device in network.devices
foreach subnet in getAllIfaceSubnets(device)
select { Device: device.name, Subnet: subnet }
```

.



- Reference
- [NQE Language](#)
- [Guides](#)
- CSV Data

## CSV Data

NQE queries can access data formatted as comma-separated values (CSV).

Specifically, CSV data can be embedded within an NQE query using a multi-line CSV literal, as in:



```
deviceRegions =
""csv
device,region,mgmtIp
core-01,nyc,112.2.3.1
core-02,nyc,112.2.4.1
edge-12,atl,112.3.3.1
"";
```

The CSV literal must start with the characters `""csv` followed by a newline, and is terminated by a newline followed by `""`. Within this region, the first non-empty line must contain the column headers, while every subsequent non-empty line is a data row. The number of column headers must be equal to the number of items in each row. The items in a row, as well as the header, must be separated by comma ( `,` ) characters, and newline characters must be used to terminate each row. Row items can be encapsulated with double-quotes ( `"` ) in order to include an item that contains spaces, newlines, or commas.

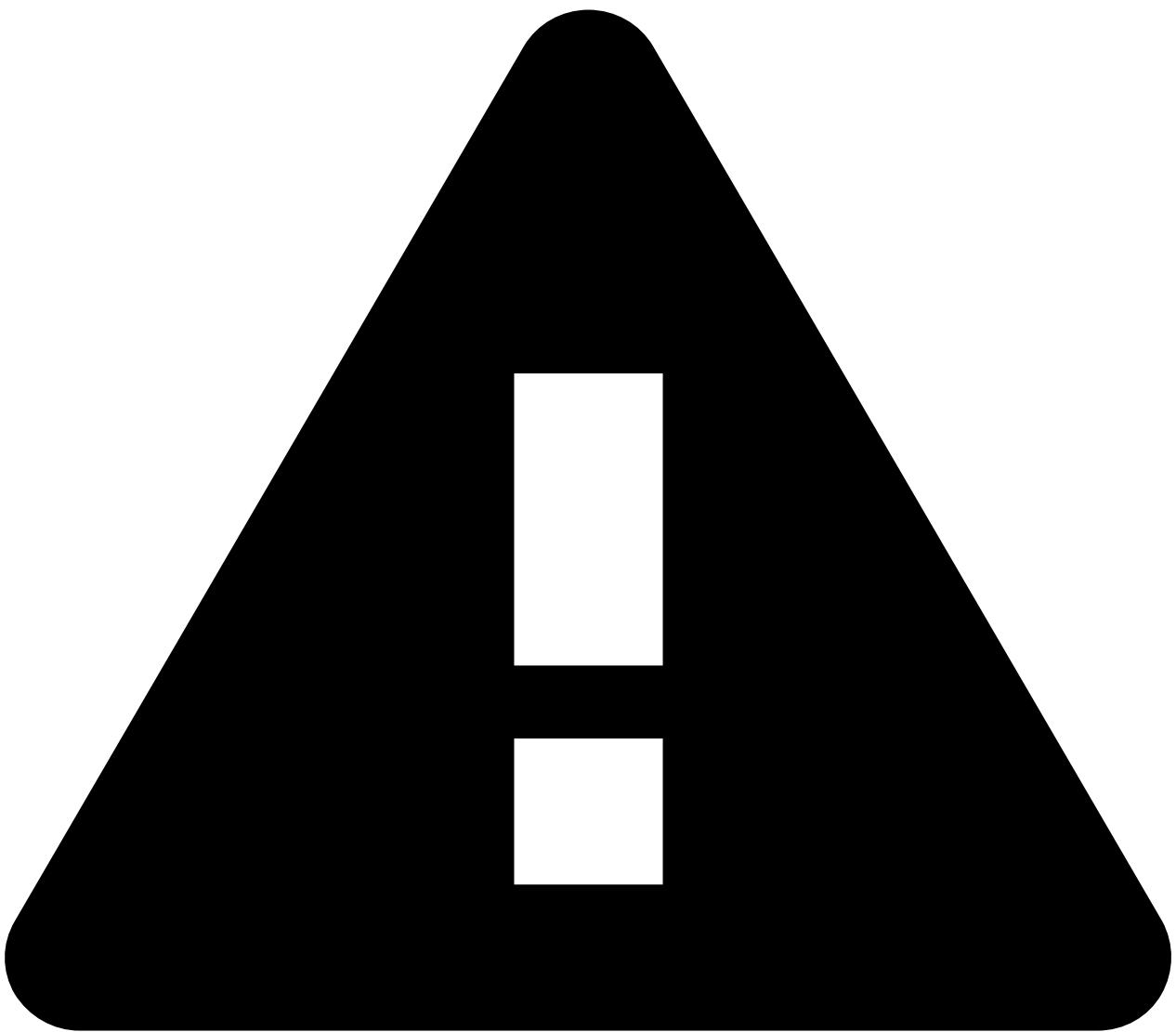
The CSV data is represented in NQE as an unordered collection (Bag) of records, where each record has fields corresponding to the columns of the CSV data. For example, the above CSV data is equivalent to the following:

```
deviceRegions =
[{ device: "core-01", region: "nyc", mgmtIp:
ipAddress("112.2.3.1") },
 { device: "core-02", region: "nyc", mgmtIp:
ipAddress("112.2.4.1") },
 { device: "edge-12", region: "atl", mgmtIp:
ipAddress("112.3.3.1") }
];
```

Since the data is represented as NQE data, it can be queried like any other data. For example, the following selects the device name and ip for devices in the `"nyc"` region:

```
foreach entry in deviceRegions
where entry.region == "nyc"
select { deviceName: entry.device, ip: entry.mgmtIp }
```

NQE infers the type of each column in the CSV data. For example, in the above CSV, the `device` and `region` columns are treated as `String` type data, while the `mgmtIp` column is inferred to be `IpAddress` type data. More generally, if all values for a given column of the CSV data have a numeric format, then the column is interpreted as a `Integer` type and the corresponding field of the NQE records has type `Integer`. Similarly, if the values of a column are all formatted as IP addresses, IP subnets, MAC addresses, or boolean values ( `true` or `false` , case-insensitive), respectively, then the column will be interpreted as `IpAddress` , `IpSubnet` , `MacAddress` , or `Bool` type, respectively. If a column's values contain a mixture of formats, then the column will be treated as `String` type.



caution

Note that the name of a column that is not a valid NQE identifier is mangled into one that is. For example, an NQE identifier cannot contain a dash, so in the following CSV data, the column name `mgmt-ip` is mangled to `mgmt_ip`.

```
deviceRegions =
"""CSV
device,region,mgmt-ip
core-01,nyc,112.2.3.1
edge-12,atl,112.2.4.1
""";
```

.



- Reference
- [NQE Language](#)
- [Guides](#)
- JSON Data

## JSON Data

NQE queries can access data formatted as JSON.

Specifically, JSON data can be embedded within an NQE query using a multi-line JSON literal, as in:

```

deviceRegions =
"""json
[
 {"device":"core-01", "region":"nyc", "mgmtIp":
"112.2.3.1"},
 {"device":"core-02", "region":"nyc", "mgmtIp":
"112.2.4.1"},
 {"device":"edge-12", "region":"atl", "mgmtIp":
"112.3.3.1"}
]
""";

```

The JSON literal must start with the characters `"""json` followed by a newline, and is terminated by a newline followed by `"""`. Within this region, any JSON object is allowed.

```

deviceRegions =
[{ device: "core-01", region: "nyc", mgmtIp:
ipAddress("112.2.3.1") },
 { device: "core-02", region: "nyc", mgmtIp:
ipAddress("112.2.4.1") },
 { device: "edge-12", region: "atl", mgmtIp:
ipAddress("112.3.3.1") }
];

```

Since the data is represented as NQE data, it can be queried like any other data. For example, the following selects the device name and IP for devices in the `"nyc"` region:

```

foreach entry in deviceRegions
where entry.region == "nyc"
select { deviceName: entry.device, ip: entry.mgmtIp }

```

NQE infers the type for the JSON data. For example, in the above CSV, the `device` and `region` columns are treated as `String` type data, while the `mgmtIp` column is

inferred to be `IpAddress` type data. More generally, the type inference works as follows:

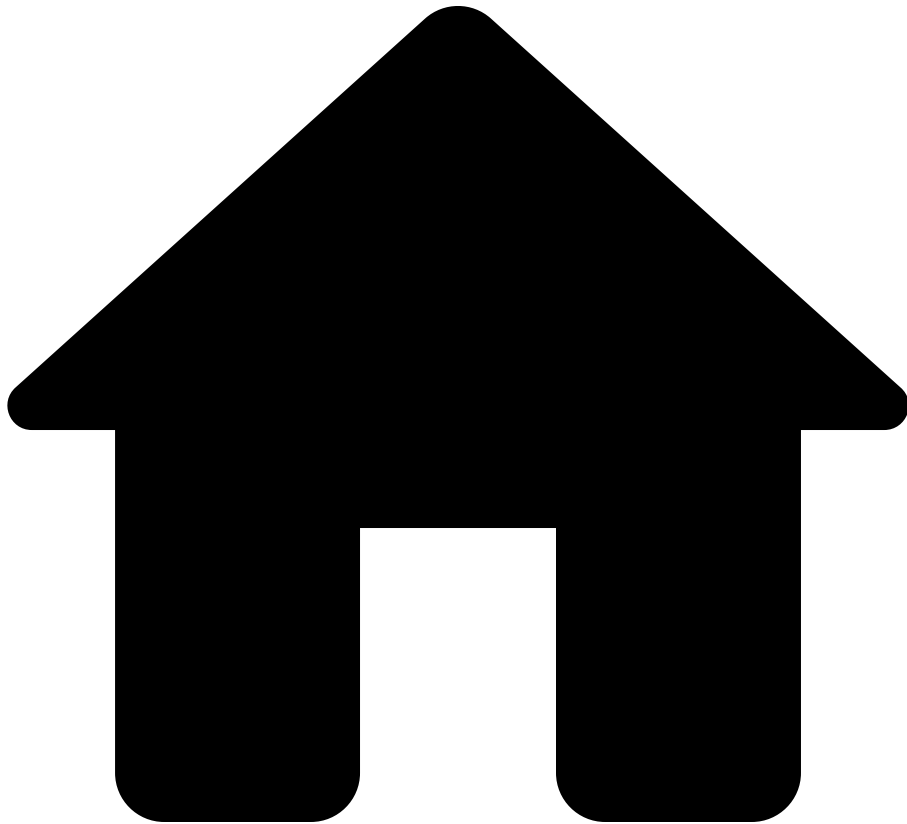
- JSON leaf values (string, numerals, boolean values) are given the most specific NQE type that is appropriate. For example, numerals are interpreted as `Integer` type and `true` and `false` are interpreted as `Boolean` type. Strings that are legal IP addresses, subnets or mac addresses are interpreted as `IpAddress`, `IpSubnet` and `MacAddress` types, respectively.
- JSON objects are interpreted as NQE records, with fields corresponding to the JSON object's properties, where the field types are determined by applying inference on the corresponding property value.
- JSON arrays are interpreted as NQE lists, where the item type of the list is determined by inferring the types of all the elements in the JSON array, and then *unifying* them to determine the most-specific common type that can be given to all the JSON array elements. For example, in a JSON array `["abc", "1.2.3.4"]`, the first element is inferred to be an NQE `String`, while the second is inferred to be an NQE `IpAddress`. Since these leaf types are not identical, they are interpreted as `String` which is the most specific type that represents both elements. If no element type can be determined for the JSON array, either because the types of the elements are incompatible or because there are no elements in the array, then no type is inferred for the array. If the entire JSON value is an array, then this situation is reported as an error in the query.

The *unification* process applied to determine the common NQE type for a collection of JSON objects works as follows:

- If all elements being unified have the same type, then that type is the result.
- If all elements are represented as JSON strings, then the result is `String`.
- If all elements are JSON objects, then the result is an NQE record with fields taken from the properties of the element objects. For a given property, if the values for the property among all element objects can be unified, then the property is included in the NQE record with the unified type; otherwise the property is not included in the NQE record. If, for a given property, some element objects do not have the object present, then the corresponding field in the NQE record will be considered optional.
- If all elements are JSON arrays, then the result is an NQE `List` type with an item type resulting from unifying the item types of the element JSON arrays. If there is no unifying item type, then there is no unifying type for the collection of arrays.

Note: If the inferred type is not the desired type, then the function `extractJson` can be used instead. This function allows users to specify the type of data they expect to extract from a string containing a JSON object.

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- Block Patterns

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## Block Patterns

Block patterns enable users to define a multi-command pattern of configuration. Block patterns can be used with NQE functions to determine if they match a device configuration or to extract data from a device configuration.



Block patterns can be defined within an NQE query, via the block pattern literal syntax, presented in the following sections. The following functions provide additional methods of creating and modifying block patterns:

- [blockPattern](#)
- [mergeBlocks](#)
- [withOption](#)

The functions that match block patterns against configuration are

- [blockMatches](#)
- [hasBlockMatch](#)
- [blockDiff](#)

⚠ There are *deprecated* versions of these methods that end in the suffix `_alpha1`.

## Creating Block Patterns

Block patterns can be defined within an NQE query using block pattern literal syntax. For example, the following NQE variable definition defines a block pattern:

```
pattern = ```
interface
 zone-member security
 ip address {ip:string}
parameter-map
 log dropped-packets
```;
```

A block pattern literal starts with triple backticks (``), then continues with any number of lines, and then ends with triple backticks on a newline.

Each line is a line pattern (see [the data extraction guide](#) for more details on these line patterns).

The indentation of the lines defines the hierarchical (tree) structure among the lines. For example, in the above pattern, `zone-member security` and `ip address {ip:string}` are children of the `interface` line.

A block pattern can also be created from an arbitrary `String` value by using the [blockPattern](#) function.

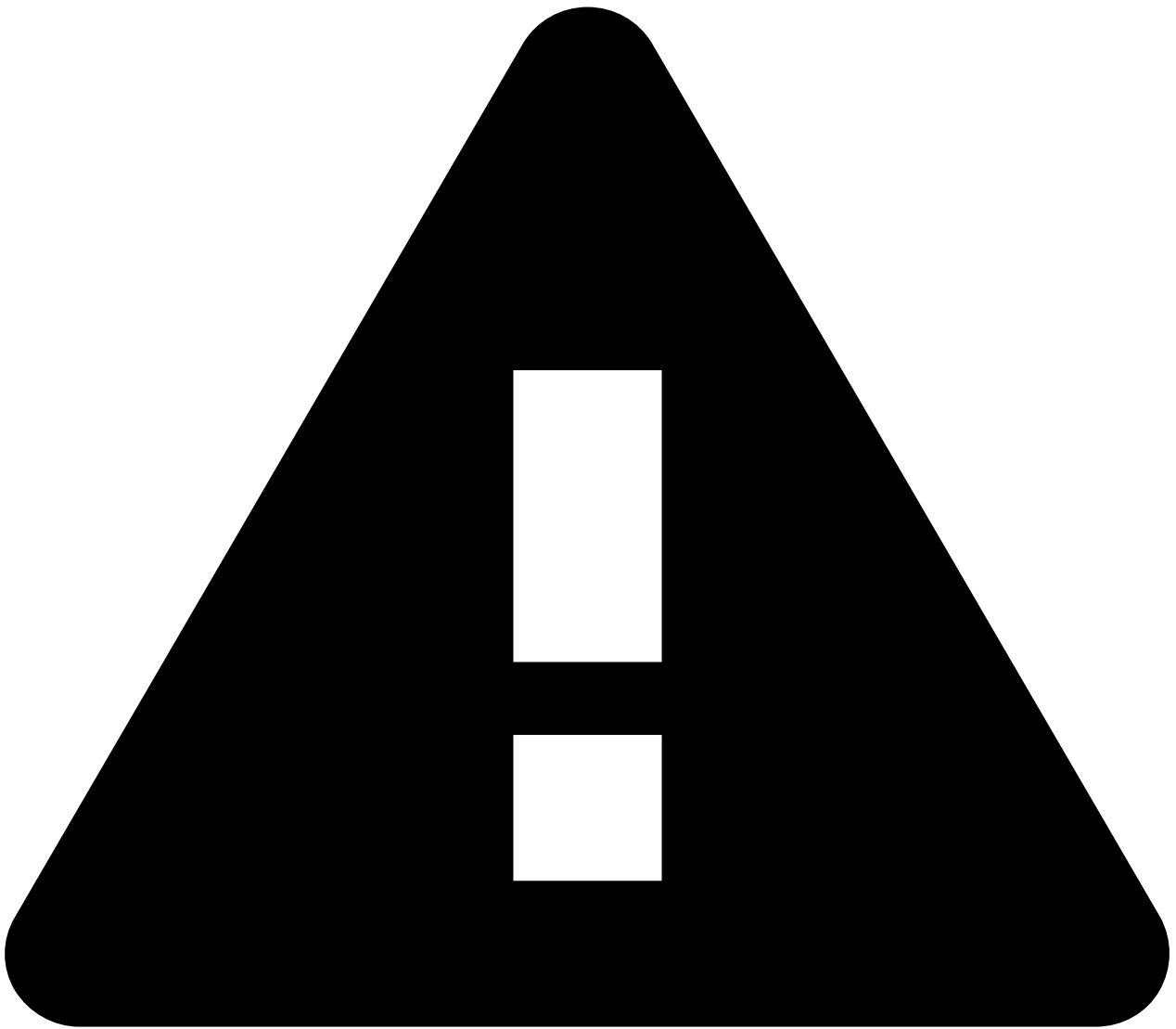
Block Pattern Matching

A block pattern matches a configuration if all the lines of the pattern match lines of configuration in a way that respects the hierarchical structure of the pattern. For example a successful match of the above pattern to a device configuration would need to match `interface` to a top-level line of configuration and would match `zone-member security` and `ip address {ip:string}` lines to child lines of the configuration line that matched `interface`. For example, the above block pattern would match the following configuration:

```
parameter-map type inspect SampleObject
  log dropped-packets
interface GigabitEthernet2
  ip address 1.2.3.4 255.255.255.248
  zone-member security abcde
```

Note that a block pattern is *unordered*: a configuration can match a pattern while having its lines appear in an order different from the order of the lines in the block pattern, as long as the hierarchy is respected. For example, the above block would also match the following configuration, where lines appear in different orders:

```
interface GigabitEthernet2
  zone-member security abcde
  ip address 1.2.3.4 255.255.255.248
parameter-map type inspect SampleObject
  log dropped-packets
```



caution

The block pattern syntax uses indentation to provide hierarchical structure and uses curly braces to denote non-literal pattern elements, such as `{string}` and `{vlan:number}`. Some device OSes do not use indentation for structure and use curly braces in their configuration syntax. For example, the following is a JunOS configuration fragment:

```
system {  
  syslog {  
    host 1.2.3.4 {  
      facility-override kernel;  
    }  
  }  
}
```

```
}  
}
```

To use block patterns to match this sort of configuration, the curly braces must be removed (because otherwise they would be interpreted as keywords in the pattern syntax), and the indentation must correspond to the structure of the original curly braces. For example, to match the above configuration, we can use the following block pattern:

```
pattern = ```  
system  
  syslog  
    host 1.2.3.4  
      facility-override kernel;  
```;
```

## Unmatched configuration [↗](#)

By default, block patterns match as long as the lines can be matched to lines of configuration. In other words, the device configuration may have extra lines of configuration that are not relevant to the block pattern. For example, the above block pattern matches a configuration such as the following, which has some extra lines of configuration in the matched interface configuration block:

```
parameter-map type inspect SampleObject
 log dropped-packets
interface GigabitEthernet2
 encapsulation dot1Q 75
 ip address 1.2.3.4 255.255.255.248
 zone-member security abcde
 no ip redirects
```

If you prefer to match a particular block of configuration only if it contains the pattern lines and *no other lines of configuration*, then use the `{%unmatched-config: forbid%}` line. For example, the above block pattern can be modified as

follows to match `interface` blocks that have only the `zone-member` and `ip address` lines and nothing else:

```
pattern = ```
interface
 zone-member security
 ip address {ip:string}
 {%unmatched-config:forbid%}
parameter-map
 log dropped-packets
```;
```

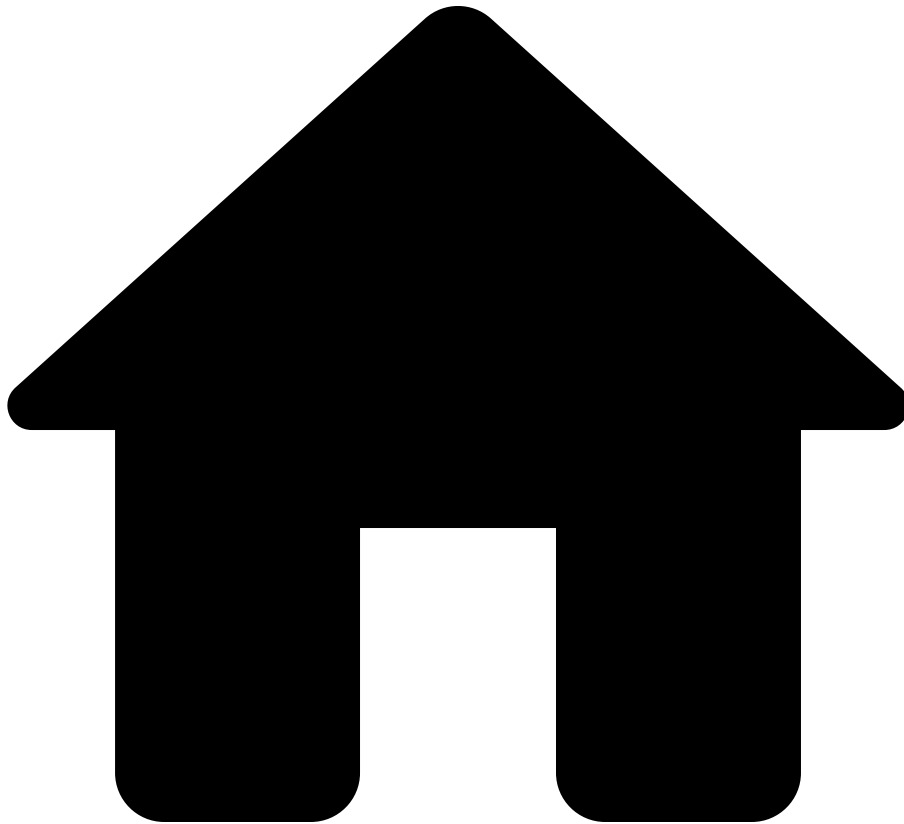
Note that this `{%unmatched-config:forbid%}` line applies this option in the block in which it appears and any blocks at a deeper level (underneath the `interface` parent of the `{%unmatched-config:forbid%}` line). Include the `{%unmatched-config:forbid%}` line at the top-level to apply this option to the whole block pattern.

Negated Block Patterns

The above block patterns match configurations if the configurations contain certain lines. Block patterns also support the ability to specify that a configuration should *not* contain a specific line or block matching some pattern. For example, the following pattern matches an interface that has an ip address but which does *not* have a `zone-member` configured:

```
pattern = ```
interface
  {%-%} zone-member security
  ip address {ip:string}
```;
```

•



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- [NQE Language](#)
- [Guides](#)
- Regular Expressions

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## Regular Expressions

NQE provides regular expression support similar to that found in other languages, like Python, Perl, Java, and SQL. In the networking context, regular expressions are useful for checking whether certain strings match specific formats, or extracting data from textual data sources, such as config files and state tables.

## Examples

The following query redacts all IP addresses found in configuration commands:

```
ipRegex = re`\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}`;

foreach device in network.devices
 foreach command in device.outputs.commands
 where command.commandType == CommandType.CONFIG
 select {
 Device: device.name,
 RedactedConfig: replaceRegexMatches(command.response,
ipRegex, "<REDACTED>")
 }
```

The following query gets only the devices following a naming convention, such as atl123:

```
validNameRegex = re`(?<Location>\w{1,4})(?
<ID:Integer>\d{3})`; // location code followed by 3-digit ID

foreach device in network.devices
where hasMatch(device.name, validNameRegex)
select { Device: device.name }
```

The above example can be modified slightly to extract the location and ID:

```
validNameRegex = re`(?<Location>\w{1,4})(?<ID:Integer>\d{3})`;

foreach device in network.devices
let result = match(device.name, validNameRegex) // Save match
result
where isPresent(result)
select {
 Device: device.name,
 Location: result.data.Location, // Use 'Location' capture
```

```
group
 ID: result.data.ID // Use 'ID' capture group
}
```

The following sections will explain the constructs used in these examples.

## Creating Regular Expressions

Regular expressions can be defined using regex literal syntax or the [regex](#) function:

```
personRegex1 = re`Name: \w+, Location: \d{1,3}\.\d{1,3}\.
\d{1,3}\.\d{1,3}`;
personRegex2 = regex("Name: \w+, Location: \d{1,3}\.\d{1,3}\.
\d{1,3}\.\d{1,3}");
```

The `regex` function is useful when the pattern is only known when the query is executed, but the resulting regular expression does not support data extraction.

See the [Regex Syntax](#) section for details of the regular expression syntax.

## Data Extraction

Data can be extracted from a piece of text by matching the text against a regular expression.

A regular expression matches a string if it adheres to the structure specified by the regular expression. For instance, `personRegex`, below, specifies text with any number of word characters, followed by a space, then any number of additional word characters.

```
peopleList =
["Name: Jasim, Location: 0.0.1.2"
, "Name: Kathy, Location: 8.8.8.8"
, "Name: Jane, Location: right here"
, "Name: Jim, Location: 123.456.7.8"
];

personRegex = re`Name: \w+, Location: \d{1,3}\.\d{1,3}\.
```



```

\d{1,3}\.\d{1,3}`;

foreach person in peopleList
select { result: hasMatch(person, personRegex) }

```

The above query results in 4 rows, only `Jane` 's row is `false` .

## Capturing matched text

The above example can be modified slightly to work with multiple people *in a single piece of text*:

```

// We have all people in a single string
peopleText =
"""
Name: Jasim, Location: 0.0.1.2
Name: Kathy, Location: 8.8.8.8
Name: Jane, Location: right here
Name: Jim, Location: 123.456.7.8
""";

personRegex = re`Name: \w+, Location: \d{1,3}\.\d{1,3}\.
\d{1,3}\.\d{1,3}`;

foreach personMatch in regexMatches(peopleText,
personRegex) // Save match results
select { result: personMatch.string } // Use matched
substring

```

The above query results in 3 rows, with `Jane` 's row omitted since it does not match.

Each `personMatch` match result is a record with fields:

- `string: String` *The substring that is matched.*
- `start: Integer` *Zero-based character offset in `text` where the match begins.*
- `end: Integer` *The first character offset index after the match ends.*
- `data: T` *The capture groups. This is a record value.*

For more details, see the [regexMatches](#) function.

## Using capture groups

The above example can be further modified to *extract* the name and location of each person:

```
peopleText =
"""
Name: Jasim, Location: 0.0.1.2
Name: Kathy, Location: 8.8.8.8
Name: Jane, Location: right here
Name: Jim, Location: 123.456.7.8
""";

// Use groups to capture interesting sub-parts of a regular
expression
personRegex = re`Name: (\w+), Location: (\d{1,3}\.\d{1,3}\.
\d{1,3}\.\d{1,3})`;

foreach personMatch in regexMatches(peopleText, personRegex)
let person = personMatch.data
select { Name: person["1"], Location: person["2"] } // Use
capture group values, by positional index
```

The above query results in the same 3 rows.

## Using named capture groups

Capture groups can also be given meaningful names, which can often increase the clarity of the query:

```
peopleText =
"""
Name: Jasim, Location: 0.0.1.2
Name: Kathy, Location: 8.8.8.8
Name: Jane, Location: right here
```

```

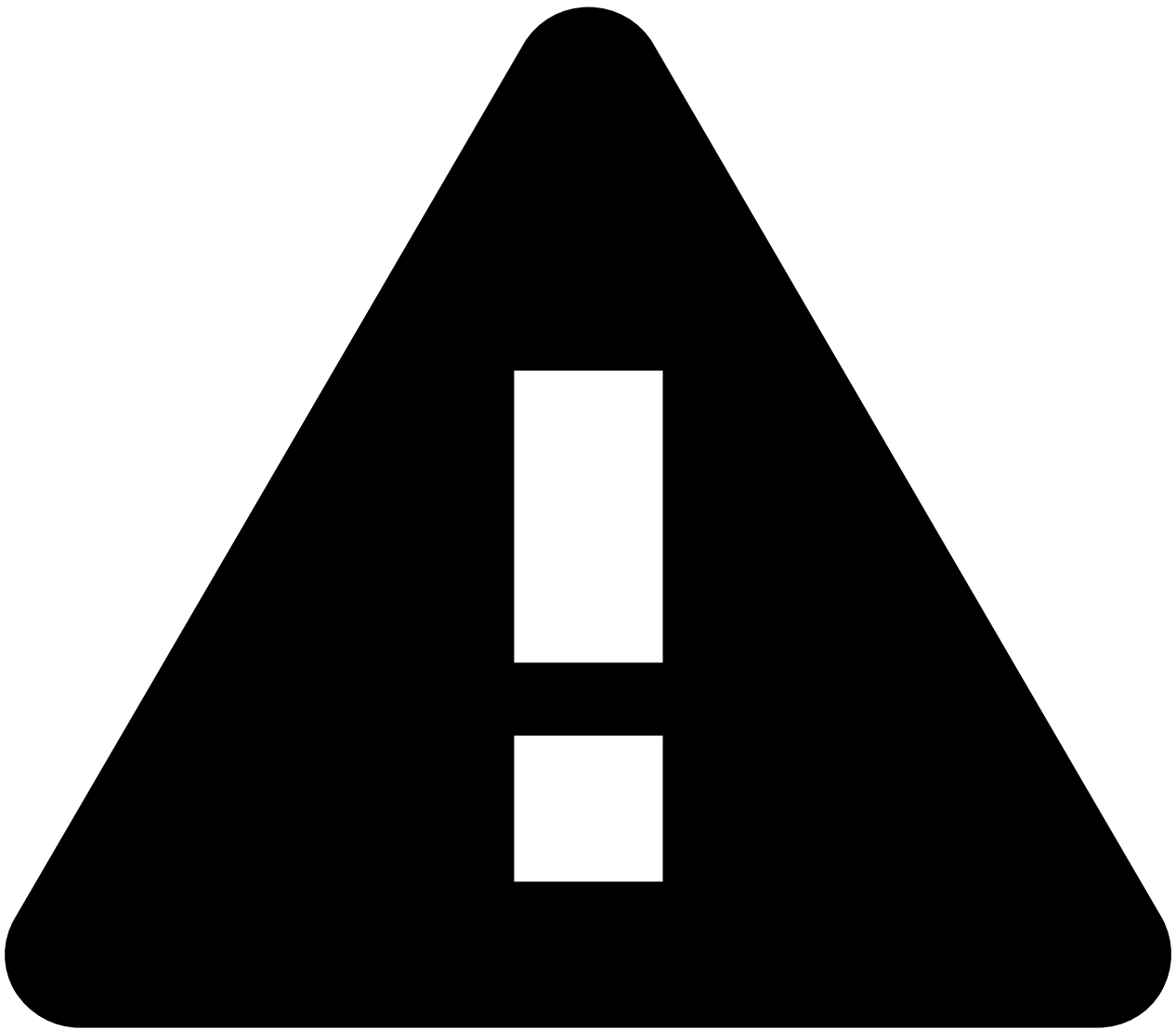
 Name: Jim, Location: 123.456.7.8
 """;

// Use names for each interesting sub-part of the regular
expression
personRegex = re`Name: (?<Name>\w+), Location: (?
<Location>\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3})`;

foreach personMatch in regexMatches(peopleText, personRegex)
let person = personMatch.data
select { Name: person.Name, Location: person.Location } // Use
capture group values, by name

```

In the above example, the name is captured using `(?<Name>\w+)`. As a result, the match value in `personMatch.data` has a `Name` field that contains the matched name. The `select` statement now references the name as `person.Name` rather than `person["1"]` --which is still a valid way to reference the name value: Named groups can be accessed both by name or by positional index. This query results in the same 3 rows as the previous one, without explicit capture group names.



## caution

Since matches of *unnamed* capture groups are accessed via a record field named according to the group's position in the regex, the field containing the matched data may change if the regex is adjusted. It is recommended to use *named capture groups*.

Named capture groups can be repeated if separated by the `|` operator. Referencing the match data by name retrieves the value wherever it was matched, whereas referencing by index retrieves the value at that particular location. For example:

```
foreach text in ["Jane is 32 years old"]
let person = match(text, re`(?<age:Integer>\\d+)|(?: (?<name>\\w+) is (?<age:Integer>\\d+) years old)`)
```

```
select {
 "The value of capture named 'age': person.age,
 "The captured value of the first occurrence of 'age':
person["1"],
 "The captured value of the second occurrence of 'age':
person["3"],
 "Aside: 'name' is at index 2": person.name == person["2"]
}
```

The above query results in a single row whose columns have values: `32`, `null`, `32`, and finally `true`.

The above query uses a *shy group* to ensure we have a choice of two regular expressions, where the second expression is `(?<name>\\w+)` is `(?<age:Integer>\\d+) years old`.

## Using shy groups

The shy group construction `(?:x)` is like a capture group `(x)` but it does not capture: Text matching the inner regular expression `x` is not captured into a field of the match.

Shy groups are useful for enforcing order of operations. For example, the following uses a shy group to apply a quantifier to the choice `[A-C] | [D-F]`.

```
initialsRegex = re`(?:<initial>(?:[A-C]|[D-F]){2} (?
<age:Integer>\\d+))`;
foreach person in ["AF 12", "CD 34", "ZD 34"]
select { isValid: hasMatch(person, initialsRegex) }
```

This query results in 3 rows: `true`, `true`, `false`.

## Using type annotations

Named capture groups can also be given type annotations, which ensure that a capture is not only of the right shape but also of the right datatype.

```

peopleText =
"""
 Name: Jasim, Location: 0.0.1.2
 Name: Kathy, Location: 8.8.8.8
 Name: Jane, Location: right here
 Name: Jim, Location: 123.456.7.8
""";

// The only change from the previous example is the type
// annotation on the 'Location' capture group.
personRegex = re`Name: (?<Name>\w+), Location: (?
<Location:IpAddress>\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3})`;

foreach personMatch in regexMatches(peopleText, personRegex)
let person = personMatch.data
select { Name: person.Name, Location: person.Location }

```

In this example, the second capture group in `personRegex` is now declared as `<Location:IpAddress>` which says that only valid IP addresses should be matched, that matching data will be captured into `Location`, and that the captured data will be converted to NQE's `IpAddress` type. As such, this query results in only two rows, since `Jane`'s row is not of the right shape and `Jim`'s row is of the right shape but is not a valid `IpAddress`.

The supported capture group types are `String`, `Integer`, `Float`, `IpAddress`. Below are further examples.

### Example

```
match("2357", re`(?<value>\d+)`)
```

```
match("2357", re`(?<value:String>\d+)`)
```

```
match("2357", re`(?<value:Integer>\d+)`)
```

```
match("3x", re`(?<value:Integer>\d+)`)
```

```
match("3.14", re`(?<value:Float>\d+\.\d*)`)
```

```
match("3 14", re`(?<value:Float>\d+\.\d*)`)
```

Example
<code>match("2.4.6.8", re`(?&lt;value:IpAddress&gt;\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3})`)</code>
<code>match("a.b.c.d", re`(?&lt;value:IpAddress&gt;\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3})`)</code>

## Regex Syntax

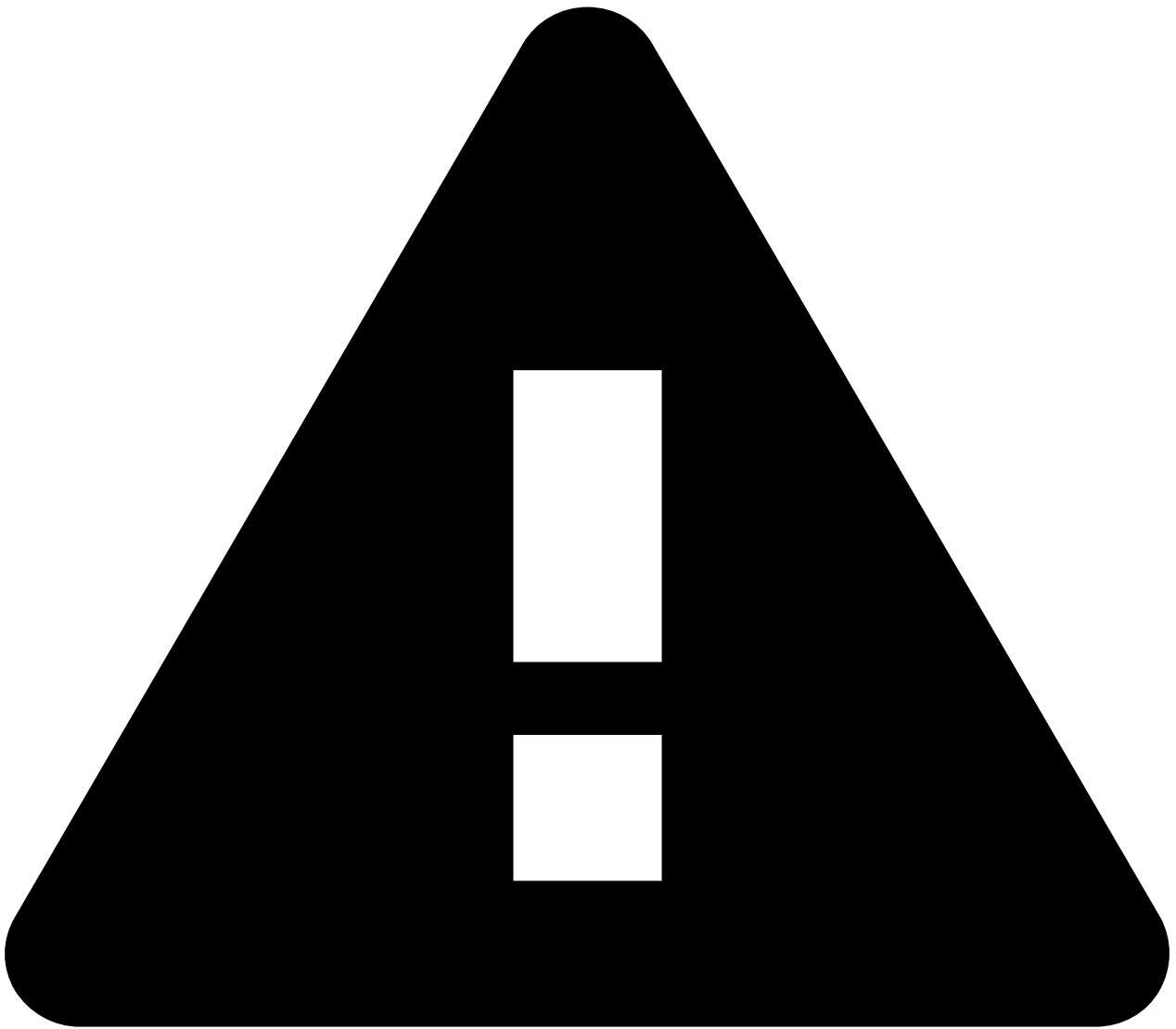
A regular expression literal starts with `re``, then continues with any number of lines, and then ends with a final backtick ```. The following table summarizes the available regular expression constructs:

Syntax	Description
<code>c</code>	The single character <code>c</code>
<code>.</code>	Any single character
<code>[abc]</code>	Any of the characters <code>a</code> , <code>b</code> , or <code>c</code>
<code>[^abc]</code>	None of the characters <code>a</code> , <code>b</code> , nor <code>c</code> ,
<code>\d</code>	Any single digit; equivalent to <code>[0-9]</code>
<code>\D</code>	Any non-digit; equivalent to <code>[^0-9]</code>
<code>\t</code>	A horizontal tab character; equivalent to <code>\011</code>
<code>\v</code>	A vertical tab character; equivalent to <code>\013</code>
<code>\n</code>	A new line character; equivalent to <code>\012</code>
<code>\r</code>	A carriage return
<code>\R</code>	Any Unicode linebreak sequence: matches <code>\r\n</code> , <code>\n</code> , <code>\r</code> , <code>\v</code> , <code>\f</code> , and Unicode line/paragraph separators
<code>\f</code>	A form feed character; equivalent to <code>\014</code>
<code>\S</code>	Any non-whitespace character; equivalent to <code>[^\t\v\n\f\r ]</code>
<code>\s</code>	Any whitespace character; equivalent to <code>[\t\v\n\f\r ]</code>

Syntax	Description
<code>\w</code>	Any word-constituent character; equivalent to <code>[0-9A-Za-z_]</code>
<code>\W</code>	Any non-word-constituent character; equivalent to <code>[^0-9A-Za-z_]</code>
<code>^</code>	Assert at beginning of text
<code>\$</code>	Assert at end of text
<code>\b</code>	Assert at ASCII word boundary: Matches the empty string, but only at the beginning or end of a word
<code>\B</code>	Assert not at ASCII word boundary
<code>xy</code>	Regex <code>x</code> followed by regex <code>y</code>
<code>x y</code>	Matches <code>x</code> if possible, otherwise try to match <code>y</code> . Ordered Choice
<code>(?&lt;name&gt;x)</code>	Named capturing group (submatch)
<code>(?&lt;name:Type&gt;x)</code>	Named and typed capturing group (submatch)
<code>(x)</code>	Numbered capturing group (submatch)
<code>(?:x)</code>	Non-capturing group: Text matching the inner regular expression <code>x</code> is not captured into a field of the match. This is useful for enforcing precedence.
<code>x*</code>	As many instances of <code>x</code> as possible, possibly zero instances
<code>x*?</code>	Like <code>x*</code> , but matches the shortest possible string so the overall regex matches
<code>x+</code>	As many instances of <code>x</code> as possible, asserting at least one instance
<code>x+?</code>	Like <code>x+</code> , but matches the shortest possible string so the overall regex matches



Syntax	Description
<code>x?</code>	Zero or one instances of <code>x</code> , prefer one
<code>x??</code>	Like <code>x?</code> , but matches the shortest possible string so the overall regex matches
<code>x{n,m}</code>	As many instances of <code>x</code> as possible, asserting at least <code>n</code> instances and no more than <code>m</code> instances
<code>x{n,m}?</code>	Like <code>x{n,m}</code> , but matches the shortest possible string so the overall regex matches
<code>x{n,}</code>	As many instances of <code>x</code> as possible, asserting at least <code>n</code> instances
<code>x{n,}?</code>	Like <code>x{n,}?</code> , but matches the shortest possible string so the overall regex matches
<code>x{n}</code>	Exactly <code>n</code> instances of <code>x</code>
<code>x{n}?</code>	Like <code>x{n}?</code> , but matches the shortest possible string so the overall regex matches



caution

There is *no* support for back-references such as `\1` .

## Regex Validity [🔗](#)

The following kinds of regular expressions are considered invalid:

- Quantifying with an upper bound greater than 1 involving capture groups, such as `re`(\w+)\{1,3}`` and `re`(<name>\w+)\{1,3}`` .
  - Fix by switching to a non-capturing group construction as in `re`(?:\w+)\{1,3}`` .

- Nesting quantifiers, such as `re`\d{3}*`` or `re`\d{3}+``.
  - NQE disallows such nestings for performance reasons.
  - Fix by using a shy capture group as in `re`(?:\d{3})*`` or `re`(?:\d{3})+``.
  - Note that you can nest quantifiers where the outer quantifier is `?`; these are *reluctant quantifiers*. For example, `re`\d{3}?`` is a valid regular expression.
- Using the same name in multiple capture groups with different types, such as `re`(<age:String>\d+) | (<age:Integer>\d+)``.
  - Fix by making the declared capture group `age` have the same type, as in `re`(<age:String>\d+) | (<age:String>\d+)`` or `re`(<age:Integer>\d+) | (<age:Integer>\d+)``.

## See Also🔗🔗

### Functions🔗🔗

Function	Description
<a href="#">hasMatch</a>	Checks whether a string matches the structure specified by a regex or not.
<a href="#">match</a>	Gets a record of information about text that matches the structure specified by a regex, or otherwise <code>null</code> .
<a href="#">regexMatches</a>	Gets the list of all substring matches of a given regex.
<a href="#">replaceRegexMatches</a>	Replaces all substring matches of a given regex using the provided replacement template.
<a href="#">regex</a>	Converts a string into a regex.

### Related Types🔗🔗

The [Pattern Type](#) allows *word-based pattern matching* and [replaceMatches](#) allows character-based matching via globs. However, globs only support a very narrow set of meta-characters. Regular expressions bridge the expressive power not covered by [Pattern Type](#) and globs.



## ❖❖❖❖ interface admin UP operational not UP

[Find every interface whose admin status is UP but operational status is not UP](#)

## ❖❖❖❖ Aruba access points

[Find all hosts that are Aruba wireless access points](#)

## ❖❖❖❖ XEROX hosts

[Find all XEROX hosts](#)

## ❖❖❖❖ summary routes not in allowed list

[Find IPv4 summary routes that are not in a specific allowed list](#)

## ❖❖❖❖ loopback subnets bigger than /32

[Find loopback interfaces assigned subnets bigger than /32](#)

## ❖❖❖❖ interface MTU mismatch

[Find connected interfaces that have mismatched MTU settings](#)

## ❖❖❖❖ IP Uniqueness

[IP Uniqueness](#)

## ❖❖❖❖ Device Group

[All devices in a particular Device Group](#)

## ❖❖❖❖ Cisco NX-OS disallowed features

[Find disallowed features that are enabled on Cisco NX-OS devices](#)

## ❖❖❖❖ Cisco NX-OS static null routes

[Find all static null routes on Cisco NX-OS devices](#)

## ❖❖❖❖ Cisco NX-OS interface slower than 1Gbps

[Find all interface speeds slower than 1Gbps on Cisco NX-OS devices](#)

## ❖❖❖❖ Devices with uptime over 6 months

[The following query shows devices that have been up for at least 6 months. For every device, the output includes the](#)

.



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- [NQE Language](#)
- [Examples](#)
- interface admin UP operational not UP

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# interface admin UP operational not UP

## Find every interface whose admin status is UP but operational status is not UP [↗](#)

The following check iterates through all interfaces on all devices and returns the device name and interface name for each qualified interface.

```
foreach device in network.devices
foreach interface in device.interfaces
select {
 deviceName: device.name,
 interfaceName: interface.name,
 adminStatus: interface.adminStatus,
 operStatus: interface.operStatus,
 violation: interface.adminStatus == AdminStatus.UP &&
interface.operStatus != OperStatus.UP
}
```

Please refer to the **Data Model** for possible values for `AdminStatus` and `OperStatus`.



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- Aruba access points

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# Aruba access points

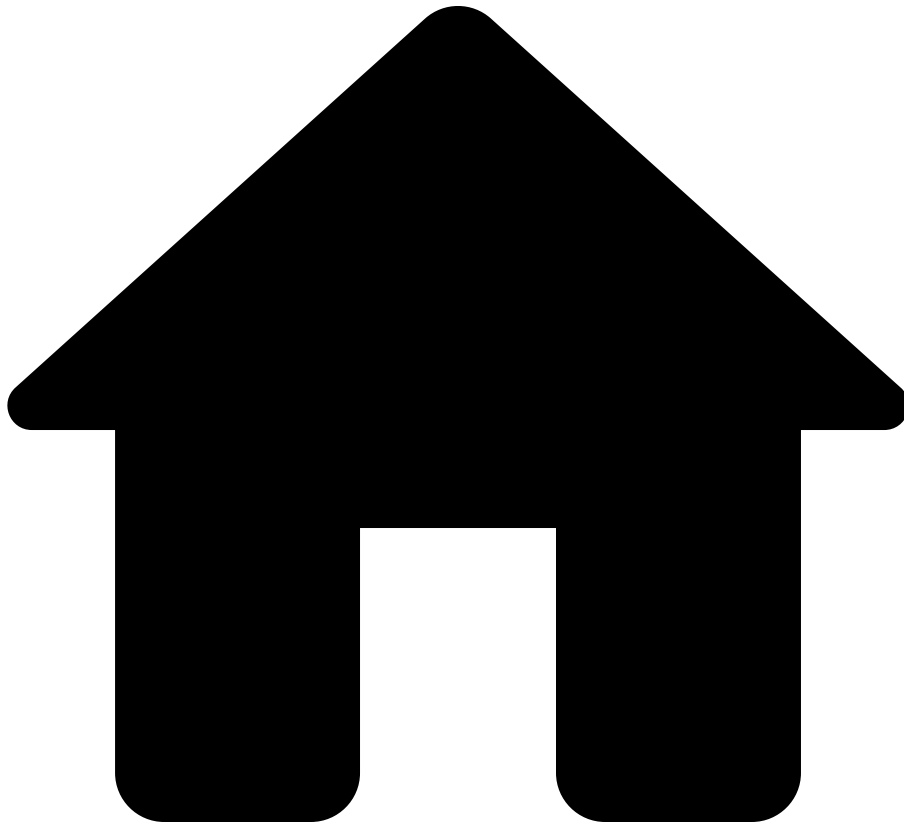
## Find all hosts that are Aruba wireless access points[â](#)

The following query finds any host in the network (that is, a device attached to a collected network device) whose mac addresses indicates that it is manufactured by Aruba.

```
foreach device in network.devices
foreach host in device.hosts
where isPresent(host.macAddress)
let vendors = ouiVendors(host.macAddress)
where Vendor.ARUBA in vendors
select {
 deviceName: device.name,
 addresses: host.addresses,
 macAddress: host.macAddress,
 interfaces: host.interfaces,
 vendors
}
```

Please refer to the **Documentation** for more information on the `ouiVendors(mac)` function and refer to the *Data Model* for possible values for the `Vendor` type.

.



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- [NQE Language](#)
- [Examples](#)
- XEROX hosts

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## XEROX hosts

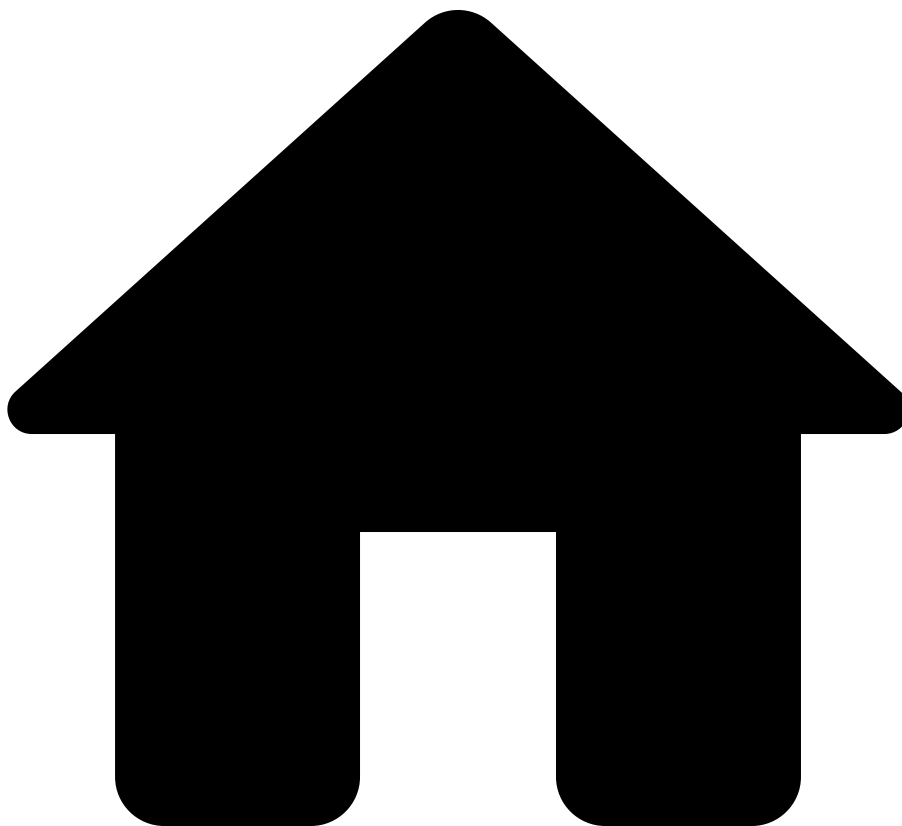
### Find all XEROX hosts[â](#)

The following query finds any host in the network (that is, a device attached to a collected network device) whose mac address is assigned to XEROX.

```
foreach device in network.devices
foreach host in device.hosts
let assigneeName = ouiAssignee(host.macAddress)
where isPresent(assigneeName)
where matches(toUpperCase(assigneeName), "*XEROX*")
select {
 deviceName: device.name,
 addresses: host.addresses,
 macAddress: host.macAddress,
 interfaces: host.interfaces,
 assigneeName
}
```

Please refer to the **Documentation** for more information on the `ouiAssignee(mac)` function.

.



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- summary routes not in allowed list

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# summary routes not in allowed list

## Find IPv4 summary routes that are not in a specific allowed list

The following check iterates through all IPv4 routes on all devices and returns those routes whose prefix length is less than 8 and whose prefix is not in a specific allowed list of summary subnets.

```
allowedSummarySubnets = [ipSubnet("224.0.0.0/4"),
ipSubnet("117.0.0.0/8")];

foreach device in network.devices
foreach vrf in device.networkInstances
where isPresent(vrf.afts.ipv4Unicast)
foreach ipEntry in vrf.afts.ipv4Unicast.ipEntries
where length(ipEntry.prefix) <= 8
select {
 device: device.name,
 vrf: vrf.name,
 prefix: ipEntry.prefix,
 violation: ipEntry.prefix not in allowedSummarySubnets
}
```

Note that the `length` function works on both lists and IP subnets, as in this example.

.



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- [NQE Language](#)
- [Examples](#)
- loopback subnets bigger than /32

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# loopback subnets bigger than /32

## Find loopback interfaces assigned subnets bigger than /32

The following check iterates through all subinterfaces and filters the results to return loopback interfaces where the assigned IP subnet is bigger than a /32.

```
foreach device in network.devices
foreach iface in device.interfaces
where iface.loopbackMode
foreach subIface in iface.subinterfaces
foreach addr in subIface.ipv4.addresses
select {
 deviceName: device.name,
 ifaceName: iface.name,
 subIfaceName: subIface.name,
 ip: addr.ip,
 prefixLength: addr.prefixLength,
 violation: addr.prefixLength < 32
}
```



.



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- [NQE Language](#)
- [Examples](#)
- interface MTU mismatch

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# interface MTU mismatch

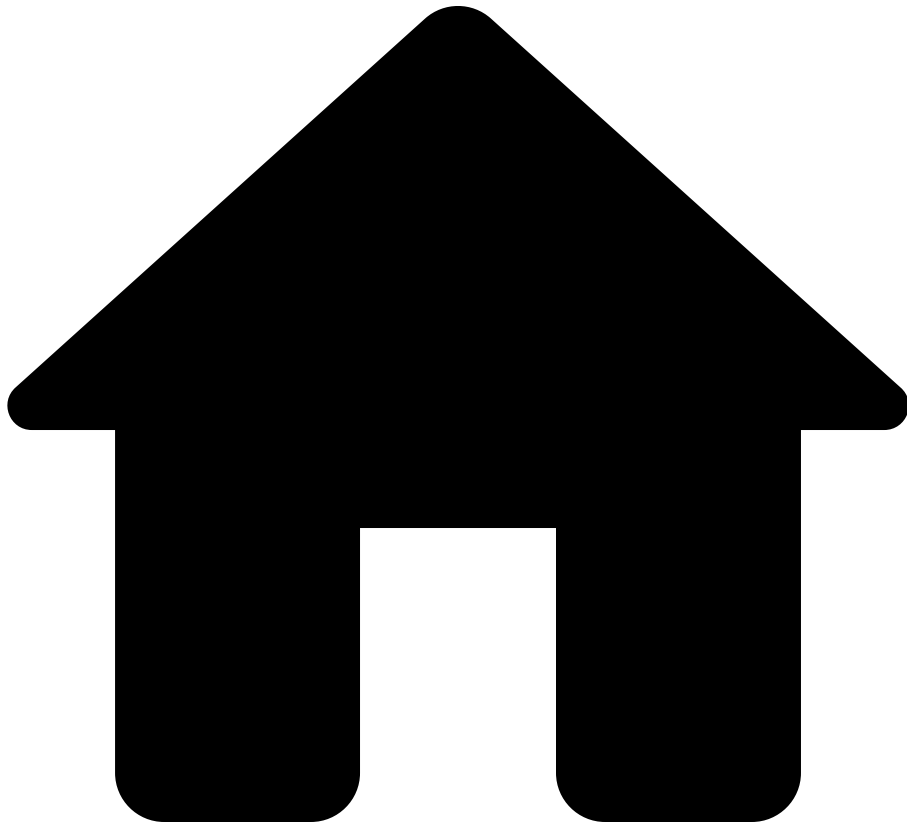
## Find connected interfaces that have mismatched MTU settings

[settings](#)

The following check iterates through all devices and interfaces, and the links from those interfaces. Then it looks up each connected interface with a second iteration, and then filters to include just those pairs of connected interfaces that have MTU values that are mismatched.

```
foreach d1 in network.devices
foreach i1 in d1.interfaces
foreach link in i1.links
foreach d2 in network.devices
where d2.name == link.deviceName
foreach i2 in d2.interfaces
where i2.name == link.ifaceName
where isPresent(i1.mtu) && isPresent(i2.mtu)
select {
 device1: d1.name,
 iface1: i1.name,
 mtu1: i1.mtu,
 device2: d2.name,
 iface2: i2.name,
 mtu2: i2.mtu,
 violation: i1.mtu != i2.mtu
}
```

.



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- [NQE Language](#)
- [Examples](#)
- IP Uniqueness

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## IP Uniqueness

### IP Uniqueness [ââ](#)

The following check finds instances of IP addresses that are assigned to multiple interfaces within a single VRF. The check iterates over devices and VRFs to get the VRF's

interfaces, and then looks up the referenced interfaces to get IP addresses on the VRF interfaces. Then, the check groups the data by the combination of VRF and IP address, with each group containing the locations at which the vrf and IP were found. The query then filters to those groups that have multiple locations. Finally, the diagnosis includes, for each VRF+IP combination that is located at multiple interfaces, the count of locations at which it was defined, as well as a comma-separated list of locations.

```
foreach device in network.devices
foreach vrf in device.networkInstances
foreach ifaceSubiface in vrf.interfaces
foreach iface in device.interfaces
foreach subIface in iface.subinterfaces
where iface.name == ifaceSubiface.ifaceName && subIface.name
== ifaceSubiface.subIfaceName
foreach address in subIface.ipv4.addresses
let location = {device: device.name, iface: iface.name}
group location as locations by {vrf: vrf.name, ip: address.ip}
as vrfIp
select {
 vrf: vrfIp.vrf,
 ip: vrfIp.ip,
 count: length(locations),
 locations: (foreach loc in locations select loc.device + ":"
+ loc.iface),
 violation: length(locations) > 1
}
```

Note that the expression `loc.device + ":" + loc.iface` in the `select` expression joins the device name, colon, and iface name (since these three expressions are all strings).

.



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- Device Group

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# Device Group

## All devices in a particular Device Group [↗](#)

```
foreach device in network.devices
where "MyDeviceGroup" in device.groupNames
select { deviceName: device.name }
```

## All devices in either of two particular groups: Union of Device Groups [↗](#)

```
foreach device in network.devices
where "MyFirstDeviceGroup" in device.groupNames ||
 "MySecondDeviceGroup" in device.groupNames
select { deviceName: device.name }
```

## All devices in both of two particular groups: Intersection of Device Groups [↗](#)

```
foreach device in network.devices
where "MyFirstDeviceGroup" in device.groupNames &&
 "MySecondDeviceGroup" in device.groupNames
select { deviceName: device.name }
```

.



- Reference
- [NQE Language](#)
- [Examples](#)
- Cisco NX-OS disallowed features

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# Cisco NX-OS disallowed features

## Find disallowed features that are enabled on Cisco NX-OS devices

This query locates instances of the `feature` config line in NXOS devices and extracts the named feature from these config lines. It checks the enabled features against a set of features that **should not** be enabled.

### Using `patternMatches`

```
disallowedFeatures = ["telnet", "scp-server"];

foreach device in network.devices
where device.platform.vendor == Vendor.CISCO &&
 device.platform.os == OS.NXOS
foreach match in patternMatches(device.files.config, `feature
{featureName:string}`)
let isDisallowed = match.data.featureName in
disallowedFeatures
select { Device: device.name,
 Feature: match.data.featureName,
 Fix: if isDisallowed then "no feature " +
match.data.featureName else "",
 violation: isDisallowed
 }
```

### Using `patternMatch`

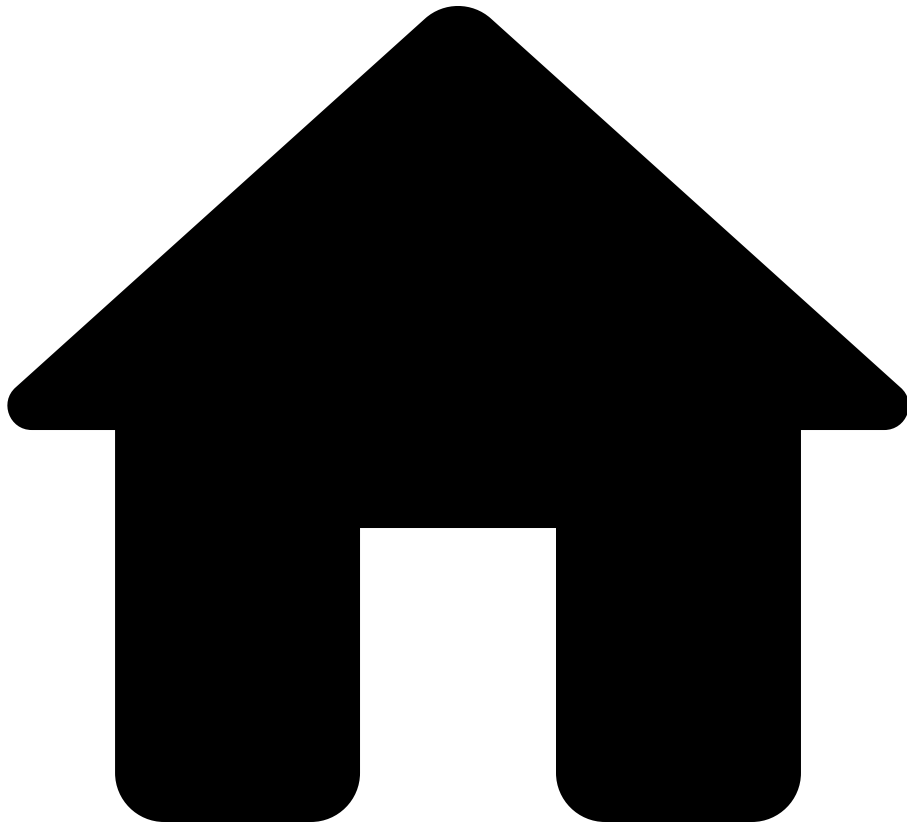
```
disallowedFeatures = ["telnet", "scp-server"];

foreach device in network.devices
where device.platform.vendor == Vendor.CISCO &&
 device.platform.os == OS.NXOS
foreach line in device.files.config
```



```
let match = patternMatch(line.text, `feature
{featureName:string}`)
where isPresent(match)
let isDisallowed = match.featureName in disallowedFeatures
select { Device: device.name,
 Feature: match.featureName,
 Fix: if isDisallowed then "no feature " +
match.featureName else "",
 violation: isDisallowed
 }
```

.



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- [NQE Language](#)
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- Cisco NX-OS static null routes

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## Cisco NX-OS static null routes

**Find all static null routes on Cisco NX-OS devices**[â](#)

This query illustrates that text matching patterns may include components such as `ipv4Subnet` that match specific types of entities, not just strings.

## Using patternMatches

```
foreach device in network.devices
where device.platform.vendor == Vendor.CISCO &&
 device.platform.os == OS.NXOS
let staticRoutes =
 (foreach match
 in patternMatches(device.files.config, `ip route
{prefix:ipv4Subnet} Null0`)
 select match.data.prefix)
select { device: device.name,
 prefixes: staticRoutes,
 violation: length(staticRoutes) > 0
 }
```

## Using patternMatch

```
foreach device in network.devices
where device.platform.vendor == Vendor.CISCO &&
 device.platform.os == OS.NXOS
let staticRoutes =
 (foreach line in device.files.config
 let match = patternMatch(line.text, `ip route
{prefix:ipv4Subnet} Null0`)
 where isPresent(match)
 select match.prefix)
select { device: device.name,
 prefixes: staticRoutes,
 violation: length(staticRoutes) > 0
 }
```

.



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- Cisco NX-OS interface slower than 1Gbps

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# Cisco NX-OS interface slower than 1Gbps

## Find all interface speeds slower than 1Gbps on Cisco NX-OS devices[â](#)

The following query shows how to find information in nested config lines.

### Using patternMatches[â](#)

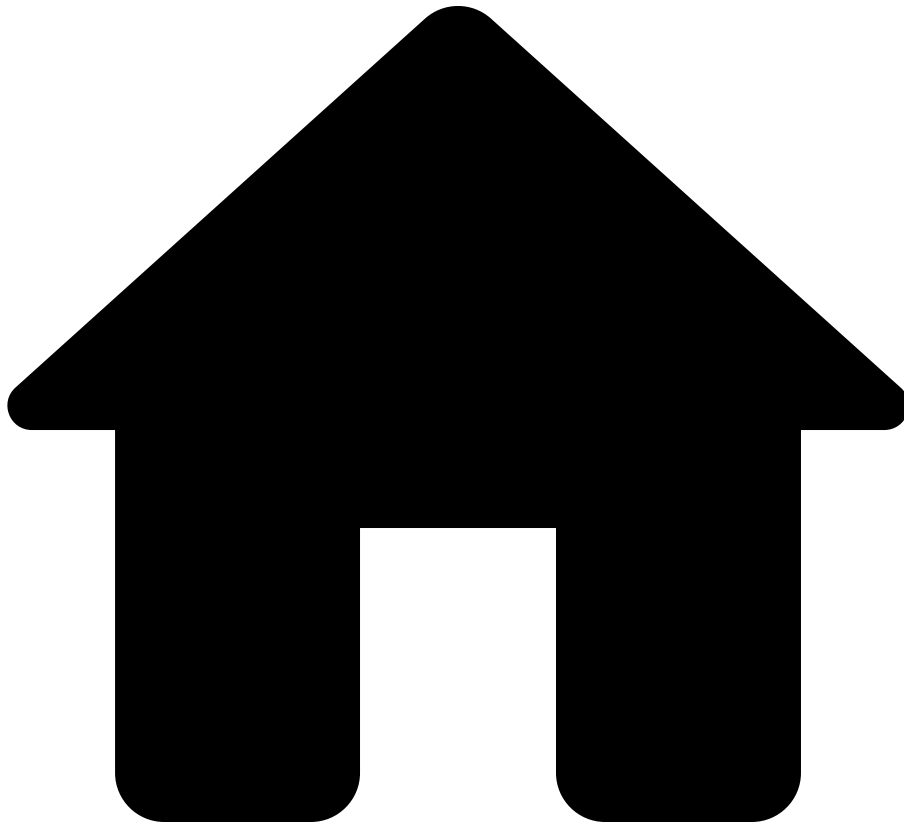
```
foreach device in network.devices
where device.platform.vendor == Vendor.CISCO &&
 device.platform.os == OS.NXOS
foreach ifaceMatch in patternMatches(device.files.config,
`interface {iface:string}`)
foreach speedMatch in patternMatches(ifaceMatch.line.children,
`speed {speed:number}`)
select {
 device: device.name,
 iface: ifaceMatch.data.iface,
 speed: speedMatch.data.speed,
 violation: speedMatch.data.speed < 1000
}
```

### Using patternMatch[â](#)

```
foreach device in network.devices
where device.platform.vendor == Vendor.CISCO &&
 device.platform.os == OS.NXOS
foreach line in device.files.config
let ifaceMatch = patternMatch(line.text, `interface
{iface:string}`)
where isPresent(ifaceMatch)
foreach childLine in line.children
let speedMatch = patternMatch(childLine.text, `speed
{speed:number}`)
```

```
where isPresent(speedMatch)
select {
 device: device.name,
 iface: ifaceMatch.iface,
 speed: speedMatch.speed,
 violation: speedMatch.speed < 1000
}
```

.



- Reference
- [NQE Language](#)
- [Examples](#)
- Devices with uptime over 6 months

## Devices with uptime over 6 months

The following query shows devices that have been up for at least 6 months. For every device, the output includes the device name, OS, snapshot status, uptime in seconds, and a formatted uptime string.

```
import "Time/Utils";
```

```
secondsIn6Months = 60 * 60 * 24 * 365 / 2;

formatStatus(deviceSnapshotResult) =
 when deviceSnapshotResult is
 collectionFailed(collectionError) ->
 "Collecting - " +
 replace(toString(collectionError),
"DeviceCollectionError.", "");
 completed -> "Completed";
 processingFailed(processingError) ->
 "Processing - " +
 replace(toString(processingError),
"DeviceProcessingError.", "");

foreach device in network.devices
let uptimeSeconds = device.system.uptimeSeconds
where uptimeSeconds > secondsIn6Months
select {
 deviceName: device.name,
 os: device.platform.os,
 Status: formatStatus(device.snapshotInfo.result),
 "Uptime Seconds": uptimeSeconds,
 Uptime: if isPresent(uptimeSeconds) then
formatSeconds(uptimeSeconds) else ""
}
```



## ðᳵᳵᳵᳵᳵᳵ AclAction

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ AclEntry

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ AdminStatus

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ AfiSafi

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ AfiSafiNeighbor

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ AfiSafiNeighborAdjRib

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ AfiSafiType

[Identity type for AFI,SAFI tuples for BGP-4](#)

## ðŸŒŽ Afts

The abstract forwarding tables (AFTs) that are associated with the network instance. An AFT is instantiated per-protocol running within the network-instance - such that one exists for IPv4 Unicast, IPv6 Unicast, MPLS, L2 forwarding entries, etc. A forwarding entry within the FIB has a set of next-hops, which may be a reference to an entry within another table - e.g., where a Layer 3 next-hop has an associated Layer 2 forwarding entry.

## ðŸŒŽ Aggregation

[Fields](#)

## ðŸŒŽ AllowedBgpPeersDevicePolicy

[Fields](#)

## ðŸŒŽ

## AllowedCertificationAuthorityEnrollmentUrlsDevicePolicy

[Fields](#)

## ðŸŒŽ AllowedServicesDevicePolicy

[Fields](#)

## ðŸŒŽ AnycastConfigurationDetail

[Fields](#)

## 🔗 AnycastService

[Fields](#)

## 🔗 AnycastServiceConfiguration

[Fields](#)

## 🔗 ArcsightConfig

[Fields](#)

## 🔗 ArpEntry

[Fields](#)

## 🔗 AsPath

[Fields](#)

## 🔗 AsType

[Options](#)

## 🔗 AttrSet

[Path attributes that may be in use by multiple routes in the table.](#)

## ðᳵᳵᳵᳵᳵ Backend

[Options](#)

## ðᳵᳵᳵᳵᳵ BackendBucket

[Fields](#)

## ðᳵᳵᳵᳵᳵ BackendServer

[Fields](#)

## ðᳵᳵᳵᳵᳵ Bgp

[Fields](#)

## ðᳵᳵᳵᳵᳵ BgpNeighbor

[Fields](#)

## ðᳵᳵᳵᳵᳵ BgpNeighborStatistics

[Fields](#)

## ðᳵᳵᳵᳵᳵ BgpOrigin

[Options](#)

## ðᳵᳵᳵᳵiᳵ BgpRib

[Defines a data model for representing BGP routing table contents.](#)

## ðᳵᳵᳵiᳵ BgpSessionState

[Options](#)

## ðᳵᳵᳵiᳵ BluecatConfig

[Fields](#)

## ðᳵᳵᳵiᳵ BluecatConfigItem

[Fields](#)

## ðᳵᳵᳵiᳵ BluecatConfigItemData

[Fields](#)

## ðᳵᳵᳵiᳵ BluecatOutputs

[Fields](#)

## ðᳵᳵᳵiᳵ BluecatRoute

[Fields](#)

## ðᳵᳵᳵᳵᳵ BluecatServerDataItem

[Fields](#)

## ðᳵᳵᳵᳵᳵ BluecatServerItem

[Fields](#)

## ðᳵᳵᳵᳵᳵ BluecatServers

[Fields](#)

## ðᳵᳵᳵᳵᳵ BluecatServices

[Fields](#)

## ðᳵᳵᳵᳵᳵ Bridge

[Bridge domain related data for a bridge interface.](#)

## ðᳵᳵᳵᳵᳵ CisaExploitationEntry

[An entry from CISA's Known Exploited Vulnerabilities \(KEV\) catalog.](#)

## ðᳵᳵᳵᳵᳵ CliCommandResponse

[Fields](#)

## ðᳵᳵᳵᳵᳵ CliSource

[Fields](#)

## ðᳵᳵᳵᳵᳵ CloudAccount

[Defines a data model for representing NQE cloud account with all related data.](#)

## ðᳵᳵᳵᳵᳵ CloudFirewall

[Defined within a VPC](#)

## ðᳵᳵᳵᳵᳵ CloudFirewallRule

[Fields](#)

## ðᳵᳵᳵᳵᳵ CloudFirewallRuleDefinition

[Fields](#)

## ðᳵᳵᳵᳵᳵ CloudIface

[Fields](#)

## ðᳵᳵᳵᳵᳵ CloudRoute

[Fields](#)

## ծရိုင်း CloudRouteNextHop

[Options](#)

## ծရိုင်း CloudRouteType

[Options](#)

## ծရိုင်း CloudType

[Options](#)

## ծရိုင်း ClusterHaInfo

[Fields](#)

## ծရိုင်း Command

[Fields](#)

## ծရိုင်း CommandType

[Options](#)

## ծရိုင်း ComputeInstance

[Fields](#)



## ðᳵᳵᳵᳵᳵᳵ ConfigLine

[Information about a single line in a device config file.](#)

## ðᳵᳵᳵᳵᳵᳵ ConnectsToAccessLayerSwitchInterfacePolicy

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ ConnectsToCoreLayerInterfacePolicy

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ ConnectsToCustomerEdgeDeviceInterfacePolicy

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ ConnectsToDodinDeviceInterfacePolicy

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ ConnectsViaFiberOpticsInterfacePolicy

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ ContainerType

[Options](#)

## ðŸŒŽi CVE

[Fields](#)

## ðŸŒŽi CVEApplicationCriteria

[Fields](#)

## ðŸŒŽi CVEDatabase

[Fields](#)

## ðŸŒŽi CVEFinding

[Fields](#)

## ðŸŒŽi CVEFindingBasis

[Options](#)

## ðŸŒŽi CVEHardwareCriteria

[Fields](#)

## ðŸŒŽi CVEOsCriteria

[Fields](#)

## ðŸŒŽ CveProductCriteria

[Fields](#)

## ðŸŒŽ DataConnector

[Fields](#)

## ðŸŒŽ DcGateway

[Fields](#)

## ðŸŒŽ DefaultInterfaceParameters

[Fields](#)

## ðŸŒŽ DefaultStigParameters

[Fields](#)

## ðŸŒŽ DefaultVrfParameters

[Fields](#)

## ðŸŒŽ Device

[Fields](#)

## ❌❌❌❌ DeviceCollectionError

[Options](#)

## ❌❌❌❌ DeviceDiscoveryIfaceInfo

[Fields](#)

## ❌❌❌❌ DeviceDiscoveryNeighbor

[Information about a CDP or LLDP neighbor. Note that the neighbor may not be a collected device. For example, it may be an end host.](#)

## ❌❌❌❌ DeviceHost

[Fields](#)

## ❌❌❌❌ DeviceHostType

[Options](#)

## ❌❌❌❌ DeviceL3IfaceName

[Fields](#)

## ❌❌❌❌ DeviceParameters

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ DevicePart

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ DevicePartSupport

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ DevicePartType

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ DevicePolicy

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ DeviceProcessingError

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ DeviceSecurityZone

[A security zone on a device.](#)

## ðᳵᳵᳵᳵᳵᳵ DeviceSecurityZoneReference

[Reference of a security zone on a firewall](#)

## ðŸ“œ DeviceSnapshotInfo

[Fields](#)

## ðŸ“œ DeviceSnapshotResult

[Options](#)

## ðŸ“œ DeviceType

[Classification of networking devices. Common types include ROUTER, SWITCH, and FIREWALL.](#)

## ðŸ“œ DirectConnection

[Fields](#)

## ðŸ“œ DnsActivityConfigurationDetail

[Fields](#)

## ðŸ“œ DnsActivityService

[Fields](#)

## ðŸ“œ DnsActivityServiceConfiguration

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ DnsResolverConfigurationDetail

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ DnsResolverService

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ DnsResolverServiceConfiguration

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ DuplexMode

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ EdgeServicePointConfigurationDetail

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ EdgeServicePointService

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ EdgeServicePointServiceConfiguration

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ EncapsulationHeaderType

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ Ethernet

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ ExternalSources

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ FhrpState

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ Files

[Information about config data sources for a device.](#)

## ðᳵᳵᳵᳵᳵᳵ FirewallConfigurationDetail

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ FirewallPolicy

[Fields](#)



## ðŸŒŽ FirewallRule

[Fields](#)

## ðŸŒŽ FirewallService

[Fields](#)

## ðŸŒŽ FirewallServiceConfiguration

[Fields](#)

## ðŸŒŽ GatewayConfigurationDetail

[Fields](#)

## ðŸŒŽ GatewayConfigurationDetailData

[Fields](#)

## ðŸŒŽ GatewayContainer

[Fields](#)

## ðŸŒŽ GatewayService

[Fields](#)

## ðᳵᳵᳵᳵᳵ GatewayServiceConfiguration

[Fields](#)

## ðᳵᳵᳵᳵᳵ GlobalAccelerator

[Fields](#)

## ðᳵᳵᳵᳵᳵ GlobalAcceleratorListener

[Fields](#)

## ðᳵᳵᳵᳵᳵ GlobalAcceleratorListenerEndpoint

[Options](#)

## ðᳵᳵᳵᳵᳵ GlobalAcceleratorListenerEndpointGroup

[Fields](#)

## ðᳵᳵᳵᳵᳵ GlobalAcceleratorStatus

[Options](#)

## ðᳵᳵᳵᳵᳵ Ha

[Fields](#)

## ձևաչափ HaOperationMode

[Options](#)

## ձևաչափ HeaderRegion

[Fields](#)

## ձևաչափ HeaderRewrite

[Fields](#)

## ձևաչափ Hop

[Represents a hop from a device to a neighboring device. If the hop is the last hop in a path, egressIface, nextDevice, and nextIface are set to null.](#)

## ձևաչափ HttpApiEndpointResult

[Fields](#)

## ձևաչափ HttpApiEndpointStatus

[Options](#)

## ձևաչափ HttpEndpoint

[Fields](#)

## ծရိုင်း HttpPage

[Fields](#)

## ծရိုင်း HttpSource

[Fields](#)

## ծရိုင်း Iface

[Fields](#)

## ծရိုင်း IfaceAcls

[Access control list details for an interface.](#)

## ծရိုင်း IfaceFhrp

[Fields](#)

## ծရိုင်း IfaceFhrpGroup

[Fields](#)

## ծရိုင်း IfaceFhrpInfo

[Fields](#)

## ၎်း၎်း၎်း၎်း IfaceIpv4Info

[The IPv4 addresses configured on an interface.](#)

## ၎်း၎်း၎်း၎်း IfaceIpv6Info

[The IPv6 addresses configured on an interface.](#)

## ၎်း၎်း၎်း၎်း IfaceSubIface

[The names of the interface and subinterface that are associated with the network instance.](#)

## ၎်း၎်း၎်း၎်း IfaceType

[Options](#)

## ၎်း၎်း၎်း၎်း InetGateway

[Fields](#)

## ၎်း၎်း၎်း၎်း Instancelface

[Fields](#)

## ၎်း၎်း၎်း၎်း InterfaceConnectivityParameters

[Fields](#)

## ❖❖❖❖❖ InterfaceConnectivityPolicy

[Fields](#)

## ❖❖❖❖❖ InterfacePolicy

[Fields](#)

## ❖❖❖❖❖ InterfacesService

[Fields](#)

## ❖❖❖❖❖ InterfacesServiceConfiguration

[Fields](#)

## ❖❖❖❖❖ InterfacesServiceConfigurationDetail

[Fields](#)

## ❖❖❖❖❖ IntRange

[Fields](#)

## ❖❖❖❖❖ IpEntry

[Fields](#)

## ðŸŒ˜i, IpUnicast

[Fields](#)

## ðŸŒ˜i, Ipv4Address

[Fields](#)

## ðŸŒ˜i, Ipv6Address

[Fields](#)

## ðŸŒ˜i, IsExternalInterfacePolicy

[Fields](#)

## ðŸŒ˜i, IsHostGatewayInterfacePolicy

[Fields](#)

## ðŸŒ˜i, IsProviderEdgeDevicePolicy

[Fields](#)

## ðŸŒ˜i, IsUntrustedInterfacePolicy

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ KeepAliveStatus

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ Layer

[Network layer in the OSI model.](#)

## ðᳵᳵᳵᳵᳵᳵ LifecycleData

[Firewall hit count and timestamps for their creation, last modification, and last used. Present for systems that were succesfully collected and have one of these operating systems: SRX, PAN\\_OS, CHECKPOINT, FORTINET and ASA, if reported by the operating system. At least one of the fields will be non-null.](#)

## ðᳵᳵᳵᳵᳵᳵ LoadBalancer

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ LoadBalancerRule

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ Location

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ MacEntry



[Fields](#)

## ðŸŒ˜i MacEntryInterfaceInfo

[Fields](#)

## ðŸŒ˜i MacEntrySourceType

[Options](#)

## ðŸŒ˜i MacEntryType

[Options](#)

## ðŸŒ˜i ManagementSubnetsDevicePolicy

[Fields](#)

## ðŸŒ˜i MetadataMatches

[Fields](#)

## ðŸŒ˜i MstInstance

[Fields](#)

## ðŸŒ˜i Mstp

[Multi Spanning-tree protocol configuration and operation data.](#)

## ճՃՃՃի՝ MulticastMode

[Options](#)

## ճՃՃՃի՝ NamedVlanCollection

[Fields](#)

## ճՃՃՃի՝ NatEntry

[Fields](#)

## ճՃՃՃի՝ NatGateway

[Fields](#)

## ճՃՃՃի՝ NatType

[Options](#)

## ճՃՃՃի՝ Network

[Fields](#)

## ճՃՃՃի՝ NetworkAcl

[Fields](#)

## NetworkEndpoint

[Fields](#)

## NetworkExtensions

[Contains user-defined extensions to the standard Network data model. By default, this type has no fields. In any Forward Instance, this type will have one field for each Data File defined.](#)

## NetworkInstance

[Fields](#)

## NetworkInstanceFdb

[Fields](#)

## NetworkInstanceProtocol

[Fields](#)

## NetworkInstanceType

[Options](#)

## NetworkObject

[Fields](#)

## NextHop

[Fields](#)

## NextHopType

[Options](#)

## NtpConfigurationDetail

[Fields](#)

## NtpServer

[Fields](#)

## NtpService

[Fields](#)

## NtpServiceConfiguration

[Fields](#)



## ձևաչափ Ospf

[Fields](#)

## ձևաչափ OspfArea

[Fields](#)

## ձևաչափ OspfAreaType

[Options](#)

## ձևաչափ OspfNeighbor

[Fields](#)

## ձևաչափ OspfRemotePeer

[Fields](#)

## ձևաչափ OspfRole

[Options](#)

## ձևաչափ OspfRouterType

[Options](#)

## OsSupport

[Contains dates for some of the OS support milestones of this device.](#)

## OutboundRule

[Fields](#)

## Outputs

[Fields](#)

## PaginatedHttpEndpoint

[Fields](#)

## PaginatedHttpEndpointStatus

[Options](#)

## PanosFiles

[Fields](#)

## Path

[Represents a path \(series of hops\) from one device to another.](#)

## ðᳵᳵᳵᳵᳵᳵ PbrAction

[PBR action data will be added in upcoming releases](#)

## ðᳵᳵᳵᳵᳵᳵ PeerLink

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ PeerStatus

[Options](#)

## ðᳵᳵᳵᳵᳵᳵ PeerType

[Labels a peer or peer group as explicitly internal or external.](#)

## ðᳵᳵᳵᳵᳵᳵ PhysicalLink

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ Platform

[Fields](#)

## ðᳵᳵᳵᳵᳵᳵ PortSpeed

[Type to specify available Ethernet link speeds.](#)



## ðŸŒ˜i PublicUnallocatedIp

[A public unallocated IPv4 address and related metadata.](#)

## ðŸŒ˜i PublicUnallocatedIpSubnet

[A public unallocated IPv4 subnet and related metadata.](#)

## ðŸŒ˜i QRadarConfig

[Fields](#)

## ðŸŒ˜i RapidPvst

[Rapid per vlan Spanning-tree protocol configuration and operation data.](#)

## ðŸŒ˜i RapidPvstVlan

[Rapid Spanning-tree protocol configuration and operation data.](#)

## ðŸŒ˜i RefObj

[Fields](#)

## ðŸŒ˜i RefObjType

[Options](#)

## RequiresDefaultVlanInterfacePolicy

[Fields](#)

## RequiresMulticastInterfacePolicy

[Fields](#)

## Route

[Fields](#)

## RoutedVlan

[Fields](#)

## RouteTable

[Fields](#)

## RoutingProtocolType

[Type for routing protocols, including those which may install prefixes into the RIB](#)

## Rstp

[Rapid Spanning-tree protocol configuration and operation data.](#)

## ծရိုင်း RuleContainer

[The origin of a firewall rule.](#)

## ծရိုင်း SecurityGroup

[Fields](#)

## ծရိုင်း SecurityRule

[Fields](#)

## ծရိုင်း SecurityRuleAction

[Options](#)

## ծရိုင်း ServiceEndpoint

[Fields](#)

## ծရိုင်း Severity

[Options](#)

## ծရိုင်း SnmpAgentService

[Fields](#)

## SnmpAgentServiceSystem

[Fields](#)

## SnmpAgentV1

[Fields](#)

## SnmpAgentV2C

[Fields](#)

## SnmpAgentV3

[Fields](#)

## SnmpConfigurationDetail

[Fields](#)

## SnmpService

[Fields](#)

## SnmpServiceConfiguration

[Fields](#)

## ၎်း၎်း၎်း၎်း SnmpSource

[Fields](#)

## ၎်း၎်း၎်း၎်း SnmpTrapServer

[Fields](#)

## ၎်း၎်း၎်း၎်း SnmpTrapService

[Fields](#)

## ၎်း၎်း၎်း၎်း SshConfigurationDetail

[Fields](#)

## ၎်း၎်း၎်း၎်း SshService

[Fields](#)

## ၎်း၎်း၎်း၎်း SshServiceConfiguration

[Fields](#)

## ၎်း၎်း၎်း၎်း SshTacacsConfig

[Fields](#)

## ðᳵᳵᳵᳵᳵ StigDatabase

[Fields](#)

## ðᳵᳵᳵᳵᳵ StigGuide

[Fields](#)

## ðᳵᳵᳵᳵᳵ StigParameterUsage

[Fields](#)

## ðᳵᳵᳵᳵᳵ StigPolicy

[Fields](#)

## ðᳵᳵᳵᳵᳵ StigPolicyCondition

[The condition \(a set of constraints\) under which a policy applies to a device in the network. The condition is true for a device when all constraint fields on this set are satisfied.](#)

## ðᳵᳵᳵᳵᳵ StigRule

[Fields](#)

## ðᳵᳵᳵᳵᳵ StigRuleCollection

[Fields](#)

## StigSeverity

[Options](#)

## Stp

[Top-level container for spanning tree configuration and state data.](#)

## SubInterface

[Subinterface data for logical interfaces associated with a given interface.](#)

## SubInterfaceVlan

[VLAN id \(single or Q-in-Q\) for a subinterface. The id is scoped to the subinterface, and could be repeated on different subinterfaces.](#)

## Subnet

[Fields](#)

## SwitchedSubInterfaceVlans

[Fields](#)

## SwitchedVlan

[Type for VLAN interface-specific data on Ethernet interfaces. These are for standard L2, switched-style VLANs.](#)

## SyslogConfigurationDetail

[Fields](#)

## SyslogServer

[Fields](#)

## SyslogService

[Fields](#)

## SyslogServiceConfiguration

[Fields](#)

## System

[Fields](#)

## TeTunnel

[Configuraton parameters relating to a TE-tunnel interface.](#)

## TeTunnelNextHop



[Fields](#)

## ðᳵᳵᳵᳵᳵ TransitGateway

[Fields](#)

## ðᳵᳵᳵᳵᳵ TransitGatewayAttachment

[Fields](#)

## ðᳵᳵᳵᳵᳵ TransitGatewayAttachmentType

[Options](#)

## ðᳵᳵᳵᳵᳵ TransitGatewayRoute

[Fields](#)

## ðᳵᳵᳵᳵᳵ TransitGatewayRouteTable

[Fields](#)

## ðᳵᳵᳵᳵᳵ Tunnel

[Configuraton parameters relating to a tunnel interface.](#)

## ðᳵᳵᳵᳵᳵ Url

[Fields](#)

## ðŸŒ˜ Vendor

[Options](#)

## ðŸŒ˜ VendorCveInfo

[Fields](#)

## ðŸŒ˜ VirtualCircuitIdInterfacePolicy

[Fields](#)

## ðŸŒ˜ Vlan

[Fields](#)

## ðŸŒ˜ VlanIdOrRange

[Options](#)

## ðŸŒ˜ VlanIdSet

[Fields](#)

## ðŸŒ˜ VlanIdSetOrAll

[Options](#)

## ծំծំծំi VlanModeType

[VLAN interface mode \(trunk or access\).](#)

## ծំծំծំi VlanPair

[Fields](#)

## ծំծំծំi VlanRange

[Fields](#)

## ծំծំծំi Vpc

[Fields](#)

## ծំծំծំi VpcData

[Fields](#)

## ծំծំծំi VpcDomain

[Fields](#)

## ծំծំծំi VpcPeering

[Fields](#)

## ծရိုင်း VpcRole

[Options](#)

## ծရိုင်း VpnConnection

[Fields](#)

## ծရိုင်း VpnGateway

[Fields](#)

## ծရိုင်း VrfPolicy

[Fields](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AclAction

On this page

# AclAction

## Options

name	type	description
DENY	--	--
PBR	<a href="#">PbrAction</a>	--
PERMIT	--	--

## Used by

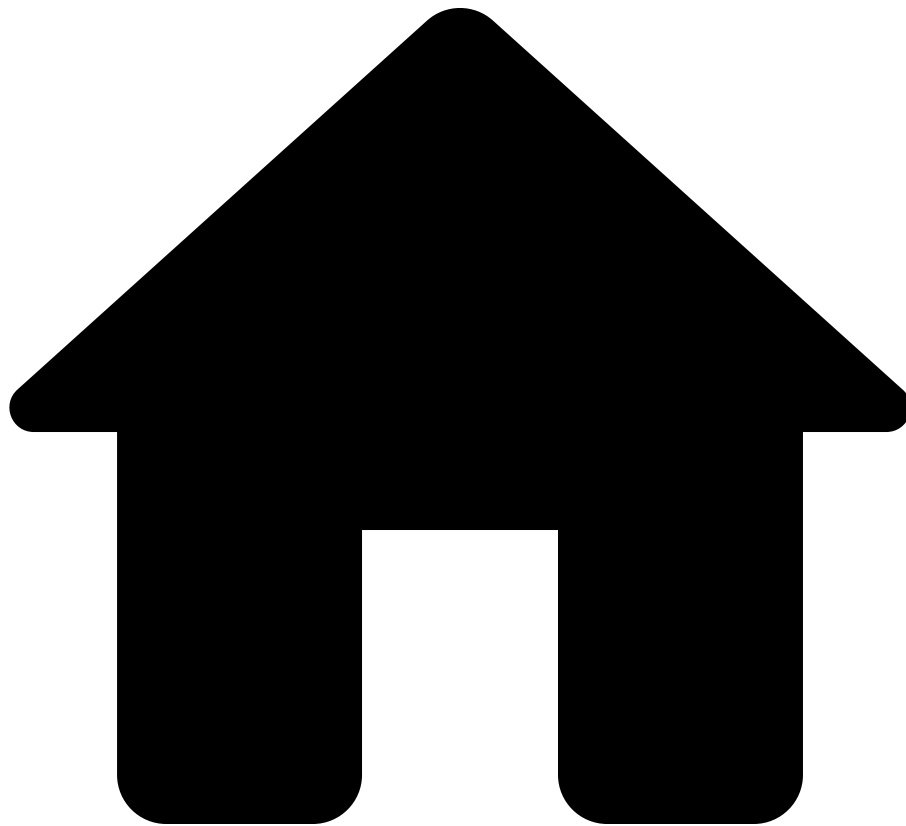
**network** **devices** **aclEntries** **action**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - action: [AclAction](#)

.



- Reference
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- AclEntry

On this page

# AclEntry

## Fields

name	type	nullable	description
headerMatches	<a href="#">HeaderRegion</a>	false	--
metadataMatches	<a href="#">MetadataMatches</a>	false	--
action	<a href="#">AclAction</a>	false	--
name	String	false	--
uuid	String	true	The UUID of this entry's security rule. Present for operating systems CHECKPOINT, FORTINET, and PAN_OS.
description	String	true	A user-defined description of the ACL entry. Only present on PanOS devices.
implicitRule	Bool	false	Indicates whether an entry is added to the Forward Networks model to capture some aspects of device behavior (e.g. security-level in ASA), as opposed to entries for ACL rules explicitly configured on the



name	type	nullable	description
			device. This is false for default rules (see defaultRule).
defaultRule	Bool	false	Whether this is a default rule.
lifecycleData	<a href="#">LifecycleData</a>	true	Stores lifecycle metadata, including timestamps for creation, last use, and modification, along with a hit count, if reported by the operating system. Only applicable to user-configured ACL entries (i.e. not for entries with implicitRule is true or defaultRule is true. Present for systems that were successfully collected and have one of these operating systems : SRX, PAN_OS, CHECKPOINT, FORTINET and ASA, if reported by the operating system.

## Used by[â](#)

**network** [â](#) **devices** [â](#) **aclEntries**[â](#)

**See also**[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >

.



- Reference
- [NQE Language](#)
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- AdminStatus

On this page

# AdminStatus

## Options

name	description
DOWN	Not ready to pass packets.
UP	Ready to pass packets.

## Used by

**network** **devices** **interfaces** **adminStatus**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - adminStatus: [AdminStatus](#)

**network** **devices** **interfaces** **subinterfaces** **adminStatus**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - adminStatus: [AdminStatus](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Afisafi

On this page

# AfiSafi

## Fields

name	type	nullable	description
afiSafiName	AfiSafiType	false	The afi-safi type.
neighbors	Bag< AfiSafiNeighbor >	false	List of neighbors (peers) of the local BGP speaker.

## Used by

network devices bgpRib afiSafis

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - bgpRib: BgpRib
      - afiSafis: Bag< AfiSafi >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AfisafiNeighbor

On this page

# AfiSafiNeighbor

## Fields

name	type	nullable	description
neighborAddress	IpAddress	false	The address of the neighbor (peer).
adjRibInPost	<a href="#">AfiSafiNeighborAdjRib</a>	true	This is a per-neighbor table containing the routes received from the neighbor that are eligible for best-path selection after local input policy rules have been applied. Only available on JunOs, IOS, IOS-XE, NX-OS, and IOS-XR. Note that on some platforms, these routes may not show import policy modifications.



name	type	nullable	description
adjRibOutPost	<a href="#">AfiSafiNeighborAdjRib</a>	true	Per-neighbor table containing paths eligible for sending (advertising) to the neighbor after output policy rules have been applied. Only available on JunOs, IOS, IOS-XE, NX-OS, and IOS-XR.

Used by

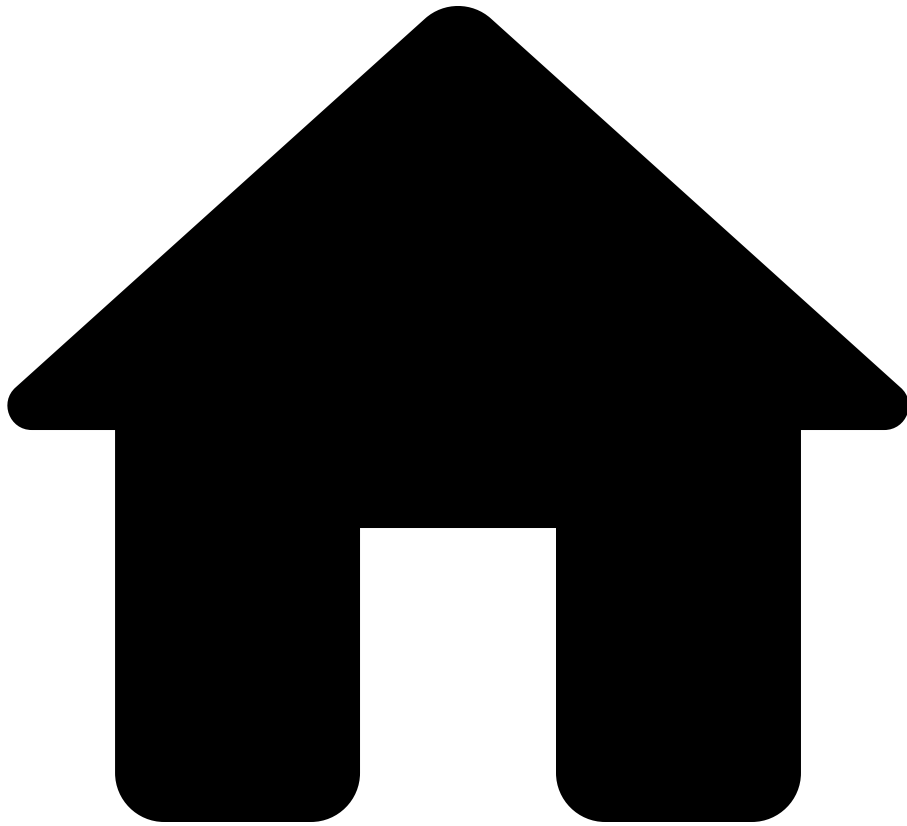
network devices bgpRib afiSafis neighbors

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AfisafiNeighborAdjRib

On this page

# AfiSafiNeighborAdjRib

## Fields

name	type	nullable	description
routes	Bag< <a href="#">Route</a> >	false	List of routes in the table, keyed by vrf, route distinguisher and route prefix.

## Used by

network devices bgpRib afiSafis neighbors adjRibInPost

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibInPost: [AfiSafiNeighborAdjRib](#)

**network** **devices** **bgpRib** **afiSafis** **neighbors** **adjRibOutPost**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibOutPost: [AfiSafiNeighborAdjRib](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AfISafIType

On this page

## AfISafIType

Identity type for AFI,SAFI tuples for BGP-4

## Options

name	description
IPV4_ANY	IPv4 unicast and multicast, applies to Juniper only (AFI, SAFI = 1, 1/2)
IPV4_LABELED_UNICAST	Labeled IPv4 unicast (AFI,SAFI = 1,4)
IPV4_MCAST_VPN	IPv4 MCAST-VPN (AFI,SAFI = 1,5)
IPV4_MDT	BGP Multicast Distribution Tree (MDT) for IPv4 (AFI, SAFI = 1, 66)
IPV4_MULTICAST	IPv4 multicast (AFI,SAFI = 1,2)
IPV4_UNICAST	IPv4 unicast (AFI,SAFI = 1,1)
IPV6_ANY	IPv6 unicast and multicast, applies to Juniper only (AFI, SAFI = 1, 1/2)
IPV6_LABELED_UNICAST	Labeled IPv6 unicast (AFI,SAFI = 2,4)
IPV6_MCAST_VPN	IPv6 MCAST-VPN (AFI,SAFI = 2,6)
IPV6_MDT	BGP Multicast Distribution Tree (MDT) for IPv4 (AFI, SAFI = 2, 66)
IPV6_MULTICAST	IPv6 multicast (AFI,SAFI = 2,2)
IPV6_UNICAST	IPv6 unicast (AFI,SAFI = 2,1)
L2VPN_EVPN	BGP MPLS Based Ethernet VPN (AFI,SAFI = 25,70)
L2VPN_VPLS	BGP-signalled VPLS (AFI,SAFI = 25,65)
L3VPN_IPV4_ANY	Unicast and multicast IPv4 MPLS L3VPN, applies to Juniper only (AFI,SAFI = 1,128/129)
L3VPN_IPV4_MULTICAST	Multicast IPv4 MPLS L3VPN (AFI,SAFI = 1,129)
L3VPN_IPV4_UNICAST	Unicast IPv4 MPLS L3VPN (AFI,SAFI = 1,128)
L3VPN_IPV6_ANY	Unicast and multicast IPv6 MPLS L3VPN, applies to Juniper only (AFI,SAFI = 2,128/129)

name	description
L3VPN_IPV6_MULTICAST	Multicast IPv6 MPLS L3VPN (AFI,SAFI = 2,129)
L3VPN_IPV6_UNICAST	Unicast IPv6 MPLS L3VPN (AFI,SAFI = 2,128)
LINK_STATE	Link state (AFI, SAFI = 16388, 71)
NSAP_UNICAST	NSAP NLRI for unicast forwarding (AFI,SAFI = 3,1)
RTFILTER_UNICAST	IPv4 Route Target constrains (AFI,SAFI = 1,132)
SRTE_POLICY_IPV4	Segment Routing Traffic Engineering (SRTE) Policy for IPv4 (AFI,SAFI = 1,73)
SRTE_POLICY_IPV6	Segment Routing Traffic Engineering (SRTE) Policy for IPv6 (AFI,SAFI = 2,73)

## Used by[â](#)

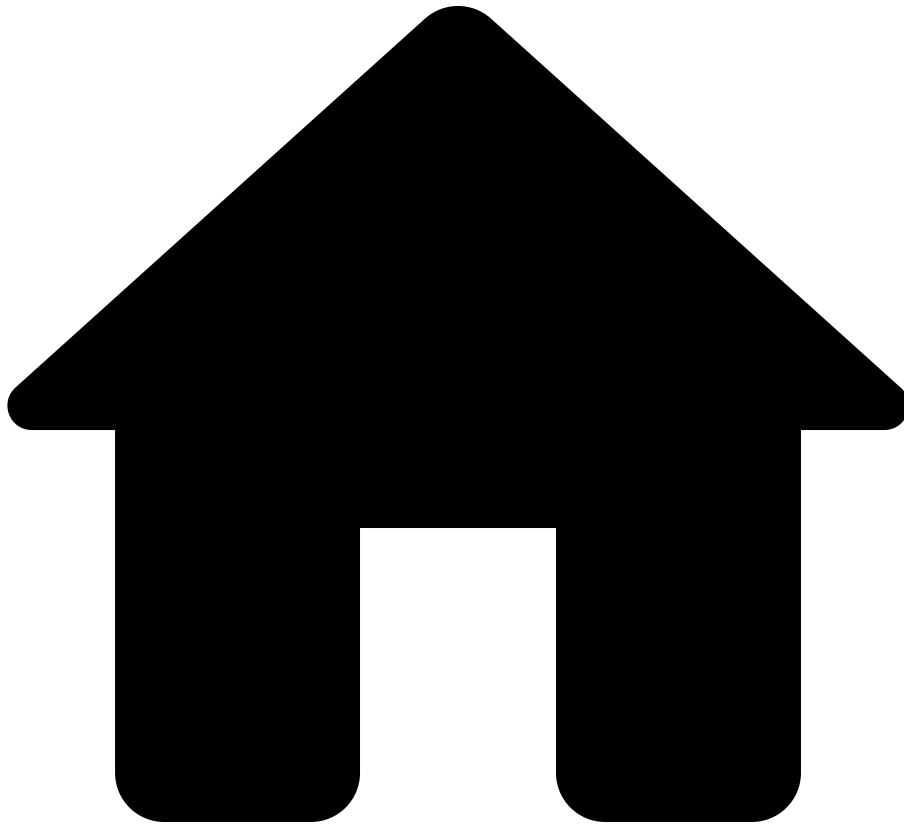
**network** [â](#) **devices** [â](#) **bgpRib** [â](#) **afiSafis** [â](#) **afiSafiName**[â](#)

**See also**[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - afiSafiName: [AfiSafiType](#)

- 



- Reference
- [NQE Language](#)
- [Data Model](#)
- Afts

On this page

## Afts

The abstract forwarding tables (AFTs) that are associated with the network instance. An AFT is instantiated per-protocol running within the network-instance - such that one exists for IPv4 Unicast, IPv6 Unicast, MPLS, L2 forwarding entries, etc. A forwarding entry within the FIB has a set of next-hops, which may be a reference to an entry within



another table - e.g., where a Layer 3 next-hop has an associated Layer 2 forwarding entry.

Fields

name	type	nullable	description
ipv4Unicast	<a href="#">IpUnicast</a>	true	The abstract forwarding table for IPv4 unicast. Entries within this table are uniquely keyed on the IPv4 unicast destination prefix which is matched by ingress packets. The data set represented by the IPv4 Unicast AFT is the set of entries from the IPv4 unicast RIB that have been selected for installation into the FIB of the device.
ipv6Unicast	<a href="#">IpUnicast</a>	true	The abstract forwarding table for IPv6 unicast. Entries within this table are uniquely keyed on the IPv6 unicast destination prefix which is matched by ingress packets. The data set represented by the IPv6 Unicast AFT is the set of entries from the IPv6 unicast RIB that have been selected for installation into the FIB of the device.

## Used by[â](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Aggregation

On this page

# Aggregation

## Fields

name	type	nullable	description
memberNames	Bag< String >	false	List of actual, current member interfaces for the aggregate, expressed as references to existing interfaces.
configuredMemberNames	Bag< String >	false	List of configured member interfaces for the aggregate.
switchedVlan	<a href="#">SwitchedVlan</a>	true	--

## Used by

network devices interfaces aggregation

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - aggregation: [Aggregation](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AllowedBgpPeersDevicePolicy

On this page

# AllowedBgpPeersDevicePolicy

## Fields

name	type	nullable	description
condition	<a href="#">StigPolicyCondition</a>	false	--
value	Bag< <a href="#">IpSubnet</a> >	false	The list of IP addresses or subnets of allowed BGP neighbors. Allowed BGP neighbors must be DOD devices. BGP peering with commercial ISPs is disallowed.

## Used by

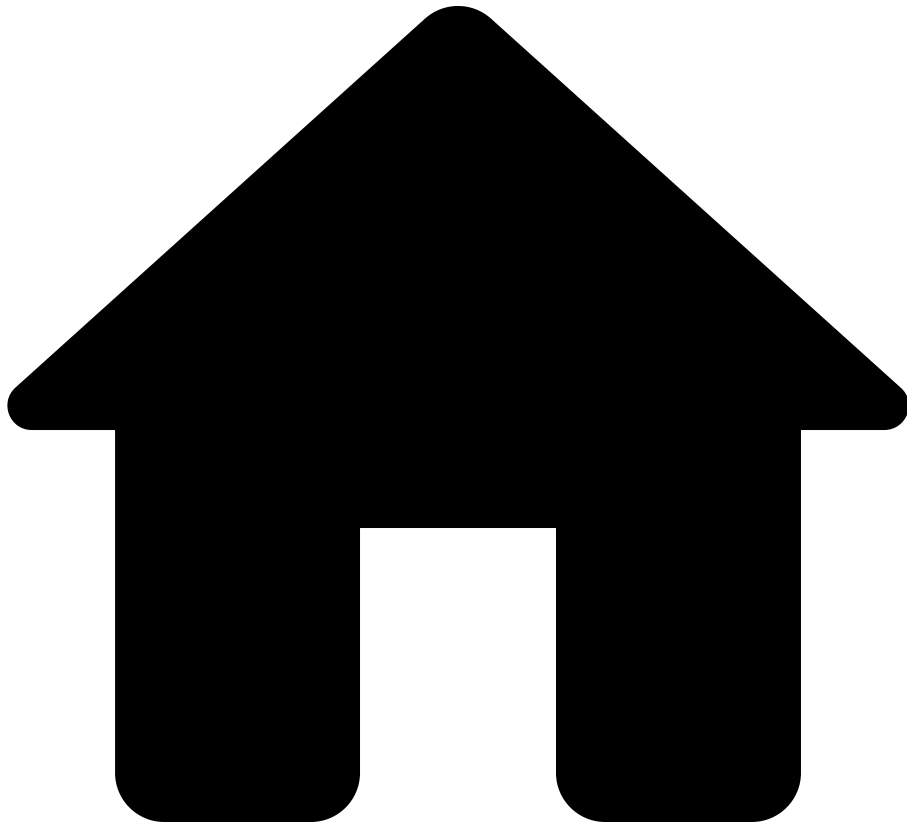
network [StigDatabase](#) policy [devicePolicy](#) [allowedBgpPeers](#)

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - devicePolicy: [DevicePolicy](#)
        - allowedBgpPeers:  
Bag< [AllowedBgpPeersDevicePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AllowedCertificationAuthorityEnrollmentUrlsDevicePolicy

On this page

# AllowedCertificationAuthorityEnrollmentUrlsDevice

## Fields

name	type	nullable	description
condition	<a href="#">StigPolicyCondition</a>	false	--
value	Bag< String >	false	The list of allowed Certification Authority (CA) enrollment URLs.

## Used by

network stigDatabase policy devicePolicy  
allowedCertificationAuthorityEnrollmentUrls

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - devicePolicy: [DevicePolicy](#)
        - allowedCertificationAuthorityEnrollmentUrls:  
Bag< [AllowedCertificationAuthorityEnrollmentUrlsDevice](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AllowedServicesDevicePolicy

On this page

# AllowedServicesDevicePolicy

## Fields

name	type	nullable	description
condition	<a href="#">StigPolicyCondition</a>	false	--
value	Bag< String >	false	The list of services that are allowed to run on the device.

## Used by

network [stigDatabase](#) [policy](#) [devicePolicy](#) [allowedServices](#)

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - devicePolicy: [DevicePolicy](#)
        - allowedServices:  
Bag< [AllowedServicesDevicePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AnycastConfigurationDetail

On this page

# AnycastConfigurationDetail

## Fields

name	type	nullable	description
enable	Bool	false	--

## Used by

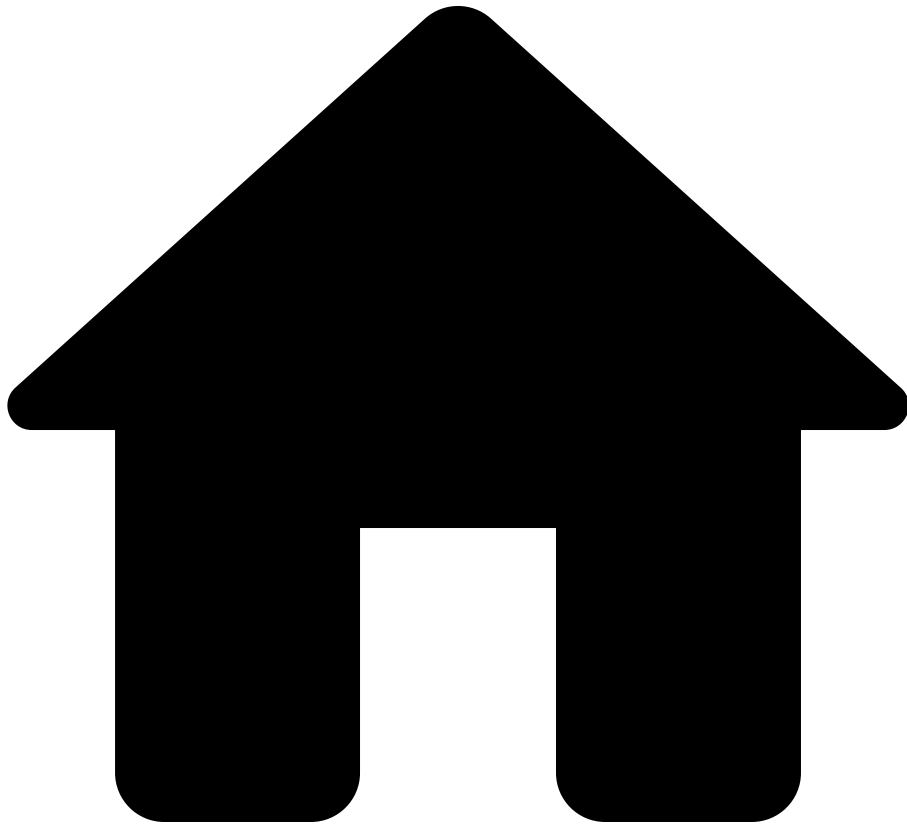
network devices outputs bluecat config items data services anycast configurations anycastConfiguration

## See also

Data model path

- network: Network
  - devices: Bag< Device >
    - outputs: Outputs
      - bluecat: BluecatOutputs
        - config: BluecatConfig
          - items: Bag< BluecatConfigItem >
            - data: BluecatConfigItemData
              - services: BluecatServices
                - anycast: AnycastService
                  - configurations: Bag< AnycastServiceConfiguration >
                    - anycastConfiguration: AnycastConfigurationDetail

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AnycastService

On this page

# AnycastService

## Fields

| name           | type                                               | nullable | description |
|----------------|----------------------------------------------------|----------|-------------|
| configurations | Bag< <a href="#">AnycastServiceConfiguration</a> > | false    | --          |

## Used by

network devices outputs bluecat config items data  
services anycast

### See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - anycast: [AnycastService](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AnycastServiceConfiguration

On this page

# AnycastServiceConfiguration

## Fields

| name                 | type                                       | nullable | description |
|----------------------|--------------------------------------------|----------|-------------|
| anycastConfiguration | <a href="#">AnycastConfigurationDetail</a> | false    | --          |

## Used by

network devices outputs bluecat config items data  
services anycast configurations

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - anycast: [AnycastService](#)
                  - configurations: Bag< [AnycastServiceConfiguration](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ArcsightConfig

On this page

# ArcsightConfig

## Fields

| name   | type | nullable | description |
|--------|------|----------|-------------|
| enable | Bool | false    | --          |

Used by [100](#)

network & devices & outputs & bluecat & config & items & data  
& services & syslog & configurations & syslogConfiguration &  
arcsight&

**See also**[ââ](#)

## Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ArpEntry

On this page

# ArpEntry

## Fields

| name             | type       | nullable | description                                  |
|------------------|------------|----------|----------------------------------------------|
| ip               | IpAddress  | false    | The IPv4 address of the neighbor node.       |
| linkLayerAddress | MacAddress | false    | The link-layer address of the neighbor node. |

## Used by

network devices interfaces subinterfaces ipv4 neighbors

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - subinterfaces: Bag< SubInterface >
        - ipv4: IfaceIpv4Info
          - neighbors: Bag< ArpEntry >

**network** **devices** **interfaces** **subinterfaces** **ipv6** **neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
          - neighbors: Bag< [ArpEntry](#) >

**network** **devices** **interfaces** **routedVlan** **ipv4** **neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - neighbors: Bag< [ArpEntry](#) >

**network** **devices** **interfaces** **routedVlan** **ipv6** **neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - neighbors: Bag< [ArpEntry](#) >

**network** **devices** **interfaces** **bridge** **ipv4** **neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)
          - neighbors: Bag< [ArpEntry](#) >

**network** **devices** **interfaces** **bridge** **ipv6** **neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - neighbors: Bag< [ArpEntry](#) >

**network** **devices** **interfaces** **tunnel** **ipv4** **neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - neighbors: Bag< [ArpEntry](#) >

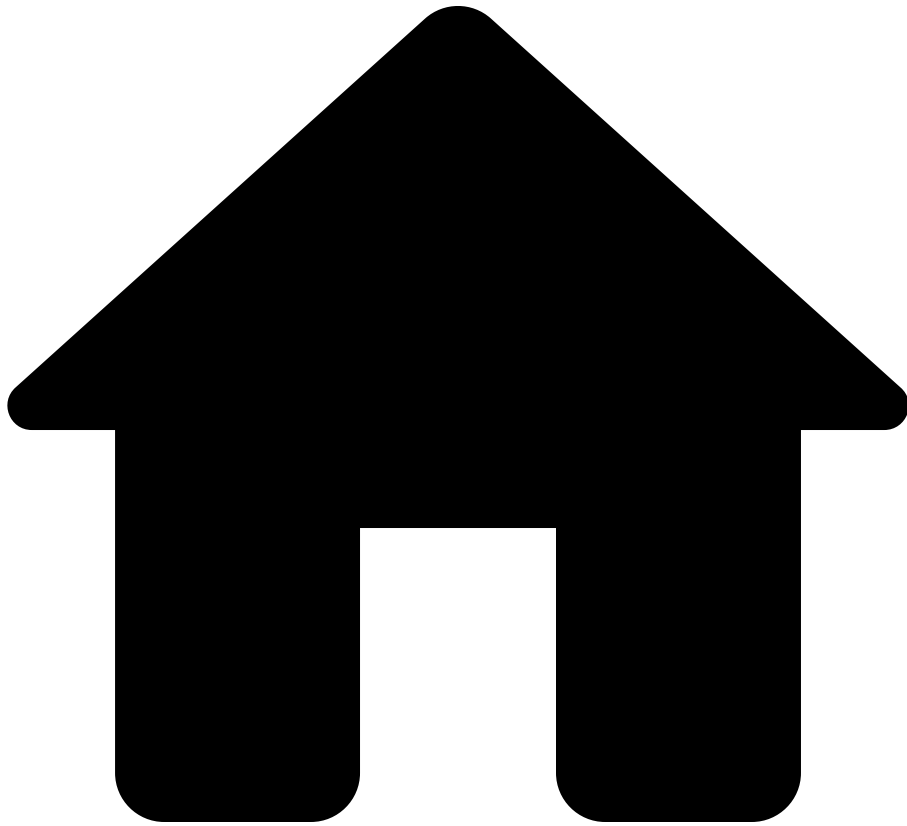
**network** **devices** **interfaces** **tunnel** **ipv6** **neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - neighbors: Bag< [ArpEntry](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AsPath

On this page



# AsPath

## Fields

| name    | type           | nullable | description                            |
|---------|----------------|----------|----------------------------------------|
| asType  | AsType         | false    | The type of AS-PATH.                   |
| members | Bag< Integer > | false    | List of the AS numbers in the AS-PATH. |

## Used by

network devices bgpRib afiSafis neighbors  
adjRibInPost routes pathAttributes asPath

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - bgpRib: BgpRib
      - afiSafis: Bag< AfiSafi >
        - neighbors: Bag< AfiSafiNeighbor >
          - adjRibInPost: AfiSafiNeighborAdjRib
            - routes: Bag< Route >
              - pathAttributes: Bag< AttrSet >
                - asPath: AsPath

**network** **devices** **bgpRib** **afiSafis** **neighbors** **adjRibOutPost** **routes** **pathAttributes** **asPath**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibOutPost: [AfiSafiNeighborAdjRib](#)
            - routes: Bag< [Route](#) >
              - pathAttributes: Bag< [AttrSet](#) >
                - asPath: [AsPath](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AsType

On this page

# AsType

## Options

| name   | description                                                                         |
|--------|-------------------------------------------------------------------------------------|
| AS_SEQ | Ordered set of autonomous systems that a route in the UPDATE message has traversed. |

## Used by

network devices bgpRib afiSafis neighbors  
adjRibInPost routes pathAttributes asPath asType

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - bgpRib: BgpRib
      - afiSafis: Bag< AfiSafi >
        - neighbors: Bag< AfiSafiNeighbor >
          - adjRibInPost: AfiSafiNeighborAdjRib
            - routes: Bag< Route >
              - pathAttributes: Bag< AttrSet >
                - asPath: AsPath
                  - asType: AsType

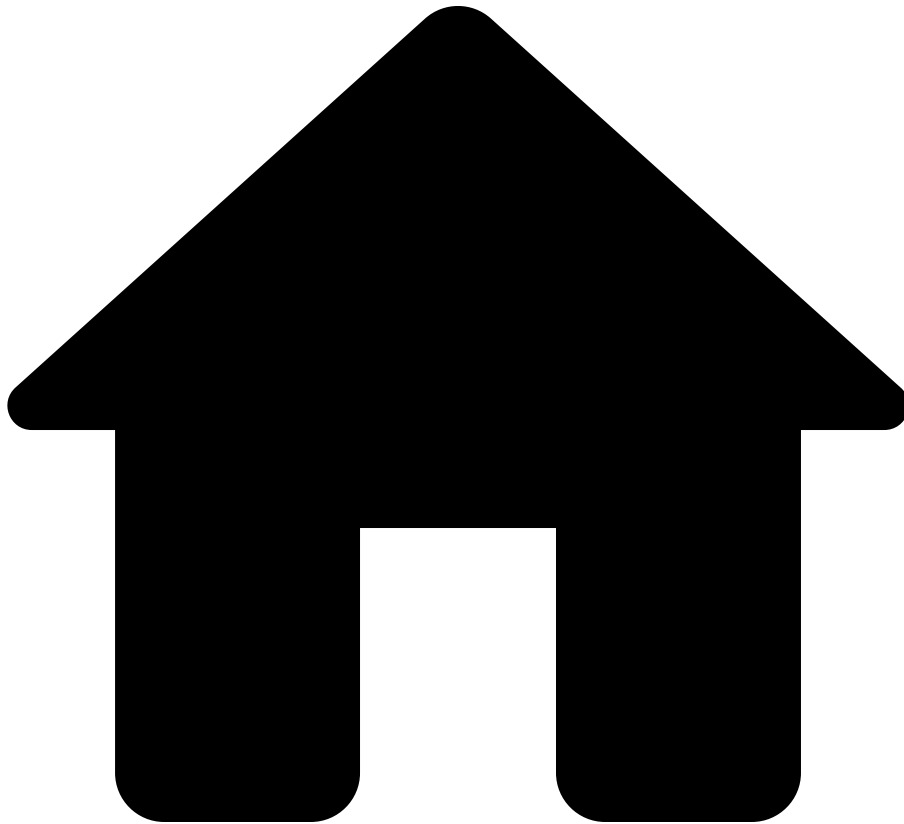
**network** **devices** **bgpRib** **afiSafis** **neighbors** **adjRibOutPost** **routes** **pathAttributes** **asPath** **asType**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibOutPost: [AfiSafiNeighborAdjRib](#)
            - routes: Bag< [Route](#) >
              - pathAttributes: Bag< [AttrSet](#) >
                - asPath: [AsPath](#)
                  - asType: [AsType](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- AttrSet

On this page

## AttrSet

Path attributes that may be in use by multiple routes in the table.

## Fields

| name         | type                      | nullable | description                                                                                                                                                                           |
|--------------|---------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| origin       | <a href="#">BgpOrigin</a> | false    | BGP attribute defining the origin of the path information.                                                                                                                            |
| nextHop      | IpAddress                 | false    | The IP address of the router that should be used as the next hop to the destination.                                                                                                  |
| med          | Integer                   | true     | The multi-exit discriminator attribute used in BGP route selection process.                                                                                                           |
| localPref    | Integer                   | true     | Local preference attribute sent to internal peers to indicate the degree of preference for externally learned routes. The route with the highest local preference value is preferred. |
| originatorId | IpAddress                 | true     | BGP attribute that provides the id as an IPv4 address of the originator of the announcement.                                                                                          |
| asPath       | <a href="#">AsPath</a>    | false    | BGP AS-PATH attribute.                                                                                                                                                                |
| activeRoute  | Bool                      | true     | Whether path is active.                                                                                                                                                               |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **bgpRib** [â](#) **afiSafis** [â](#) **neighbors** [â](#)  
**adjRibInPost** [â](#) **routes** [â](#) **pathAttributes**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibInPost: [AfiSafiNeighborAdjRib](#)
            - routes: Bag< [Route](#) >
              - pathAttributes: Bag< [AttrSet](#) >

**network** [â](#) **devices** [â](#) **bgpRib** [â](#) **afiSafis** [â](#) **neighbors** [â](#)  
**adjRibOutPost** [â](#) **routes** [â](#) **pathAttributes**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibOutPost: [AfiSafiNeighborAdjRib](#)
            - routes: Bag< [Route](#) >
              - pathAttributes: Bag< [AttrSet](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Backend

On this page

# Backend

## Options[â](#)

| name          | type                          | description |
|---------------|-------------------------------|-------------|
| backendBucket | <a href="#">BackendBucket</a> | --          |
| server        | <a href="#">BackendServer</a> | --          |
| url           | <a href="#">Url</a>           | --          |

## Used by[â](#)

**network** [â](#) **cloudAccounts** [â](#) **vpcs** [â](#) **loadBalancers** [â](#)  
**loadBalancerRules** [â](#) **backends**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - loadBalancers: Bag< [LoadBalancer](#) >
        - loadBalancerRules: Bag< [LoadBalancerRule](#) >
          - backends: Bag< [Backend](#) >

**network** **cloudAccounts** **externalLoadBalancers**  
**loadBalancerRules** **backends**

See also

Data model path

- network:
  - cloudAccounts: Bag<
    - externalLoadBalancers: Bag<
      - loadBalancerRules: Bag<
        - backends: Bag<

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BackendBucket

On this page

# BackendBucket

## Fields[â](#)

| name | type   | nullable | description |
|------|--------|----------|-------------|
| name | String | false    | --          |

## Used by[â](#)

**network** [â](#) **cloudAccounts** [â](#) **vpcs** [â](#) **loadBalancers** [â](#)  
**loadBalancerRules** [â](#) **backends**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - loadBalancers: Bag< [LoadBalancer](#) >
        - loadBalancerRules: Bag< [LoadBalancerRule](#) >
          - backends: Bag< [Backend](#) >

**network** [â](#) **cloudAccounts** [â](#) **externalLoadBalancers** [â](#)  
**loadBalancerRules** [â](#) **backends**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - externalLoadBalancers: Bag< [LoadBalancer](#) >
      - loadBalancerRules: Bag< [LoadBalancerRule](#) >
        - backends: Bag< [Backend](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BackendServer

On this page

# BackendServer

## Fields

| name         | type            | nullable | description |
|--------------|-----------------|----------|-------------|
| isUp         | Bool            | false    | --          |
| backendIps   | Bag<IpAddress > | false    | --          |
| backendPorts | Bag<Integer >   | false    | --          |

## Used by

network [cloudAccounts](#) [vpcs](#) [loadBalancers](#) [loadBalancerRules](#) [backends](#)

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - loadBalancers: Bag< [LoadBalancer](#) >
        - loadBalancerRules: Bag< [LoadBalancerRule](#) >
          - backends: Bag< [Backend](#) >

**network** **cloudAccounts** **externalLoadBalancers**  
**loadBalancerRules** **backends**

See also

Data model path

- network:
  - cloudAccounts: Bag<
    - externalLoadBalancers: Bag<
      - loadBalancerRules: Bag<
        - backends: Bag<



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Bgp

On this page

# Bgp

## Fields

| name      | type                               | nullable | description                                                                                                                                                                                                              |
|-----------|------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| neighbors | Bag< <a href="#">BgpNeighbor</a> > | false    | List of BGP neighbors configured on the local system, uniquely identified by peer IPv4 or IPv6 address.                                                                                                                  |
| routerId  | IpAddress                          | true     | An identifier for the BGP sessions in this network instance.                                                                                                                                                             |
| asNumber  | Integer                            | true     | AS number of this network instance. Present for devices with OS ACOS, APIC, ARISTA_EOS, ASA, CUMULUS, EXTREME_NOS, FXOS, HP_COMWARE, HP_PROVISION, IOS, IOS_XE, IOS_XR, JUNOS, NXOS, SG, SILVER_PEAK, T128, and VIPTELA. |

## Used by[↗](#)

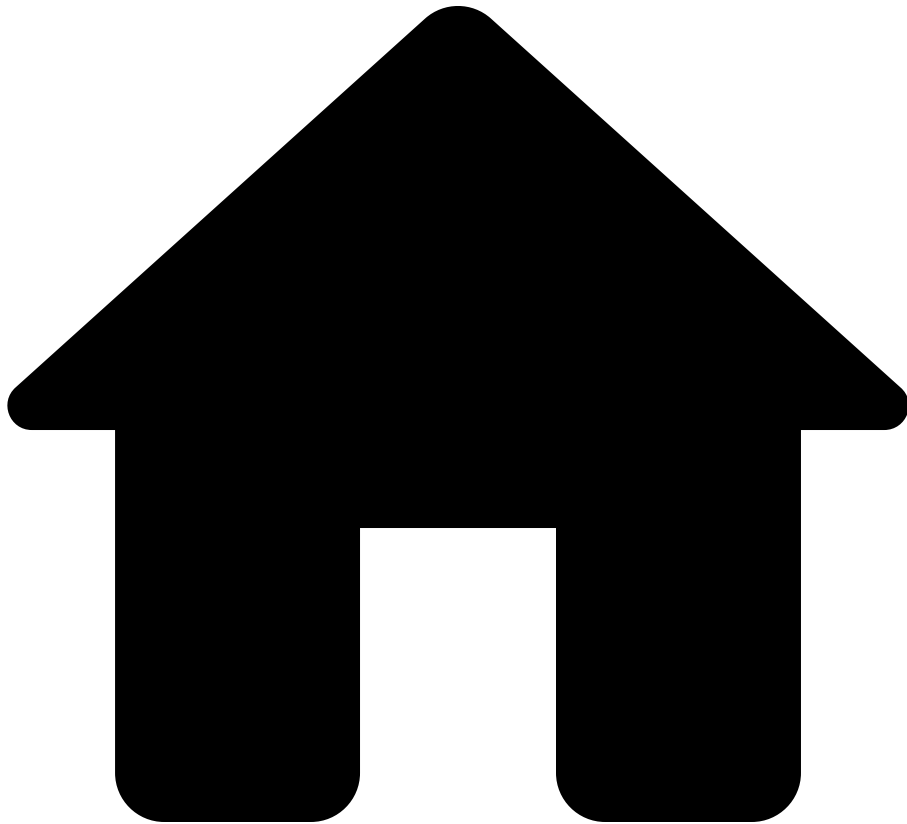
**network** [↗](#) **devices** [↗](#) **networkInstances** [↗](#) **protocols** [↗](#) **bgp**[↗](#)

See also[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - bgp: [Bgp](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BgpNeighbor

On this page

# BgpNeighbor

## Fields

| name            | type      | nullable | description                                                                                                                                         |
|-----------------|-----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| neighborAddress | IpAddress | false    | Address of the BGP peer, either in IPv4 or IPv6                                                                                                     |
| peerDeviceName  | String    | true     | Device name of the BGP peer if one could be discovered in the snapshot, otherwise absent.                                                           |
| peerVrf         | String    | true     | The VRF under which this connection was configured on the peer's device. Absent if peerDeviceName is absent.                                        |
| peerRouterId    | IpAddress | true     | Router ID of the peer. Present for devices with OS ACOS, APIC, ARISTA_EOS, ASA, CUMULUS, EXTREME_NOS, F5, FORTINET, FXOS, HP_COMWARE, HP_PROVISION, |

| name         | type                            | nullable | description                                                                                                                                                                                           |
|--------------|---------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                                 |          | IOS, IOS_XE, IOS_XR, JUNOS, NXOS, PAN_OS, SG, SILVER_PEAK, T128, and VIPTELA.                                                                                                                         |
| sessionState | <a href="#">BgpSessionState</a> | true     | Operational state of the BGP peer. Absent if enabled is false or if the data is not available.                                                                                                        |
| enabled      | Bool                            | false    | Whether the BGP peer is enabled. In cases where enabled is false, the local system does not initiate connections to the neighbor, and does not respond to TCP connections attempts from the neighbor. |
| description  | String                          | true     | Description of the neighbor.                                                                                                                                                                          |
| peerAS       | Integer                         | false    | AS number of the peer.                                                                                                                                                                                |
| localAS      | Integer                         | true     | The secondary AS number of the local device                                                                                                                                                           |

| name       | type                                  | nullable | description                                                                                                            |
|------------|---------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------|
|            |                                       |          | if a local AS is configured for an eBGP peer.<br>Absent if no local AS is configured, or the peer is not an eBGP peer. |
| peerType   | <a href="#">PeerType</a>              | true     | Explicitly designate the neighbor as internal (iBGP) or external (eBGP).                                               |
| statistics | <a href="#">BgpNeighborStatistics</a> | true     | Absent if the session state is not ESTABLISHED.                                                                        |

## Used by

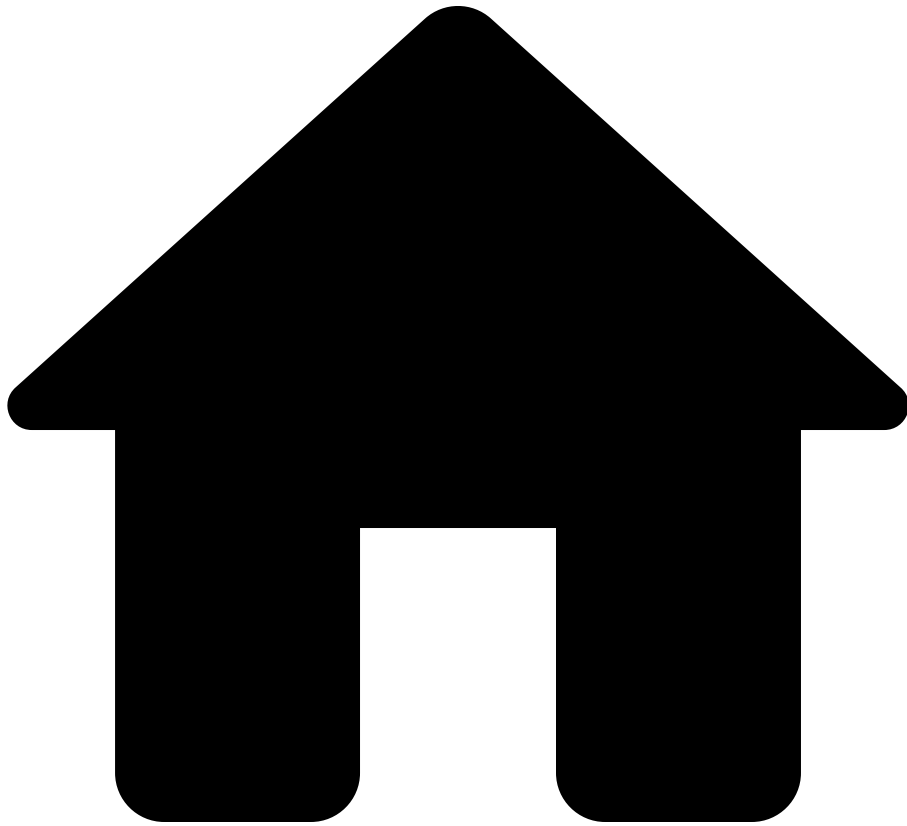
network devices networkInstances protocols bgp neighbors

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - bgp: [Bgp](#)
          - neighbors: Bag< [BgpNeighbor](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BgpNeighborStatistics

On this page



# BgpNeighborStatistics

## Fields

| name                     | type    | nullable | description                             |
|--------------------------|---------|----------|-----------------------------------------|
| sessionDurationInSeconds | Integer | false    | Duration of the BGP session in seconds. |
| advertisedPrefixes       | Integer | false    | Total number of advertised prefixes.    |
| receivedPrefixes         | Integer | false    | Total number of received prefixes.      |

## Used by

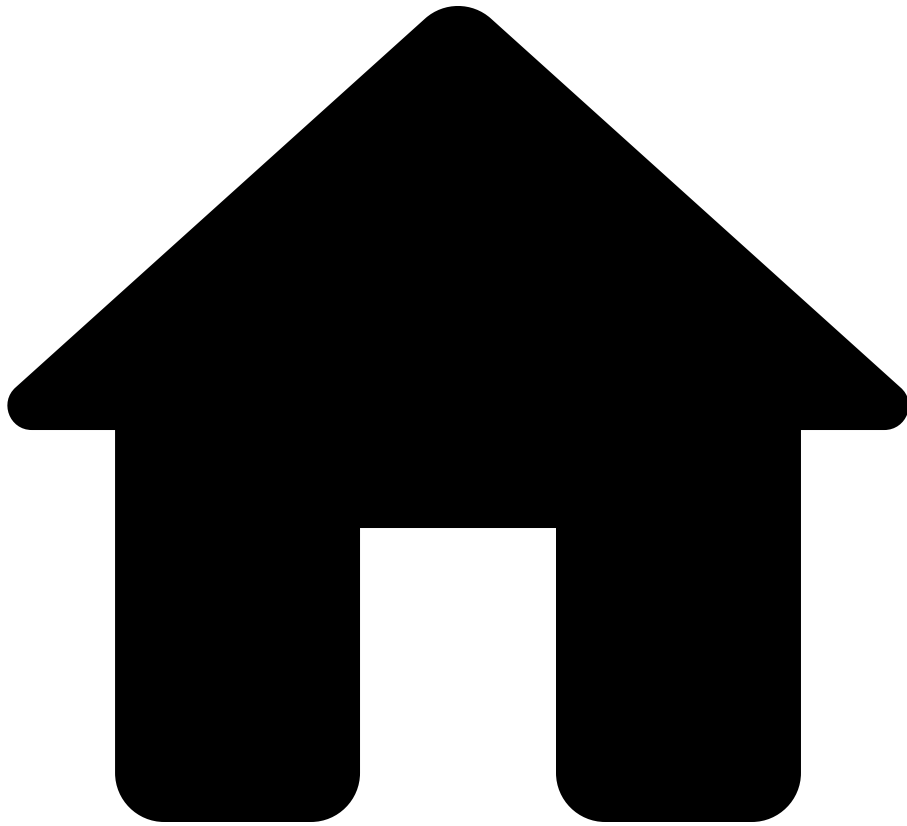
network devices networkInstances protocols bgp neighbors statistics

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - bgp: [Bgp](#)
          - neighbors: Bag< [BgpNeighbor](#) >
            - statistics: [BgpNeighborStatistics](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BgpOrigin

On this page

# BgpOrigin

## Options[â](#)

| name       | description                               |
|------------|-------------------------------------------|
| EGP        | Origin of the NLRI is EGP.                |
| IGP        | Origin of the NLRI is internal.           |
| INCOMPLETE | Origin of the NLRI is neither IGP or EGP. |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **bgpRib** [â](#) **afiSafis** [â](#) **neighbors** [â](#)  
**adjRibInPost** [â](#) **routes** [â](#) **pathAttributes** [â](#) **origin**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibInPost: [AfiSafiNeighborAdjRib](#)
            - routes: Bag< [Route](#) >
              - pathAttributes: Bag< [AttrSet](#) >
                - origin: [BgpOrigin](#)

**network** **devices** **bgpRib** **afiSafis** **neighbors** **adjRibOutPost** **routes** **pathAttributes** **origin**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibOutPost: [AfiSafiNeighborAdjRib](#)
            - routes: Bag< [Route](#) >
              - pathAttributes: Bag< [AttrSet](#) >
                - origin: [BgpOrigin](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BgpRib

On this page

## BgpRib

Defines a data model for representing BGP routing table contents.

## Fields

| name     | type                           | nullable | description       |
|----------|--------------------------------|----------|-------------------|
| afiSafis | Bag< <a href="#">AfiSafi</a> > | false    | Address families. |

## Used by

**network** **devices** **bgpRib**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BgpSessionState

On this page

# BgpSessionState

## Options

| name        | description                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| ACTIVE      | Neighbor is down, and the local system is awaiting a connection from the remote peer.                              |
| CONNECT     | Neighbor is down, and the session is waiting for the underlying transport session to be established.               |
| ESTABLISHED | Neighbor is up - the BGP session with the peer is established.                                                     |
| IDLE        | Neighbor is down, and in the Idle state of the FSM.                                                                |
| OPENCONFIRM | Neighbor is in the process of being established. The local system is awaiting a NOTIFICATION or KEEPALIVE message. |
| OPENSENT    | Neighbor is in the process of being established. The local system has sent an OPEN message.                        |

## Used by

**network** **devices** **networkInstances** **protocols** **bgp** **neighbors** **sessionState**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - bgp: [Bgp](#)
          - neighbors: Bag< [BgpNeighbor](#) >
            - sessionState: [BgpSessionState](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BluecatConfig

On this page

# BluecatConfig

## Fields

| name  | type                                     | nullable | description |
|-------|------------------------------------------|----------|-------------|
| items | Bag< <a href="#">BluecatConfigItem</a> > | false    | --          |

## Used by

network devices outputs bluecat config

### See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BluecatConfigItem

On this page

# BluecatConfigItem

## Fields

| name     | type                                  | nullable | description |
|----------|---------------------------------------|----------|-------------|
| serverId | Integer                               | false    | --          |
| data     | <a href="#">BluecatConfigItemData</a> | true     | --          |

## Used by

network devices outputs bluecat config items

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BluecatConfigItemData

On this page

# BluecatConfigItemData

## Fields

| name     | type                            | nullable | description |
|----------|---------------------------------|----------|-------------|
| services | <a href="#">BluecatServices</a> | false    | --          |
| version  | String                          | false    | --          |

## Used by

network devices outputs bluecat config items data

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BluecatOutputs

On this page

# BluecatOutputs

## Fields

| name    | type                           | nullable | description |
|---------|--------------------------------|----------|-------------|
| config  | <a href="#">BluecatConfig</a>  | true     | --          |
| servers | <a href="#">BluecatServers</a> | true     | --          |

## Used by

network devices outputs bluecat

### See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BluecatRoute

On this page

# BluecatRoute

## Fields

| name    | type      | nullable | description |
|---------|-----------|----------|-------------|
| cidr    | Integer   | false    | --          |
| gateway | IpAddress | false    | --          |
| network | String    | false    | --          |

Used by an

network → devices → outputs → bluecat → config → items → data  
→ services → interfaces → configurations → interfacesConfiguration  
→ routes

**See also** [ââ](#)

## Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BluecatServerDataItem

On this page

# BluecatServerDataItem

## Fields

| name       | type    | nullable | description |
|------------|---------|----------|-------------|
| id         | Integer | false    | --          |
| name       | String  | false    | --          |
| properties | String  | false    | --          |

## Used by

network devices outputs bluecat servers items data

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - outputs: Outputs
      - bluecat: BluecatOutputs
        - servers: BluecatServers
          - items: Bag< BluecatServerItem >
            - data: Bag< BluecatServerDataItem >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BluecatServerItem

On this page

# BluecatServerItem

## Fields

| name     | type                                         | nullable | description |
|----------|----------------------------------------------|----------|-------------|
| configId | Integer                                      | false    | --          |
| data     | Bag< <a href="#">BluecatServerDataItem</a> > | false    | --          |

## Used by

network devices outputs bluecat servers items

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - servers: [BluecatServers](#)
          - items: Bag< [BluecatServerItem](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- BluecatServers

On this page

# BluecatServers

## Fields

| name  | type                                     | nullable | description |
|-------|------------------------------------------|----------|-------------|
| items | Bag< <a href="#">BluecatServerItem</a> > | false    | --          |

## Used by

network devices outputs bluecat servers

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - servers: [BluecatServers](#)



.



- Reference
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- [Data Model](#)
- BluecatServices

On this page

# BluecatServices

## Fields

| name             | type                                    | nullable | description |
|------------------|-----------------------------------------|----------|-------------|
| dnsActivity      | <a href="#">DnsActivityService</a>      | false    | --          |
| interfaces       | <a href="#">InterfacesService</a>       | false    | --          |
| anycast          | <a href="#">AnycastService</a>          | false    | --          |
| edgeServicePoint | <a href="#">EdgeServicePointService</a> | false    | --          |
| firewall         | <a href="#">FirewallService</a>         | false    | --          |
| ssh              | <a href="#">SshService</a>              | false    | --          |
| syslog           | <a href="#">SyslogService</a>           | false    | --          |
| ntp              | <a href="#">NtpService</a>              | false    | --          |
| snmp             | <a href="#">SnmpService</a>             | false    | --          |
| dnsResolver      | <a href="#">DnsResolverService</a>      | false    | --          |
| gateway          | <a href="#">GatewayService</a>          | false    | --          |

## Used by[â](#)

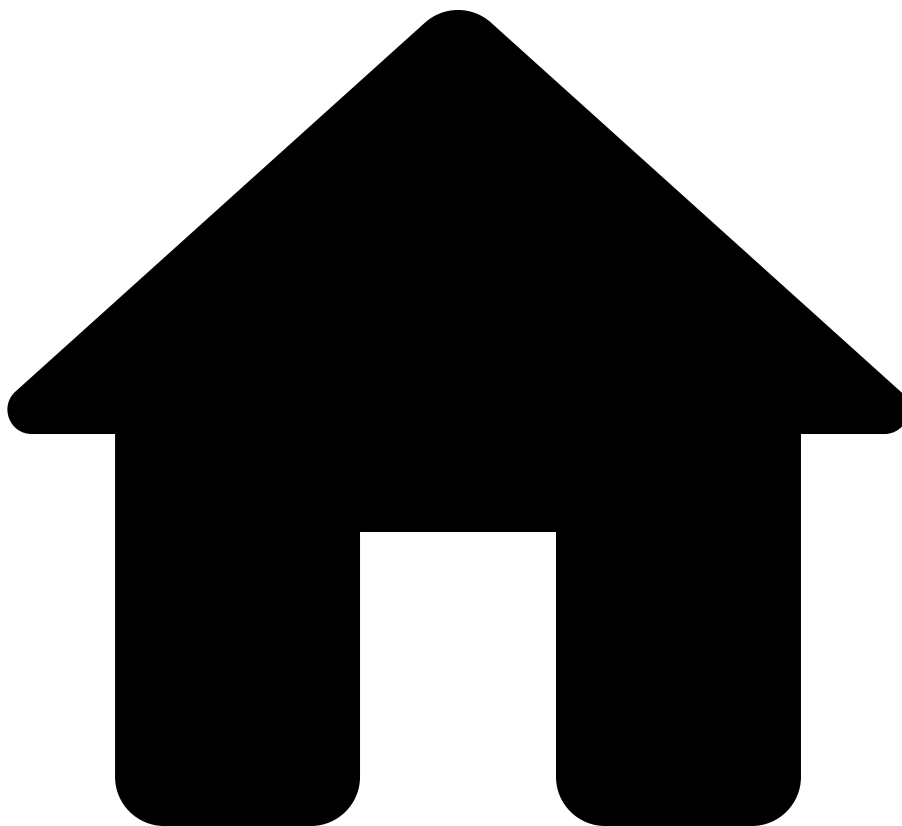
network [â](#) devices [â](#) outputs [â](#) bluecat [â](#) config [â](#) items [â](#) data [â](#) services[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Bridge

On this page

## Bridge

Bridge domain related data for a bridge interface.

## Fields

| name                | type                          | nullable | description                                                                                                                                     |
|---------------------|-------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| memberNames         | Bag< String >                 | false    | The members of the bridge interface.                                                                                                            |
| ipv4                | <a href="#">IfaceIpv4Info</a> | false    | The IPv4 addresses configured on the bridge interface.                                                                                          |
| ipv6                | <a href="#">IfaceIpv6Info</a> | false    | The IPv6 addresses configured on the bridge interface.                                                                                          |
| vlan                | Integer                       | true     | The VLAN of the bridge interface. This VLAN will be pushed on packets while sending them out of this bridge interface as part of L3 forwarding. |
| networkInstanceName | String                        | false    | The network instance (i.e. VRF) that this interface is part of.                                                                                 |

## Used by[↗](#)

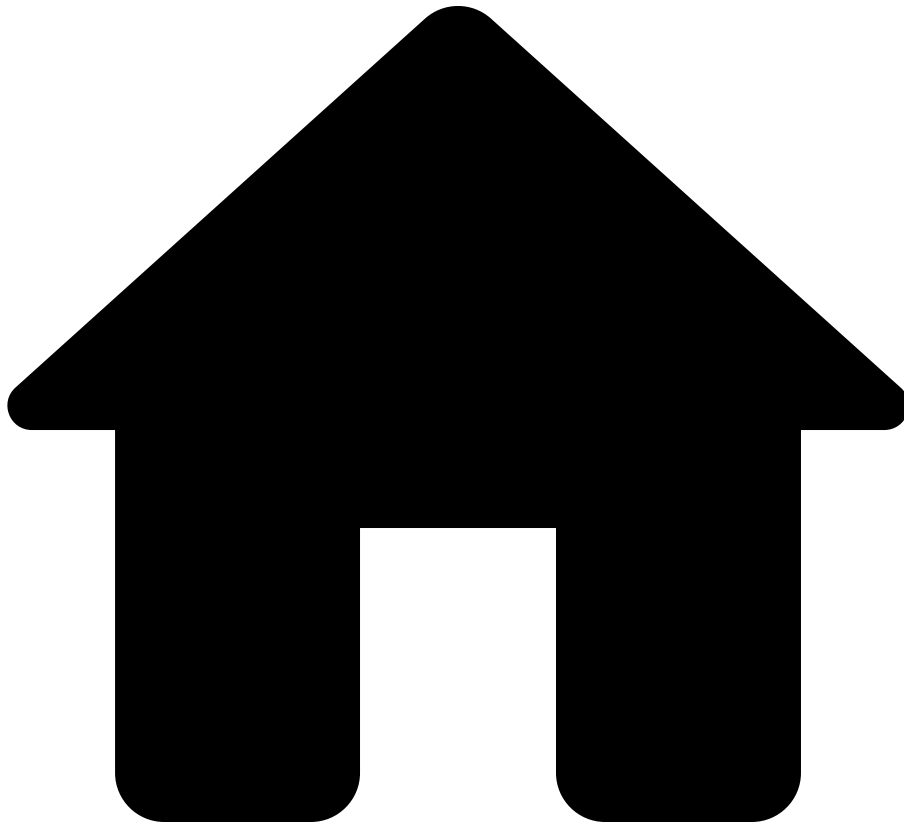
**network** [↗](#) **devices** [↗](#) **interfaces** [↗](#) **bridge**[↗](#)

**See also**[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CisaExploitationEntry

On this page

## CisaExploitationEntry

An entry from CISA's Known Exploited Vulnerabilities (KEV) catalog.

## Fields

| name | type   | nullable | description                                                                     |
|------|--------|----------|---------------------------------------------------------------------------------|
| url  | String | false    | The URL for this entry in CISA's Known Exploited Vulnerabilities (KEV) catalog. |

## Used by

network cveDatabase cves cisaKevEntry

### See also

Data model path

- network: Network
  - cveDatabase: CveDatabase
    - cves: Bag< Cve >
      - cisaKevEntry: CisaExploitationEntry



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CliCommandResponse

On this page

# CliCommandResponse

## Fields

| name     | type   | nullable | description |
|----------|--------|----------|-------------|
| command  | String | false    | --          |
| response | String | false    | --          |

## Used by

network endpoints cliCommandResponses

See also

Data model path

- network: Network
  - endpoints: Bag< NetworkEndpoint >
    - cliCommandResponses: Bag< CliCommandResponse >

network externalSources cliSources commandResponses

See also

Data model path

- network: Network
  - externalSources: ExternalSources
    - cliSources: Bag< CliSource >
      - commandResponses: Bag< CliCommandResponse >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CliSource

On this page

# CliSource

## Fields

| name             | type                                      | nullable | description                                                       |
|------------------|-------------------------------------------|----------|-------------------------------------------------------------------|
| name             | String                                    | false    | Unique name for the CLI source; configured in Forward Enterprise. |
| snapshotInfo     | <a href="#">DeviceSnapshotInfo</a>        | false    | Metadata about the inclusion of this source in the snapshot.      |
| profileName      | String                                    | false    | The name of the profile from which this source was derived.       |
| commandResponses | Bag< <a href="#">CliCommandResponse</a> > | false    | --                                                                |

## Used by[↗](#)

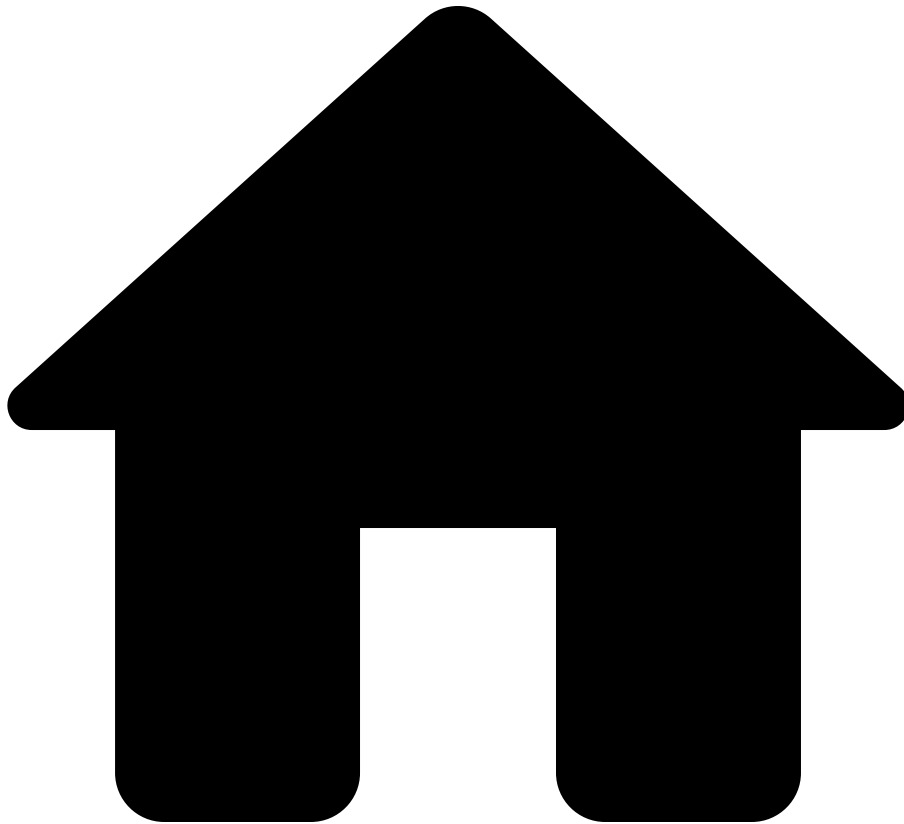
**network** [↗](#) **externalSources** [↗](#) **cliSources**[↗](#)

**See also**[↗](#)

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - cliSources: Bag< [CliSource](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CloudAccount

On this page

## CloudAccount

Defines a data model for representing NQE cloud account with all related data.

Fields

| name         | type                      | nullable | description                                                                                       |
|--------------|---------------------------|----------|---------------------------------------------------------------------------------------------------|
| name         | String                    | false    | The account's name, as collected from the user.                                                   |
| cloudSetupId | String                    | false    | The ID of the cloud setup, which is displayed to the user. ID was manually generated by the user. |
| id           | String                    | false    | The internal ID of the account. Other values may be present.                                      |
| email        | String                    | true     | The email address associated with the account, if it is collected.                                |
| collected    | Bool                      | false    | Whether the account's information was collected.                                                  |
| cloudType    | <a href="#">CloudType</a> | false    | --                                                                                                |

| name                     | type                                             | nullable | description                                                    |
|--------------------------|--------------------------------------------------|----------|----------------------------------------------------------------|
| vpcs                     | Bag< <a href="#">VpcData</a> >                   | false    | --                                                             |
| transitGateways          | Bag< <a href="#">TransitGateway</a> >            | false    | --                                                             |
| externalLoadBalancers    | Bag< <a href="#">LoadBalancer</a> >              | false    | Th                                                             |
| dcGateways               | Bag< <a href="#">DcGateway</a> >                 | false    | --                                                             |
| firewallRules            | Bag< <a href="#">FirewallRule</a> >              | false    | --                                                             |
| firewallPolicy           | <a href="#">FirewallPolicy</a>                   | false    | De<br>in<br>Re<br>in<br>CL                                     |
| globalAccelerators       | Bag< <a href="#">GlobalAccelerator</a> >         | false    | Th                                                             |
| organizationalUnitIds    | Bag< String >                                    | false    | ID<br>ac<br>fie<br>an                                          |
| publicUnallocatedIps     | Bag< <a href="#">PublicUnallocatedIp</a> >       | false    | Th<br>ac<br>wh<br>pr<br>19<br>ac<br>se<br>CL<br>AV<br>wi<br>cl |
| publicUnallocatedSubnets | Bag< <a href="#">PublicUnallocatedIpSubnet</a> > | false    | Th<br>su<br>wh<br>pr<br>19                                     |



| name | type | nullable | de                               |
|------|------|----------|----------------------------------|
|      |      |          | ac<br>se<br>Cl<br>AW<br>wi<br>cl |

Used by[â](#)

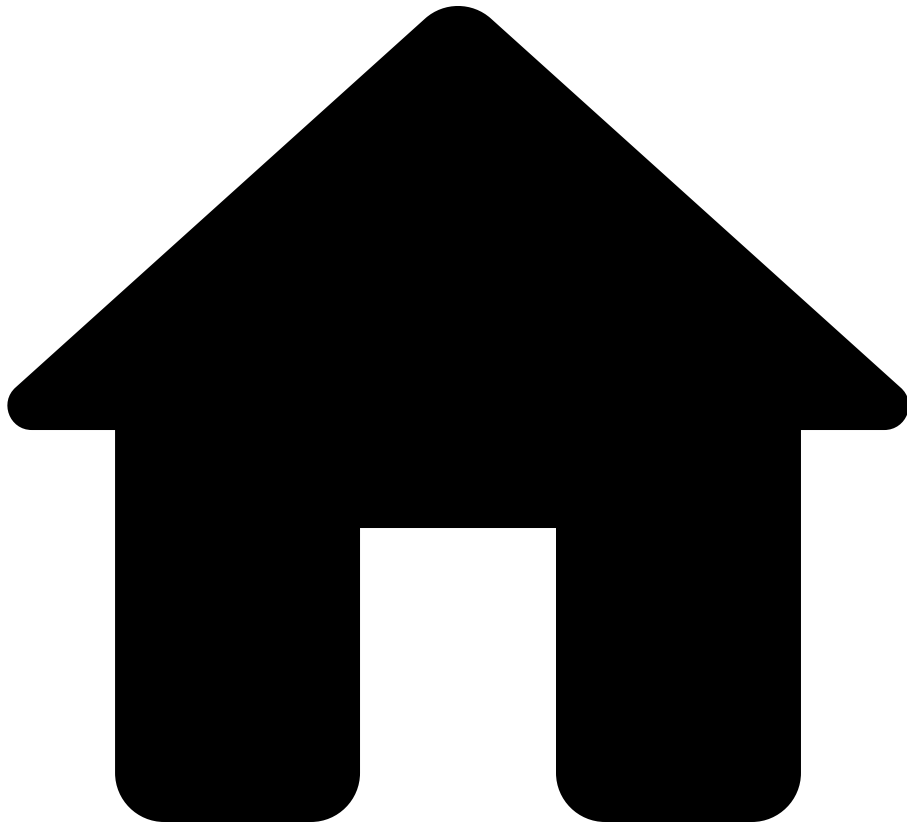
network [â](#) cloudAccounts[â](#)

See also[â](#)

Data model path

- network: `Network`
  - cloudAccounts: Bag<`CloudAccount`>

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CloudFirewall

On this page

# CloudFirewall

Defined within a VPC

# Fields

| name           | type                                      | nullable | description                                                                                                                                                   |
|----------------|-------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name           | String                                    | false    | An optional name for the cloud-provided firewall. Empty if none was configured.                                                                               |
| id             | String                                    | false    | --                                                                                                                                                            |
| resourceGroups | Bag<String>                               | false    | Logical groups to which this firewall belongs. The firewall can act on virtual machines, interfaces, groups of interfaces, physical devices, or routed VLANs. |
| rules          | List< <a href="#">CloudFirewallRule</a> > | false    | Evaluate in priority order, first match wins.                                                                                                                 |

## Used by[â](#)

**network** [â](#) **cloudAccounts** [â](#) **vpcs** [â](#) **firewalls**[â](#)

**See also**[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >

.



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- [Data Model](#)
- CloudFirewallRule

On this page

# CloudFirewallRule

## Fields

| name       | type                                        | nullable | description                                                              |
|------------|---------------------------------------------|----------|--------------------------------------------------------------------------|
| priority   | Integer                                     | false    | The priority of the rule.<br>Unique within the containing CloudFirewall. |
| name       | String                                      | false    | An optional name of this rule. Empty if none was configured.             |
| definition | <a href="#">CloudFirewallRuleDefinition</a> | false    | The definition of the firewall rule; i.e. how it is configured.          |
| match      | <a href="#">HeaderRegion</a>                | false    | The effective policy of this rule, i.e. what it actually does.           |
| action     | <a href="#">SecurityRuleAction</a>          | false    | --                                                                       |

## Used by[â](#)

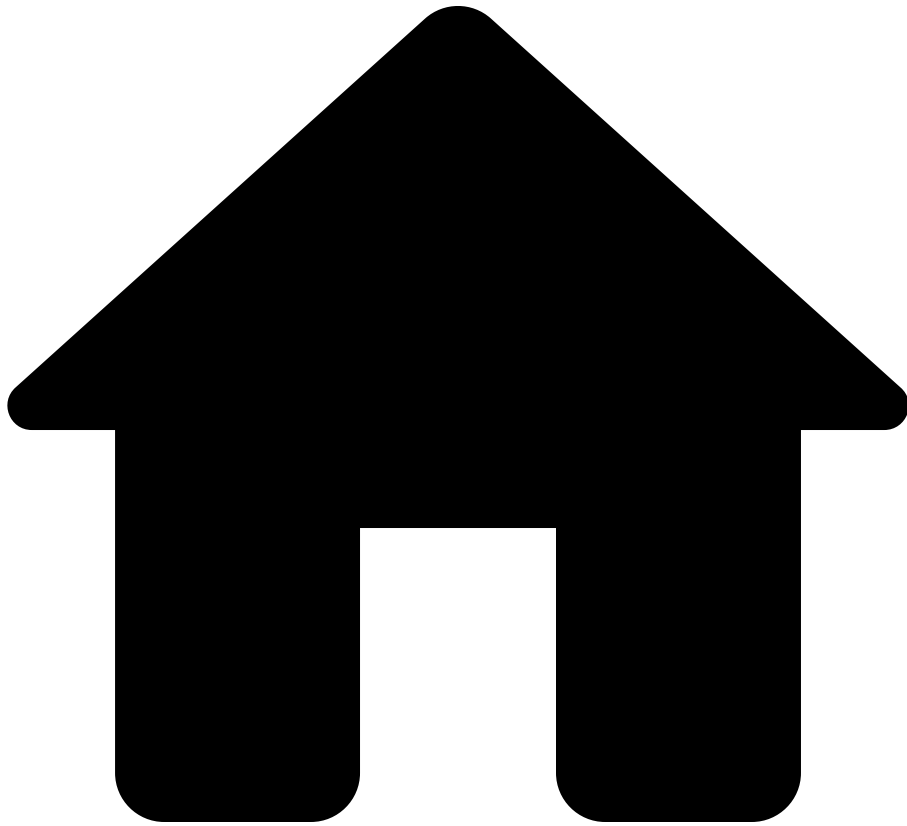
**network** [â](#) **cloudAccounts** [â](#) **vpcs** [â](#) **firewalls** [â](#) **rules**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >

.



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- [Data Model](#)
- CloudFirewallRuleDefinition

On this page



# CloudFirewallRuleDefinition

## Fields

| name               | type                            | nullable | description                                                                                                                                                   |
|--------------------|---------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| origins            | Bag< String >                   | false    | The names of the policy, collection, or group that causes this rule to be included in this firewall. Absent if the rule is directly included in the firewall. |
| sourceObjects      | Bag< String >                   | false    | --                                                                                                                                                            |
| destinationObjects | Bag< String >                   | false    | --                                                                                                                                                            |
| sourcePorts        | Bag< <a href="#">IntRange</a> > | false    | Source port matches for this rule.                                                                                                                            |
| destinationPorts   | Bag< <a href="#">IntRange</a> > | false    | Destination port matches for this rule.                                                                                                                       |
| sourceUsers        | Bag< String >                   | false    | Source user matches for this rule.                                                                                                                            |
| services           | Bag< String >                   | false    | --                                                                                                                                                            |

## Used by[â](#)

**network** [â](#) **cloudAccounts** [â](#) **vpcs** [â](#) **firewalls** [â](#) **rules** [â](#) **definition**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - definition: [CloudFirewallRuleDefinition](#)

.



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- [Data Model](#)
- Cloudiface

On this page

# Cloudiface

## Fields

| name              | type             | nullable | description                                                                                                                                                                                                            |
|-------------------|------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name              | String           | true     | Never null for GCP.                                                                                                                                                                                                    |
| description       | String           | true     | Present on AWS.                                                                                                                                                                                                        |
| id                | String           | false    | --                                                                                                                                                                                                                     |
| tags              | Bag< String >    | false    | --                                                                                                                                                                                                                     |
| isUp              | Bool             | false    | --                                                                                                                                                                                                                     |
| computeInstanceId | String           | true     | The ID of the compute instance attached to this interface. Absent if no compute instance is attached to this interface. When present, this interface is also included in the compute instances's instancelfaces field. |
| macAddress        | MacAddress       | true     | --                                                                                                                                                                                                                     |
| ipAddresses       | Bag< IPAddress > | false    | IP addresses allocated to this interface, where public means non-private as defined in RFC 1918. For the unallocated (public and IPv4) subnets, see the publicUnallocatedSubnets field of CloudAccount.                |
| securityGroupIds  | Bag< String >    | false    | --                                                                                                                                                                                                                     |

## Used by[â](#)

**network** [â](#) **cloudAccounts** [â](#) **vpcs** [â](#) **subnets** [â](#) **ifaces**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - subnets: Bag< [Subnet](#) >
        - ifaces: Bag< [CloudIface](#) >

.



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- [Data Model](#)
- CloudRoute

On this page

# CloudRoute

## Fields

| name          | type                              | nullable | description                            |
|---------------|-----------------------------------|----------|----------------------------------------|
| prefixes      | Bag< IpSubnet >                   | false    | --                                     |
| tags          | Bag< String >                     | false    | --                                     |
| routeType     | <a href="#">CloudRouteType</a>    | false    | --                                     |
| nextHop       | <a href="#">CloudRouteNextHop</a> | true     | Route with null nextHop drops traffic. |
| priority      | Integer                           | true     | GCP only.                              |
| subnetIdMatch | String                            | false    | --                                     |
| inactive      | Bool                              | true     | --                                     |

## Used by

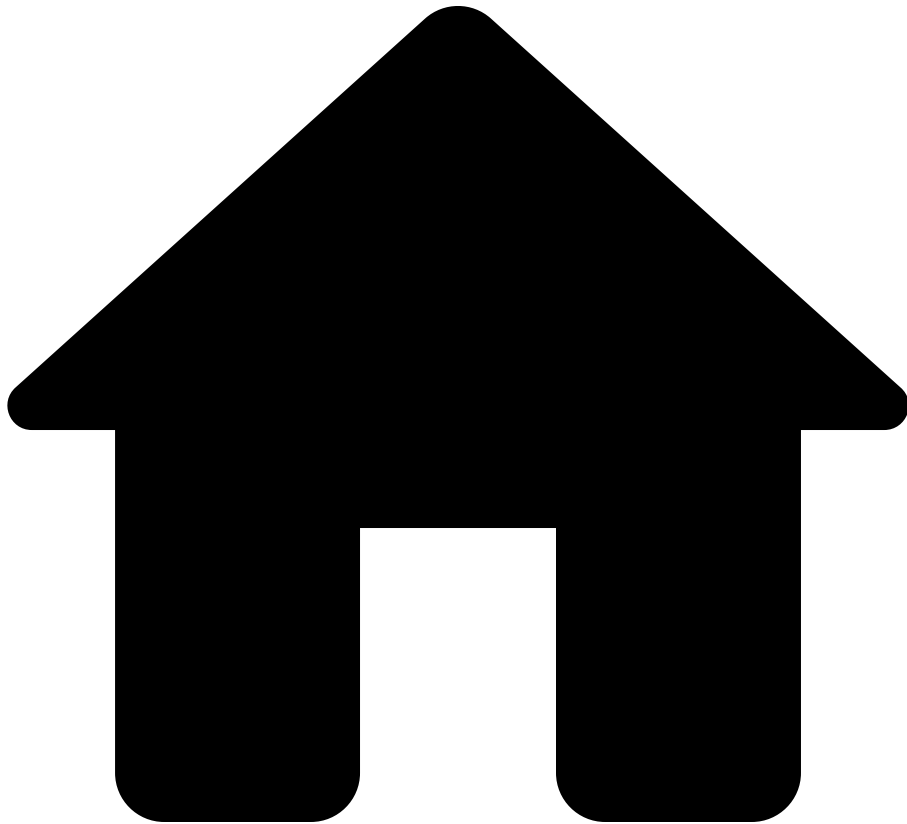
network cloudAccounts vpcs routeTables routes

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - routeTables: Bag< [RouteTable](#) >
        - routes: Bag< [CloudRoute](#) >

.



- Reference
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- CloudRouteNextHop

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# CloudRouteNextHop

## Options

| name      | type      | description                                                                                                                 |
|-----------|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| internet  | --        | --                                                                                                                          |
| ipAddress | IpAddress | For GCP, the IP address may refer to the address of an interface in a compute instance within the VPC containing the route. |
| local     | --        | --                                                                                                                          |
| refObj    | RefObj    | --                                                                                                                          |
| unknown   | String    | The ID of some AWS object that was not collected for some reason and is therefore unreachable in the model.                 |

## Used by

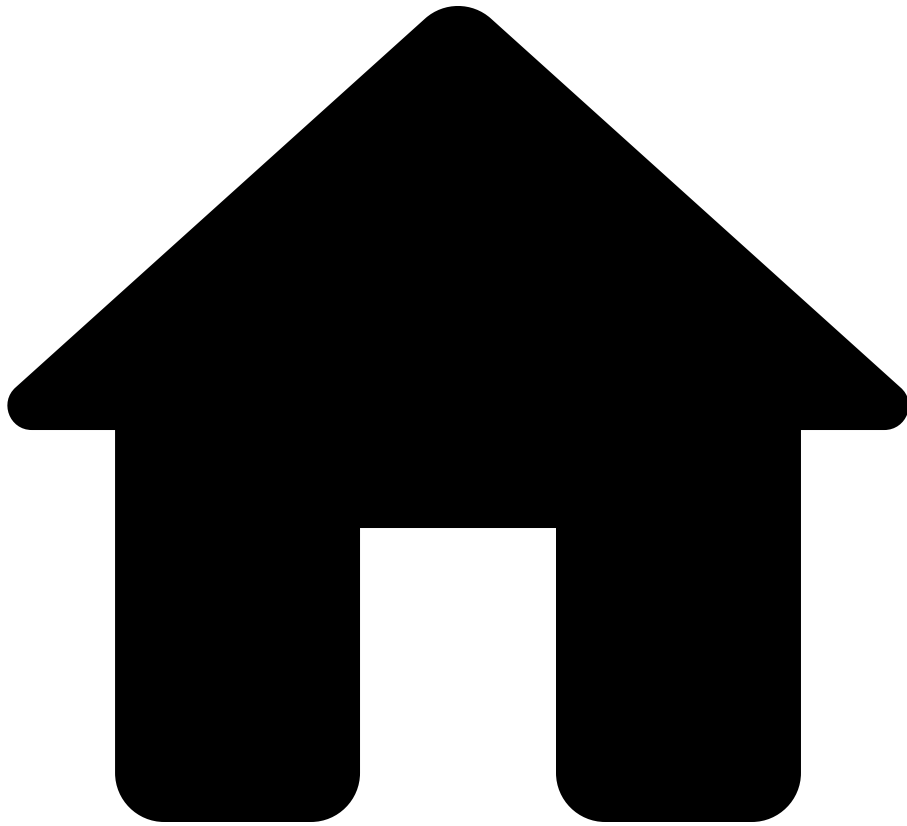
network cloudAccounts vpcs routeTables routes  
nextHop

## See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - vpcs: Bag< VpcData >
      - routeTables: Bag< RouteTable >
        - routes: Bag< CloudRoute >
          - nextHop: CloudRouteNextHop

.



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# CloudRouteType

## Options

| name   | description |
|--------|-------------|
| BGP    | --          |
| STATIC | --          |

## Used by

network cloudAccounts vpcs routeTables routes routeType

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - vpcs: Bag< VpcData >
      - routeTables: Bag< RouteTable >
        - routes: Bag< CloudRoute >
          - routeType: CloudRouteType

**network** **cloudAccounts** **transitGateways** **attachments**  
**routeTables** **routes** **routeType**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - transitGateways: Bag< [TransitGateway](#) >
      - attachments: Bag< [TransitGatewayAttachment](#) >
        - routeTables: Bag< [TransitGatewayRouteTable](#) >
          - routes: Bag< [TransitGatewayRoute](#) >
            - routeType: [CloudRouteType](#)

**network** **cloudAccounts** **transitGateways** **routeTables**  
**routes** **routeType**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - transitGateways: Bag< [TransitGateway](#) >
      - routeTables: Bag< [TransitGatewayRouteTable](#) >
        - routes: Bag< [TransitGatewayRoute](#) >
          - routeType: [CloudRouteType](#)

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# CloudType

## Options[â](#)

| name  | description |
|-------|-------------|
| AWS   | --          |
| AZURE | --          |
| GCP   | --          |

## Used by[â](#)

**network** [â](#) **cloudAccounts** [â](#) **cloudType**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - cloudType: [CloudType](#)

**network** [â](#) **cloudAccounts** [â](#) **vpcs** [â](#) **cloudType**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - cloudType: [CloudType](#)

.



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- ClusterHaInfo

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# ClusterHaInfo

## Fields

| name          | type                            | nullable | description  |
|---------------|---------------------------------|----------|--------------|
| peerDevices   | Bag< String >                   | false    | Peer devices |
| operationMode | <a href="#">HaOperationMode</a> | false    | --           |

## Used by

network devices ha clusterHa

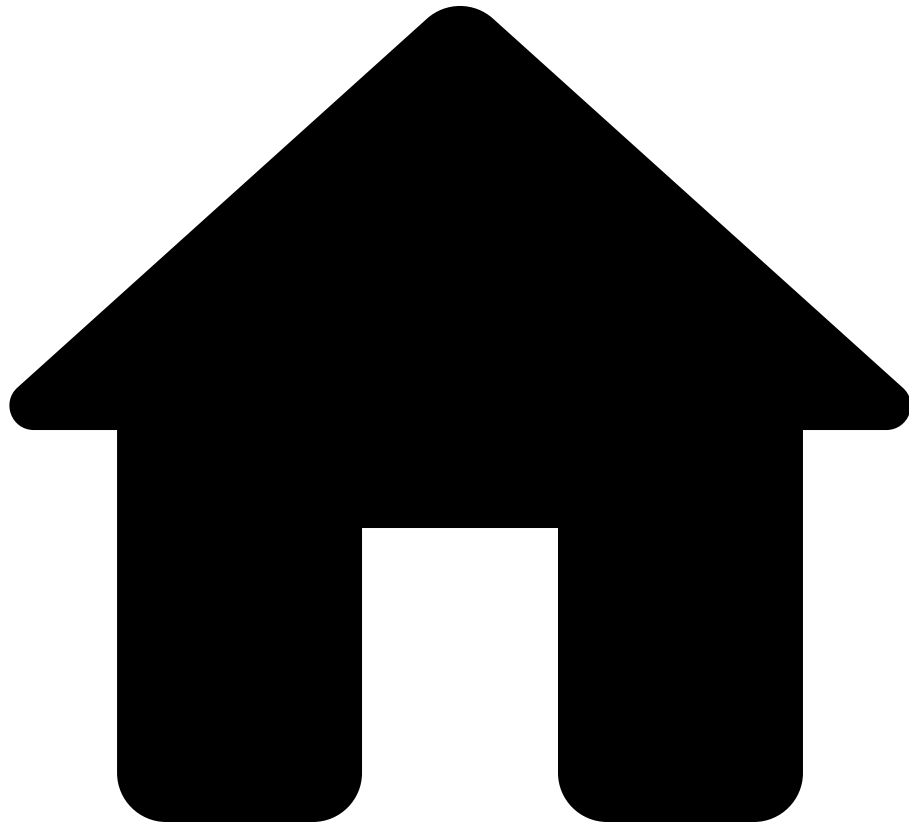
### See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - ha: [Ha](#)
      - clusterHa: [ClusterHaInfo](#)



.



- Reference
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- [Data Model](#)
- Command

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# Command

## Fields

| name        | type                        | nullable | description                                                                         |
|-------------|-----------------------------|----------|-------------------------------------------------------------------------------------|
| commandType | <a href="#">CommandType</a> | false    | --                                                                                  |
| commandText | String                      | true     | The command that was used to collect this source. Only present for CUSTOM commands. |
| response    | String                      | false    | The text value returned in response to the command.                                 |

## Used by

network devices outputs commands

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - commands: Bag< [Command](#) >

.



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- CommandType

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# CommandType

## Options

| name            | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A10_AFLEX_RULES | ACOS: 'show aflex {}'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| A10_BW_LISTS    | ACOS: 'show bw-list {} detail'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| ACL_ADDRESSES   | PAN_OS: 'show running security-policy'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ACL_CONFIG      | AVI_VANTAGE: 'GET /api/networksecuritypolicy', 'show networksecuritypolicy'<br>BLUECOAT: 'show security'<br>F5: 'bigpipe db PacketFilter show', 'bigpipe db PacketFilter.Allow.ARP show', 'bigpipe db PacketFilter.Allow.Important.ICMP show', 'bigpipe db PacketFilter.DefaultAction show', 'bigpipe db PacketFilter.DefaultLog show', 'bigpipe db PacketFilter.Established show', 'bigpipe db PacketFilter.SendICMPErrors show', 'bigpipe packet filter allow trusted list', 'bigpipe packet filter list all', 'list net packet-filter-trusted;', 'list net packet-filter;', 'list sys db packetfilter*'<br>HP_PROVISION: 'show access-list config'<br>LINUX: 'iptables -L -n -v'<br>OTHER: 'show access inbound rules' |

| name                | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACL_CONFIG_IPV6     | LINUX: 'ip6tables -L -v -n'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ACL_STATE           | ACOS: 'show access-list'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| APPLICATION_PROFILE | AVI_VANTAGE: 'GET /api/applicationprofile', 'show applicationprofile'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| AP_ESSID            | ARUBA_WIFI: 'show ap essid'<br>IOS_XE: 'show ap wlan summary'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| AP_USERS            | ARUBA_WIFI: 'show user rows 1 {}'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| ARP_TABLE           | ACOS: 'show arp'<br>ARISTA_EOS: 'show ip arp'<br>ARUBA_WIFI: 'show arp'<br>ASA: 'show arp'<br>AVI_VANTAGE: 'GET /api/serviceengine/{name}/arptable', 'show serviceengine {} arptable'<br>BLUECOAT: 'show arp-table'<br>CHECKPOINT: 'arp -an', 'show arp dynamic all', 'show arp static all'<br>CUMULUS: 'arp -n', 'ip -4 neigh show'<br>DELL_OS10: 'show arp', 'show ip arp'<br>DELL_OS6: 'show arp', 'show ip arp'<br>DELL_OS9: 'show arp', 'show ip arp'<br>DELL_SONIC: 'show ip arp', 'show ipv6 neighbors vrf {}'<br>F5: 'bigpipe arp show', 'show net arp;', 'bash -c 'arp -n -i mgmt';'<br>FORTINET: 'get system arp'<br>FXOS: 'show arp'<br>HP_COMWARE: 'display arp all'<br>HP_PROVISION: 'show arp'<br>HUAWEI: 'display arp', 'display arp all'<br>IOS: 'show ip arp'<br>IOS_XE: 'show ip arp' |

| name    | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>IOS_XR: 'show arp'</p> <p>JUNOS: 'show arp no-resolve'</p> <p>LINUX: 'arp -n', 'ip -4 neigh show'</p> <p>NETSCALER: 'show arp -format TEXT'</p> <p>NETSCREEN: 'get arp'</p> <p>NXOS: 'show ip arp vrf {}', 'show ipv6 neighbor vrf {}'</p> <p>OTHER: 'show interfaces {} arp'</p> <p>PAN_OS: 'show arp all', 'show arp management dns no'</p> <p>SRX: 'show arp no-resolve'</p> <p>VERSA: 'show arp all'</p> <p>VIPTELA: 'show arp'</p>                                                                                                                                                                                                                            |
| ARP_VRF | <p>APIC: 'fabric {} show ip arp vrf all'</p> <p>ARISTA_EOS: 'show ip arp vrf {}', 'show ipv6 neighbors vrf {}'</p> <p>ARUBA_AOS_CX: 'show arp all-vrfs'</p> <p>DELL_OS10: 'show arp vrf {}', 'show ip arp vrf {}'</p> <p>DELL_OS6: 'show arp vrf {}', 'show ip arp vrf {}'</p> <p>DELL_OS9: 'show arp vrf {}', 'show ip arp vrf {}'</p> <p>DELL_SONIC: 'show ip arp vrf {}'</p> <p>HP_COMWARE: 'display arp vpn-instance {}', 'display ip vpn-instance', 'display ipv6 neighbors vpn-instance {}'</p> <p>IOS: 'show ip arp vrf {}', 'show ipv6 neighbor vrf {}'</p> <p>IOS_XE: 'show ip arp vrf {}', 'show ipv6 neighbor vrf {}'</p> <p>IOS_XR: 'show arp vrf {}'</p> |

| name                     | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | NXOS: 'show ip arp vrf {', 'show ipv6 neighbor vrf {', 'show ip arp vrf all'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| AVI_CLOUD                | AVI_VANTAGE: 'GET /api/cloud', 'show cloud'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| AVI_SERVICE_ENGINE_GROUP | AVI_VANTAGE: 'GET /api/serviceenginegroup', 'show serviceenginegroup'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| AVI_VIRTUAL_SERVICE      | AVI_VANTAGE: 'GET /api/virtualservice', 'show virtualservice'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| BGP_PEERS                | ACOS: 'show ip bgp neighbors'<br>APIC: 'fabric { show bgp sessions vrf all'<br>ARISTA_EOS: 'show ip bgp neighbors vrf {', 'show ipv6 bgp peers vrf {'<br>ARUBA_AOS_CX: 'show bgp ipv4 unicast neighbors'<br>ASA: 'show bgp ipv6 unicast neighbors', 'show bgp neighbors'<br>AVI_VANTAGE: 'GET /api/serviceengine/{name}/bgp', 'show serviceengine { bgp'<br>CUMULUS: 'net show bgp vrf all neighbor'<br>DELL_OS10: 'show ip bgp neighbors', 'show ip bgp vrf { neighbors'<br>DELL_OS6: 'show ip bgp neighbors', 'show ip bgp vrf { neighbors'<br>DELL_OS9: 'show ip bgp neighbors', 'show ip bgp vrf { neighbors'<br>DELL_SONIC: 'show bgp all neighbors'<br>F5: 'imish -e 'show ip bgp neighbors"<br>FORTINET: 'get router info bgp |

| name | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p>neighbors'</p> <p>HP_COMWARE: 'display bgp peer {}',<br/>'display bgp peer {} {}', 'display bgp {}<br/>all peer {}', 'display bgp {} peer {}'</p> <p>HP_PROVISION: 'show ip bgp<br/>neighbor'</p> <p>HUAWEI: 'display bgp peer', 'display<br/>bgp vpnv4 all peer', 'display bgp<br/>vpnv6 all peer'</p> <p>IOS: 'show bgp vpnv4 unicast all<br/>neighbors', 'show bgp vpnv6 unicast<br/>all neighbors', 'show ip bgp ipv6<br/>unicast neighbors', 'show ip bgp<br/>neighbors', 'show ip bgp vpnv4 vrf {}<br/>neighbors', 'show ip bgp vpnv6<br/>unicast vrf {} neighbors'</p> <p>IOS_XE: 'show bgp ipv6 unicast<br/>neighbors', 'show bgp neighbors',<br/>'show bgp vpnv4 unicast all<br/>neighbors', 'show bgp vpnv6 unicast<br/>all neighbors', 'show bgp vpnv6<br/>unicast vrf {} neighbors', 'show bgp<br/>vrf {} neighbors'</p> <p>IOS_XR: 'show bgp vrf {} neighbors',<br/>'show bgp vrf {} {} neighbors {}<br/>advertised-count'</p> <p>JUNOS: 'show bgp neighbor'</p> <p>NXOS: 'show bgp vrf {} all<br/>neighbors', 'show bgp sessions vrf<br/>all'</p> <p>PAN_OS: 'show advanced-routing<br/>bgp peer detail', 'show routing<br/>protocol bgp peer'</p> <p>SRX: 'show bgp neighbor'</p> <p>VERSA: 'show bgp neighbor detail'</p> |



| name               | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BGP_SUMMARY        | ARISTA_EOS: 'show bgp instance vrf all'<br>ARUBA_AOS_CX: 'show bgp all', 'show bgp ipv4 unicast summary'<br>ASA: 'show bgp summary'<br>DELL_OS10: 'show ip bgp summary', 'show ip bgp vrf {} summary'<br>DELL_OS6: 'show ip bgp summary', 'show ip bgp vrf {} summary'<br>DELL_OS9: 'show ip bgp summary', 'show ip bgp vrf {} summary'<br>DELL_SONIC: 'show bgp {} unicast summary'<br>FORTINET: 'get router info bgp summary'<br>HP_COMWARE: 'display bgp peer', 'display bgp peer {}', 'display bgp peer {} {}', 'display bgp {} all peer {}', 'display bgp {} peer {}'<br>IOS: 'show bgp all summary', 'show ip bgp vpnv4 vrf {} summary', 'show ip bgp vpnv6 unicast vrf {} summary'<br>IOS_XE: 'show bgp all summary', 'show bgp vpnv6 unicast vrf {} summary', 'show bgp vrf {} all summary'<br>IOS_XR: 'show bgp vrf {} summary'<br>NXOS: 'show bgp vrf all all summary'<br>VIPTELA: 'show bgp summary' |
| BLOCKED_CLIENT_IPS | BLUECOAT: 'show attack-detection client blocked'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| BLUECAT_CONFIG     | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| BLUECOAT_ADN       | BLUECOAT: 'show advanced-url / ADN/configuration'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| name                    | description                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BRIDGE_DOMAIN           | IOS: 'show bridge-domain'<br>IOS_XE: 'show bridge-domain'<br>VIPTELA: 'show bridge table'                                                                                                                                                                                                                                                                                             |
| BRIDGE_INTERFACES       | APIC: 'show interface bridge-domain'<br>ASA: 'show bridge-group'<br>CHECKPOINT: 'brctl showstp {}', 'for<br>iface in \$(brctl show   tail -n +2  <br>grep -v '^[:space:]')   awk '{print \$1}';<br>do echo "brctl showstp \$iface" &&<br>brctl showstp \$iface; done'<br>FXOS: 'show bridge-group'<br>IOS: 'show bridge'<br>IOS_XE: 'show bridge'<br>VIPTELA: 'show bridge interface' |
| BUNDLE_STATE            | HP_COMWARE: 'display irf'<br>IOS: 'show switch virtual'<br>IOS_XE: 'show switch virtual'                                                                                                                                                                                                                                                                                              |
| CDP                     | APIC: 'fabric {} show cdp neighbors<br>detail'<br>ARUBA_AOS_CX: 'show cdp<br>neighbor-info', 'show cdp neighbor-<br>info {}'<br>ASA: 'show cdp neighbors detail'<br>FXOS: 'show cdp neighbors detail'<br>IOS: 'show cdp neighbors detail'<br>IOS_XE: 'show cdp neighbors detail'<br>IOS_XR: 'show cdp neighbors detail'<br>NXOS: 'show cdp neighbors detail'                          |
| CISCO_ACI_EPM_ENDPOINTS | APIC: 'fabric {} show system internal<br>epm endpoint all'<br>NXOS: 'show system internal epm<br>endpoint all'                                                                                                                                                                                                                                                                        |
| CISCO_ACI_FABRIC_NODES  | NXOS: 'acidiag fnvread'                                                                                                                                                                                                                                                                                                                                                               |

| name                            | description                                                                                                                                                                              |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CISCO_ACI_FABRIC_VRFS           | APIC: 'fabric {} show system internal<br>epm vrf all'<br>NXOS: 'show system internal epm<br>vrf all'                                                                                     |
| CISCO_ACI_NODE_TYPE             | APIC: 'fabric {} show coop internal<br>info global'                                                                                                                                      |
| CISCO_ACI_SPINE_INTERFACE_MODE  | APIC: 'fabric {} show ip interface  <br>grep -E -A 1 'mode: etep anycast-v(4 <br>6),external\$"<br>NXOS: 'show ip interface   grep -E -A<br>1 'mode: etep anycast-v(4 <br>6),external\$" |
| CISCO_ACI_SPINE_IP_ENDPOINTS    | APIC: 'fabric {} show coop internal<br>info ip-db'<br>NXOS: 'show coop internal info ip-<br>db'                                                                                          |
| CISCO_ACI_SPINE_TUNNELS_NEXTHOP | APIC: 'fabric {} show coop internal<br>info tun-nh'<br>NXOS: 'show coop internal info tun-<br>nh'                                                                                        |
| CISCO_ACI_TUNNELS_ENDPOINTS     | APIC: 'fabric {} show isis dteps vrf<br>all'<br>NXOS: 'show isis dteps vrf all'                                                                                                          |
| CISCO_ACI_ZONING_FILTER         | APIC: 'fabric {} show zoning-filter'<br>NXOS: 'show zoning-filter'                                                                                                                       |
| CISCO_ACI_ZONING_RULE           | APIC: 'fabric {} show zoning-rule'<br>NXOS: 'show zoning-rule'                                                                                                                           |
| CISCO_APIC_CONTROLLER_DETAIL    | APIC: 'show controller detail'                                                                                                                                                           |
| CISCO_APIC_EXT_L3OUT_INTERFACES | APIC: 'show external-l3 interfaces<br>detail'                                                                                                                                            |

| name                        | description                                                                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CISCO_APIC_SWITCH           | APIC: 'show switch detail'                                                                                                                                                            |
| CISCO_APIC_VLAN_LEAF        | APIC: 'fabric {} show vlan extended'<br>NXOS: 'show vlan extended'                                                                                                                    |
| CISCO_APIC_VLAN_STATUS_LEAF | APIC: 'fabric {} show system internal<br>epm vlan all detail'<br>NXOS: 'show system internal epm<br>vlan all detail'                                                                  |
| CISCO_APIC_VPC_MAP          | APIC: 'show vpc map'                                                                                                                                                                  |
| CISCO_ASA_SHUN_TABLE        | ASA: 'show shun'<br>FXOS: 'show shun'                                                                                                                                                 |
| CISCO_IOS_VFI_INTERFACES    | IOS: 'show platform software<br>ethernet fp active vfi detail'<br>IOS_XE: 'show platform software<br>ethernet fp active vfi detail'                                                   |
| CISCO_IOS_VPLS_VFI_LIST     | IOS: 'show l2vpn vfi'<br>IOS_XE: 'show l2vpn vfi'                                                                                                                                     |
| CISCO_ITD                   | NXOS: 'show itd'                                                                                                                                                                      |
| CLUSTER_INTERFACES          | CHECKPOINT: 'cphaprob -a if', 'show<br>cluster members interfaces all'                                                                                                                |
| CLUSTER_STATE               | ARUBA_WIFI: 'show lc-cluster group-<br>membership'<br>AVI_VANTAGE: 'GET /api/cluster/<br>runtime', 'show cluster status'<br>CHECKPOINT: 'cphaprob state',<br>'show cluster state'     |
| CONFIG                      | ACOS: 'show running-config'<br>APIC: 'show running-config tenant',<br>'show running-config tenant {}'<br>ARISTA_EOS: 'show running-config'<br>ARUBA_AOS_CX: 'show running-<br>config' |

| name | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p>ARUBA_WIFI: 'show running-config'</p> <p>ASA: 'show running-config'</p> <p>BLUECOAT: 'show configuration'</p> <p>CHECKPOINT: 'clish -c "show configuration"', 'show configuration'</p> <p>CUMULUS: 'net show configuration'</p> <p>DELL_OS10: 'show running-config', 'show running-configuration'</p> <p>DELL_OS6: 'show running-config', 'show running-configuration'</p> <p>DELL_OS9: 'show running-config', 'show running-configuration'</p> <p>DELL_SONIC: 'show running-configuration'</p> <p>F5: 'bigpipe failover list all', 'bigpipe interface list all', 'bigpipe mgmt list', 'bigpipe self list', 'bigpipe stp list', 'bigpipe system list', 'bigpipe trunk list all', 'bigpipe vlan list', 'bigpipe vlangroup list', 'list apm log-setting', 'list apm policy', 'list apm resource network-access', 'list asm policy &lt;name&gt; policy-builder', 'list cli global-settings;', 'list cm device', 'list cm traffic-group;', 'list ltm dns dnssec key', 'list ltm dns dnssec zone', 'list ltm dns nameserver', 'list ltm dns zone', 'list ltm eviction-policy', 'list ltm global-setting', 'list ltm message-routing sip profile router', 'list ltm message-routing sip profile session', 'list ltm policy', 'list ltm pool', 'list ltm profile', 'list ltm profile client-ssl all session-ticket;', 'list ltm profile client-ssl;', 'list ltm profile dns', 'list ltm profile fastl4 all</p> |

| name | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | reassemble-fragments;', 'list ltm<br>profile ftp', 'list ltm profile http', 'list<br>ltm profile http2', 'list ltm profile http<br>all proxy-type;', 'list ltm profile one-<br>connect', 'list ltm profile server-ssl;',<br>'list ltm profile smtp', 'list ltm profile<br>socks;', 'list ltm profile tcp', 'list ltm<br>profile tcp all fast-open;', 'list ltm<br>profile udp', 'list ltm virtual', 'list net<br>dag-globals dag-ipv6-prefix-len;', 'list<br>net ipsec ike-peer', 'list net ipsec;',<br>'list net stp-globals all-properties;',<br>'list net stp;', 'list security dos device-<br>config', 'list security firewall', 'list<br>security firewall policy', 'list security<br>log profile global-network', 'list<br>security protocol-inspection profile',<br>'list sys aom', 'list sys daemon-log-<br>settings;', 'list sys db<br>connection.syncookies.algorithm<br>configsync.allowmanagement<br>crypto.hwacceleration dns.queueing<br>ipv6.strictcompliance<br>pvasyncookies.enabled<br>statemirror.secure tmm.ffwd.enable<br>tm.tcpprogressive<br>tm.tcplargereceiveoffload<br>tm.tcpsegmentationoffload<br>tm.minipfragsize<br>security.commoncriteria<br>vlangroup.flow.allocate<br>systemauth.disablebash<br>systemauth.disablerootlogin<br>ui.logaccess*;', 'list sys db<br>liveinstall.checksig;', 'list sys db<br>tm.tcpudptxchecksum', 'list sys |

| name | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p>global-settings;', 'list sys httpd ssl-ciphersuite;', 'list sys httpd;', 'list sys iconcontrol-soap', 'list sys log-config destination;', 'list sys log-config;', 'list sys management-ip;', 'list sys management-route;', 'list sys ntp;', 'list sys software update;', 'list sys sshd all-properties;', 'list sys syslog;', 'list sys ucs;', 'show running-config', 'show running-config auth;', 'show running-config net interface;', 'show running-config net route-domain;', 'show running-config net self;', 'show running-config net trunk;', 'show running-config net vlan-group;', 'show running-config net vlan;',</p> <p>FORTINET: 'show application list', 'show endpoint-control fctems', 'show firewall address', 'show firewall address6', 'show firewall addrgrp', 'show firewall addrgrp6', 'show firewall central', 'show firewall ipmacbinding setting', 'show firewall ipmacbinding table', 'show firewall ippool', 'show firewall ippool6', 'show firewall policy', 'show firewall policy46', 'show firewall policy6', 'show firewall policy64', 'show firewall schedule group', 'show firewall schedule onetime', 'show firewall schedule recurring', 'show firewall service category', 'show firewall service custom', 'show firewall service group', 'show firewall vip', 'show firewall vip46', 'show firewall vip6', 'show firewall vip64',</p> |

| name | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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|      | 'show firewall vipgrp', 'show firewall vipgrp46', 'show firewall vipgrp6',<br>'show firewall vipgrp64', 'show full-configuration system csf', 'show full-configuration vpn ssl settings', 'show full-configuration vpn ssl web portal',<br>'show log setting', 'show router access-list', 'show router access-list6', 'show router aspath-list', 'show router auth-path', 'show router bgp',<br>'show router community-list', 'show router extcommunity-list', 'show router isis', 'show router ospf', 'show router ospf6', 'show router policy',<br>'show router policy6', 'show router prefix-list', 'show router prefix-list6', 'show router rip', 'show router ripng',<br>'show router route-map', 'show router setting', 'show router static', 'show router static6', 'show system admin', 'show system central-management', 'show system dhcp server', 'show system dhcp6 server',<br>'show system dns', 'show system dns-server', 'show system external-resource', 'show system geoip-override', 'show system global',<br>'show system gre-tunnel', 'show system ha', 'show system interface', 'show system ipsec-aggregate',<br>'show system lldp network-policy', 'show system ntp', 'show system settings', 'show system snmp sysinfo', 'show system ssh-config',<br>'show system switch-interface', 'show system virtual-switch', 'show |



| name | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p>system zone', 'show user group', 'show user ldap', 'show user local', 'show user radius', 'show vpn ssl web portal', 'show webfilter profile'</p> <p>FXOS: 'show running-config'</p> <p>HP_COMWARE: 'display current-configuration'</p> <p>HP_PROVISION: 'show run', 'show running-config'</p> <p>HUAWEI: 'display current-configuration'</p> <p>IOS: 'show running-config', 'show running-config view full'</p> <p>IOS_XE: 'show running-config', 'show running-config view full'</p> <p>IOS_XR: 'show running-config', 'show running-config inheritance'</p> <p>JUNOS: 'show configuration groups   display omit', 'show configuration   display omit   display inheritance   except ##\$'</p> <p>NETSCALER: 'show ns runningConfig -withDefaults', 'show simpleacl -format INPUT'</p> <p>NETSCREEN: 'get config'</p> <p>NXOS: 'show running-config'</p> <p>OTHER: 'show running-config'</p> <p>PAN_OS: 'show config effective-running', 'show config running'</p> <p>SRX: 'show configuration groups   display omit', 'show configuration   display omit   display inheritance   except ##\$'</p> <p>VERSA: 'show configuration'</p> <p>VIPTELA: 'show running-config', 'show running-config   include host-</p> |

| name                 | description                                                                                                                                                                                                                                                                                                                                                              |
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|                      | name', 'show running-config   include system-ip'                                                                                                                                                                                                                                                                                                                         |
| CUSTOM               | --                                                                                                                                                                                                                                                                                                                                                                       |
| DEFAULT_GATEWAYS     | BLUECOAT: 'show ip-default-gateway'                                                                                                                                                                                                                                                                                                                                      |
| DEVICE_HARDWARE_INFO | ARUBA_AOS_CX: 'show aruba-central'<br>CUMULUS: 'cat /etc/lsb-release'<br>HP_PROVISION: 'display device manuinfo'<br>HUAWEI: 'display device elabel', 'display elabel'<br>IOS_XR: 'show inventory rack'<br>LINUX: 'uname -so'<br>OTHER: 'show version'<br>PAN_OS: 'show chassis inventory'<br>VERSA: 'show orgs org {} sd-wan detail {}'<br>VIPTELA: 'show system status' |
| DNS_CACHE            | ASA: 'show dns'<br>FXOS: 'show dns'<br>NETSCALER: 'show dns addrec {}'<br>NETSCREEN: 'get dns host cache'<br>PAN_OS: 'request system fqdn show', 'show dns-proxy fqdn all'<br>SRX: 'show host routing-instance {}', 'show host {}', 'show security dns-cache'                                                                                                            |
| DOMAIN_NAME          | APIC: 'show dns-domain'<br>CHECKPOINT: 'domainname', 'show domainname'<br>FORTINET: 'get system dns   grep domain'                                                                                                                                                                                                                                                       |

| name               | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| EVPN               | ARISTA_EOS: 'show bgp evpn instance'<br>IOS: 'show evpn peers vrf {}'<br>IOS_XE: 'show evpn peers vrf {}'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| F5_AFM_CONFIG      | F5: 'list security', 'list security bot-defense', 'list security dos profile', 'list security http profile', 'list security log', 'list sys db tm.fw.defaultaction'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| F5_AFM_STATE       | F5: 'show security', 'show sys clock'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| F5_CM_MODULE_STATE | F5: 'show cm failover-status;', 'show cm sync-status;'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| F5_LTM_CONFIG      | F5: 'bigpipe class list all', 'bigpipe ltm list all', 'bigpipe nat list all', 'bigpipe pool list all', 'bigpipe snat list all', 'bigpipe snat translation list all', 'bigpipe snatpool list all', 'bigpipe virtual address list all', 'bigpipe virtual list all', 'list ltm global-settings all-properties recursive;', 'show running-config ltm data-group all-properties recursive;', 'show running-config ltm data-group internal;', 'show running-config ltm nat all-properties recursive;', 'show running-config ltm pool all-properties recursive;', 'show running-config ltm snat all-properties recursive;', 'show running-config ltm snat-translation all-properties recursive;', 'show running-config ltm snatpool all-properties recursive;', 'show running-config ltm virtual all-properties recursive;', 'show running- |

| name                     | description                                                                                                                                                                                                                                                                                                                                                                              |
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|                          | config ltm virtual-address all-properties recursive;                                                                                                                                                                                                                                                                                                                                     |
| F5_LTM_IRULES            | F5: 'bigpipe rule list all', 'show running-config ltm rule recursive;'                                                                                                                                                                                                                                                                                                                   |
| F5_LTM_POOL_STATE        | F5: 'bigpipe pool all members show', 'show ltm pool recursive members;'                                                                                                                                                                                                                                                                                                                  |
| F5_VCMP_GUESTS           | F5: 'show vcmp guest detail all-properties;'                                                                                                                                                                                                                                                                                                                                             |
| F5_VCMP_GUESTS_ADDRS     | F5: 'list vcmp guest hostname management-ip;'                                                                                                                                                                                                                                                                                                                                            |
| F5_VIRTUAL_SERVER_STATE  | F5: 'bigpipe virtual show', 'show ltm virtual recursive;'                                                                                                                                                                                                                                                                                                                                |
| FABRICPATH_TABLE         | NXOS: 'show fabricpath route'                                                                                                                                                                                                                                                                                                                                                            |
| FAILOVER                 | ACOS: 'show ha detail', 'show ha mac'<br>ASA: 'show failover'<br>BLUECOAT: 'show failover'<br>F5: 'bigpipe failover show all'<br>FORTINET: 'get system ha', 'get system ha status'<br>FXOS: 'show failover'<br>JUNOS: 'show chassis routing-engine'<br>NETSCALER: 'show HA node'<br>OTHER: 'show failover'<br>PAN_OS: 'show high-availability all'<br>SRX: 'show chassis cluster status' |
| FIREWALL_DYNAMIC_OBJECTS | CHECKPOINT: '([ -f '<FW-dir>/CTX/CTX<context-id>/database/<dynamic_objects.db>' ] && cat '<FW-dir>/CTX/CTX<context-id>/database/<dynamic_objects.db>'                                                                                                                                                                                                                                    |

| name                      | description                                                                                                                                                                                                                                                                                                                                                                                              |
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|                           | echo ' '), '([ -f '<FW-dir>/database/<br><dynamic_objects.db>' ] && cat<br>'<FW-dir>/database/<br><dynamic_objects.db>'    echo ' '),<br>'fw showuptables', 'fw6<br>showuptables'                                                                                                                                                                                                                        |
| FIREWALL_IDENTITY_MONITOR | CHECKPOINT: 'pdp monitor all'                                                                                                                                                                                                                                                                                                                                                                            |
| FIREWALL_IDENTITY_ROLES   | CHECKPOINT: '([ -f '<FW-dir>/CTX/<br>CTX<context-id>/database/<br><identity_roles.C>' ] && cat '<FW-<br>dir>/CTX/CTX<context-id>/<br>database/<identity_roles.C>'    echo '<br>' )   gzip -c   uuencode -m -, '([ -f '<FW-<br>dir>/database/<identity_roles.C>' ]<br>&& cat '<FW-dir>/database/<br><identity_roles.C>'    echo ' ')   gzip -c<br>  uuencode -m -, 'cat {}/database/<br>identity_roles.C' |
| FIREWALL_OBJECTS          | CHECKPOINT: '([ -f '<FW-dir>/CTX/<br>CTX<context-id>/database/<br><objects.C>' ] && cat '<FW-dir>/CTX/<br>CTX<context-id>/database/<br><objects.C>'    echo ' ')   gzip -c  <br>uuencode -m -, '([ -f '<FW-dir>/<br>database/<objects.C>' ] && cat '<FW-<br>dir>/database/<objects.C>'    echo '<br>' )   gzip -c   uuencode -m -, 'cat {}/<br>database/objects.C'                                       |
| FIREWALL_POLICIES         | CHECKPOINT: '([ -f '<FW-dir>/CTX/<br>CTX<context-id>/database/<br><rules.C>' ] && cat '<FW-dir>/CTX/<br>CTX<context-id>/database/<br><rules.C>'    echo ' ')   gzip -c  <br>uuencode -m -, '([ -f '<FW-dir>/                                                                                                                                                                                             |

| name                               | description                                                                                                                                                                                                                                                                                                                                                                                                                                |
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|                                    | database/<rules.C>' ] && cat '<FW-dir>/database/<rules.C>'    echo ' ')   gzip -c   uuencode -m '-', 'cat {}/' database/rules.C'                                                                                                                                                                                                                                                                                                           |
| FORWARDING_MODES                   | SRX: 'show security flow status'                                                                                                                                                                                                                                                                                                                                                                                                           |
| HAIPE_BATTERY_CHANGE_DATE          | --                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| HAIPE_DVROW_STATUS                 | --                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| HAIPE_FIREFLY_EXPIRATION           | --                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| HAIPE_FIREFLY_NATIONAL_SHORT_TITLE | --                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| HAIPE_FIREFLY_TABLE                | --                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| HAIPE_STATION_IDENTIFIER           | --                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| HAIPE_SYSTEM_DATE                  | --                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| HAIPE_SYSTEM_UPTIME                | --                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| HEALTH_CHECKS_STATS                | BLUECOAT: 'show health-checks statistics'                                                                                                                                                                                                                                                                                                                                                                                                  |
| HOSTNAME                           | APIC: 'fabric {} show hostname'<br>ARISTA_EOS: 'show hostname'<br>ARUBA_AOS_CX: 'show hostname'<br>ASA: 'show hostname'<br>BLUECOAT: 'show appliance-name',<br>'show hostname'<br>CHECKPOINT: 'hostname', 'show<br>host names'<br>CUMULUS: 'hostname', 'hostname --<br>fqdn'<br>DELL_SONIC: 'show system'<br>F5: 'show running-config sys global-<br>settings hostname   grep hostname'<br>IOS: 'show hostname'<br>IOS_XE: 'show hostname' |

| name                | description                                                                                                                                                                                                       |
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|                     | LINUX: 'hostname', 'hostname --fqdn'<br>NETSCALER: 'show ns hostname'<br>NXOS: 'show hostname'                                                                                                                    |
| HSRP_STATE          | IOS: 'show standby'<br>IOS_XE: 'show standby'<br>IOS_XR: 'show hsrp brief'<br>NXOS: 'show hsrp detail'                                                                                                            |
| IFINDEX2_IPV4       | --                                                                                                                                                                                                                |
| IFINDEX_IPV4        | --                                                                                                                                                                                                                |
| IGMP_GROUPS         | ARISTA_EOS: 'show ip igmp groups'<br>IOS: 'show ip igmp group'<br>IOS_XE: 'show ip igmp group'<br>NXOS: 'show ip igmp local-groups'                                                                               |
| IGMP_INTERFACES     | ARISTA_EOS: 'show ip igmp interface'<br>IOS: 'show ip igmp interface'<br>IOS_XE: 'show ip igmp interface'<br>JUNOS: 'show igmp interface'<br>NXOS: 'show ip igmp interface vrf all'<br>SRX: 'show igmp interface' |
| IGMP_VRF_GROUPS     | ARISTA_EOS: 'show ip igmp vrf {} groups'<br>IOS: 'show ip igmp vrf {} groups'<br>IOS_XE: 'show ip igmp vrf {} groups'<br>NXOS: 'show ip igmp groups vrf all'                                                      |
| IGMP_VRF_INTERFACES | ARISTA_EOS: 'show ip igmp vrf {} interface'<br>IOS: 'show ip igmp vrf {} interface'<br>IOS_XE: 'show ip igmp vrf {} interface'<br>IOS_XR: 'show igmp vrf {} interface'                                            |

| name       | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| INTERFACES | <p>ACOS: 'show interfaces', 'show interfaces management', 'show trunk'</p> <p>APIC: 'fabric {} show interface', 'ifconfig'</p> <p>ARISTA_EOS: 'show interfaces'</p> <p>ARUBA_AOS_CX: 'show interface'</p> <p>ARUBA_WIFI: 'show interface', 'show interface gigabitethernet {}', 'show interface port-channel {}', 'show interface vlan {}', 'show port status'</p> <p>ASA: 'show interface detail'</p> <p>AVI_VANTAGE: 'GET /api/serviceengine/{name}', 'show serviceengine {}'</p> <p>BLUECOAT: 'show advanced-url /Diagnostics/Hardware/Info', 'show interface all'</p> <p>CHECKPOINT: 'ifconfig -a', 'show interface {}', 'show interfaces all'</p> <p>CUMULUS: 'ifconfig -a', 'ip addr'</p> <p>DELL_OS10: 'show interface', 'show interfaces status'</p> <p>DELL_OS6: 'show interface', 'show interfaces status'</p> <p>DELL_OS9: 'show interface', 'show interfaces status'</p> <p>DELL_SONIC: 'show interface {}'</p> <p>F5: 'bigpipe interface show all', 'bigpipe trunk show all', 'show net interface all-properties;', 'show net trunk;'</p> <p>FORTINET: 'get system interface'</p> <p>FXOS: 'show interface detail'</p> <p>HP_COMWARE: 'display interface'</p> <p>HP_PROVISION: 'show interfaces all'</p> |



| name                     | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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|                          | HUAWEI: 'display interface'<br>IOS: 'show interfaces'<br>IOS_XE: 'show interfaces'<br>IOS_XR: 'show interfaces'<br>JUNOS: 'show interfaces', 'show interfaces   display no-interface-alias'<br>LINUX: 'ifconfig -a', 'ip addr'<br>NETSCALER: 'show interface -format TEXT'<br>NETSCREEN: 'get interface', 'get interface {}'<br>NXOS: 'show interface', 'show interface {}'<br>OTHER: 'show interfaces brief'<br>PAN_OS: 'show interface hardware'<br>SRX: 'show interfaces', 'show interfaces   display no-interface-alias'<br>VERSA: 'show interfaces detail' |
| INTERFACES_MAC           | APIC: 'fabric {} show interface mac-address'<br>F5: 'show sys mac-address;'<br>FORTINET: 'diagnose netlink interface list "{}"<br>HUAWEI: 'display system', 'display system mac-address all'<br>NXOS: 'show interface mac-address'                                                                                                                                                                                                                                                                                                                              |
| INTERFACES_QUEUEING_INFO | NXOS: 'show queuing interface', 'show queuing interface {}'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| INTERFACES_STATUS        | APIC: 'fabric {} show ip interface brief'<br>ARISTA_EOS: 'show interfaces status'                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| name             | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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|                  | CHECKPOINT: 'for f in /sys/class/net/*; do ethtool \$(basename \$f); done'<br>DELL_OS10: 'show int switchport', 'show ip interface'<br>DELL_OS6: 'show int switchport', 'show ip interface'<br>DELL_OS9: 'show int switchport', 'show ip interface'<br>DELL_SONIC: 'show ip interfaces'<br>FORTINET: 'diagnose hardware deviceinfo nic', 'diagnose hardware deviceinfo nic "{}"<br>HP_PROVISION: 'show interfaces status'<br>HUAWEI: 'display interface brief'<br>IOS: 'show ip interface brief'<br>IOS_XE: 'show ip interface brief'<br>JUNOS: 'show interfaces terse', 'show interfaces terse   display no-interface-alias'<br>NXOS: 'show interface brief', 'show ip interface brief'<br>SRX: 'show interfaces terse', 'show interfaces terse   display no-interface-alias'<br>VERSA: 'show interfaces brief'<br>VIPTELA: 'show interface' |
| INTERFACES_TRUNK | IOS: 'show interface trunk'<br>IOS_XE: 'show interface trunk'<br>NXOS: 'show interface trunk'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| INTERFACES_VRF   | APIC: 'fabric {} show vrf all interface'<br>NXOS: 'show vrf all interface'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| INVENTORY        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| name | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|      | <p> APIC: 'fabric {} show inventory'<br/> ARISTA_EOS: 'show inventory'<br/> ARUBA_AOS_CX: 'show system inventory'<br/> ARUBA_WIFI: 'show inventory'<br/> ASA: 'show chassis detail', 'show inventory'<br/> CHECKPOINT: 'clish -c "show asset system"', 'show asset all', 'system_info; dmidecode'<br/> CUMULUS: 'cat /etc/machine-id', 'cat /var/lib/dbus/machine-id'<br/> DELL_OS10: 'show inventory', 'show switch', 'show system id'<br/> DELL_OS6: 'show inventory', 'show switch', 'show system id'<br/> DELL_OS9: 'show inventory', 'show switch', 'show system id'<br/> FXOS: 'show inventory'<br/> HP_COMWARE: 'display device manuinfo'<br/> HP_PROVISION: 'show modules'<br/> HUAWEI: 'display device esn', 'display esn', 'display sn'<br/> IOS: 'show inventory'<br/> IOS_XE: 'show inventory'<br/> IOS_XR: 'show inventory'<br/> JUNOS: 'show chassis hardware detail'<br/> LINUX: 'cat /etc/machine-id', 'cat /var/lib/dbus/machine-id'<br/> NETSCALER: 'show hardware'<br/> NETSCREEN: 'get chassis'<br/> NXOS: 'show inventory'<br/> OTHER: 'show hardware all' </p> |

| name            | description                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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|                 | PAN_OS: 'show system info'<br>SRX: 'show chassis hardware detail'                                                                                                                                                                                                                                                                                                                                                                                                        |
| IPSEC_STATE     | ARUBA_WIFI: 'show crypto ipsec sa'<br>F5: 'show net ipsec ike-sa all-properties', 'show net ipsec traffic-selector'<br>HUAWEI: 'display ipsec sa'                                                                                                                                                                                                                                                                                                                        |
| IPV4_INTERFACES | HP_COMWARE: 'display ip interface'<br>HP_PROVISION: 'show ip'<br>PAN_OS: 'show interface logical'                                                                                                                                                                                                                                                                                                                                                                        |
| IPV4_MCAST_RP   | ARISTA_EOS: 'show ip pim rp detail', 'show ip pim vrf {} rp detail'<br>IOS: 'show ip pim rp mapping', 'show ip pim vrf {} rp mapping'<br>IOS_XE: 'show ip pim rp mapping', 'show ip pim vrf {} rp mapping'<br>IOS_XR: 'show pim vrf {} ipv4 rp mapping'<br>NXOS: 'show ip pim rp vrf all'                                                                                                                                                                                |
| IPV4_VRRP_STATE | ACOS: 'show vrrp-a detail', 'show vrrp-a hostid', 'show vrrp-a mac'<br>ARISTA_EOS: 'show vrrp vrf all'<br>ARUBA_AOS_CX: 'show vrrp'<br>CHECKPOINT: 'clish -c "show vrrp interfaces"', 'show vrrp interfaces'<br>DELL_OS10: 'show vrrp', 'show vrrp {}'<br>DELL_OS6: 'show vrrp', 'show vrrp {}'<br>DELL_OS9: 'show vrrp', 'show vrrp {}'<br>DELL_SONIC: 'show vrrp'<br>HUAWEI: 'display vrrp', 'display vrrp verbose'<br>IOS: 'show vrrp all'<br>IOS_XE: 'show vrrp all' |

| name                     | description                                                                                                                                                                                                                                                                                             |
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|                          | IOS_XR: 'show vrrp brief'<br>JUNOS: 'show vrrp detail'<br>NETSCALER: 'show vrID'<br>NXOS: 'show vrrp detail'<br>SRX: 'show vrrp detail'<br>VERSA: 'show vrrp'<br>VIPTELA: 'show vrrp'                                                                                                                   |
| IPV6_IGMP_VRF_INTERFACES | IOS: 'show ipv6 mld interface'<br>IOS_XE: 'show ipv6 mld interface'<br>IOS_XR: 'show pim vrf {} ipv6 interface'<br>NXOS: 'show ipv6 mld interface vrf all'                                                                                                                                              |
| IPV6_INTERFACES          | HP_COMWARE: 'display ipv6 interface', 'display ipv6 interface verbose'<br>HP_PROVISION: 'show ipv6'<br>HUAWEI: 'display ipv6 interface'                                                                                                                                                                 |
| IPV6_MCAST_RP            | ARISTA_EOS: 'show ipv6 pim rp detail', 'show ipv6 pim vrf {} rp detail'<br>IOS: 'show ipv6 pim range-list', 'show ipv6 pim vrf {} range-list'<br>IOS_XE: 'show ipv6 pim range-list', 'show ipv6 pim vrf {} range-list'<br>IOS_XR: 'show pim vrf {} ipv6 rp mapping'<br>NXOS: 'show ipv6 pim rp vrf all' |
| IPV6_VRRP_STATE          | DELL_OS10: 'show vrrp ipv6 {}'<br>DELL_OS6: 'show vrrp ipv6 {}'<br>DELL_OS9: 'show vrrp ipv6 {}'<br>HUAWEI: 'display vrrp6', 'display vrrp6 verbose'<br>NETSCALER: 'show vrID6'                                                                                                                         |

| name                   | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| IP_IGMP_SNOOP_TRACKING | ARISTA_EOS: 'show ip igmp snooping groups local detail'<br>NXOS: 'show ip igmp snooping explicit-tracking'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| L2CIRCUITS_STATE       | JUNOS: 'show l2circuit connections'<br>SRX: 'show l2circuit connections'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| L2VPN_BRIDGE_DOMAIN    | IOS_XR: 'show l2vpn bridge-domain detail'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| L2VPN_XCONNECT_STATE   | IOS_XR: 'show l2vpn xconnect detail'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| LICENSE_INFO           | ACOS: 'show license', 'show license-info'<br>IOS_XR: 'show license udi'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| LLDP                   | ACOS: 'show lldp neighbors'<br>APIC: 'fabric {} show lldp neighbors', 'fabric {} show lldp neighbors detail'<br>ARISTA_EOS: 'show lldp neighbors detail'<br>ARUBA_AOS_CX: 'show lldp neighbor-info', 'show lldp neighbor-info detail'<br>ARUBA_WIFI: 'show lldp neighbor'<br>CUMULUS: 'net show lldp json'<br>DELL_OS10: 'show lldp neighbors', 'show lldp neighbors detail', 'show lldp remote-device all'<br>DELL_OS6: 'show lldp neighbors', 'show lldp neighbors detail', 'show lldp remote-device all'<br>DELL_OS9: 'show lldp neighbors', 'show lldp neighbors detail', 'show lldp remote-device all'<br>DELL_SONIC: 'show lldp neighbor'<br>F5: 'show net lldp-neighbors all' |

| name      | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | <p>properties;'</p> <p>FORTINET: 'diag lldp rx neighbor details', 'diag lldprx neighbor details'</p> <p>HP_COMWARE: 'display lldp neighbor-information', 'display lldp neighbor-information verbose'</p> <p>HP_PROVISION: 'show lldp info remote-device all'</p> <p>HUAWEI: 'display lldp neighbor'</p> <p>IOS: 'show lldp neighbors', 'show lldp neighbors detail'</p> <p>IOS_XE: 'show lldp neighbors', 'show lldp neighbors detail'</p> <p>IOS_XR: 'show lldp neighbors', 'show lldp neighbors detail'</p> <p>JUNOS: 'show lldp neighbors', 'show lldp neighbors interface {}'</p> <p>NETSCALER: 'show lldp neighbors'</p> <p>NXOS: 'show lldp neighbors', 'show lldp neighbors detail'</p> <p>PAN_OS: 'show lldp neighbors all'</p> <p>SRX: 'show lldp neighbors', 'show lldp neighbors interface {}'</p> <p>VERSA: 'show lldp neighbor detail'</p> |
| MAC_TABLE | <p>ACOS: 'show mac-address-table'</p> <p>ARISTA_EOS: 'show mac address-table'</p> <p>ARUBA_AOS_CX: 'show mac-address-table'</p> <p>ARUBA_WIFI: 'show datapath bridge'</p> <p>ASA: 'show mac-address-table'</p> <p>BLUECOAT: 'show bridge fwtable {}'</p> <p>CHECKPOINT: 'brctl showmacs {}', 'for iface in \$(brctl show   tail -n +2   grep -v '^[:space:]')   awk '{print \$1}');</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| name | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p>do echo "brctl showmacs \$iface" &amp;&amp;<br/>brctl showmacs \$iface; done'</p> <p>CUMULUS: 'bridge fdb', 'net show<br/>bridge macs'</p> <p>DELL_OS10: 'show mac address-<br/>table', 'show mac-address-table'</p> <p>DELL_OS6: 'show mac address-<br/>table', 'show mac-address-table'</p> <p>DELL_OS9: 'show mac address-<br/>table', 'show mac-address-table'</p> <p>DELL_SONIC: 'show mac address-<br/>table'</p> <p>F5: 'bigpipe vlan all fdb show', 'show<br/>net fdb;'</p> <p>FORTINET: 'diagnose netlink brctl<br/>name host {}'</p> <p>FXOS: 'show mac-address-table'</p> <p>HP_COMWARE: 'display mac-<br/>address'</p> <p>HP_PROVISION: 'show mac-address'</p> <p>HUAWEI: 'display mac-address'</p> <p>IOS: 'show mac address-table', 'show<br/>mac address-table module {}', 'show<br/>mac-address-table', 'show mac-<br/>address-table module {}'</p> <p>IOS_XE: 'show mac address-table',<br/>'show mac address-table module {}',<br/>'show mac-address-table', 'show<br/>mac-address-table module {}'</p> <p>IOS_XR: 'show l2vpn forwarding<br/>bridge-domain mac-address<br/>location ?', 'show l2vpn forwarding<br/>bridge-domain mac-address<br/>location {}'</p> <p>JUNOS: 'show bridge mac-table',<br/>'show ethernet-switching table'</p> |



| name                         | description                                                                                                                                                                                                                                                                                                 |
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|                              | NETSCALER: 'show bridgetable<br>-format TEXT'<br>NETSCREEN: 'get mac-learn'<br>NXOS: 'show mac address-table',<br>'show mac-address-table'<br>OTHER: 'support show ether-relay'<br>PAN_OS: 'show mac all'<br>SRX: 'show bridge mac-table', 'show<br>ethernet-switching table'<br>VIPTELA: 'show bridge mac' |
| MGMT_SERVICES                | BLUECOAT: 'show management-<br>services'                                                                                                                                                                                                                                                                    |
| MIB_MAC_ADDRESSES            | HP_PROVISION: 'walkMIB<br>ifPhysAddress'                                                                                                                                                                                                                                                                    |
| MLAG_INTERFACE_STATE         | ARISTA_EOS: 'show mlag interfaces<br>detail'<br>DELL_OS10: 'show vlt brief', 'show vlt<br>{', 'show vpc brief'<br>DELL_OS6: 'show vlt brief', 'show vlt<br>{', 'show vpc brief'<br>DELL_OS9: 'show vlt brief', 'show vlt<br>{', 'show vpc brief'<br>HUAWEI: 'display dfs-group 1 peer-<br>link'             |
| MLAG_STATE                   | ARISTA_EOS: 'show mlag detail'<br>CUMULUS: 'net show clag'<br>HUAWEI: 'display dfs-group 1 m-lag',<br>'display dfs-group 1 m-lag brief'                                                                                                                                                                     |
| MPLS_BYPASS_LSP_TUNNEL_STATE | JUNOS: 'show mpls lsp bypass<br>detail'<br>SRX: 'show mpls lsp bypass detail'                                                                                                                                                                                                                               |
| MPLS_PSEUDOWIRE_STATE        |                                                                                                                                                                                                                                                                                                             |

| name                       | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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|                            | IOS: 'show mpls l2transport pwid'<br>IOS_XE: 'show mpls l2transport pwid'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| MPLS_TE_TUNNELS_STATE      | ARISTA_EOS: 'show tunnel fib'<br>HUAWEI: 'display mpls lsp verbose'<br>IOS: 'show mpls traffic-eng tunnels'<br>IOS_XE: 'show mpls traffic-eng tunnels'                                                                                                                                                                                                                                                                                                                                                                                                                        |
| MPLS_VIRTUAL_CIRCUIT_STATE | HUAWEI: 'display mpls l2vc'<br>IOS: 'show mpls l2transport vc detail'<br>IOS_XE: 'show mpls l2transport vc detail'<br>NXOS: 'show mpls l2transport vc detail'                                                                                                                                                                                                                                                                                                                                                                                                                 |
| NAT_ADDRESSES              | PAN_OS: 'show running nat-policy'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| NDP_TABLE                  | ACOS: 'show ipv6 neighbor'<br>ARISTA_EOS: 'show ipv6 neighbors'<br>ARUBA_AOS_CX: 'show ipv6 neighbors all-vrfs'<br>ARUBA_WIFI: 'show ipv6 neighbors'<br>ASA: 'show ipv6 neighbor'<br>AVI_VANTAGE: 'GET /api/serviceengine/{name}/ndtable',<br>'show serviceengine {} ndtable'<br>BLUECOAT: 'show ndp'<br>CHECKPOINT: 'ip -6 neigh show',<br>'show neighbor table'<br>CUMULUS: 'ip -6 neigh show'<br>DELL_OS10: 'show ipv6 neighbors',<br>'show ipv6 neighbors vrf {}'<br>DELL_OS6: 'show ipv6 neighbors',<br>'show ipv6 neighbors vrf {}'<br>DELL_OS9: 'show ipv6 neighbors', |

| name                  | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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|                       | 'show ipv6 neighbors vrf {}'<br>DELL_SONIC: 'show ipv6 neighbors'<br>F5: 'bigpipe ndp show', 'show net ndp;'<br>FORTINET: 'diagnose ipv6 neighbor-cache list'<br>FXOS: 'show ipv6 neighbor'<br>HP_COMWARE: 'display ipv6 neighbors all'<br>HP_PROVISION: 'show ipv6 neighbors'<br>IOS: 'show ipv6 neighbors'<br>IOS_XE: 'show ipv6 neighbors'<br>IOS_XR: 'show ipv6 neighbors'<br>JUNOS: 'show ipv6 neighbors'<br>LINUX: 'ip -6 neigh show'<br>NETSCALER: 'show nd6 -format TEXT'<br>NETSCREEN: 'get ndp'<br>NXOS: 'show ipv6 adjacency vrf all'<br>PAN_OS: 'show neighbor all', 'show neighbor interface all'<br>SRX: 'show ipv6 neighbors'<br>VERSA: 'show ipv6 neighbors all' |
| NETSCREEN_NSRRP_STATE | NETSCREEN: 'get nsrrp cluster', 'get nsrrp vsd-group'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| NETWORK_PROFILE       | AVI_VANTAGE: 'GET /api/networkprofile', 'show networkprofile'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| NHRP_STATE            | IOS: 'show ip nhrp', 'show ipv6 nhrp'<br>IOS_XE: 'show ip nhrp', 'show ipv6 nhrp'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| OTV                   | NXOS: 'show otv', 'show otv adjacency', 'show otv route'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| name                              | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| PACKET_FILTER_CONFIG              | HP_COMWARE: 'display packet-filter all', 'display packet-filter interface'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| PANOS_CONFIG_PUSHED_SHARED_POLICY | PAN_OS: 'show config pushed-shared-policy vsys {}'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| PANOS_CONFIG_PUSHED_TEMPLATE      | PAN_OS: 'show config pushed-template'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| PANOS_EXTERNAL_IP_LIST            | PAN_OS: 'request system external-list show type ip name "{}"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| PARTITION_CONFIG                  | F5: 'list auth partition'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| POLICY_ORDER                      | BLUECOAT: 'show policy order'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| POLICY_TRAFFIC_SELECTORS          | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| PORT_CHANNEL_MEMBERS              | APIC: 'fabric {} show port-channel extended'<br>ARISTA_EOS: 'show port-channel active-ports'<br>ARUBA_AOS_CX: 'show lacp interfaces'<br>ASA: 'show port-channel summary'<br>DELL_OS10: 'show interfaces port-channel', 'show interfaces port-channel brief', 'show port-channel summary'<br>DELL_OS6: 'show interfaces port-channel', 'show interfaces port-channel brief', 'show port-channel summary'<br>DELL_OS9: 'show interfaces port-channel', 'show interfaces port-channel brief', 'show port-channel summary'<br>FORTINET: 'diag netlink aggregate list', 'diag netlink aggregate name |

| name                          | description                                                                                                                                                                                                                                                                           |
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|                               | "{}", 'diag netlink redundant list', 'diag netlink redundant name "{}"<br>IOS: 'show etherchannel summary'<br>IOS_XE: 'show etherchannel summary'<br>NETSCALER: 'show channel'<br>NXOS: 'show port-channel summary', 'show port-channel extended'<br>VERSA: 'show aggregate-ethernet' |
| PORT_LABELS                   | OTHER: 'show port-label {}'                                                                                                                                                                                                                                                           |
| PORT_MIRRORING_STATE          | F5: 'bigpipe mirror list', 'list net port-mirror;'                                                                                                                                                                                                                                    |
| PORT_SECURITY_STATE           | IOS: 'show port-security address'<br>IOS_XE: 'show port-security address'<br>NXOS: 'show port-security address'                                                                                                                                                                       |
| PREDEFINED_APPS               | SRX: 'show configuration groups junos-defaults applications'                                                                                                                                                                                                                          |
| PROXY_POLICIES                | BLUECOAT: 'show policy', 'show policy executable'                                                                                                                                                                                                                                     |
| PROXY_SERVICES                | BLUECOAT: 'show proxy-services'                                                                                                                                                                                                                                                       |
| REGISTERED_IP_ADDRESSES       | PAN_OS: 'show object registered-ip all'                                                                                                                                                                                                                                               |
| RIVERBED_HOST_LABELS          | OTHER: 'show host-label {} detailed'                                                                                                                                                                                                                                                  |
| RIVERBED_IN_PATH_NEIGHBORS    | OTHER: 'show in-path neighbor'                                                                                                                                                                                                                                                        |
| RIVERBED_IN_PATH_RULES        | OTHER: 'show in-path rules'                                                                                                                                                                                                                                                           |
| RIVERBED_PATH_SELECTION_RULES | OTHER: 'show path-selection rules', 'show topology uplink {} site {} path-selection state'                                                                                                                                                                                            |
| RIVERBED_STEELHEAD_NEIGHBORS  | OTHER: 'show steelhead name all'                                                                                                                                                                                                                                                      |

| name                     | description                                                                                                                                    |
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| ROOT_CONTEXT_CONFIG      | ASA: 'show running-config'<br>NETSCALER: 'show simpleacl<br>-format INPUT'                                                                     |
| RSVP_SESSION_STATE       | JUNOS: 'show rsvp session detail'<br>SRX: 'show rsvp session detail'                                                                           |
| SDWAN_CONFIG             | IOS_XE: 'show sdwan running-<br>config', 'show sdwan running-config  <br>include system-ip'<br>VERSA: 'show orgs org {} sd-wan<br>detail'      |
| SLA_IP_TRACKING_STATE    | IOS: 'show ip sla summary'<br>IOS_XE: 'show ip sla summary'<br>IOS_XR: 'show track'<br>VERSA: 'show orgs org {} sd-wan sla-<br>monitor status' |
| SLB_CS_VIRTUAL_SERVERS   | NETSCALER: 'show cs vserver'                                                                                                                   |
| SLB_GSLB_SERVICES        | NETSCALER: 'show gslb service'                                                                                                                 |
| SLB_GSLB_VIRTUAL_SERVERS | NETSCALER: 'show gslb vserver'                                                                                                                 |
| SLB_SERVERS              | ACOS: 'show slb server'                                                                                                                        |
| SLB_SERVER_DETAILS       | ACOS: 'show slb server {} detail'                                                                                                              |
| SLB_SERVICES             | NETSCALER: 'show service'                                                                                                                      |
| SLB_SERVICE_GROUPS       | ACOS: 'show slb service-group'<br>NETSCALER: 'show serviceGroup<br>-includeMembers'                                                            |
| SLB_VIRTUAL_SERVERS      | ACOS: 'show slb virtual-server'<br>NETSCALER: 'show lb vserver'                                                                                |
| SNMP_SYS_DESCR           | --                                                                                                                                             |
| SNMP_SYS_NAME            | --                                                                                                                                             |

| name                | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| SNMP_TARGET_ADDRESS | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| SSH_RSA_KEY_LENGTH  | IOS_XR: 'show crypto key mypubkey<br>rsa   include Size'<br>JUNOS: 'show security pki local-<br>certificate system-generated   find<br>bits'<br>NXOS: 'show ssh key   include<br>bitcount'<br>SRX: 'show security pki local-<br>certificate system-generated   find<br>bits'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| STP_STATE           | ARISTA_EOS: 'show spanning-tree'<br>ARUBA_AOS_CX: 'show spanning-<br>tree', 'show spanning-tree mst'<br>CUMULUS: 'net show bridge<br>spanning-tree detail'<br>DELL_OS10: 'show spanning-tree',<br>'show spanning-tree 0 brief', 'show<br>spanning-tree msti {} brief', 'show<br>spanning-tree pvst vlan {} brief',<br>'show spanning-tree rstp brief',<br>'show spanning-tree vlan {}'<br>DELL_OS6: 'show spanning-tree',<br>'show spanning-tree 0 brief', 'show<br>spanning-tree msti {} brief', 'show<br>spanning-tree pvst vlan {} brief',<br>'show spanning-tree rstp brief',<br>'show spanning-tree vlan {}'<br>DELL_OS9: 'show spanning-tree',<br>'show spanning-tree 0 brief', 'show<br>spanning-tree msti {} brief', 'show<br>spanning-tree pvst vlan {} brief',<br>'show spanning-tree rstp brief',<br>'show spanning-tree vlan {}'<br>DELL_SONIC: 'show spanning-tree' |

| name             | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                  | <p>F5: 'bigpipe stp show all', 'show net stp;'</p> <p>FORTINET: 'get system stp list'</p> <p>HP_COMWARE: 'display stp brief'</p> <p>HP_PROVISION: 'show spanning-tree', 'show spanning-tree instance {}', 'show spanning-tree vlan {}'</p> <p>HUAWEI: 'display stp brief'</p> <p>IOS: 'show spanning-tree'</p> <p>IOS_XE: 'show spanning-tree'</p> <p>JUNOS: 'show spanning-tree interface', 'show spanning-tree interface routing-instance {}'</p> <p>NXOS: 'show spanning-tree'</p> <p>SRX: 'show spanning-tree interface', 'show spanning-tree interface routing-instance {}'</p> |
| SWITCHPORT_STATE | <p>APIC: 'fabric {} show interface switchport'</p> <p>ARISTA_EOS: 'show interfaces switchport'</p> <p>IOS: 'show interface switchport'</p> <p>IOS_XE: 'show interface switchport'</p> <p>NXOS: 'show interface switchport', 'show interface {} switchport'</p>                                                                                                                                                                                                                                                                                                                       |
| SYSTEM_SETTINGS  | <p>ARUBA_AOS_CX: 'show system'</p> <p>ASA: 'show firewall'</p> <p>CUMULUS: 'net show system'</p> <p>DELL_OS10: 'show system'</p> <p>DELL_OS6: 'show system'</p> <p>DELL_OS9: 'show system'</p> <p>FORTINET: 'get system global', 'get system settings'</p> <p>FXOS: 'show firewall'</p> <p>IOS_XE: 'show sdwan control local-</p>                                                                                                                                                                                                                                                    |



| name           | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|                | properties'<br>NETSCREEN: 'get system'<br>PAN_OS: 'show system setting<br>jumbo-frame'<br>VERSA: 'show system details'<br>VIPTELA: 'show control local-<br>properties'                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| TUNNELS_STATUS | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| UPTIME         | CHECKPOINT: 'show uptime',<br>'uptime -p'<br>F5: 'bash -c uptime', 'uptime'<br>FORTINET: 'get sys per status'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| VERSION        | ACOS: 'show hardware', 'show<br>version'<br>APIC: 'fabric {} show version', 'show<br>version'<br>ARISTA_EOS: 'show version'<br>ARUBA_AOS_CX: 'show version'<br>ARUBA_WIFI: 'show version'<br>ASA: 'show version'<br>AVAYA: 'show version'<br>AVI_VANTAGE: 'GET /api/<br>serviceengine', 'show serviceengine<br>{} summary'<br>BLUECOAT: 'show advanced-url /<br>Diagnostics/Hardware/Info', 'show<br>version'<br>CHECKPOINT: 'POST /web_api/<br>show-api-versions', 'cat /etc/cp-<br>release && cpinfo -y all', 'cpinfo -y<br>all', 'show version all'<br>CLOUD_GENIX: 'dump overview'<br>CUMULUS: 'cat /etc/lsb-release',<br>'cat /etc/os-release' |

| name              | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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|                   | DELL_OS10: 'show version'<br>DELL_OS6: 'show version'<br>DELL_OS9: 'show version'<br>DELL_SONIC: 'show version'<br>F5: 'bigpipe platform show', 'bigpipe<br>version show', 'show sys hardware;',<br>'show sys license', 'show sys<br>version;'<br>FORTINET: 'get system status'<br>FXOS: 'show version system'<br>HP_COMWARE: 'display version'<br>HP_PROVISION: 'show version'<br>HUAWEI: 'display version'<br>IOS: 'show version'<br>IOS_XE: 'show version'<br>IOS_XR: 'show version'<br>JUNOS: 'show version', 'show<br>version invoke-on other-routing-<br>engine'<br>LINUX: 'cat /etc/lsb-release', 'cat /<br>etc/os-release'<br>NETSCALER: 'show ns info'<br>NXOS: 'show version'<br>OTHER: 'show info'<br>PENSANDO: 'GET /configs/cluster/<br>v1/version'<br>SILVER_PEAK: 'GET /gms/rest/<br>gmsserver/info', 'show version'<br>SRX: 'show version', 'show version<br>invoke-on other-routing-engine'<br>T128: 'show system version', 'show<br>system version router {} node {}<br>force'<br>VIPTELA: 'show version' |
| VIRTUAL_ADDRESSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| name             | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                  | PAN_OS: 'show high-availability virtual-address'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| VIRTUAL_CONTEXTS | ASA: 'show context'<br>CHECKPOINT: 'fw vsx stat -l'<br>F5: 'show sys provision'<br>NETSCALER: 'show ns partition -format OLD'<br>PAN_OS: 'set system setting target-vsys ?', 'show plugins vm_series aws gwlb'                                                                                                                                                                                                                                                                                                                                                                                                                               |
| VLAN_LIST        | ACOS: 'show vlans'<br>ARISTA_EOS: 'show vlan'<br>ARUBA_AOS_CX: 'show vlan'<br>ARUBA_WIFI: 'show vlan'<br>DELL_OS10: 'show vlan'<br>DELL_OS6: 'show vlan'<br>DELL_OS9: 'show vlan'<br>F5: 'bigpipe vlan show', 'bigpipe vlangroup show', 'show net vlan-group;', 'show net vlan;<br>HP_PROVISION: 'show vlans'<br>HUAWEI: 'display mux-vlan', 'display vlan', 'display vlan brief'<br>IOS: 'show vlan'<br>IOS_XE: 'show vlan'<br>JUNOS: 'show bridge domain', 'show vlans extensive'<br>NXOS: 'show vlan'<br>OTHER: 'show in-path interfaces', 'show in-path'<br>PAN_OS: 'show vlan all'<br>SRX: 'show bridge domain', 'show vlans extensive' |
| VNIC_STATE       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| name              | description                                                                                                                                                                                                                                                                             |
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|                   | AVI_VANTAGE: 'GET /api/serviceengine/{name}/vnicdb', 'show serviceengine {} vnicdb'                                                                                                                                                                                                     |
| VPC_CONSISTENCY   | NXOS: 'show vpc', 'show vpc consistency-parameters global', 'show vpc consistency-parameters vpc {}'                                                                                                                                                                                    |
| VPLS_FORWARDING   | HUAWEI: 'display vpls services all'<br>JUNOS: 'show route forwarding-table family vpls'<br>SRX: 'show route forwarding-table family vpls'                                                                                                                                               |
| VPLS_MAC_TABLE    | JUNOS: 'show vpls mac-table'<br>SRX: 'show vpls mac-table'                                                                                                                                                                                                                              |
| VPN_TUNNELS       | ASA: 'show ipsec sa'<br>FORTINET: 'get vpn ipsec tunnel details', 'get vpn ipsec tunnel summary', 'show vpn ipsec phase1-interface', 'show vpn ipsec phase2-interface'<br>PAN_OS: 'show global-protect-gateway gateway', 'show vpn flow'<br>SRX: 'show security ipsec next-hop-tunnels' |
| VRF_MCAST_ROUTING | ARISTA_EOS: 'show ip mroute', 'show ip mroute vrf {}'<br>IOS: 'show ip mroute', 'show ip mroute vrf {}'<br>IOS_XE: 'show ip mroute', 'show ip mroute vrf {}'<br>IOS_XR: 'show mrib vrf {} ipv4 route'<br>JUNOS: 'show pim rps instance {}'<br>NXOS: 'show ip mroute vrf all', 'show     |

| name            | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | ip mroute detail vrf all'<br>SRX: 'show pim rps instance {}'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| VXLAN_ESI       | JUNOS: 'show ethernet-switching vxlan-tunnel-end-point esi', 'show l2-learning vxlan-tunnel-end-point esi'<br>SRX: 'show ethernet-switching vxlan-tunnel-end-point esi', 'show l2-learning vxlan-tunnel-end-point esi'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| VXLAN_MAC_TABLE | ARISTA_EOS: 'show vxlan address-table'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| VXLAN_STATE     | ARISTA_EOS: 'show vxlan vtep', 'show vxlan vtep detail'<br>ARUBA_AOS_CX: 'show interface vxlan'<br>CUMULUS: 'net show evpn rmac vni all;net show evpn vni json'<br>DELL_OS10: 'show nve vxlan-vni'<br>DELL_OS6: 'show nve vxlan-vni'<br>DELL_OS9: 'show nve vxlan-vni'<br>DELL_SONIC: 'show vxlan remote vni', 'show vxlan tunnel'<br>HUAWEI: 'display vxlan tunnel verbose'<br>IOS: 'show nve vni'<br>IOS_XE: 'show nve vni'<br>JUNOS: 'show ethernet-switching vxlan-tunnel-end-point remote', 'show l2-learning vxlan-tunnel-end-point remote'<br>NXOS: 'show nve ethernet-segment', 'show nve interface', 'show nve peers detail', 'show nve vni'<br>SRX: 'show ethernet-switching vxlan-tunnel-end-point remote', 'show l2- |

| name       | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | learning vxlan-tunnel-end-point remote'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| WCCP_STATE | ASA: 'show wccp', 'show wccp vrf {}',<br>'show wccp vrf {} {} detail', 'show wccp {} detail'<br>BLUECOAT: 'show wccp configuration'<br>IOS: 'show ip wccp', 'show ip wccp vrf {}', 'show ip wccp vrf {} {} detail', 'show ip wccp {} detail'<br>IOS_XE: 'show ip wccp', 'show ip wccp vrf {}', 'show ip wccp vrf {} {} detail', 'show ip wccp {} detail'<br>NXOS: 'show ip wccp', 'show ip wccp vrf {}', 'show ip wccp vrf {} {} detail', 'show ip wccp {} detail'<br>OTHER: 'show wccp interface {} service-group {} detail' |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **outputs** [â](#) **commands** [â](#) **commandType**[â](#)

**See also**[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - commands: Bag< [Command](#) >
        - commandType: [CommandType](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ComputeInstance

On this page

# ComputeInstance

## Fields

| name           | type               | nullable | description         |
|----------------|--------------------|----------|---------------------|
| name           | String             | true     | Never null for GCP. |
| id             | String             | false    | --                  |
| tags           | Bag<String>        | false    | --                  |
| isUp           | Bool               | false    | --                  |
| imageName      | String             | true     | --                  |
| instanceType   | String             | false    | --                  |
| instanceIfaces | Bag<InstanceIface> | false    | --                  |

## Used by

network cloudAccounts vpcs computeInstances

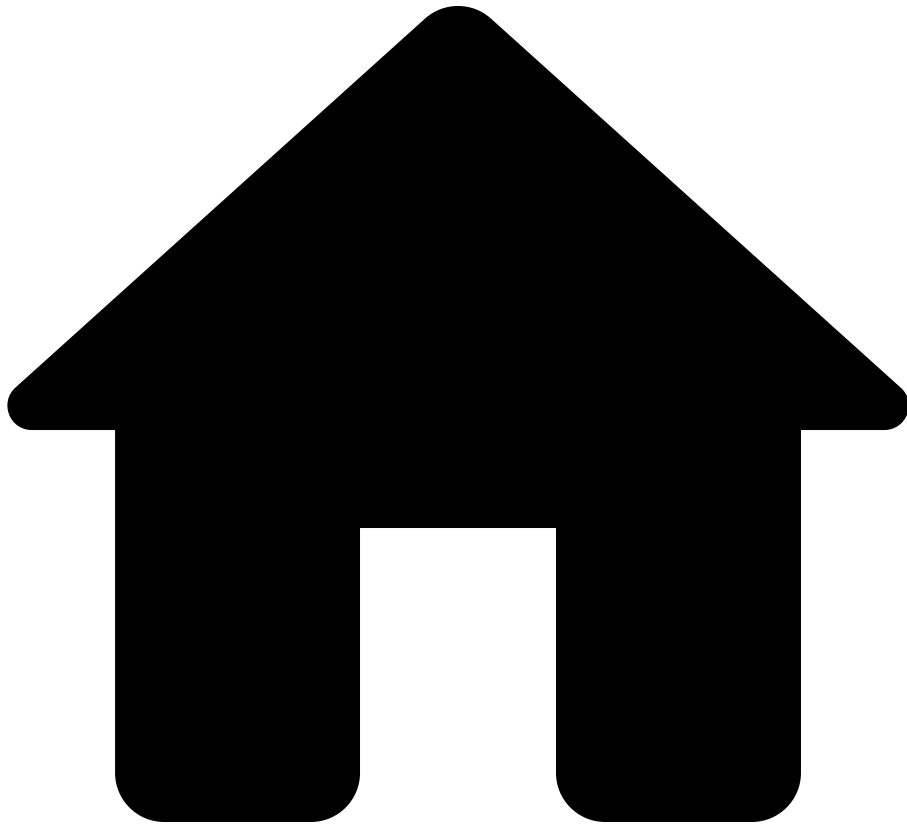
### See also

Data model path

- network: Network
  - cloudAccounts: Bag<CloudAccount>
    - vpcs: Bag<VpcData>
      - computeInstances: Bag<ComputeInstance>



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ConfigLine

On this page

## ConfigLine

Information about a single line in a device config file.

## Fields

| name       | type                               | nullable | description                                                                                                                                                                     |
|------------|------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| text       | String                             | false    | The text of the line.                                                                                                                                                           |
| lineNumber | Integer                            | false    | The line number (starting from zero) on which the line appears within the configuration.                                                                                        |
| children   | List< <a href="#">ConfigLine</a> > | false    | Lines of config that are nested directly under this line. For example, on Cisco NX-OS devices, the children are the subsequent lines indented one level further than this line. |

## Used by

network devices files config

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - files: [Files](#)
      - config: Bag< [ConfigLine](#) >

**network** **devices** **files** **panos** **configTemplate**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - files: [Files](#)
      - panos: [PanosFiles](#)
        - configTemplate: Bag< [ConfigLine](#) >

**network** **devices** **files** **panos** **configPolicy**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - files: [Files](#)
      - panos: [PanosFiles](#)
        - configPolicy: Bag< [ConfigLine](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ConnectsToAccessLayerSwitchInterfacePolicy

On this page

# ConnectsToAccessLayerSwitchInterfacePolicy

## Fields

| name      | type                                | nullable | description                                                   |
|-----------|-------------------------------------|----------|---------------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                            |
| value     | Bool                                | false    | Whether the interface is connected to an access layer switch. |

## Used by

network stigDatabase policy ifacePolicy connectivity toAccessLayerSwitch

## See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - connectivity: [InterfaceConnectivityPolicy](#)
          - toAccessLayerSwitch: Bag< [ConnectsToAccessLayerSwitchInterfacePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ConnectsToCoreLayerInterfacePolicy

On this page

# ConnectsToCoreLayerInterfacePolicy

## Fields

| name      | type                                | nullable | description                                           |
|-----------|-------------------------------------|----------|-------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                    |
| value     | Bool                                | false    | Whether the interface is connected to the core layer. |

## Used by

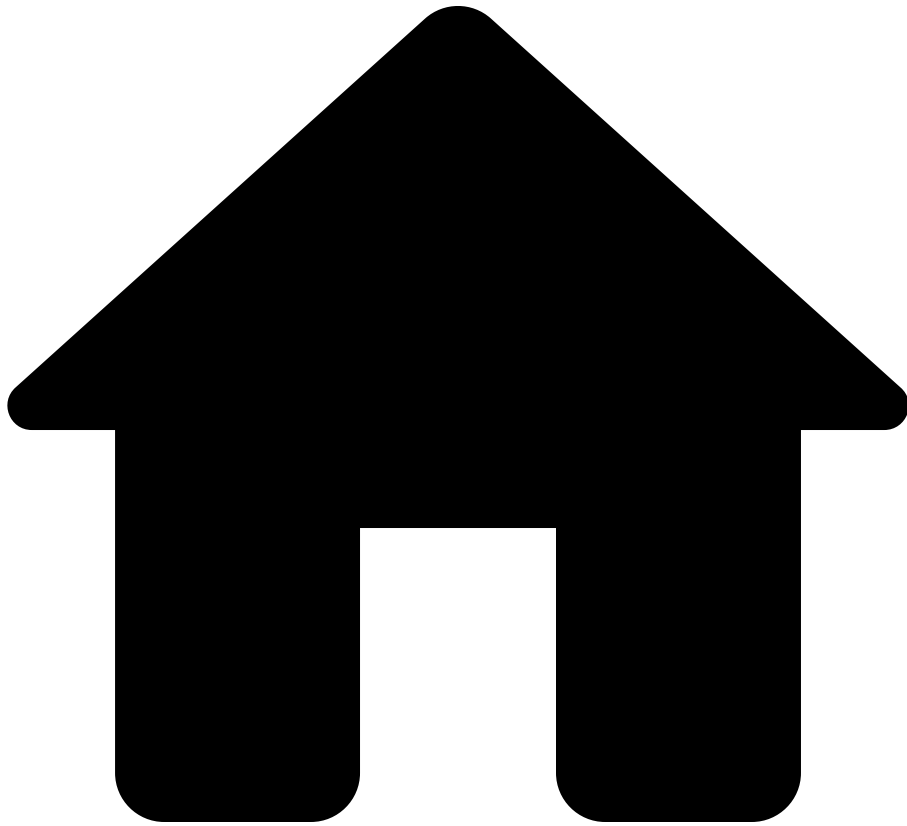
network stigDatabase policy ifacePolicy connectivity toCoreLayer

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - connectivity: [InterfaceConnectivityPolicy](#)
          - toCoreLayer: [ConnectsToCoreLayerInterfacePolicy](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ConnectsToCustomerEdgeDeviceInterfacePolicy

On this page



# ConnectsToCustomerEdgeDeviceInterfacePolicy

## Fields

| name      | type                                | nullable | description                                                                   |
|-----------|-------------------------------------|----------|-------------------------------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                                            |
| value     | Bool                                | false    | Whether the interface connects to a customer edge device (i.e. is CE-facing). |

## Used by

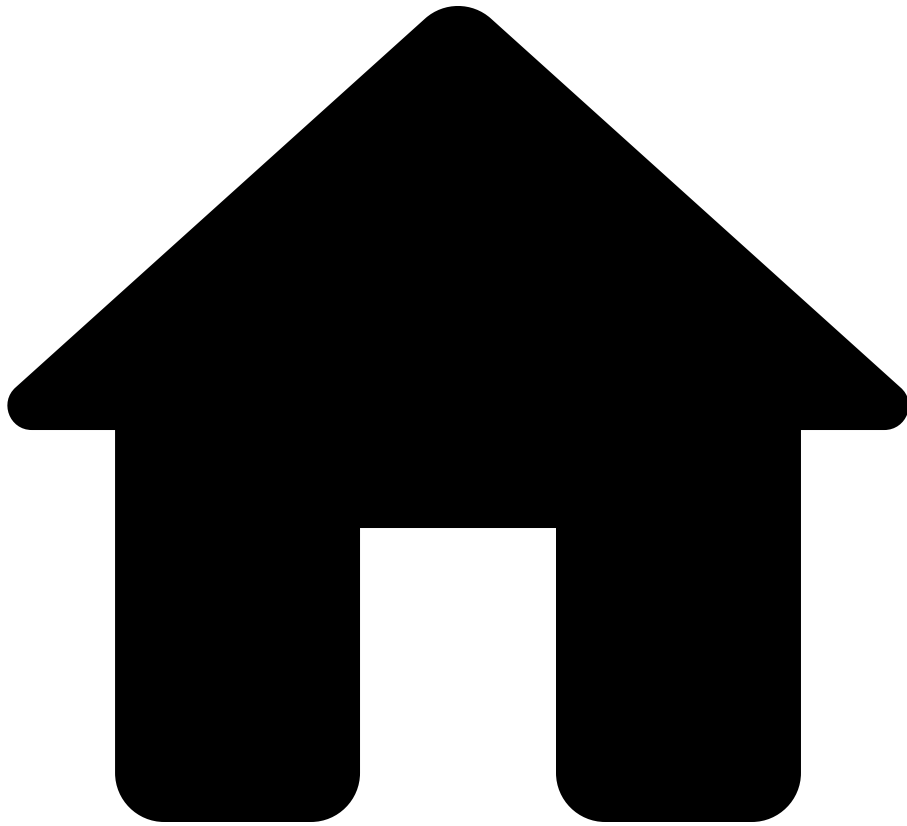
network [stigDatabase](#) [policy](#) [ifacePolicy](#) [connectivity](#) [toCustomerEdgeDevice](#)

## See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - connectivity: [InterfaceConnectivityPolicy](#)
          - toCustomerEdgeDevice: Bag< [ConnectsToCustomerEdgeDeviceInterfacePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ConnectsToDodinDeviceInterfacePolicy

On this page

# ConnectsToDodinDeviceInterfacePolicy

## Fields

| name      | type                                | nullable | description                                                |
|-----------|-------------------------------------|----------|------------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                         |
| value     | Bool                                | false    | Whether the interface connects to a DODIN backbone device. |

## Used by

network [stigDatabase](#) [policy](#) [ifacePolicy](#) [connectivity](#) [toDodinBackbone](#)

## See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - connectivity: [InterfaceConnectivityPolicy](#)
          - toDodinBackbone: [ConnectsToDodinDeviceInterfacePolicy](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ConnectsViaFiberOpticsInterfacePolicy

On this page

# ConnectsViaFiberOpticsInterfacePolicy

## Fields

| name      | type                                | nullable | description                                                      |
|-----------|-------------------------------------|----------|------------------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                               |
| value     | Bool                                | false    | Whether the interface connects to its neighbor via fiber optics. |

## Used by

network [stigDatabase](#) policy [ifacePolicy](#) connectivity [viaFiberOptics](#)

## See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - connectivity: [InterfaceConnectivityPolicy](#)
          - viaFiberOptics:
            - Bag< [ConnectsViaFiberOpticsInterfacePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ContainerType

On this page

# ContainerType

## Options

| name       | description |
|------------|-------------|
| COLLECTION | --          |
| GROUP      | --          |
| POLICY     | --          |

## Used by

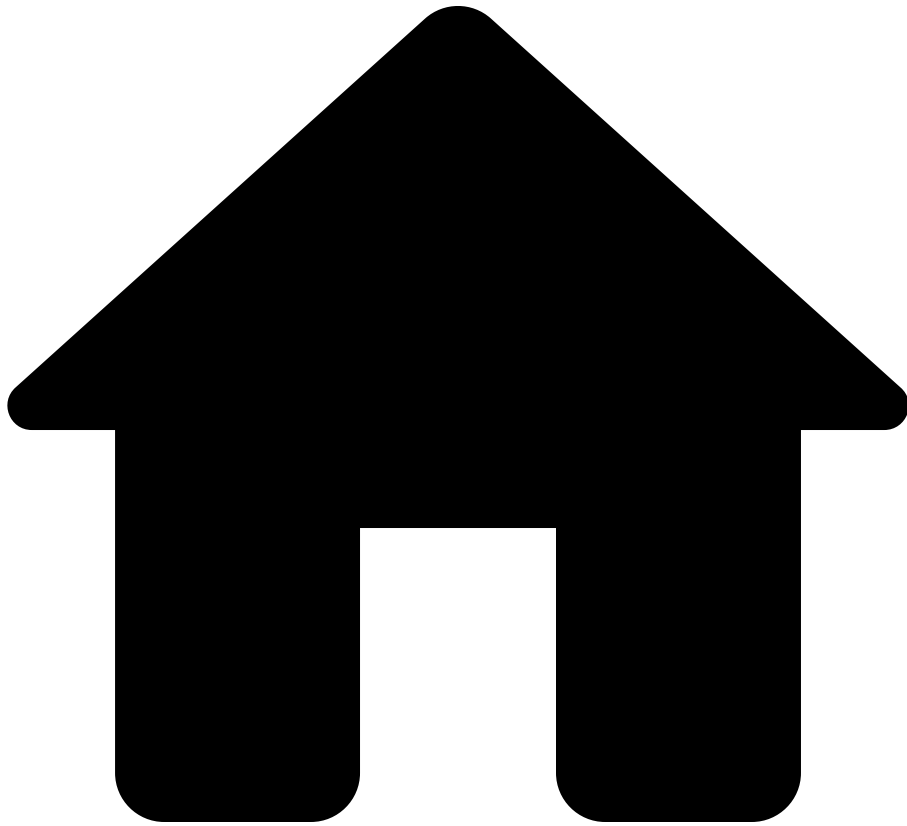
network cloudAccounts firewallPolicy ruleContainers containerType

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - firewallPolicy: FirewallPolicy
      - ruleContainers: Bag< RuleContainer >
        - containerType: ContainerType

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Cve

On this page



# Cve

## Fields

| name        | type                                      | nullable | description                                                                                                                                                                                                                                                                      |
|-------------|-------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cveId       | String                                    | false    | The identifier of the CVE, such as "CVE-2017-3857" or "CVE-2020-3204".                                                                                                                                                                                                           |
| severity    | <a href="#">Severity</a>                  | false    | The severity corresponding to baseScoreV4 if that score is known. Otherwise, the severity corresponding to baseScoreV3 if that score is known. Otherwise, the severity corresponding to baseScoreV2 if that score is known. Otherwise, no score is known and this value is NONE. |
| description | String                                    | true     | --                                                                                                                                                                                                                                                                               |
| criteria    | Bag< <a href="#">CveProductCriteria</a> > | false    | --                                                                                                                                                                                                                                                                               |

| name         | type                                  | nullable | description                                                                                                                       |
|--------------|---------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| cisaKevEntry | <a href="#">CisaExploitationEntry</a> | true     | An entry for this CVE from CISA's Known Exploited Vulnerabilities (KEV) catalog. Absent if this CVE is not in CISA's KEV catalog. |
| vendorInfos  | Bag< <a href="#">VendorCveInfo</a> >  | false    | --                                                                                                                                |

## Used by

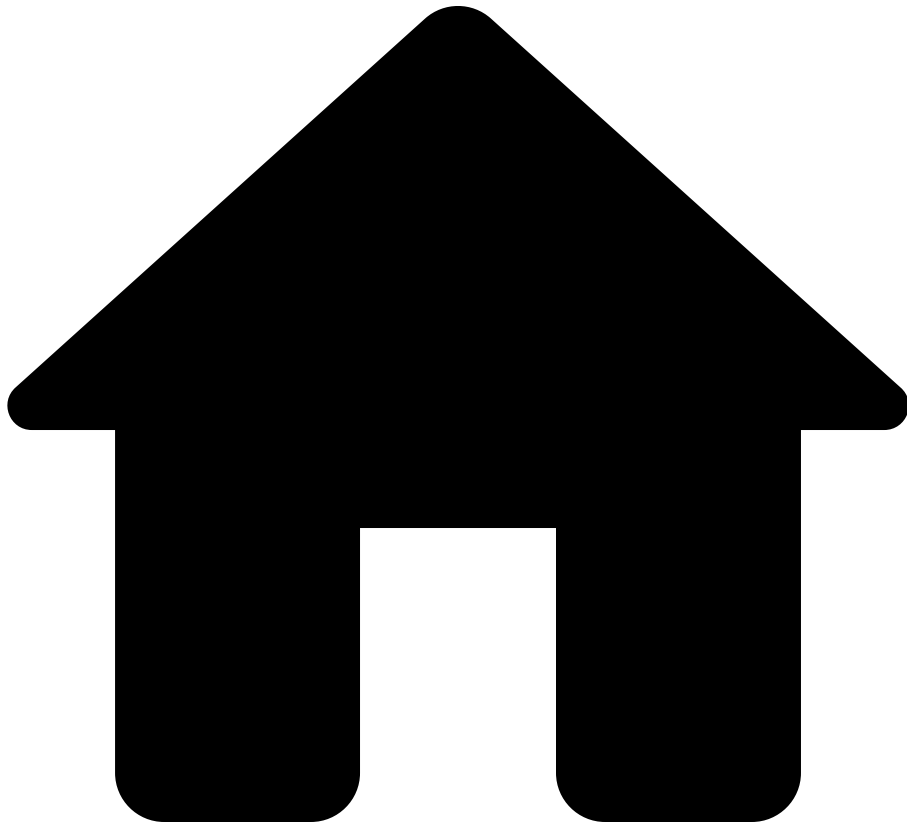
network [cveDatabase](#) [cves](#)

See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CveApplicationCriteria

On this page

# CveApplicationCriteria

## Fields

| name | type   | nullable | description |
|------|--------|----------|-------------|
| name | String | false    | --          |

## Used by

network [cveDatabase](#) [cves](#) [criteria](#) [applicationCriteria](#)

### See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - criteria: Bag< [CveProductCriteria](#) >
        - applicationCriteria: Bag< [CveApplicationCriteria](#) >

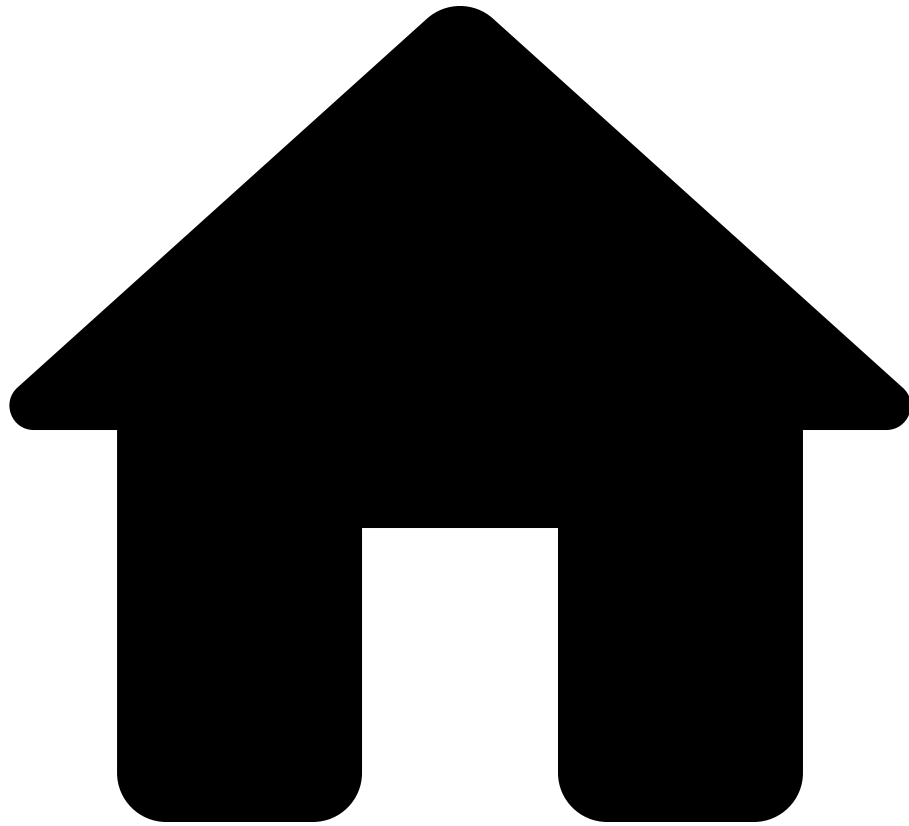
network [cveDatabase](#) [cves](#) [vendorInfos](#) [criteria](#) [applicationCriteria](#)

### See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - vendorInfos: Bag< [VendorCveInfo](#) >
        - criteria: Bag< [CveProductCriteria](#) >
          - applicationCriteria: Bag< [CveApplicationCriteria](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CveDatabase

On this page

# CveDatabase

## Fields

| name | type                       | nullable | description |
|------|----------------------------|----------|-------------|
| cves | Bag< <a href="#">Cve</a> > | false    | --          |

## Used by

network [cveDatabase](#)

## See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CveFinding

On this page

# CveFinding

## Fields

| name         | type                            | nullable | description                                           |
|--------------|---------------------------------|----------|-------------------------------------------------------|
| cveId        | String                          | false    | --                                                    |
| isVulnerable | Bool                            | false    | Whether or not this Device is vulnerable to this CVE. |
| basis        | <a href="#">CveFindingBasis</a> | false    | The grounds on which the isVulnerable was determined. |

## Used by

**network** **devices** **cveFindings**

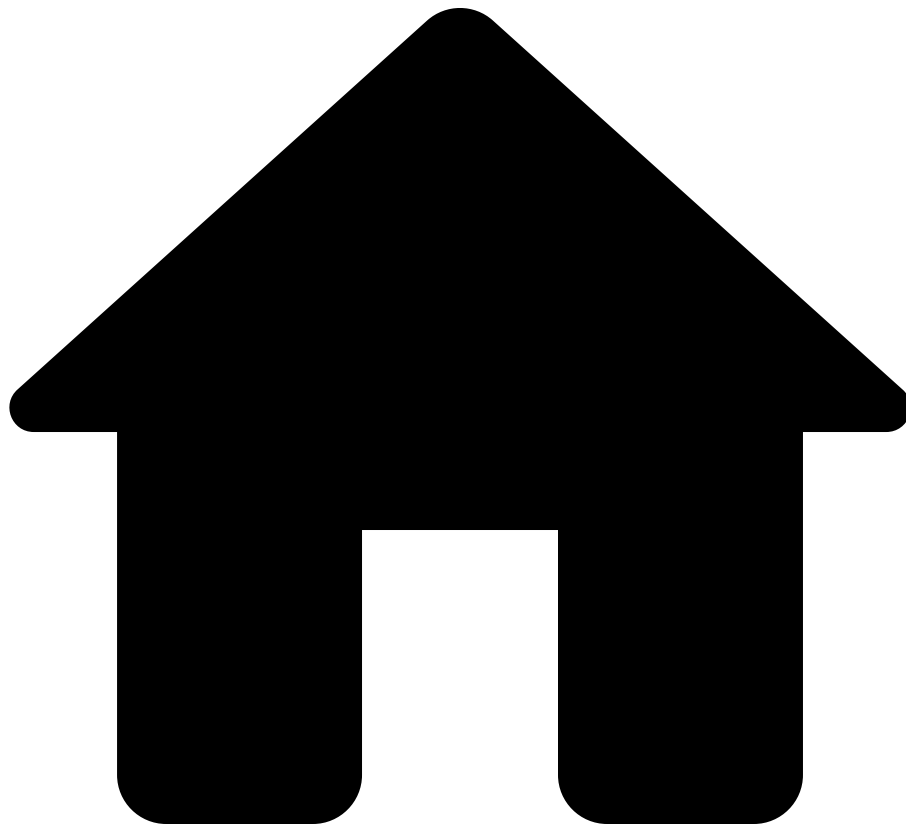
See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - cveFindings: Bag< [CveFinding](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CveFindingBasis

On this page

# CveFindingBasis

## Options

| name     | description |
|----------|-------------|
| CONFIG   | --          |
| PLATFORM | --          |

## Used by

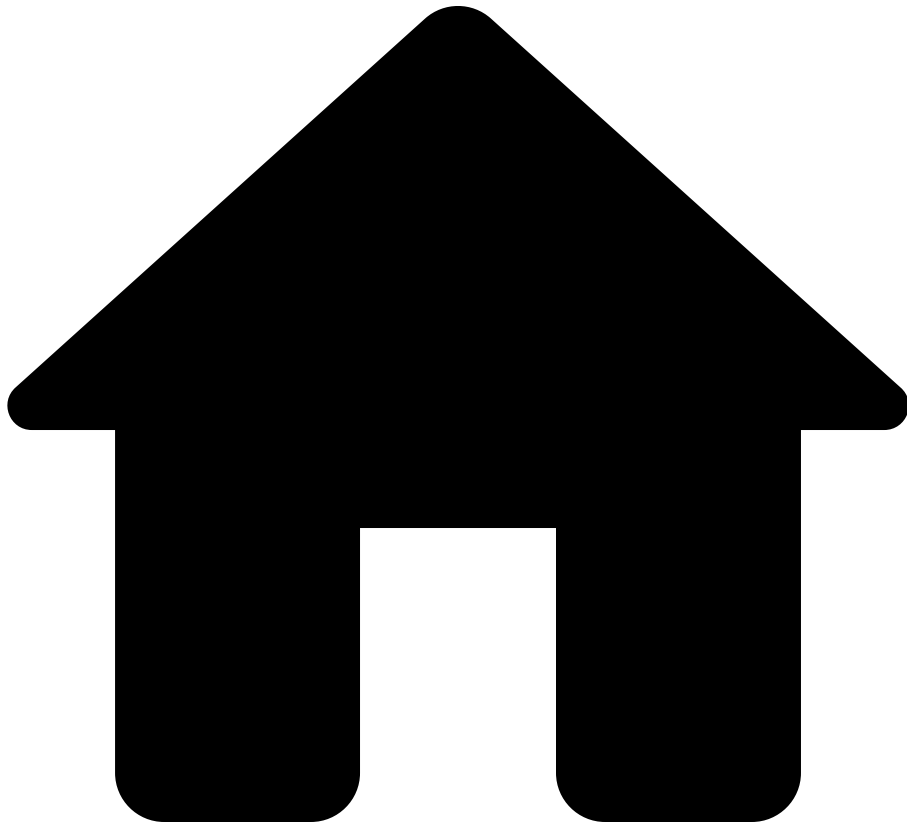
network devices cveFindings basis

### See also

Data model path

- network: Network
  - devices: Bag< Device >
    - cveFindings: Bag< CveFinding >
      - basis: CveFindingBasis

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CveHardwareCriteria

On this page

# CveHardwareCriteria

## Fields

| name | type   | nullable | description |
|------|--------|----------|-------------|
| name | String | false    | --          |

## Used by

network cveDatabase cves criteria hardwareCriteria

### See also

Data model path

- network:
  - cveDatabase:
    - cves: Bag<
      - criteria: Bag<
        - hardwareCriteria: Bag<

network cveDatabase cves vendorInfos criteria hardwareCriteria

### See also

Data model path

- network:
  - cveDatabase:
    - cves: Bag<
      - vendorInfos: Bag<
        - criteria: Bag<
          - hardwareCriteria: Bag<

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CveOsCriteria

On this page

# CveOsCriteria

## Fields

| name            | type               | nullable | description                                                                                                                                                                                                                                                                                                                     |
|-----------------|--------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| os              | <a href="#">OS</a> | false    | --                                                                                                                                                                                                                                                                                                                              |
| dependsOnConfig | Bool               | true     | Indicates whether the vulnerability analysis for this CVE (on this OS) depends on the device's configuration. For example, it is true if a device is only vulnerable to the CVE when it has certain features enabled. A null value means that it is unknown whether the CVE (on this OS) depends on the device's configuration. |

## Used by

network [cveDatabase](#) [cves](#) [criteria](#) [osCriteria](#)

## See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - criteria: Bag< [CveProductCriteria](#) >
        - osCriteria: Bag< [CveOsCriteria](#) >

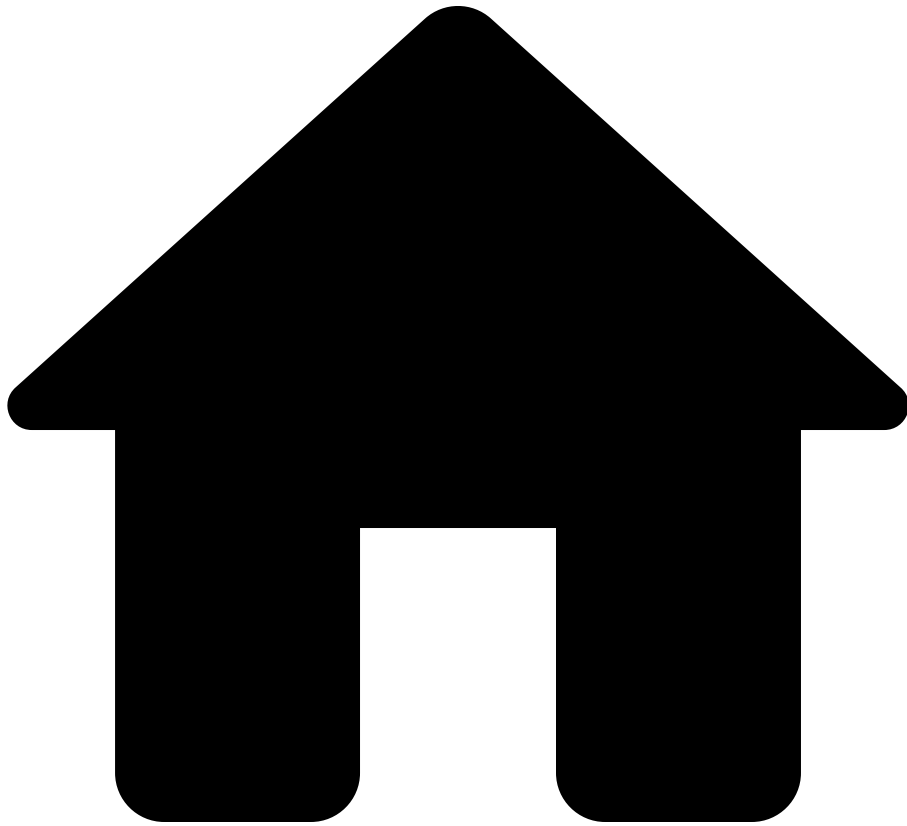
**network** **cveDatabase** **cves** **vendorInfos** **criteria** **osCriteria**[↗](#)

**See also**[↗](#)

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - vendorInfos: Bag< [VendorCveInfo](#) >
        - criteria: Bag< [CveProductCriteria](#) >
          - osCriteria: Bag< [CveOsCriteria](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- CveProductCriteria

On this page



# CveProductCriteria

## Fields

| name                | type                                          | nullable | description                                         |
|---------------------|-----------------------------------------------|----------|-----------------------------------------------------|
| vendor              | <a href="#">Vendor</a>                        | false    | --                                                  |
| hardwareCriteria    | Bag< <a href="#">CveHardwareCriteria</a> >    | false    | Empty means there is no dependency on the hardware. |
| osCriteria          | Bag< <a href="#">CveOsCriteria</a> >          | false    | Empty means there is no dependency on the OS.       |
| applicationCriteria | Bag< <a href="#">CveApplicationCriteria</a> > | false    | Empty means there is no dependency on applications. |

## Used by

**network** **cveDatabase** **cves** **criteria**

See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - criteria: Bag< [CveProductCriteria](#) >

**network** **cveDatabase** **cves** **vendorInfos** **criteria**

See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - vendorInfos: Bag< [VendorCveInfo](#) >
        - criteria: Bag< [CveProductCriteria](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DataConnector

On this page

# DataConnector

## Fields

| name        | type                                         | nullable | description                                                |
|-------------|----------------------------------------------|----------|------------------------------------------------------------|
| name        | String                                       | false    | The name of the data connector. Unique across the network. |
| httpResults | Bag< <a href="#">HttpApiEndpointResult</a> > | false    | The set of results per source for this data connector.     |

## Used by

network [dataConnectors](#)

See also

Data model path

- network: [Network](#)
  - dataConnectors: Bag< [DataConnector](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DcGateway

On this page

# DcGateway

## Fields

| name              | type                                    | nullable | description |
|-------------------|-----------------------------------------|----------|-------------|
| name              | String                                  | true     | --          |
| id                | String                                  | false    | --          |
| tags              | Bag< String >                           | false    | --          |
| directConnections | Bag< <a href="#">DirectConnection</a> > | false    | --          |

## Used by

**network** contains **cloudAccounts** which contains **dcGateways**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - dcGateways: Bag< [DcGateway](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DefaultInterfaceParameters

On this page

# DefaultInterfaceParameters

## Fields

| name                       | type                                            | nullable |
|----------------------------|-------------------------------------------------|----------|
| requiresDefaultVlan        | Bool                                            | true     |
| requiresMulticast          | Bool                                            | true     |
| isHostGateway              | Bool                                            | true     |
| pseudowireVirtualCircuitId | Integer                                         | true     |
| connectivity               | <a href="#">InterfaceConnectivityParameters</a> | false    |



## Used by[↗](#)

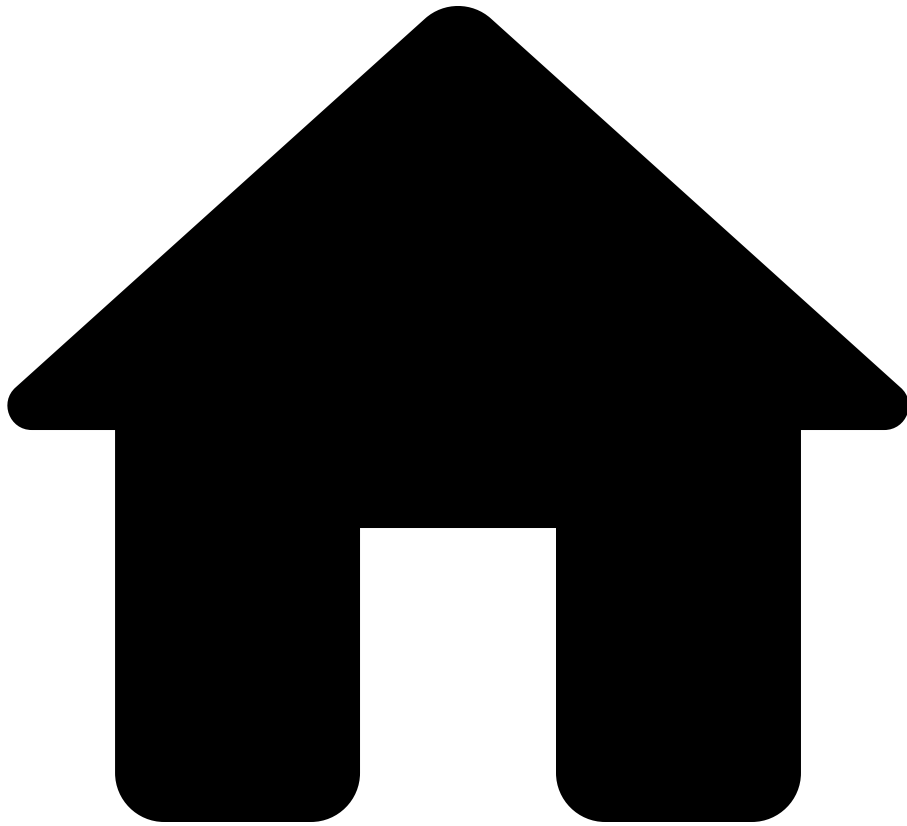
**network** [↗](#) **stigDatabase** [↗](#) **defaultParameters** [↗](#) **ifaceParameters**[↗](#)

See also[↗](#)

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - defaultParameters: [DefaultStigParameters](#)
      - ifaceParameters: [DefaultInterfaceParameters](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DefaultStigParameters

On this page

# DefaultStigParameters

## Fields

| name             | type                                       | nullable | description |
|------------------|--------------------------------------------|----------|-------------|
| deviceParameters | <a href="#">DeviceParameters</a>           | false    | --          |
| ifaceParameters  | <a href="#">DefaultInterfaceParameters</a> | false    | --          |
| vrfParameters    | <a href="#">DefaultVrfParameters</a>       | false    | --          |

## Used by

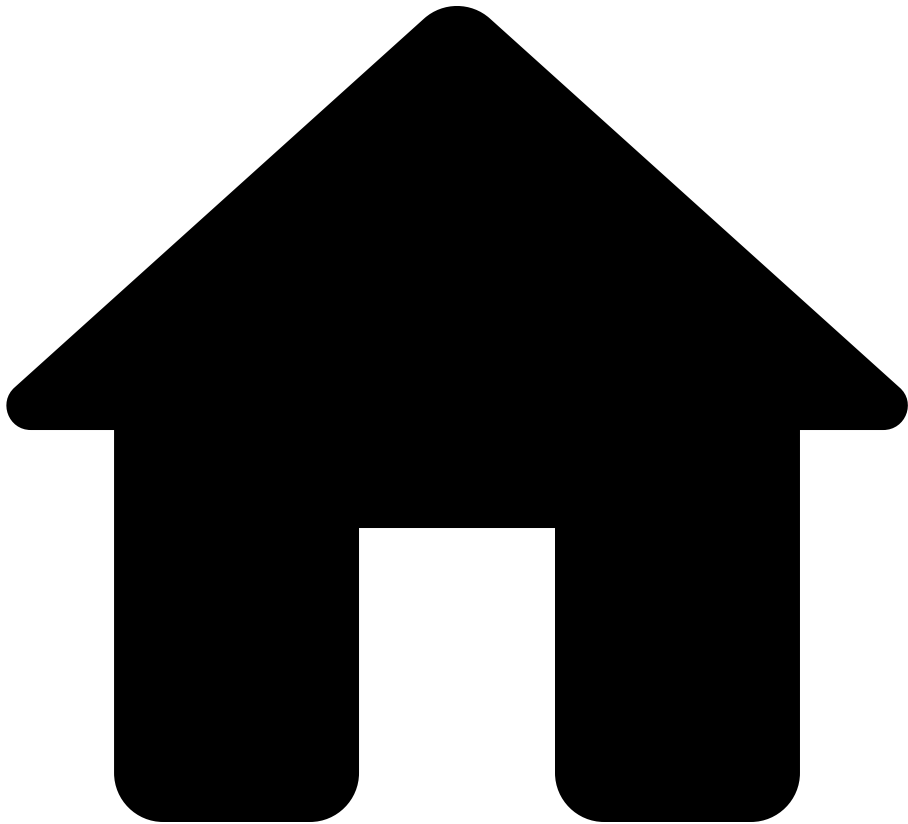
network stigDatabase defaultParameters

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - defaultParameters: [DefaultStigParameters](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DefaultVrfParameters

On this page

# DefaultVrfParameters

## Fields [↗](#)

| name | type | nullable | description |
|------|------|----------|-------------|
|------|------|----------|-------------|

## Used by[↗](#)

**network** [↗](#) **stigDatabase** [↗](#) **defaultParameters** [↗](#) **vrfParameters**[↗](#)

See also[↗](#)

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - defaultParameters: [DefaultStigParameters](#)
      - vrfParameters: [DefaultVrfParameters](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Device

On this page

# Device

## Fields

| name             | type                               | nullable | description                                                          |
|------------------|------------------------------------|----------|----------------------------------------------------------------------|
| name             | String                             | false    | --                                                                   |
| locationName     | String                             | true     | The location name, or null if no location is assigned.               |
| locationId       | String                             | true     | The location ID or null if no location is assigned.                  |
| groupNames       | Bag< String >                      | false    | Names of the device groups that contain this device.                 |
| tagNames         | Bag< String >                      | false    | Names of the device tags that contain this device.                   |
| accessLabelNames | Bag< String >                      | false    | Device access labels for this device.                                |
| snapshotInfo     | <a href="#">DeviceSnapshotInfo</a> | false    | Contains metadata about the inclusion of this device in the selected |

| name     | type                                       | nullable | description                                                                                                             |
|----------|--------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------|
|          |                                            |          | snapshot.<br>Contains information about the world and the Forward Networks system related to this device.               |
| ha       | <a href="#">Ha</a>                         | false    | --                                                                                                                      |
| system   | <a href="#">System</a>                     | false    | System-level attributes of a device, whether maintained by the device or inferred by Forward Networks about the system. |
| platform | <a href="#">Platform</a>                   | false    | --                                                                                                                      |
| vlan     | Bag< <a href="#">NamedVlanCollection</a> > | false    | Globally-defined VLANs on the device. This includes VLANs that are not associated with any interfaces.                  |



| name             | type                                   | nullable | description                                                                                                                                                                                                                                                |
|------------------|----------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| interfaces       | Bag< <a href="#">Iface</a> >           | false    | --                                                                                                                                                                                                                                                         |
| networkInstances | Bag< <a href="#">NetworkInstance</a> > | false    | --                                                                                                                                                                                                                                                         |
| bgpRib           | <a href="#">BgpRib</a>                 | true     | --                                                                                                                                                                                                                                                         |
| stp              | <a href="#">Stp</a>                    | false    | --                                                                                                                                                                                                                                                         |
| aclEntries       | List< <a href="#">AclEntry</a> >       | false    | ACL entries defined on the device. The entries are unique based on all their non-lifecycle-data attributes. Note that a device's ACL entries may match on packet attributes, such as mac source and mac dest, that are not exposed in the AclEntry record. |
| natEntries       | List< <a href="#">NatEntry</a> >       | false    | --                                                                                                                                                                                                                                                         |
| hosts            | Bag< <a href="#">DeviceHost</a> >      | false    | --                                                                                                                                                                                                                                                         |
| cveFindings      | Bag< <a href="#">CveFinding</a> >      | false    | The CVE findings for this device. Includes only                                                                                                                                                                                                            |

| name          | type                                      | nullable | description                                                                                                                           |
|---------------|-------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------|
|               |                                           |          | the CVEs that impact the device or that do not impact the device on the basis of the device's configuration.                          |
| outputs       | <a href="#">Outputs</a>                   | false    | Information about data sources for a device.                                                                                          |
| securityZones | Bag< <a href="#">DeviceSecurityZone</a> > | false    | Security zones defined on this firewall device.<br>Empty if this device is not a firewall or is a firewall but has no security zones. |
| files         | <a href="#">Files</a>                     | false    | Information about config data sources for a device.                                                                                   |

**Used by** [â](#)

**network** [â](#) **devices** [â](#)

**See also** [â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DeviceCollectionError

On this page

# DeviceCollectionError

## Options

| name                                           | description                     |
|------------------------------------------------|---------------------------------|
| APIC_CONFIG_COLLECTION_FAILED                  | --                              |
| API_SERVER_FAILED_TO_RESPOND                   | --                              |
| AUTHENTICATION_FAILED                          | --                              |
| AUTHORIZATION_FAILED                           | --                              |
| AVI_CONTROLLER_WITHOUT_HEALTHY_SERVICE_ENGINES | --                              |
| AVI_SHELL_AUTH_FAILED                          | --                              |
| BMP_COLLECTION_FAILED                          | --                              |
| BMP_COLLECTION_REFUSED                         | --                              |
| BMP_COLLECTION_TIMEOUT                         | --                              |
| BMP_CONFIGURED_WITH_LOCAL_IP                   | --                              |
| BMP_CONNECTION_FAILED                          | --                              |
| BMP_CONNECTION_TIMEOUT                         | --                              |
| BMP_NOT_CONFIGURED                             | --                              |
| BMP_SESSION_ALREADY_ESTABLISHED                | --                              |
| CERTIFICATE_CHECK_FAILED                       | --                              |
| COLLECTION_CANCELED                            | --                              |
| COLLECTION_NOT_FOUND                           | --                              |
| COLLECTION_TIMED_OUT                           | --                              |
| COMMAND_DISABLED                               | --                              |
| CONFIG_COLLECTION_UNAUTHORIZED                 | Collection of the config failed |

| name                              | description                        |
|-----------------------------------|------------------------------------|
|                                   | because of an authorization error. |
| CONNECTION_REFUSED                | --                                 |
| CONNECTION_TIMEOUT                | --                                 |
| DEVICE_IN_BAD_STATE               | --                                 |
| DEVICE_IS_CHILD_CONTEXT           | --                                 |
| DEVICE_TYPE_UNDETECTED            | --                                 |
| DUPLICATE_DEVICE                  | --                                 |
| EXPERT_PASSWORD_ERROR             | --                                 |
| FILE_TRANSFER_FAILED              | --                                 |
| INCOMPLETE_SETUP                  | --                                 |
| INFINITE_LOOP_IN_COMMAND_OUTPUT   | --                                 |
| IO_ERROR                          | --                                 |
| JUMP_SERVER_CONNECTION_FAILED     | --                                 |
| JUMP_SERVER_CONNECTION_TIMEOUT    | --                                 |
| JUMP_SERVER_KEY_EXCHANGE_FAILED   | --                                 |
| JUMP_SERVER_PASSWORD_AUTH_FAILED  | --                                 |
| JUMP_SERVER_PASSWORD_EXPIRED      | --                                 |
| KEY_EXCHANGE_FAILED               | --                                 |
| MANAGER_COLLECTOR_NOT_FOUND       | --                                 |
| MISSING_FILE                      | --                                 |
| NETWORK_UNREACHABLE               | --                                 |
| NO_SPACE_LEFT_ON_COLLECTED_DEVICE | --                                 |
| NO_SPACE_LEFT_ON_DEVICE           | --                                 |

| name                                   | description                                                                  |
|----------------------------------------|------------------------------------------------------------------------------|
| OPERATION_TIMED_OUT                    | --                                                                           |
| PASSWORD_EXPIRED                       | --                                                                           |
| PRIV_PASSWORD_ERROR                    | --                                                                           |
| PROJECT_VIEW_PERMISSION_MISSING        | --                                                                           |
| PROMPT_DISCOVERY_FAILED                | --                                                                           |
| PROXY_SERVER_AUTHENTICATION_FAILED     | --                                                                           |
| PROXY_SERVER_CONNECTION_FAILED         | --                                                                           |
| PROXY_SERVER_PING_FAILED               | --                                                                           |
| PROXY_SERVER_PORT_REACHABILITY_FAILED  | --                                                                           |
| SESSION_CLOSED                         | --                                                                           |
| SESSION_LIMIT_REACHED                  | --                                                                           |
| SLOW_READ_RATE_DETECTED                | --                                                                           |
| SOME_OID_FAILED                        | This error is specific to device using SNMP for collection. Some OID failed. |
| STATE_COLLECTION_FAILED                | --                                                                           |
| T128_CONDUCTOR_WITHOUT_HEALTHY_ROUTERS | --                                                                           |
| T128_CONFIG_ABSENT                     | --                                                                           |
| TACACS_SESSION_EXPIRED                 | --                                                                           |
| TRANSIENT_SLOW_READ_RATE_DETECTED      | Slow read rate which we retry on device level. This is a transient error.    |
| UNDISCOVERED_ACI_FABRIC                | --                                                                           |
| UNEXPECTED_KEY_EXCHANGE_MESSAGE        | --                                                                           |

| name                       | description |
|----------------------------|-------------|
| UNKNOWN                    | --          |
| UNRESOLVABLE_HOSTNAME      | --          |
| UNSUPPORTED_SSL_CONNECTION | --          |
| UNSUPPORTED_VENDOR         | --          |
| UNSUPPORTED_VERSION        | --          |
| WARN_TYPE_MISMATCH         | --          |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **snapshotInfo** [â](#) **result**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - snapshotInfo: [DeviceSnapshotInfo](#)
      - result: [DeviceSnapshotResult](#)

**network** [â](#) **endpoints** [â](#) **snapshotInfo** [â](#) **result**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - snapshotInfo: [DeviceSnapshotInfo](#)
      - result: [DeviceSnapshotResult](#)



**network** **externalSources** **cliSources** **snapshotInfo** **result**

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - cliSources: Bag< [CliSource](#) >
      - snapshotInfo: [DeviceSnapshotInfo](#)
        - result: [DeviceSnapshotResult](#)

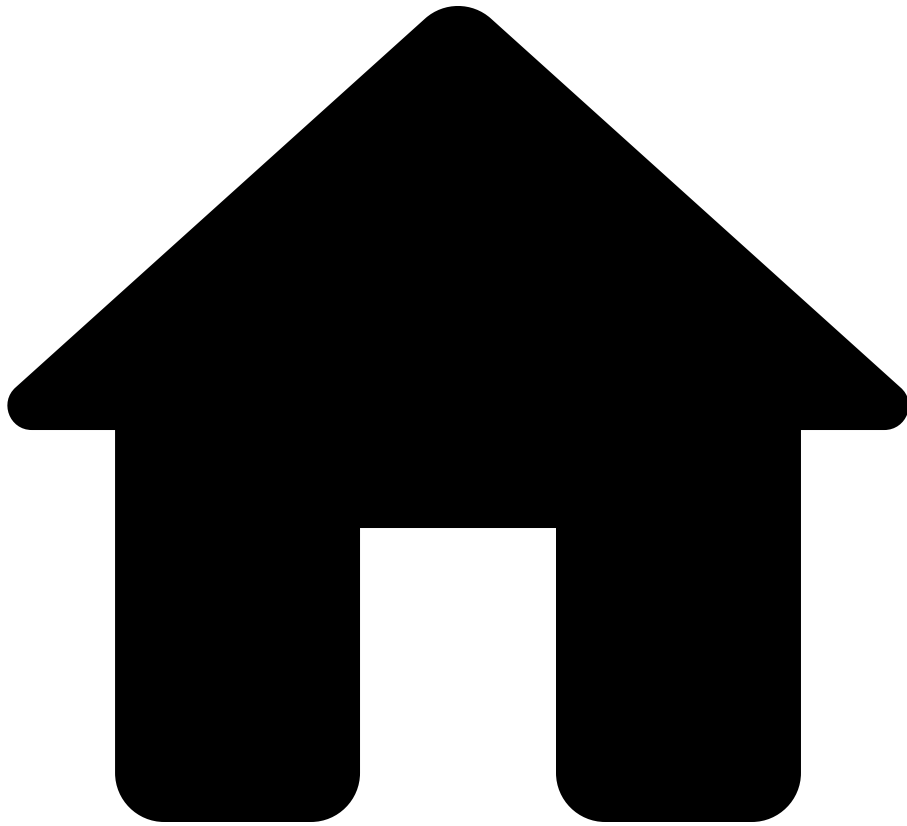
**network** **externalSources** **snmpSources** **snapshotInfo** **result**

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - snmpSources: Bag< [SnmpSource](#) >
      - snapshotInfo: [DeviceSnapshotInfo](#)
        - result: [DeviceSnapshotResult](#)

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- DeviceDiscoveryIfacelInfo

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# DeviceDiscoveryIfaceInfo

## Fields

| name      | type                                           | nullable | description |
|-----------|------------------------------------------------|----------|-------------|
| neighbors | Bag< <a href="#">DeviceDiscoveryNeighbor</a> > | false    | --          |

## Used by

**network** **devices** **interfaces** **cdp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - cdp: [DeviceDiscoveryIfaceInfo](#)

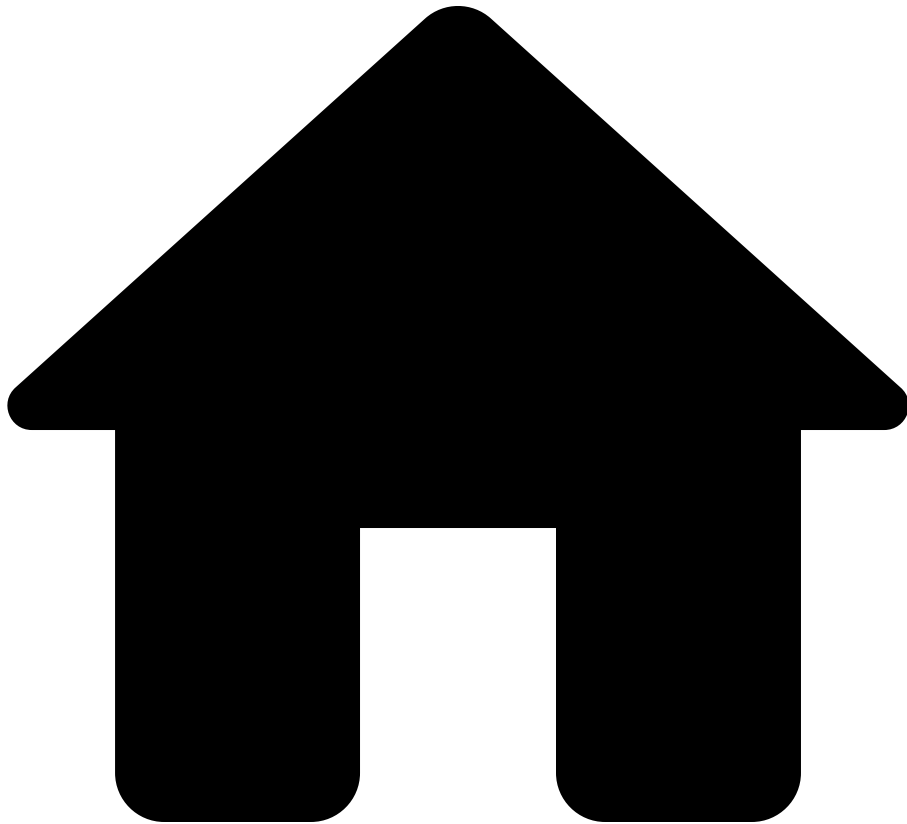
**network** **devices** **interfaces** **lldp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - lldp: [DeviceDiscoveryIfaceInfo](#)

.



- Reference
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- [Data Model](#)
- DeviceDiscoveryNeighbor

On this page

## DeviceDiscoveryNeighbor

Information about a CDP or LLDP neighbor. Note that the neighbor may not be a collected device. For example, it may be an end host.

## Fields

| name       | type   | nullable | description |
|------------|--------|----------|-------------|
| deviceName | String | false    | --          |
| portName   | String | false    | --          |

## Used by

**network devices interfaces cdp neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - cdp: [DeviceDiscoveryIfaceInfo](#)
        - neighbors: Bag< [DeviceDiscoveryNeighbor](#) >

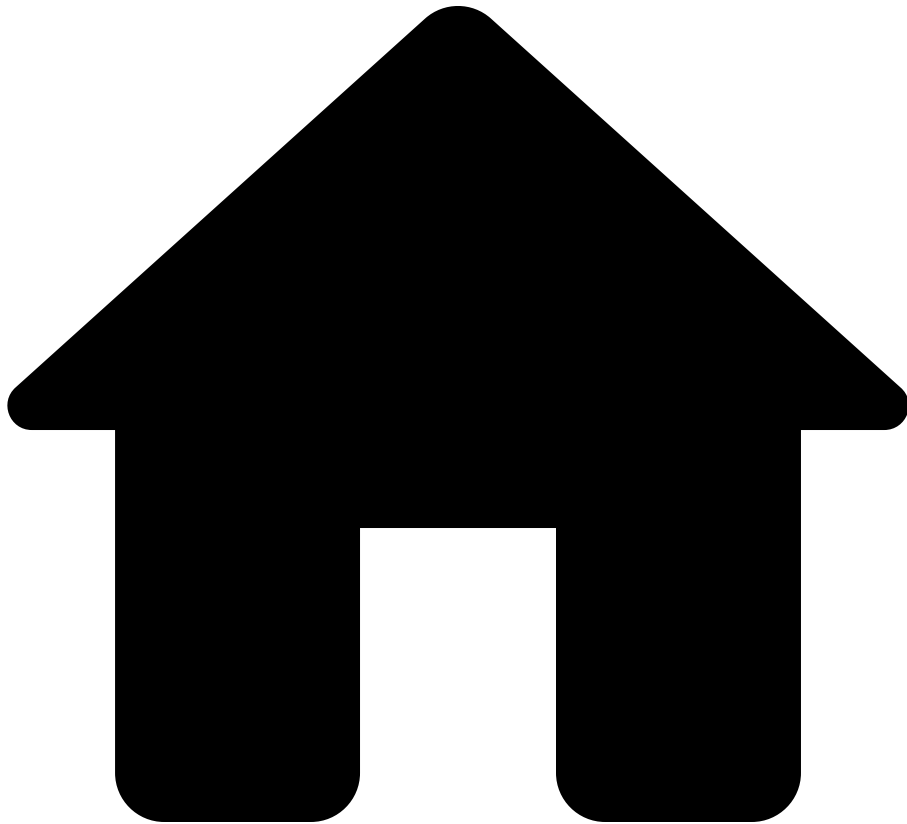
**network devices interfaces lldp neighbors**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - lldp: [DeviceDiscoveryIfaceInfo](#)
        - neighbors: Bag< [DeviceDiscoveryNeighbor](#) >

.



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- [Data Model](#)
- DeviceHost

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# DeviceHost

## Fields

| name      | type            | nullable | description                                                                                                                                                                                   |
|-----------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name      | String          | true     | The name of the host. Present for ESX hosts or cloud compute instances. Missing for hosts inferred from ARP and MAC entries.                                                                  |
| addresses | Bag< IpSubnet > | false    | The IP addresses of this host. If a host has a single address then this will contain a single address or /128 subnet. If the host has multiple addresses these are represented by one or more |

| name               | type             | nullable | description                                                                                                                                 |
|--------------------|------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                  |          | more subnets where each subnet represents all the IP addresses within the subnet.                                                           |
| colocatedAddresses | Bag< IPAddress > | false    | The /32 128 IP addresses that are configured as routes as route gateway devices where the route's nexthop is one of the host's IP addresses |
| macAddress         | MacAddress       | true     | Can be missing VMs discovered from vCenter                                                                                                  |
| interfaces         | Bag< String >    | false    | The name of interface on the device                                                                                                         |



| name      | type                             | nullable | description                                                                                                                                                                                                                                                                                      |
|-----------|----------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                  |          | which the host was observed                                                                                                                                                                                                                                                                      |
| v l a n s | Bag< <a href="#">VlanRange</a> > | false    | The VLAN to which this host's traffic is mapped, exposing a set of VLAN ranges. For access ports, the single VLAN to which untagged frames are mapped. For end hosts or trunk ports, this is the single VLAN ID to which the uncollected downstream access switch maps untagged frames. hosts th |

| name              | type                                         | nullable | description                                                                                                                                                                                                                                                                                                                                          |
|-------------------|----------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                                              |          | are actually uncollected network devices attached to a trunk. This will contain multiple VLAN tags. For ACI devices, this is the external VLANs accepted by the leaf. The VLANs which are mapped to internal interfaces. If this set is empty, then the interface is on the device and VLAN-unaware and the traffic from this host is being bridged. |
| gatewayInterfaces | Bag< <a href="#">DeviceL3InterfaceName</a> > | false    | --                                                                                                                                                                                                                                                                                                                                                   |

| name          | type                                               | nullable | descrip |
|---------------|----------------------------------------------------|----------|---------|
| hostType      | <a href="#">DeviceHostType</a>                     | false    | --      |
| securityZones | Bag< <a href="#">DeviceSecurityZoneReference</a> > | false    | --      |

Used by

network devices hosts

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - hosts: Bag< [DeviceHost](#) >

.



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- DeviceHostType

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# DeviceHostType

## Options

| name         | description                                                                              |
|--------------|------------------------------------------------------------------------------------------|
| CloudHost    | A cloud host. Currently unused, since cloud hosts are not attached to non-cloud devices. |
| EdgeNode     | --                                                                                       |
| InferredHost | --                                                                                       |
| Unknown      | --                                                                                       |

## Used by

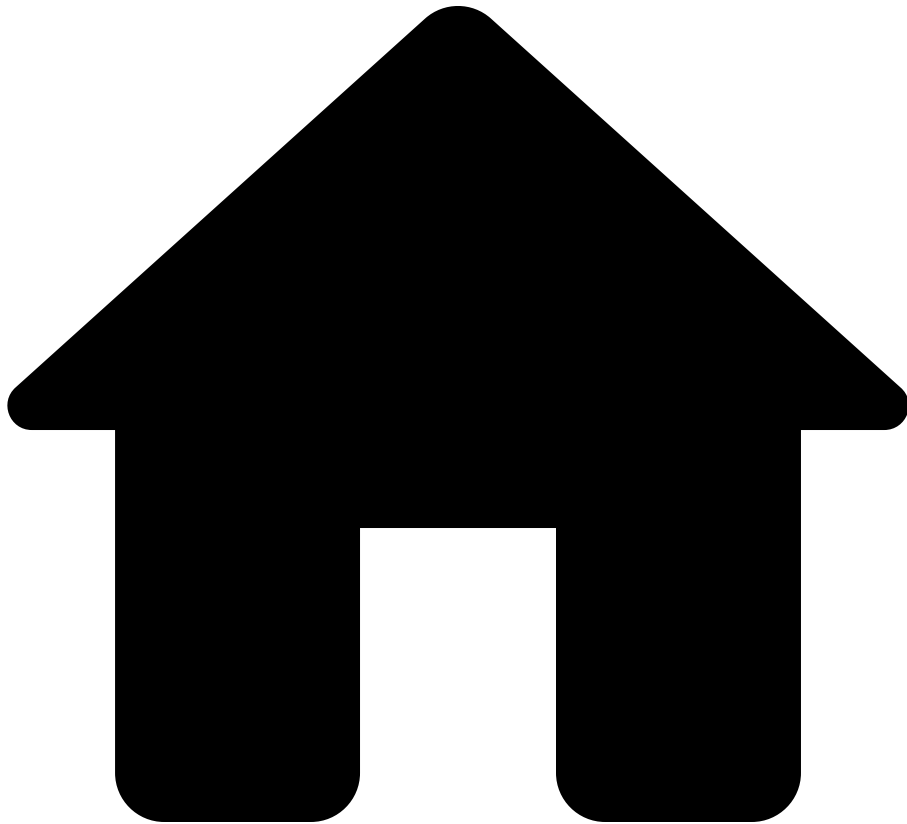
network devices hosts hostType

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - hosts: Bag< DeviceHost >
      - hostType: DeviceHostType

.



- Reference
- [NQE Language](#)
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- DeviceL3IfaceName

On this page

# DeviceL3IfaceName

## Fields

| name         | type   | nullable | description |
|--------------|--------|----------|-------------|
| deviceName   | String | false    | --          |
| ifaceName    | String | false    | --          |
| subIfaceName | String | true     | --          |

## Used by

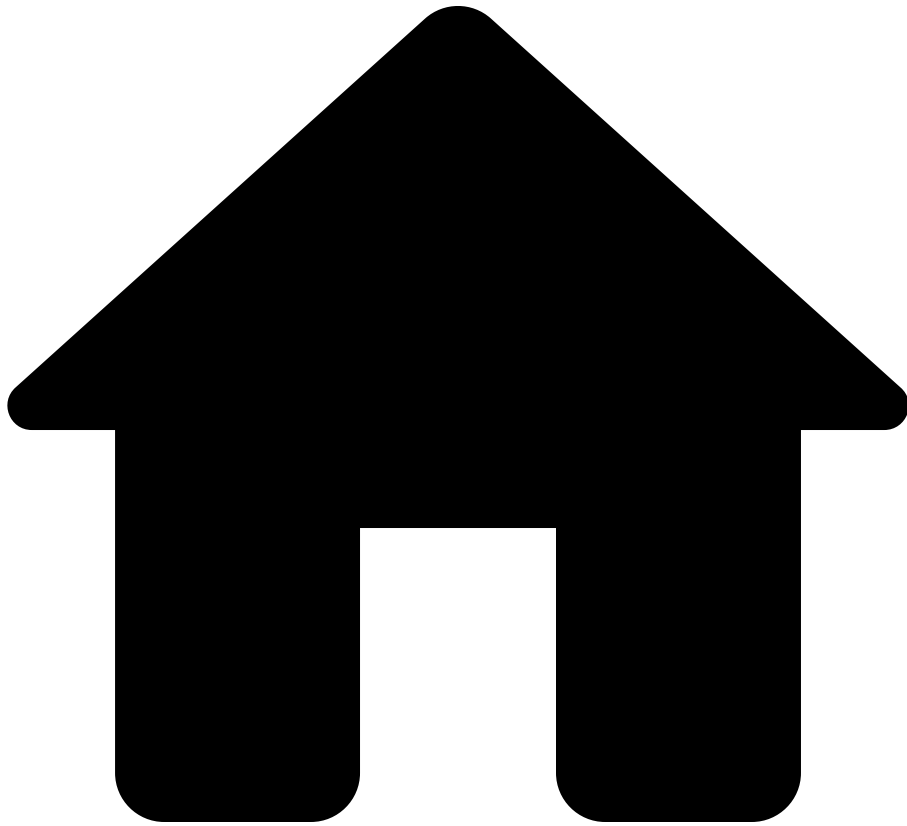
network devices hosts gatewayInterfaces

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - hosts: Bag< DeviceHost >
      - gatewayInterfaces: Bag< DeviceL3IfaceName >

.



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- DeviceParameters

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# DeviceParameters

## Fields

| name                                        | type            | nullable |
|---------------------------------------------|-----------------|----------|
| allowedServices                             | Bag< String >   | true     |
| allowedCertificationAuthorityEnrollmentUrls | Bag< String >   | true     |
| managementSubnets                           | Bag< IpSubnet > | true     |
| isProviderEdge                              | Bool            | true     |
| allowedBgpPeers                             | Bag< IpSubnet > | true     |

| name | type | nullable |  |
|------|------|----------|--|
|      |      |          |  |

Used by[↗](#)

**network** [↗](#) **stigDatabase** [↗](#) **defaultParameters** [↗](#) **deviceParameters**[↗](#)

See also[↗](#)

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - defaultParameters: [DefaultStigParameters](#)
      - deviceParameters: [DeviceParameters](#)

.



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- [Data Model](#)
- DevicePart

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# DevicePart

## Fields

| name         | type                           | nullable | description                                                                                                                                                                                                                                                                                                                             |
|--------------|--------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name         | String                         | false    | A readable name to identify the device part within the device. Note that the assigned name for the same part (with a particular serial number) may change from snapshot to snapshot if a new device part is added or order of parts is changed. If the device does not list a name for this part, then a unique name will be generated. |
| partId       | String                         | true     | Device part's model number. Cisco-like format often refer to this as 'PID', while Arista devices refer to this as 'Model'. Absent if the device part's PID is marked as 'unspecified' or 'N/A'.                                                                                                                                         |
| serialNumber | String                         | true     | Device part's serial number. Absent if the device part's serial number is marked as 'unspecified' or 'N/A'.                                                                                                                                                                                                                             |
| partType     | <a href="#">DevicePartType</a> | false    |                                                                                                                                                                                                                                                                                                                                         |

| name        | type                              | nullable | description                                                                                                                                                                                                                                                                                                                     |
|-------------|-----------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |                                   |          | The type of the device part. The value will be OTHER if this part could not be classified.                                                                                                                                                                                                                                      |
| description | String                            | true     | The description of the device part. Absent if the device part's description is missing or blank.                                                                                                                                                                                                                                |
| versionId   | String                            | true     | Device part's version number. Cisco-like format often refer to this as 'VID' while Juniper devices refer to this as 'Version'. Absent if the device part's VID is marked as 'unspecified' or 'N/A'.                                                                                                                             |
| support     | <a href="#">DevicePartSupport</a> | true     | End-of-life dates for this device part. Absent unless the vendor is CISCO, PALO_ALTO_NETWORKS, FORTINET, or F5. When the fields lastMaintenanceDate and lastVulnerabilityDate are present, the fields of DevicePartSupport satisfy the following condition:<br>lastMaintenanceDate <= lastVulnerabilityDate <= lastSupportDate. |

## Used by[↗](#)

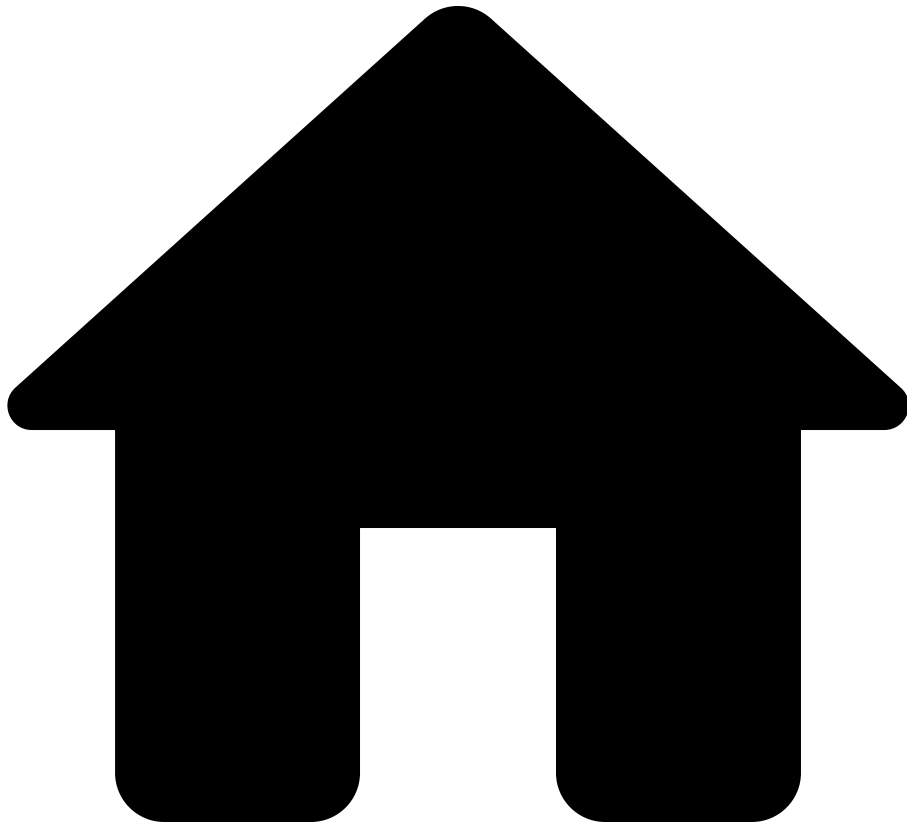
**network** [↗](#) **devices** [↗](#) **platform** [↗](#) **components**[↗](#)

**See also**[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - platform: [Platform](#)
      - components: Bag< [DevicePart](#) >

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- DevicePartSupport

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# DevicePartSupport

## Fields

| name                  | type   | nullable | description                                                                    |
|-----------------------|--------|----------|--------------------------------------------------------------------------------|
| announcementUrl       | String | false    | --                                                                             |
| lastMaintenanceDate   | Date   | true     | The last date with maintenance support for any OS compatible with this part.   |
| lastVulnerabilityDate | Date   | true     | The last date with vulnerability support for any OS compatible with this part. |
| lastSupportDate       | Date   | false    | The last date with any support for this part.                                  |

## Used by

network devices platform components support

### See also

Data model path

- network: Network
  - devices: Bag< Device >
    - platform: Platform
      - components: Bag< DevicePart >
        - support: DevicePartSupport



.



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- DevicePartType

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# DevicePartType

## Options

| name                    | description                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| APPLICATION             | --                                                                                                                                                                                                       |
| CHASSIS                 | --                                                                                                                                                                                                       |
| FABRIC_MODULE           | --                                                                                                                                                                                                       |
| FAN_MODULE              | --                                                                                                                                                                                                       |
| LINE_CARD               | --                                                                                                                                                                                                       |
| MOTHERBOARD             | --                                                                                                                                                                                                       |
| OTHER                   | --                                                                                                                                                                                                       |
| POWER_SUPPLY            | --                                                                                                                                                                                                       |
| ROUTING_ENGINE          | --                                                                                                                                                                                                       |
| STACK                   | --                                                                                                                                                                                                       |
| STACK_MODULE            | Stacking modules are additional parts that, when attached to stackable switches, allow them to stack with each other and offer flexible, unified management. Currently only supported for Cisco devices. |
| STACK_PORT              | --                                                                                                                                                                                                       |
| STACK_SWITCH            | --                                                                                                                                                                                                       |
| SUPERVISOR              | --                                                                                                                                                                                                       |
| TRANSCEIVER             | --                                                                                                                                                                                                       |
| UNMODELED_BACKUP_DEVICE | The device part for which we don't have enough information to model it. For example on Velocloud we collect from the controller, which doesn't give any visibility into the backup device                |

| name | description                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------|
|      | of an HA pair. All endpoints that return data about the devices it manages omits the backup device except for its serial number. |

## Used by[^](#)

**network** [^](#) **devices** [^](#) **platform** [^](#) **components** [^](#) **partType**[^](#)

See also[^](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - platform: [Platform](#)
      - components: Bag< [DevicePart](#) >
        - partType: [DevicePartType](#)

.



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- DevicePolicy

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# DevicePolicy

## Fields

| name                                        | type                                       |
|---------------------------------------------|--------------------------------------------|
| allowedServices                             | Bag< <a href="#">AllowedServicesDevice</a> |
| allowedCertificationAuthorityEnrollmentUrls | Bag< <a href="#">AllowedCertificationA</a> |
| managementSubnets                           | Bag< <a href="#">ManagementSubnetsDevi</a> |
| isProviderEdge                              | Bag< <a href="#">IsProviderEdgeDeviceP</a> |
| allowedBgpPeers                             | Bag< <a href="#">AllowedBgpPeersDevice</a> |

## Used by

network stigDatabase policy devicePolicy

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - devicePolicy: [DevicePolicy](#)

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- [Data Model](#)
- DeviceProcessingError

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# DeviceProcessingError

## Options

| name                    | description |
|-------------------------|-------------|
| DUPLICATE               | --          |
| LICENSE_EXHAUSTED       | --          |
| MISSING_SIGNATURE       | --          |
| PARSER_EXCEPTION        | --          |
| UNSUPPORTED_DEVICE_TYPE | --          |
| UNSUPPORTED_VENDOR      | --          |

## Used by

network devices snapshotInfo result

## See also

Data model path

- network: Network
  - devices: Bag< Device >
    - snapshotInfo: DeviceSnapshotInfo
      - result: DeviceSnapshotResult

**network endpoints snapshotInfo result**

See also

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - snapshotInfo: [DeviceSnapshotInfo](#)
      - result: [DeviceSnapshotResult](#)

**network externalSources cliSources snapshotInfo result**

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - cliSources: Bag< [CliSource](#) >
      - snapshotInfo: [DeviceSnapshotInfo](#)
        - result: [DeviceSnapshotResult](#)

**network externalSources snmpSources snapshotInfo result**

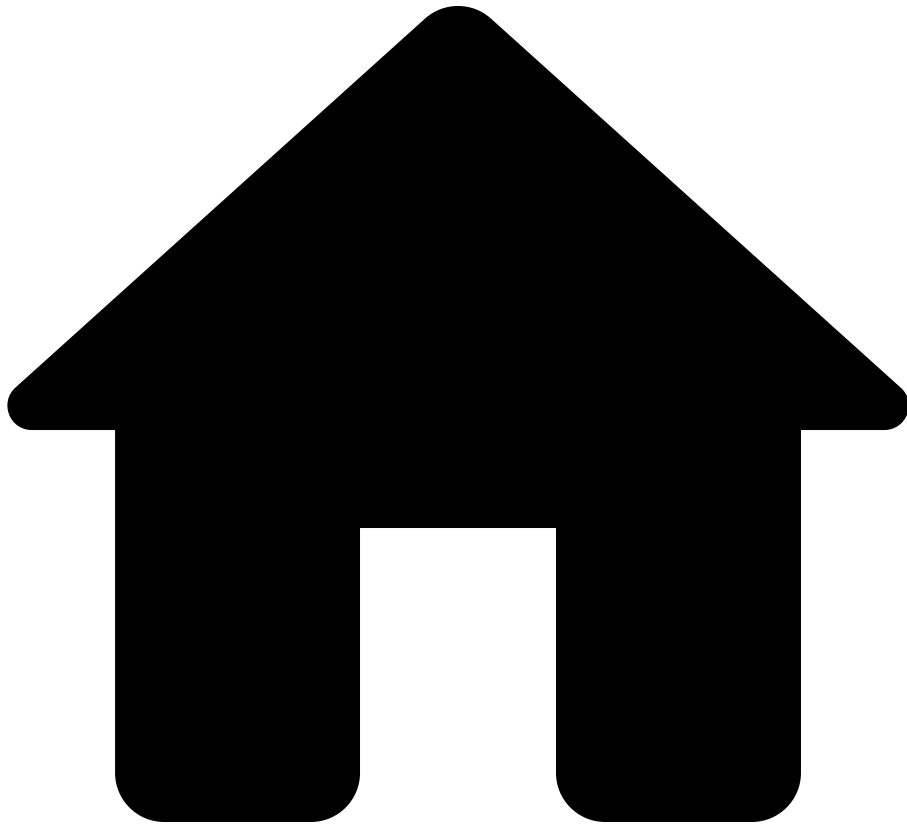
See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - snmpSources: Bag< [SnmpSource](#) >
      - snapshotInfo: [DeviceSnapshotInfo](#)
        - result: [DeviceSnapshotResult](#)



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- DeviceSecurityZone

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## DeviceSecurityZone

A security zone on a device.

## Fields

| name       | type                                 | nullable | description |
|------------|--------------------------------------|----------|-------------|
| name       | String                               | false    | --          |
| interfaces | Bag< <a href="#">IfaceSubIface</a> > | false    | --          |

## Used by

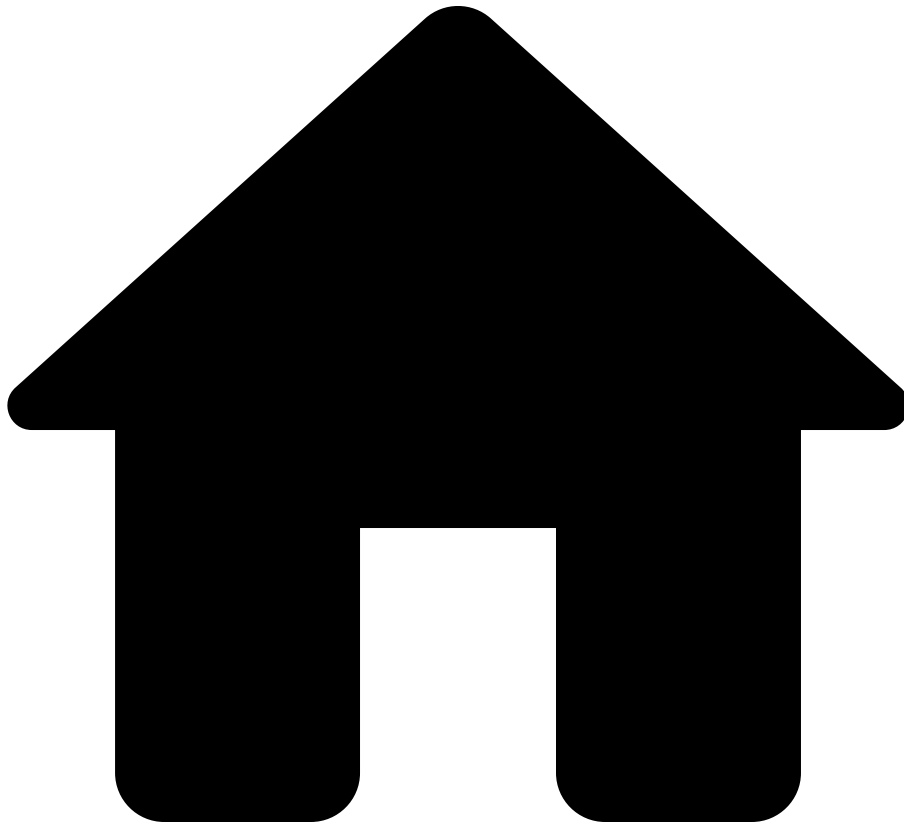
network devices securityZones

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - securityZones: Bag< [DeviceSecurityZone](#) >

.



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- DeviceSecurityZoneReference

On this page

## DeviceSecurityZoneReference

Reference of a security zone on a firewall

## Fields

| name       | type   | nullable | description                                                 |
|------------|--------|----------|-------------------------------------------------------------|
| deviceName | String | false    | Name of the firewall device where security zone is located. |
| zone       | String | false    | Name of the security zone.                                  |

## Used by

network devices hosts securityZones

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - hosts: Bag< DeviceHost >
      - securityZones: Bag< DeviceSecurityZoneReference >

.



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- [NQE Language](#)
- [Data Model](#)
- DeviceSnapshotInfo

On this page

# DeviceSnapshotInfo

## Fields

| name         | type                                 | nullable | description                                                                                                                                                                                                                                                                                                                                                  |
|--------------|--------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| result       | <a href="#">DeviceSnapshotResult</a> | false    | --                                                                                                                                                                                                                                                                                                                                                           |
| collectionIp | IpAddress                            | true     | The user-provided IP address used to collect for this device. There are some special cases. This value is (1) the DNS-resolved address if the collection address is specified as a hostname, (2) the collectionIp value of the parent physical device if this device is a partition on a parent physical device, or (3) null for devices that do not require |

| name           | type      | nullable | description                                                                                                                                                                                                                           |
|----------------|-----------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |           |          | IP address for collection, such as synthetic devices, as well as for devices whose collection IP address could not be resolved at collection time (and which were therefore not collected).                                           |
| collectionTime | Timestamp | true     | The approximate time at which collection occurred for this device in this snapshot. Absent for synthetic devices, when no collection was attempted, and when collection failed but this device was not backfilled. Present otherwise. |

| name         | type      | nullable | description                                                                                                                                                                                                                                                |
|--------------|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |           |          | There might be no collection attempt if the snapshot timeout was reached or if the collection was manually stopped.                                                                                                                                        |
| backfillTime | Timestamp | true     | If the data for this device has been backfilled from a previous snapshot, then the approximate time at which that data was collected in that snapshot. Otherwise, this device has not been backfilled and this value is absent. Absent for custom sources. |



## Used by[â](#)

**network** [â](#) **devices** [â](#) **snapshotInfo**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - snapshotInfo: [DeviceSnapshotInfo](#)

**network** [â](#) **endpoints** [â](#) **snapshotInfo**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - snapshotInfo: [DeviceSnapshotInfo](#)

**network** [â](#) **externalSources** [â](#) **cliSources** [â](#) **snapshotInfo**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - cliSources: Bag< [CliSource](#) >
      - snapshotInfo: [DeviceSnapshotInfo](#)

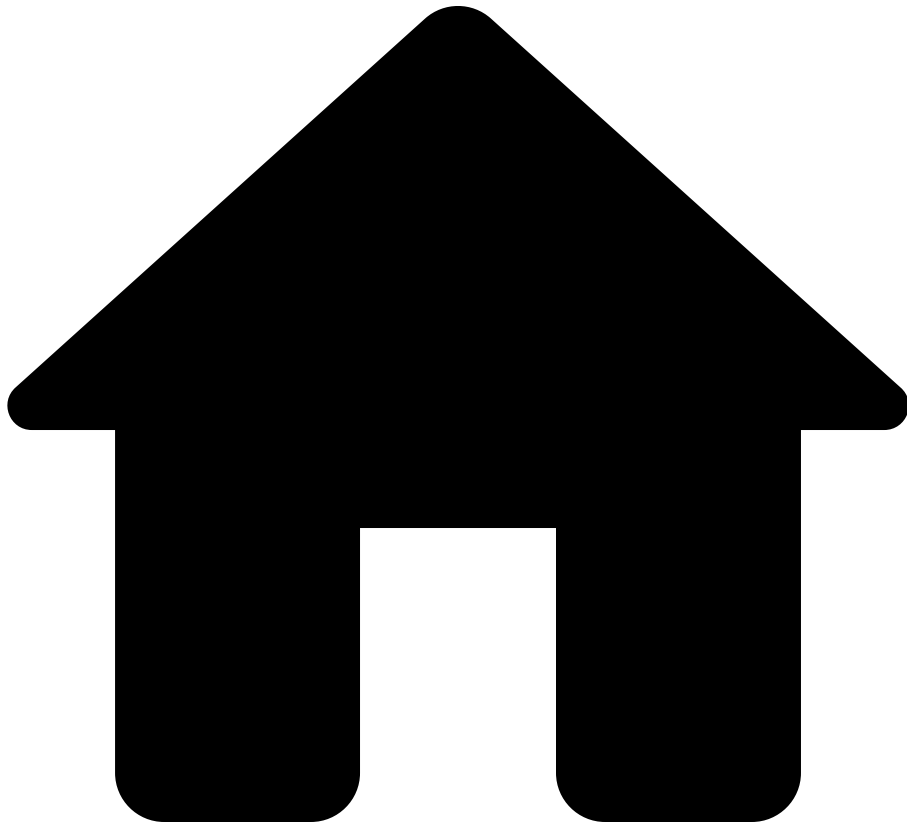
**network** [externalSources](#) [snmpSources](#) [snapshotInfo](#)

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - snmpSources: Bag< [SnmpSource](#) >
      - snapshotInfo: [DeviceSnapshotInfo](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DeviceSnapshotResult

On this page

# DeviceSnapshotResult

## Options

| name             | type                                  | description |
|------------------|---------------------------------------|-------------|
| collectionFailed | <a href="#">DeviceCollectionError</a> | --          |
| completed        | --                                    | --          |
| processingFailed | <a href="#">DeviceProcessingError</a> | --          |

## Used by

**network** **devices** **snapshotInfo** **result**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - snapshotInfo: [DeviceSnapshotInfo](#)
      - result: [DeviceSnapshotResult](#)

**network** **endpoints** **snapshotInfo** **result**

See also

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - snapshotInfo: [DeviceSnapshotInfo](#)
      - result: [DeviceSnapshotResult](#)

**network** **externalSources** **cliSources** **snapshotInfo** **result**

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - cliSources: Bag< [CliSource](#) >
      - snapshotInfo: [DeviceSnapshotInfo](#)
        - result: [DeviceSnapshotResult](#)

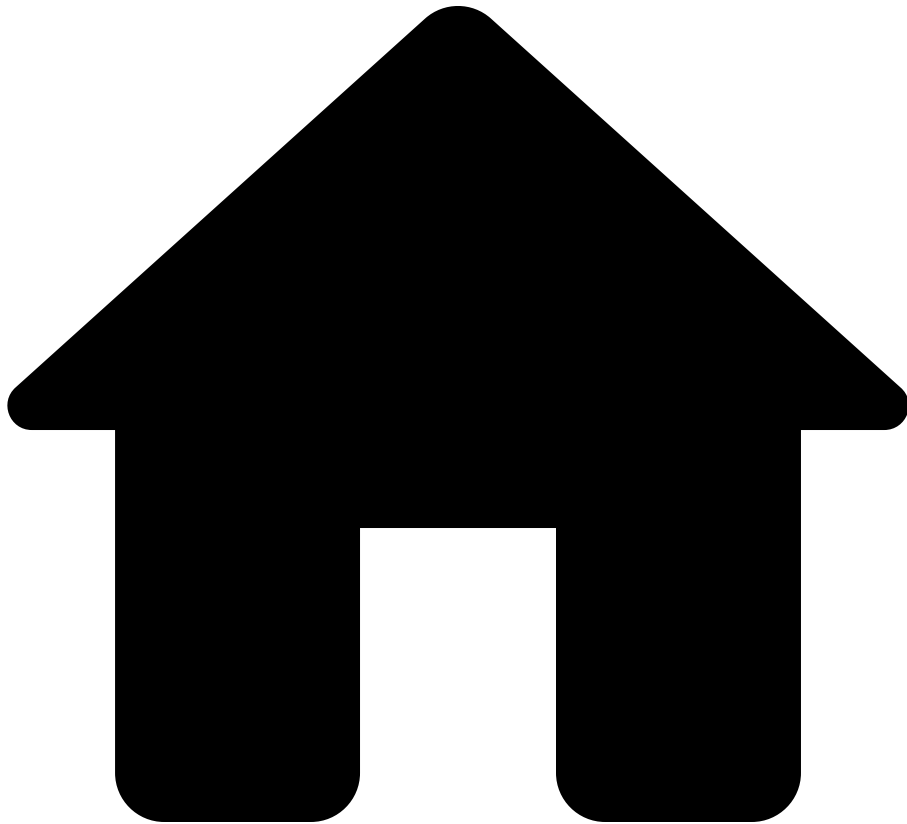
**network** **externalSources** **snmpSources** **snapshotInfo** **result**

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - snmpSources: Bag< [SnmpSource](#) >
      - snapshotInfo: [DeviceSnapshotInfo](#)
        - result: [DeviceSnapshotResult](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DeviceType

On this page

## DeviceType

Classification of networking devices. Common types include ROUTER, SWITCH, and FIREWALL.

## Options

| name          | description                                                                                                                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTROLLER    | Device for orchestrating network functions.                                                                                                                                                                                                 |
| ENCRYPTOR     | Synthetic device that encapsulates traffic in an IPSEC tunnel.                                                                                                                                                                              |
| FIREWALL      | Device that monitors and controls incoming and outgoing traffic.                                                                                                                                                                            |
| INTERNET      | Synthetic device representing access to the Internet.                                                                                                                                                                                       |
| INTRANET      | Synthetic device representing access to another intranet network.                                                                                                                                                                           |
| L2_VPN        | Synthetic device that is an extension of a L2 broadcast domain over the WAN. Emulates a distributed virtual bridge that brings all access circuits into a single broadcast domain. An L2_VPN device is also called an L2 transport service. |
| L3_VPN        | Synthetic device that models a Service Provider Layer-3 VPN MPLS network without the need to collect from the Provider Edge (PE) and Provider (P) routers. An L3_VPN device is also called an L3 transport service.                         |
| LOAD_BALANCER | Device for balancing traffic across multiple wide-area networks.                                                                                                                                                                            |
| MISSING_PEER  | Synthetic device that models traffic ignored by other synthetic devices.                                                                                                                                                                    |
| OTHER         | Catch-all value for any device whose type is not mapped to any other (known) category.                                                                                                                                                      |
| ROUTER        | Device that forwards data packets between networks (at layer 3).                                                                                                                                                                            |
| SD_WAN        | Device for enabling software-defined wide-area networks (SD-WANs).                                                                                                                                                                          |

| name              | description                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| SWITCH            | Device that switches frames (at layer 2).                                                                                                         |
| WAN_CIRCUIT       | Synthetic device that models point-to-point L2 connections; essentially a bridge that allows each side of the connection to have different VLANs. |
| WAN_OPTIMIZER     | Device for optimizing traffic across wide-area networks (WANs).                                                                                   |
| WIFI_ACCESS_POINT | Wireless access point.                                                                                                                            |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **platform** [â](#) **deviceType**[â](#)

**See also**[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - platform: [Platform](#)
      - deviceType: [DeviceType](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DirectConnection

On this page

# DirectConnection

## Fields

| name                | type          | nullable | description |
|---------------------|---------------|----------|-------------|
| name                | String        | true     | --          |
| id                  | String        | false    | --          |
| tags                | Bag< String > | false    | --          |
| isUp                | Bool          | false    | --          |
| cloudIp             | IpSubnet      | true     | --          |
| customerIp          | IpSubnet      | true     | --          |
| vlanId              | Integer       | true     | --          |
| virtualIfaceId      | String        | true     | --          |
| backupCustomerIp    | IpSubnet      | true     | --          |
| serviceProviderName | String        | true     | --          |
| peeringLocation     | String        | true     | --          |

## Used by

network [cloudAccounts](#) [dcGateways](#) [directConnections](#)

## See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - dcGateways: Bag< [DcGateway](#) >
      - directConnections: Bag< [DirectConnection](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DnsActivityConfigurationDetail

On this page

## DnsActivityConfigurationDetail

## Fields

| name      | type   | nullable | description |
|-----------|--------|----------|-------------|
| enable    | Bool   | false    | --          |
| serviceId | String | true     | --          |

Used by an

network devices outputs bluecat config items data  
services dnsActivity configurations  
dnsActivityConfiguration

**See also**[ââ](#)

## Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DnsActivityService

On this page

# DnsActivityService

## Fields

| name           | type                                                   | nullable | description |
|----------------|--------------------------------------------------------|----------|-------------|
| configurations | Bag< <a href="#">DnsActivityServiceConfiguration</a> > | false    | --          |

## Used by

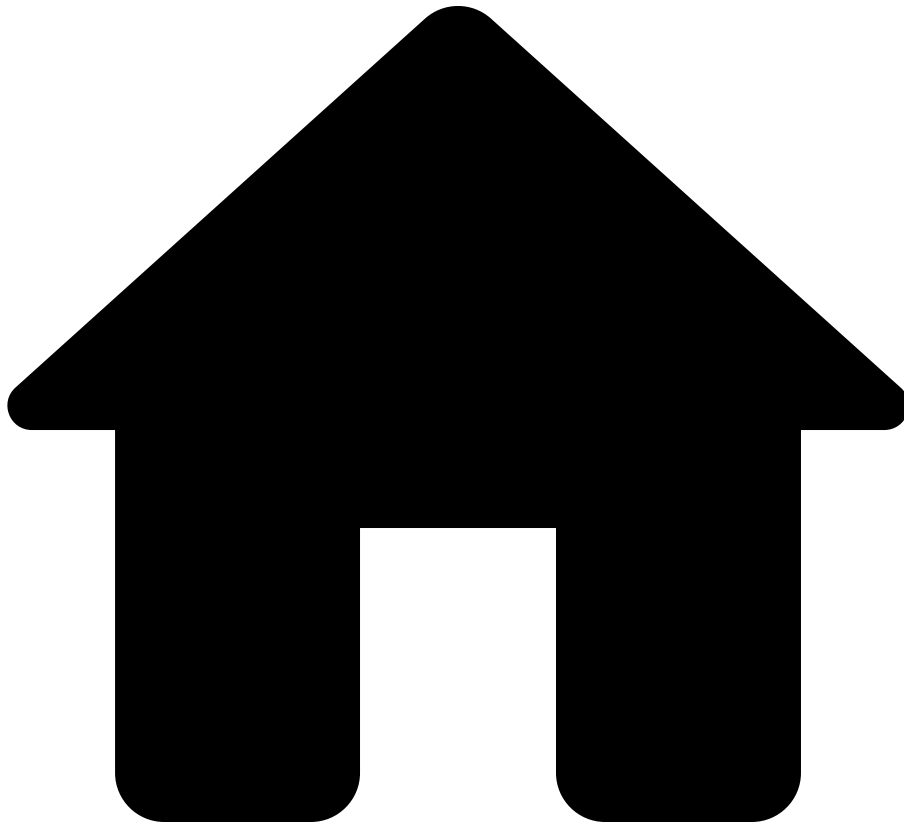
network devices outputs bluecat config items data  
services dnsActivity

### See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - dnsActivity: [DnsActivityService](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DnsActivityServiceConfiguration

On this page

# DnsActivityServiceConfiguration

## Fields

| name                     | type                                           | nullable | default |
|--------------------------|------------------------------------------------|----------|---------|
| dnsActivityConfiguration | <a href="#">DnsActivityConfigurationDetail</a> | false    | --      |

## Used by

network devices outputs bluecat config items data  
services dnsActivity configurations

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - dnsActivity: [DnsActivityService](#)
                  - configurations: Bag< [DnsActivityServiceConfigurat](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DnsResolverConfigurationDetail

On this page

# DnsResolverConfigurationDetail

## Fields

| name    | type             | nullable | description |
|---------|------------------|----------|-------------|
| servers | Bag< IpAddress > | false    | --          |

## Used by

network devices outputs bluecat config items data  
services dnsResolver configurations  
dnsResolverConfiguration

## See also

Data model path

- network: Network
  - devices: Bag< Device >
    - outputs: Outputs
      - bluecat: BluecatOutputs
        - config: BluecatConfig
          - items: Bag< BluecatConfigItem >
            - data: BluecatConfigItemData
              - services: BluecatServices
                - dnsResolver: DnsResolverService
                  - configurations: Bag< DnsResolverServiceConfiguration >
                    - dnsResolverConfiguration: DnsResolverConfigurationDetail

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DnsResolverService

On this page

# DnsResolverService

## Fields

| name           | type                                                   | nullable | description |
|----------------|--------------------------------------------------------|----------|-------------|
| configurations | Bag< <a href="#">DnsResolverServiceConfiguration</a> > | false    | --          |

## Used by

network devices outputs bluecat config items data  
services dnsResolver

### See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - dnsResolver: [DnsResolverService](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DnsResolverServiceConfiguration

On this page

# DnsResolverServiceConfiguration

## Fields

| name                     | type                                           | nullable | default |
|--------------------------|------------------------------------------------|----------|---------|
| dnsResolverConfiguration | <a href="#">DnsResolverConfigurationDetail</a> | false    | --      |

## Used by

network devices outputs bluecat config items data  
services dnsResolver configurations

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - dnsResolver: [DnsResolverService](#)
                  - configurations: Bag< [DnsResolverServiceConfiguration](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- DuplexMode

On this page

# DuplexMode

## Options

| name | description |
|------|-------------|
| FULL | --          |
| HALF | --          |

## Used by

network devices interfaces ethernet negotiatedDuplexMode

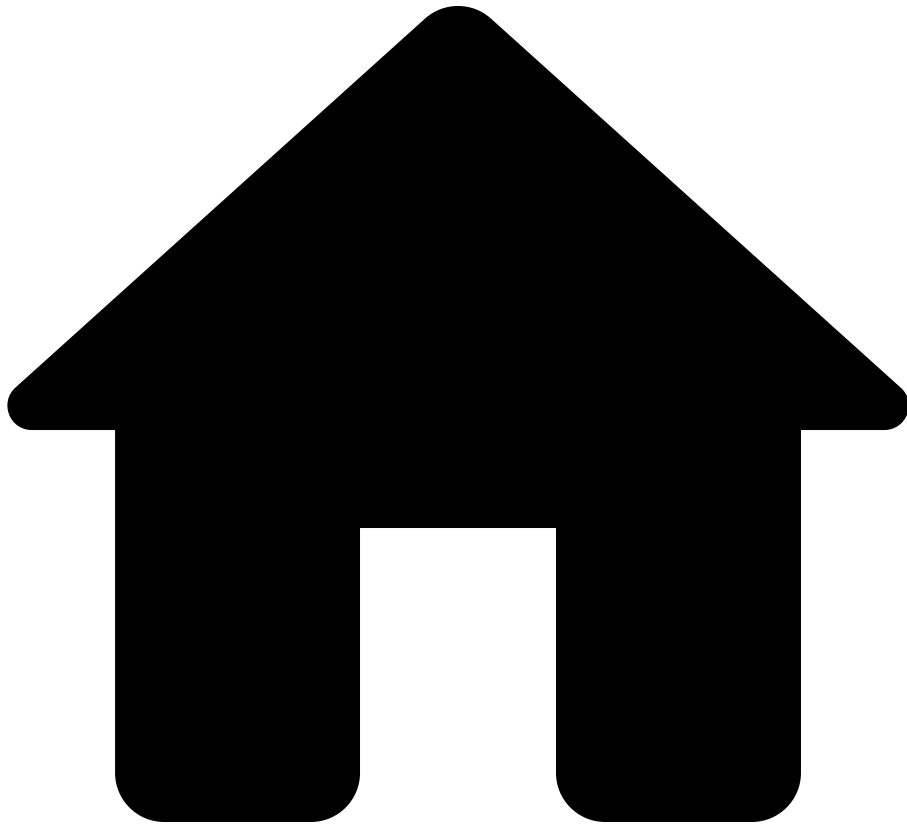
See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - ethernet: Ethernet
        - negotiatedDuplexMode: DuplexMode



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- EdgeServicePointConfigurationDetail

On this page

# EdgeServicePointConfigurationDetail

## Fields

| name   | type | nullable | description |
|--------|------|----------|-------------|
| enable | Bool | false    | --          |

Used by an

network devices outputs bluecat config items data  
services edgeServicePoint configurations  
edgeServicePointConfiguration

**See also** [ââ](#)

### Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- EdgeServicePointService

On this page

# EdgeServicePointService

## Fields

| name           | type                                                        | nullable | comment |
|----------------|-------------------------------------------------------------|----------|---------|
| configurations | Bag< <a href="#">EdgeServicePointServiceConfiguration</a> > | false    |         |

## Used by

network devices outputs bluecat config items data  
services edgeServicePoint

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - edgeServicePoint: [EdgeServicePointService](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- EdgeServicePointServiceConfiguration

On this page

# EdgeServicePointServiceConfiguration

## Fields

| name                          | type                                                |
|-------------------------------|-----------------------------------------------------|
| edgeServicePointConfiguration | <a href="#">EdgeServicePointConfigurationDetail</a> |

## Used by

network devices outputs bluecat config items data  
services edgeServicePoint configurations

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - edgeServicePoint: [EdgeServicePointService](#)
                  - configurations: Bag< [EdgeServicePointServiceConf](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- EncapsulationHeaderType

On this page

# EncapsulationHeaderType

## Options

| name | description                    |
|------|--------------------------------|
| GRE  | Generic routing encapsulation. |
| IPV4 | --                             |
| IPV6 | --                             |
| MPLS | --                             |

## Used by

**network** **devices** **networkInstances** **afts** **ipv4Unicast** **ipEntries** **nextHops** **encapsulateHeader**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv4Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >
            - nextHops: Bag< [NextHop](#) >
              - encapsulateHeader: [EncapsulationHeaderType](#)



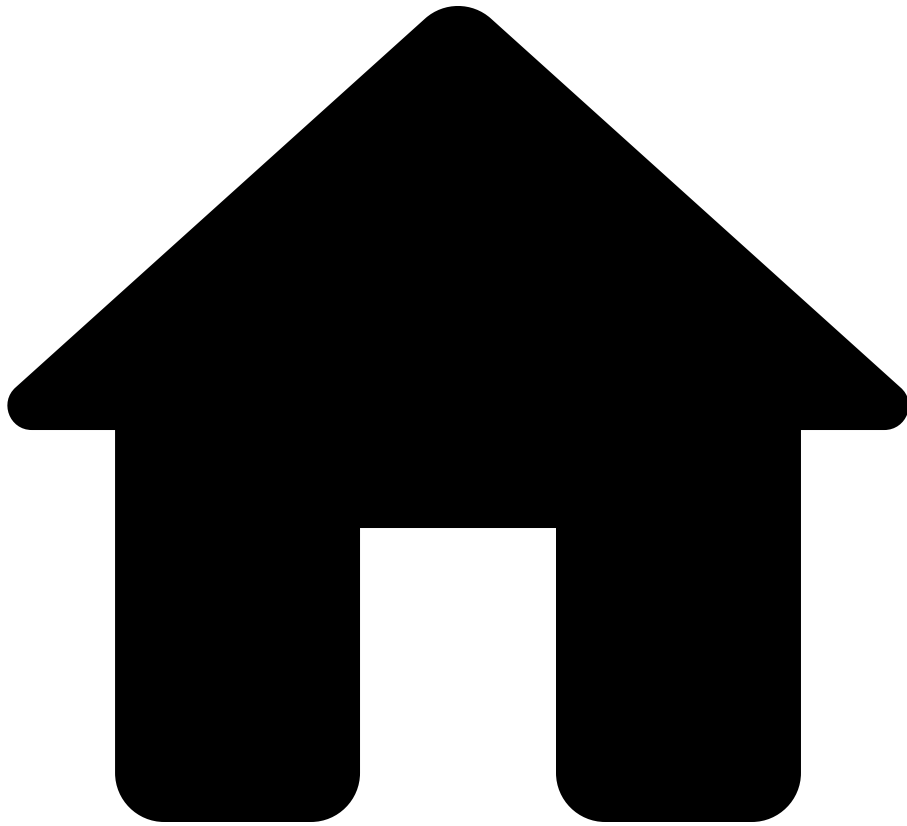
**network** **devices** **networkInstances** **afts** **ipv6Unicast** **ipEntries** **nextHops** **encapsulateHeader**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv6Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >
            - nextHops: Bag< [NextHop](#) >
              - encapsulateHeader: [EncapsulationHeaderType](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Ethernet

On this page

# Ethernet

## Fields

| name                 | type                       | nullable | description                                                                                                                                                                                                                                                                  |
|----------------------|----------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| macAddress           | MacAddress                 | true     | MAC address of the Ethernet interface.                                                                                                                                                                                                                                       |
| negotiatedDuplexMode | <a href="#">DuplexMode</a> | false    | This value shows the duplex mode that has been negotiated.                                                                                                                                                                                                                   |
| negotiatedPortSpeed  | <a href="#">PortSpeed</a>  | false    | This value shows the interface speed that has been negotiated.                                                                                                                                                                                                               |
| speedMbps            | Integer                    | true     | The effective speed in Mbps of the interface. If interface bandwidth is statically configured, then the configured bandwidth value. Otherwise, if the port speed is negotiated, then the speed that both sides settled on. If not configured or negotiated, then the highest |

| name         | type                         | nullable | description                                                                                                                                                                                                                                                                                                               |
|--------------|------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                              |          | <p>possible speed of the port. If an aggregated interface, then the combined speed of its active member interfaces.</p> <p>Absent if the speed could not be determined.</p> <p>Note that a future update will exclusively populate this value based on port speed (rather than sometimes using configured bandwidth).</p> |
| aggregateId  | String                       | true     | Specifies the logical aggregate interface to which this interface belongs.                                                                                                                                                                                                                                                |
| switchedVlan | <a href="#">SwitchedVlan</a> | true     | --                                                                                                                                                                                                                                                                                                                        |

## Used by[↗](#)

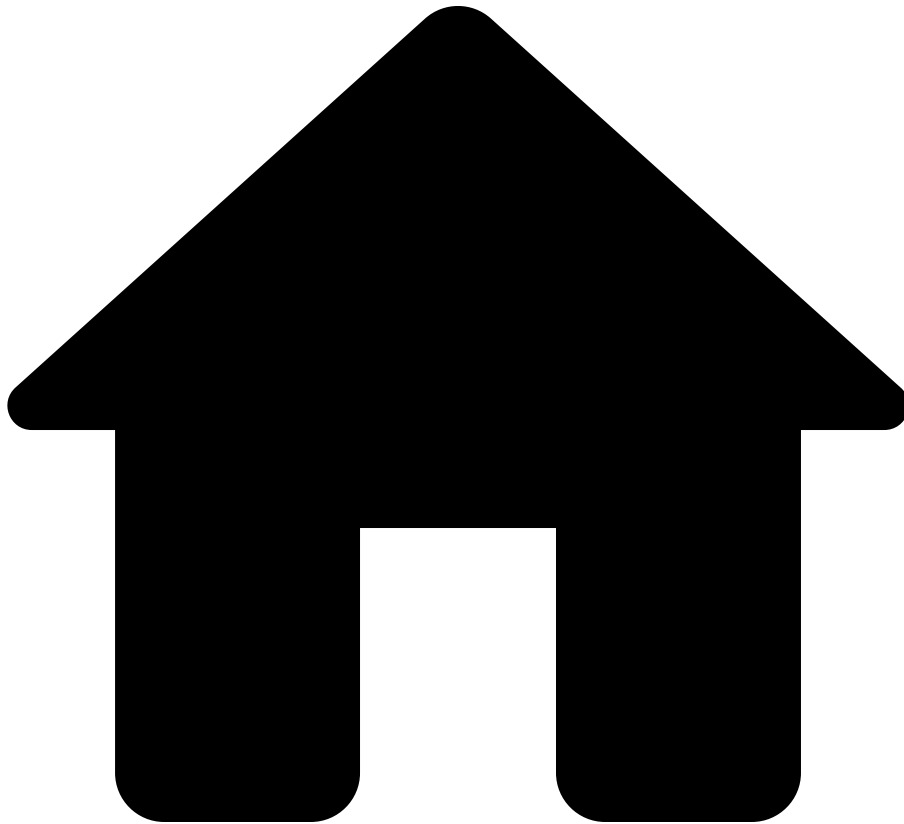
**network** [↗](#) **devices** [↗](#) **interfaces** [↗](#) **ethernet**[↗](#)

**See also**[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - ethernet: [Ethernet](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ExternalSources

On this page

# ExternalSources

## Fields

| name        | type                              | nullable | description |
|-------------|-----------------------------------|----------|-------------|
| httpSources | Bag< <a href="#">HttpSource</a> > | false    | --          |
| cliSources  | Bag< <a href="#">CliSource</a> >  | false    | --          |
| snmpSources | Bag< <a href="#">SnmpSource</a> > | false    | --          |

## Used by

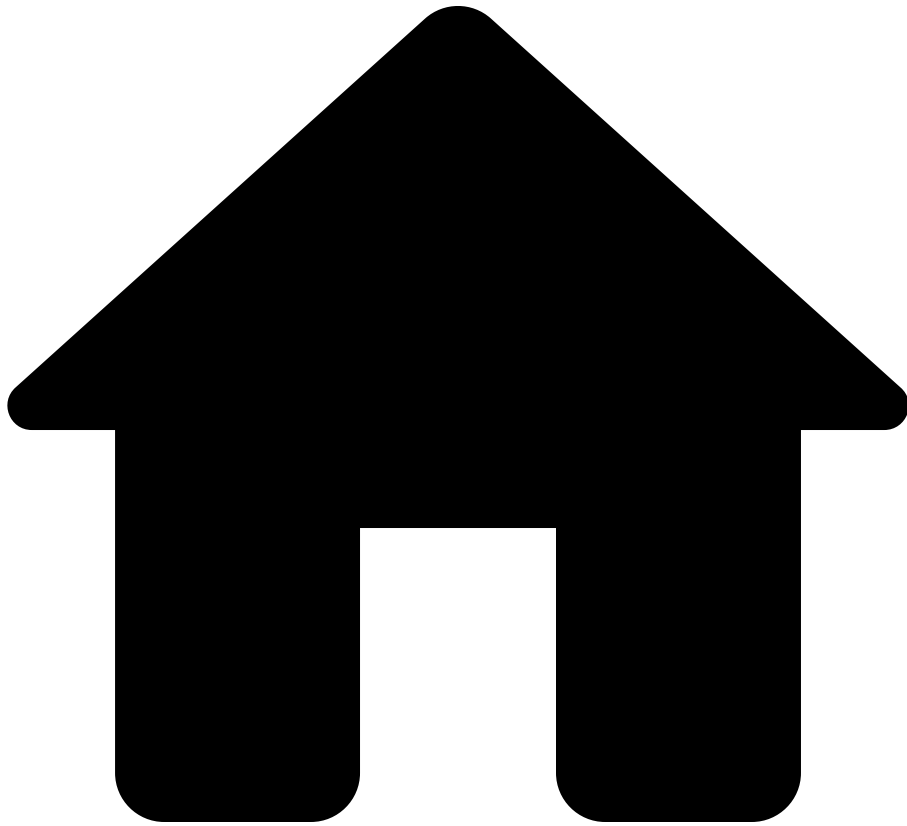
network externalSources

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- FhrpState

On this page



# FhrpState

## Options[â](#)

| name   | description                                                               |
|--------|---------------------------------------------------------------------------|
| BACKUP | --                                                                        |
| MASTER | --                                                                        |
| OTHER  | States other than MASTER and BACKUP, such as LISTEN, STOPPED, SPEAK, etc. |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **interfaces** [â](#) **subinterfaces** [â](#) **ipv4** [â](#) **fhrp** [â](#) **hsrp** [â](#) **fhrpGroups** [â](#) **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **subinterfaces** **ipv4** **fhrp** **vrrp** **fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **subinterfaces** **ipv6** **fhrp** **hsrp** **fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **subinterfaces** **ipv6** **fhrp** **vrrp** **fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **routedVlan** **ipv4** **fhrp** **hsrp** **fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **routedVlan** **ipv4** **fhrp** **vrrp** **fhrpGroups** **state**[^](#)

See also[^](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **routedVlan** **ipv6** **fhrp** **hsrp** **fhrpGroups** **state**[^](#)

See also[^](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **routedVlan** **ipv6** **fhrp** **vrrp** **fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
      - ipv6: [IfaceIpv6Info](#)
      - fhrp: [IfaceFhrp](#)
        - vrrp: [IfaceFhrpInfo](#)
          - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
            - state: [FhrpState](#)

**network** **devices** **interfaces** **bridge** **ipv4** **fhrp** **hsrp** **fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
      - ipv4: [IfaceIpv4Info](#)
      - fhrp: [IfaceFhrp](#)
        - hsrp: [IfaceFhrpInfo](#)
          - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
            - state: [FhrpState](#)

**network** **devices** **interfaces** **bridge** **ipv4** **fhrp** **vrrp**  
**fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **bridge** **ipv6** **fhrp** **hsrp**  
**fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **bridge** **ipv6** **fhrp** **vrrp**  
**fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
      - ipv6: [IfaceIpv6Info](#)
      - fhrp: [IfaceFhrp](#)
      - vrrp: [IfaceFhrpInfo](#)
        - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
          - state: [FhrpState](#)

**network** **devices** **interfaces** **tunnel** **ipv4** **fhrp** **hsrp**  
**fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
      - ipv4: [IfaceIpv4Info](#)
      - fhrp: [IfaceFhrp](#)
      - hsrp: [IfaceFhrpInfo](#)
        - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
          - state: [FhrpState](#)

**network** **devices** **interfaces** **tunnel** **ipv4** **fhrp** **vrrp** **fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

**network** **devices** **interfaces** **tunnel** **ipv6** **fhrp** **hsrp** **fhrpGroups** **state**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)



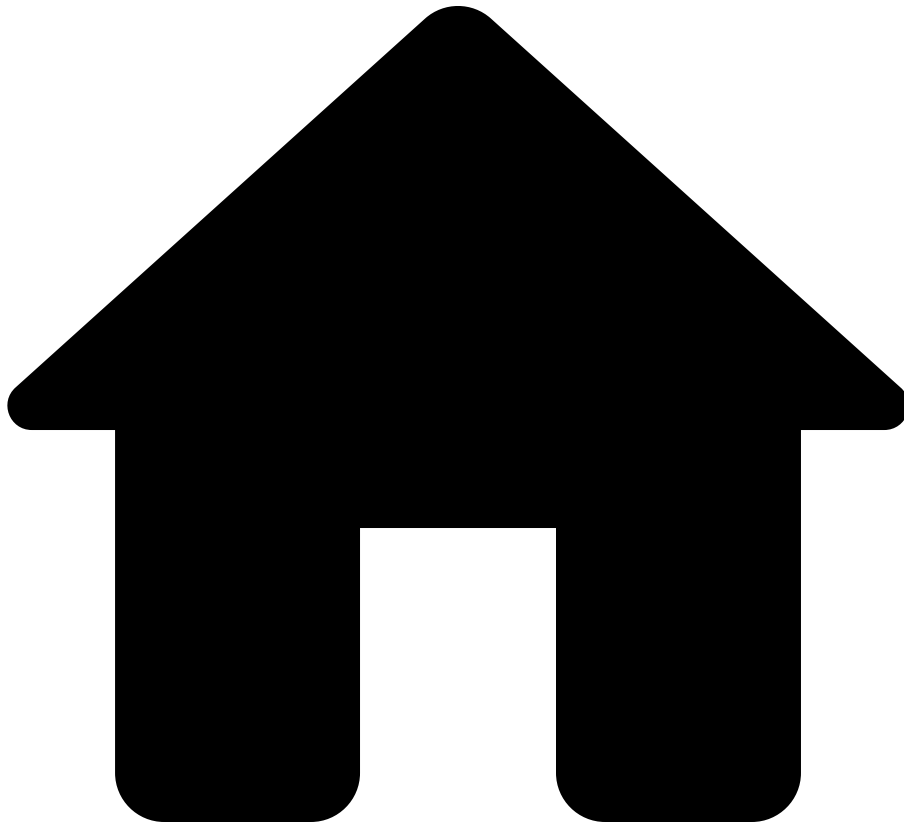
**network** **devices** **interfaces** **tunnel** **ipv6** **fhrp** **vrrp** **fhrpGroups** **state**[â](#)

**See also**[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >
                - state: [FhrpState](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Files

On this page

## Files

Information about config data sources for a device.

## Fields

| name   | type                               | nullable | description                                                                                                                                                        |
|--------|------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| config | List< <a href="#">ConfigLine</a> > | true     | Lines of config for a device.<br>Some lines from the original config are not included. See the <a href="#">parseConfigBlock</a> documentation for a detailed list. |
| panos  | <a href="#">PanosFiles</a>         | true     | PANOS-specific files                                                                                                                                               |

## Used by

network devices files

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - files: [Files](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- FirewallConfigurationDetail

On this page

# FirewallConfigurationDetail

## Fields

| name      | type | nullable | description |
|-----------|------|----------|-------------|
| allowPing | Bool | false    | --          |
| enable    | Bool | false    | --          |

## Used by

network devices outputs bluecat config items data services firewall configurations firewallConfiguration

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - outputs: Outputs
      - bluecat: BluecatOutputs
        - config: BluecatConfig
          - items: Bag< BluecatConfigItem >
            - data: BluecatConfigItemData
              - services: BluecatServices
                - firewall: FirewallService
                  - configurations: Bag< FirewallServiceConfiguration >
                    - firewallConfiguration: FirewallConfigurationDetail

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- FirewallPolicy

On this page

# FirewallPolicy

## Fields

| name           | type                                 | nullable | description |
|----------------|--------------------------------------|----------|-------------|
| networkObjects | Bag< <a href="#">NetworkObject</a> > | false    | --          |
| ruleContainers | Bag< <a href="#">RuleContainer</a> > | false    | --          |

## Used by

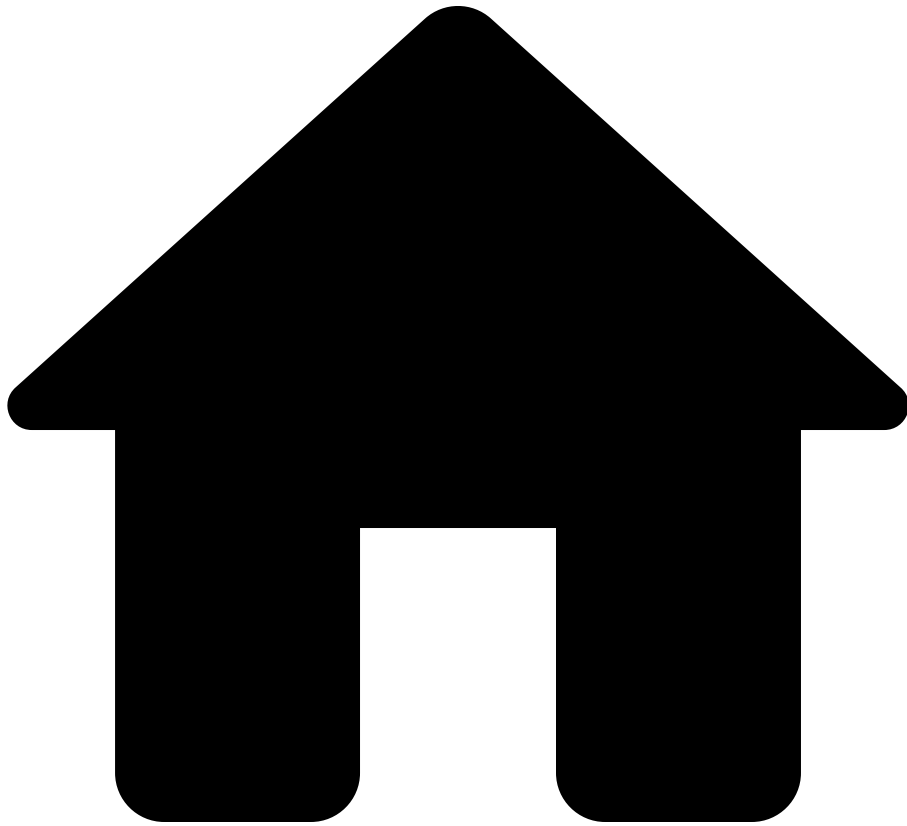
network cloudAccounts firewallPolicy

### See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallPolicy: [FirewallPolicy](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- FirewallRule

On this page



# FirewallRule

## Fields

| name                        | type                         | nullable | description |
|-----------------------------|------------------------------|----------|-------------|
| id                          | String                       | false    | --          |
| securityRule                | <a href="#">SecurityRule</a> | false    | --          |
| appliedToVpcId              | String                       | true     | --          |
| appliedToSourceInstanceTags | Bag< String >                | false    | --          |
| appliedToTargetInstanceTags | Bag< String >                | false    | --          |

## Used by

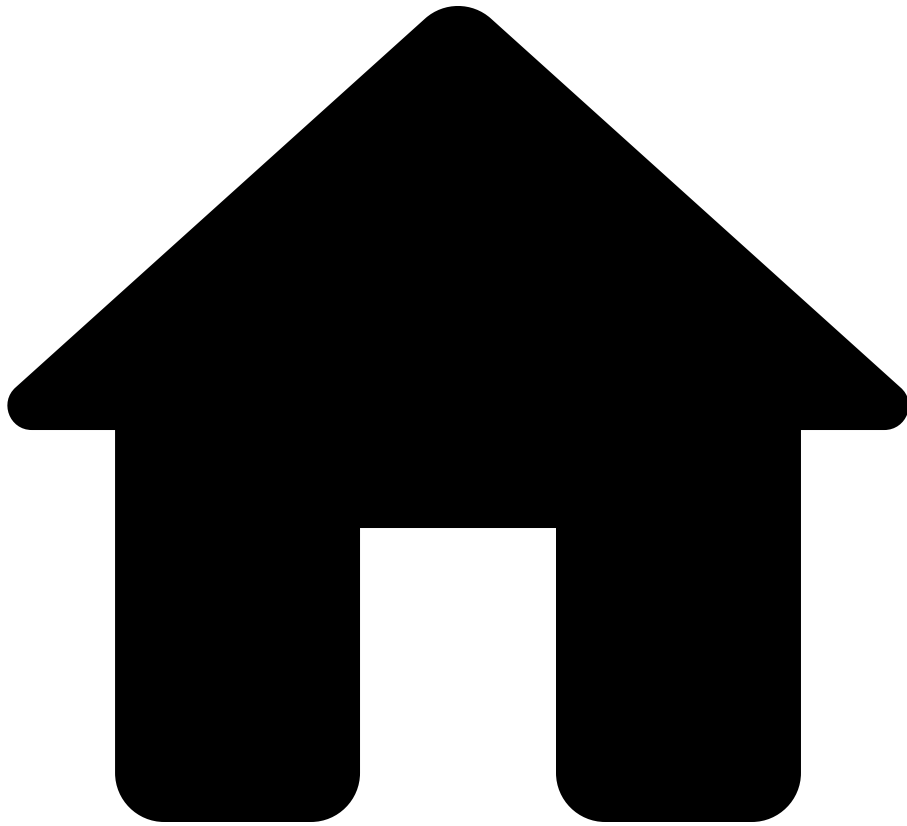
network [cloudAccounts](#) [firewallRules](#)

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- FirewallService

On this page

# FirewallService

## Fields

| name           | type                                                | nullable | description |
|----------------|-----------------------------------------------------|----------|-------------|
| configurations | Bag< <a href="#">FirewallServiceConfiguration</a> > | false    | --          |

## Used by

network devices outputs bluecat config items data  
services firewall

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - firewall: [FirewallService](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- FirewallServiceConfiguration

On this page

# FirewallServiceConfiguration

## Fields

| name                  | type                                        | nullable | description |
|-----------------------|---------------------------------------------|----------|-------------|
| firewallConfiguration | <a href="#">FirewallConfigurationDetail</a> | false    | --          |

## Used by

network devices outputs bluecat config items data  
services firewall configurations

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - firewall: [FirewallService](#)
                  - configurations:  
Bag< [FirewallServiceConfiguration](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GatewayConfigurationDetail

On this page

# GatewayConfigurationDetail

## Fields

| name        | type                                           | nullable | description |
|-------------|------------------------------------------------|----------|-------------|
| removeImage | Bool                                           | false    | --          |
| data        | <a href="#">GatewayConfigurationDetailData</a> | false    | --          |

## Used by

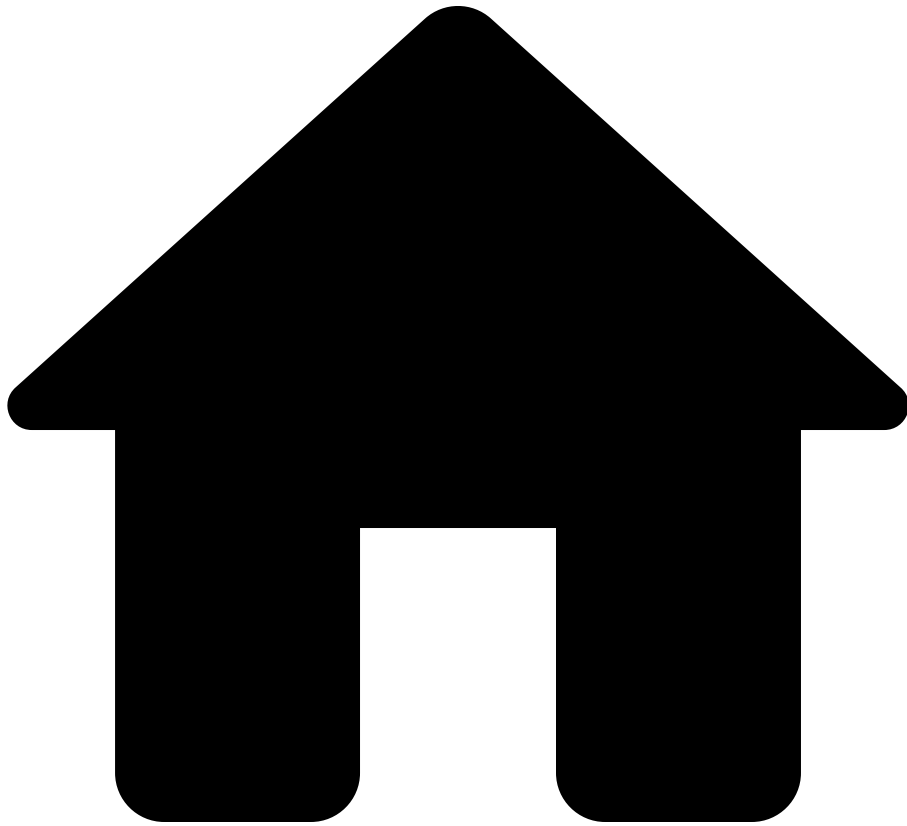
network devices outputs bluecat config items data  
services gateway configurations gatewayConfiguration

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - gateway: [GatewayService](#)
                  - configurations: Bag< [GatewayServiceConfiguration](#) >
                    - gatewayConfiguration: [GatewayConfigurationDetail](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GatewayConfigurationDetailData

On this page



# GatewayConfigurationDetailData

## Fields

| name       | type                                    | nullable | description |
|------------|-----------------------------------------|----------|-------------|
| enable     | Bool                                    | false    | --          |
| containers | Bag< <a href="#">GatewayContainer</a> > | false    | --          |
| version    | Integer                                 | false    | --          |

Used by 100

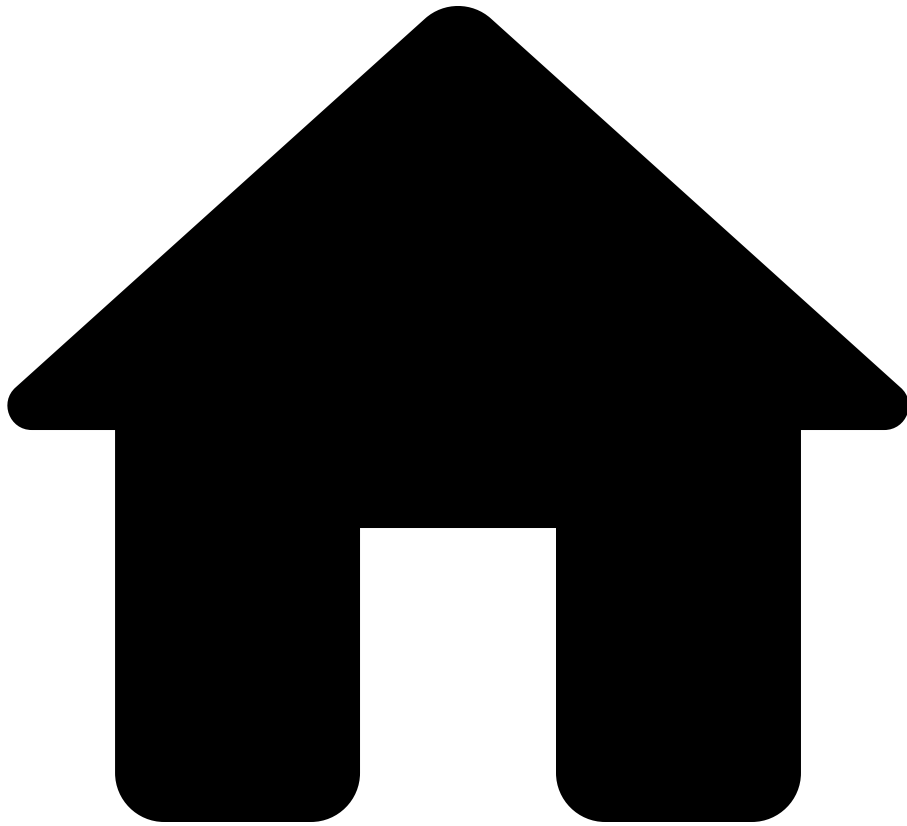
network â devices â outputs â bluecat â config â items â data  
â services â gateway â configurations â gatewayConfiguration â  
dataâ

**See also** [ââ](#)

## Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GatewayContainer

On this page

# GatewayContainer

## Fields

| name        | type          | nullable | description |
|-------------|---------------|----------|-------------|
| name        | String        | false    | --          |
| image       | String        | false    | --          |
| ports       | Bag< String > | false    | --          |
| volumes     | Bag< String > | false    | --          |
| environment | Bag< String > | false    | --          |

Used by [1000](#)

network & devices & outputs & bluecat & config & items & data  
& services & gateway & configurations & gatewayConfiguration &  
data & containers&

**See also** [ââ](#)

## Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GatewayService

On this page

# GatewayService

## Fields

| name           | type                                               | nullable | description |
|----------------|----------------------------------------------------|----------|-------------|
| configurations | Bag< <a href="#">GatewayServiceConfiguration</a> > | false    | --          |

## Used by

network devices outputs bluecat config items data  
services gateway

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - gateway: [GatewayService](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GatewayServiceConfiguration

On this page

# GatewayServiceConfiguration

## Fields

| name                 | type                                       | nullable | description |
|----------------------|--------------------------------------------|----------|-------------|
| gatewayConfiguration | <a href="#">GatewayConfigurationDetail</a> | false    | --          |

## Used by

network devices outputs bluecat config items data  
services gateway configurations

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - gateway: [GatewayService](#)
                  - configurations:  
Bag< [GatewayServiceConfiguration](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GlobalAccelerator

On this page

# GlobalAccelerator

## Fields

| name      | type                                             | nullable | description                                                                                                              |
|-----------|--------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------|
| name      | String                                           | true     | --                                                                                                                       |
| id        | String                                           | false    | --                                                                                                                       |
| dnsName   | String                                           | false    | The Domain Name System (DNS) name that Global Accelerator creates that points to an accelerator's static IPv4 addresses. |
| publicIps | Bag< IPAddress >                                 | false    | --                                                                                                                       |
| tags      | Bag< String >                                    | false    | --                                                                                                                       |
| status    | <a href="#">GlobalAcceleratorStatus</a>          | false    | --                                                                                                                       |
| listeners | Bag< <a href="#">GlobalAcceleratorListener</a> > | false    | --                                                                                                                       |

## Used by[â](#)

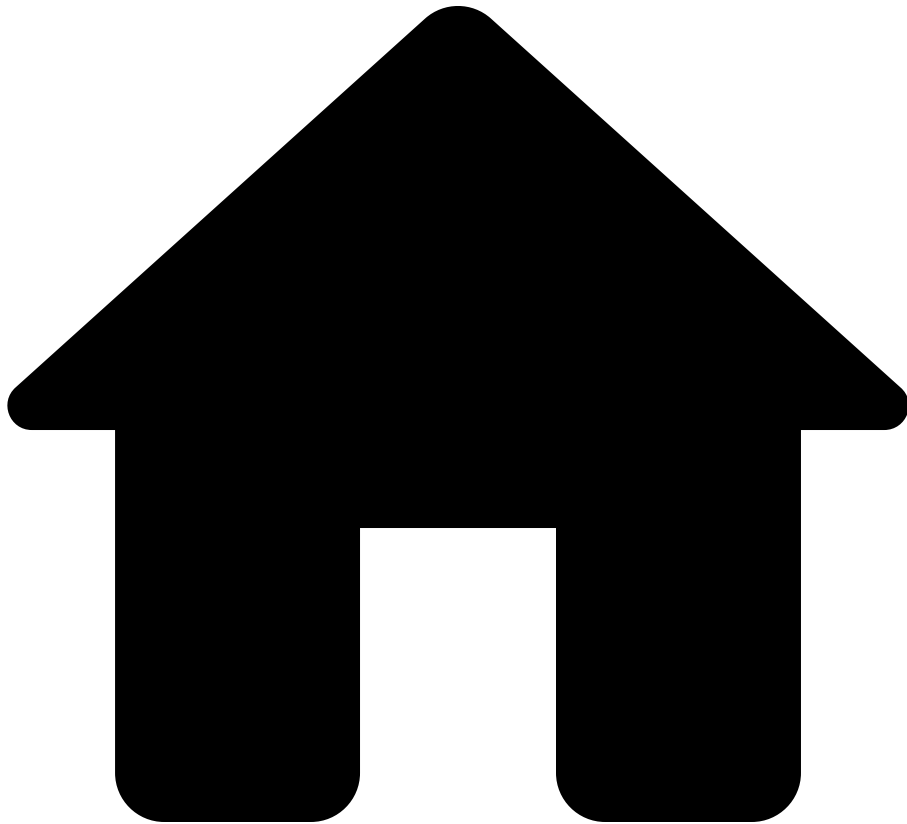
**network** [â](#) **cloudAccounts** [â](#) **globalAccelerators**[â](#)

**See also**[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - globalAccelerators: Bag< [GlobalAccelerator](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GlobalAcceleratorListener

On this page

# GlobalAcceleratorListener

## Fields

| name           | type                                        | nullable |
|----------------|---------------------------------------------|----------|
| id             | String                                      | false    |
| protocol       | String                                      | false    |
| ports          | Bag<String>                                 | false    |
| endpointGroups | Bag<GlobalAcceleratorListenerEndpointGroup> | false    |

## Used by

network cloudAccounts globalAccelerators listeners

See also

Data model path

- network: Network
  - cloudAccounts: Bag<CloudAccount>
    - globalAccelerators: Bag<GlobalAccelerator>
      - listeners: Bag<GlobalAcceleratorListener>

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GlobalAcceleratorListenerEndpoint

On this page

# GlobalAcceleratorListenerEndpoint

## Options

| name           | type      | description |
|----------------|-----------|-------------|
| instanceId     | String    | --          |
| ipAddress      | IpAddress | --          |
| loadBalancerId | String    | --          |

## Used by

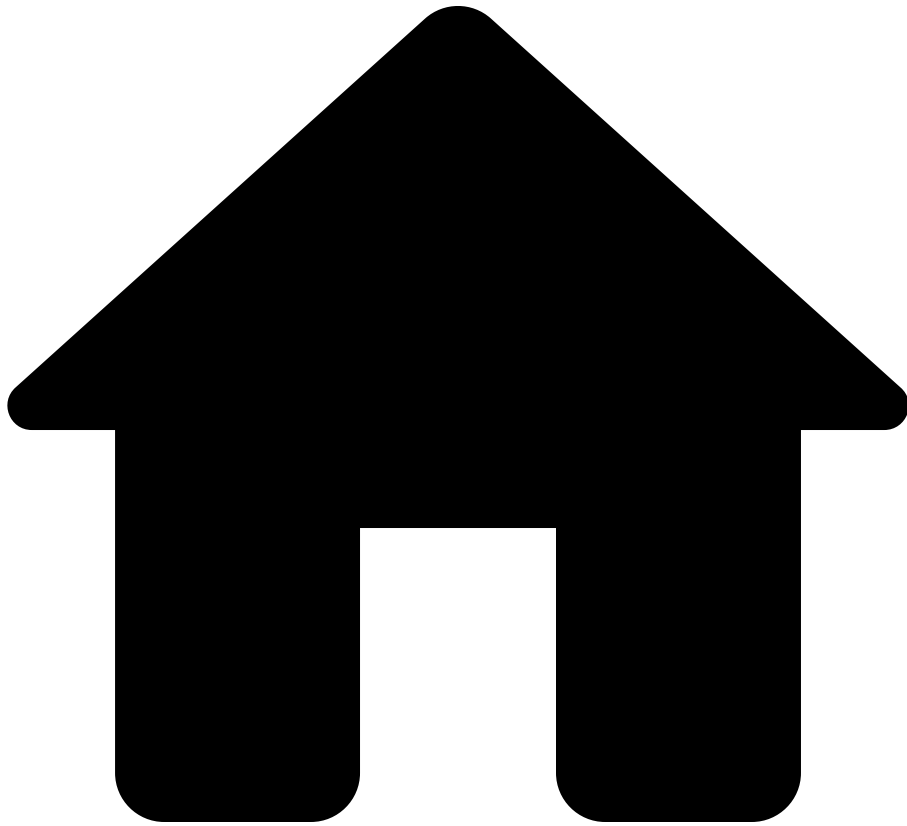
network [cloudAccounts](#) [globalAccelerators](#) [listeners](#) [endpointGroups](#) [endpoints](#)

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - globalAccelerators: Bag< [GlobalAccelerator](#) >
      - listeners: Bag< [GlobalAcceleratorListener](#) >
        - endpointGroups: Bag< [GlobalAcceleratorListenerEndpointGroup](#) >
          - endpoints: Bag< [GlobalAcceleratorListenerEndpoint](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GlobalAcceleratorListenerEndpointGroup

On this page



# GlobalAcceleratorListenerEndpointGroup

## Fields

| name          | type                                   | nullable | description |
|---------------|----------------------------------------|----------|-------------|
| name          | String                                 | true     | --          |
| id            | String                                 | false    | --          |
| tags          | Bag<String>                            | false    | --          |
| region        | String                                 | false    | --          |
| portOverrides | Bag<String>                            | false    | --          |
| isHealthy     | Bool                                   | false    | --          |
| endpoints     | Bag<GlobalAcceleratorListenerEndpoint> | false    | --          |

## Used by

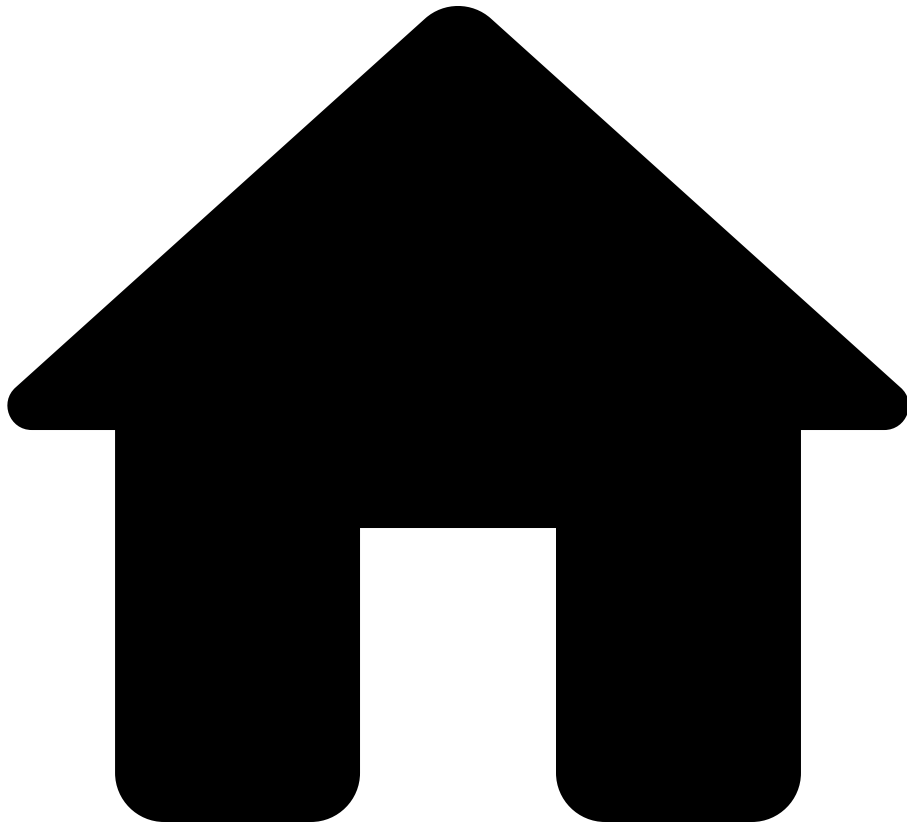
network cloudAccounts globalAccelerators listeners endpointGroups

See also

Data model path

- network: Network
  - cloudAccounts: Bag<CloudAccount>
    - globalAccelerators: Bag<GlobalAccelerator>
      - listeners: Bag<GlobalAcceleratorListener>
        - endpointGroups: Bag<GlobalAcceleratorListenerEndpointGroup>

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- GlobalAcceleratorStatus

On this page

# GlobalAcceleratorStatus

## Options

| name        | description |
|-------------|-------------|
| DEPLOYED    | --          |
| IN_PROGRESS | --          |

## Used by

network cloudAccounts globalAccelerators status

## See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - globalAccelerators: Bag< GlobalAccelerator >
      - status: GlobalAcceleratorStatus

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Ha

On this page

# Ha

## Fields

| name      | type          | nullable | description                                             |
|-----------|---------------|----------|---------------------------------------------------------|
| mlagPeer  | String        | true     | The MLAG peer device, or null if there is no MLAG peer. |
| vpc       | VpcDomain     | false    | --                                                      |
| clusterHa | ClusterHaInfo | true     | --                                                      |

## Used by

network devices ha

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - ha: Ha

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- HaOperationMode

On this page

# HaOperationMode

## Options

| name                | description                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACTIVE              | ACTIVE - failover is configured, and running status of the device is active. This state includes device that is configured to be secondary, but is active because the primary device is down. |
| ACTIVEBACKUP        | ACTIVEBACKUP - failover is configured, and running status of the device is active backup (aka active standby). Device in this state is in backup state, but is active in packet forwarding.   |
| BACKUP              | BACKUP - failover is configured, and running status of the device is standby. Device in this state does not forward packet                                                                    |
| STANDALONE          | STANDALONE - no failover is configured on this device                                                                                                                                         |
| STANDALONE_INACTIVE | E.g. the device is booting up or being initialized.                                                                                                                                           |

## Used by

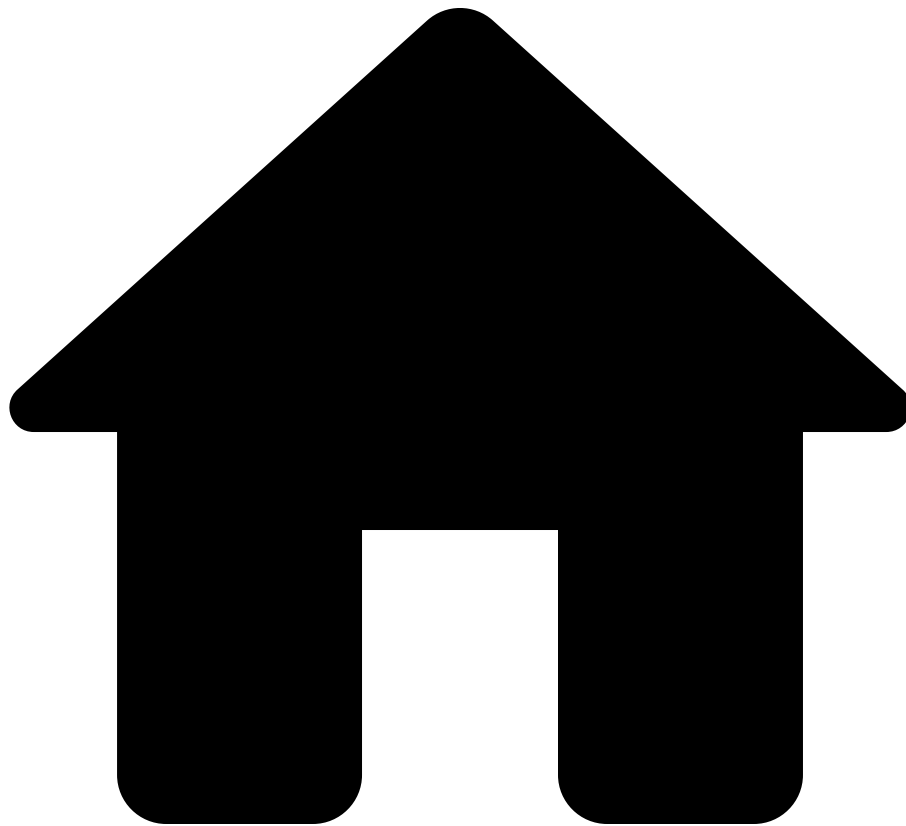
network devices ha clusterHa operationMode

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - ha: Ha
      - clusterHa: ClusterHaInfo
        - operationMode: HaOperationMode

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- HeaderRegion

On this page



# HeaderRegion

## Fields

| name             | type                            | nullable | description                                                                                                                                                        |
|------------------|---------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ipv4Src          | Bag< IpSubnet >                 | false    | --                                                                                                                                                                 |
| ipv4Dst          | Bag< IpSubnet >                 | false    | --                                                                                                                                                                 |
| ipv6Src          | Bag< IpSubnet >                 | false    | --                                                                                                                                                                 |
| ipv6Dst          | Bag< IpSubnet >                 | false    | --                                                                                                                                                                 |
| ipProtocol       | Bag< <a href="#">IntRange</a> > | false    | --                                                                                                                                                                 |
| tpSrc            | Bag< <a href="#">IntRange</a> > | false    | Source transport ports                                                                                                                                             |
| tpDst            | Bag< <a href="#">IntRange</a> > | false    | Destination transport ports                                                                                                                                        |
| icmpType         | Bag< <a href="#">IntRange</a> > | false    | --                                                                                                                                                                 |
| icmpCode         | Bag< <a href="#">IntRange</a> > | false    | --                                                                                                                                                                 |
| overApproximated | Bool                            | false    | Whether any header field is overapproximated. In this case, the value for each overapproximated header field will be the set of all possible values for the field. |
| domains          | Bag< String >                   | false    | L7 domain matches. Example members include "google.com" and ".aws.com". There are no restrictions when empty. Always                                               |

| name | type | nullable | description                          |
|------|------|----------|--------------------------------------|
|      |      |          | empty when the CloudType is not AWS. |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **aclEntries** [â](#) **headerMatches**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - headerMatches: [HeaderRegion](#)

**network** [â](#) **devices** [â](#) **natEntries** [â](#) **headerMatches**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - headerMatches: [HeaderRegion](#)

**network** **cloudAccounts** **vpcs** **securityGroups** **ingressRules**  
**match**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)

**network** **cloudAccounts** **vpcs** **securityGroups** **egressRules**  
**match**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)

**network** **cloudAccounts** **vpcs** **networkAcls** **ingressRules**  
**match**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)

**network** **cloudAccounts** **vpcs** **networkAcls** **egressRules**  
**match**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **match**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - match: [HeaderRegion](#)

**network** **cloudAccounts** **firewallRules** **securityRule** **match**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >
      - securityRule: [SecurityRule](#)
        - match: [HeaderRegion](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- HeaderRewrite

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# HeaderRewrite

## Fields

| name    | type                            | nullable | description                                                                                                                   |
|---------|---------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------|
| ipv4Src | Bag< IpSubnet >                 | false    | The possible subnets to which the src address can be rewritten. An empty list indicates that the field is not rewritten.      |
| ipv4Dst | Bag< IpSubnet >                 | false    | The possible subnets to which the dst address can be rewritten. An empty list indicates that the field is not rewritten.      |
| ipv6Src | Bag< IpSubnet >                 | false    | The possible IPv6 subnets to which the src address can be rewritten. An empty list indicates that the field is not rewritten. |
| ipv6Dst | Bag< IpSubnet >                 | false    | The possible IPv6 subnets to which the dst address can be rewritten. An empty list indicates that the field is not rewritten. |
| tpSrc   | Bag< <a href="#">IntRange</a> > | false    | The possible ports to which the src port field can be rewritten. An empty list                                                |

| name             | type                            | nullable | description                                                                                                                                                                          |
|------------------|---------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                                 |          | indicates that the field is not rewritten.                                                                                                                                           |
| tpDst            | Bag< <a href="#">IntRange</a> > | false    | The possible ports to which the dst port field can be rewritten. An empty list indicates that the field is not rewritten.                                                            |
| overApproximated | Bool                            | false    | Whether any header field rewrite is overapproximated. In this case, the value for each overapproximated rewritten header field will be the set of all possible values for the field. |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **natEntries** [â](#) **rewrites**[â](#)

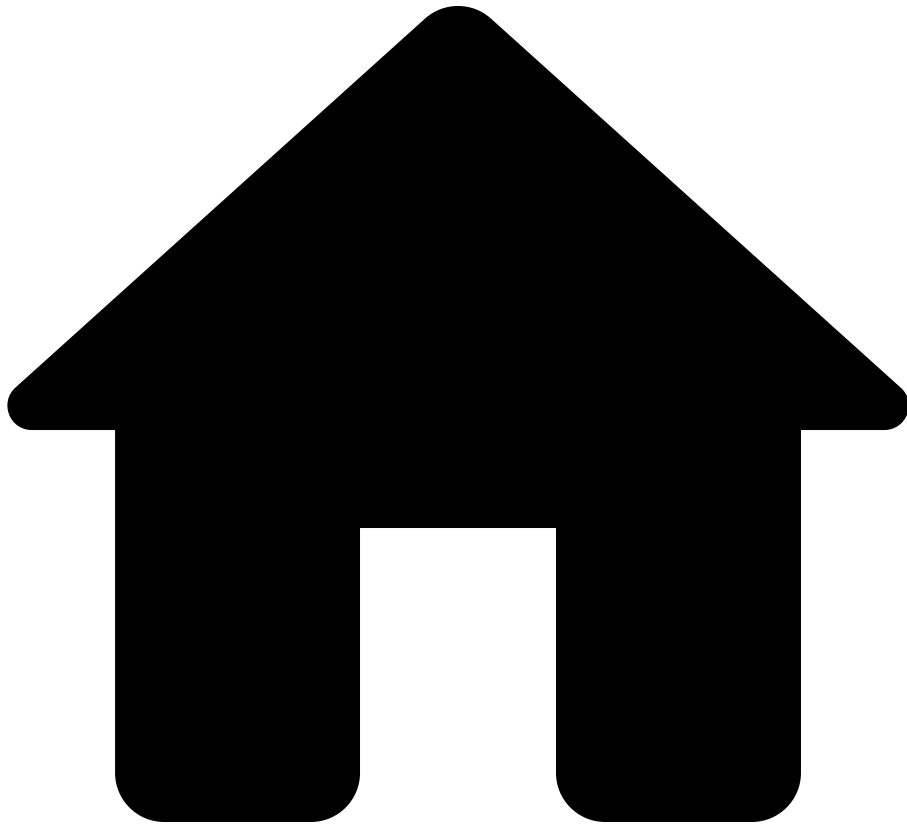
See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - rewrites: Bag< [HeaderRewrite](#) >



.



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- Hop

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## Hop

Represents a hop from a device to a neighboring device. If the hop is the last hop in a path, egressIface, nextDevice, and nextIface are set to null.

## Fields

| name             | type    | nullable | description |
|------------------|---------|----------|-------------|
| index            | Integer | false    | --          |
| deviceName       | String  | false    | --          |
| ingressIfaceName | String  | false    | --          |
| egressIfaceName  | String  | true     | --          |
| nextDeviceName   | String  | true     | --          |
| nextIfaceName    | String  | true     | --          |

## Used by

path hops

See also

Data model path

- path: Path
  - hops: Bag< Hop >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- HttpApiEndpointResult

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# HttpApiEndpointResult

## Fields

| name   | type                                  | nullable | description                                                 |
|--------|---------------------------------------|----------|-------------------------------------------------------------|
| name   | String                                | false    | --                                                          |
| status | <a href="#">HttpApiEndpointStatus</a> | false    | --                                                          |
| error  | String                                | true     | Absent if status is OK.                                     |
| pages  | Bag< <a href="#">HttpPage</a> >       | false    | Contains exactly one page if the endpoint is not paginated. |

## Used by

network endpoints httpResults

See also

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - httpResults: Bag< [HttpApiEndpointResult](#) >

network dataConnectors httpResults

See also

Data model path

- network: [Network](#)
  - dataConnectors: Bag< [DataConnector](#) >
    - httpResults: Bag< [HttpApiEndpointResult](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- HttpApiEndpointStatus

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# HttpApiEndpointStatus

## Options

| name              | description                                                                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ERROR             | The collection was terminated due to an error retrieving a page.                                                                                                                                   |
| MAX_PAGES_REACHED | The collection was terminated after the number of collected pages reached the max number of pages (defined for the endpoint) before the stopping condition (defined for the endpoint) was reached. |
| OK                | All pages of the endpoint were collected successfully.                                                                                                                                             |
| TIMED_OUT         | The collection was terminated after the maximum time allowed for collection was exceeded.                                                                                                          |

## Used by

**network** endpoints **httpResults** status

See also

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - httpResults: Bag< [HttpApiEndpointResult](#) >
      - status: [HttpApiEndpointStatus](#)

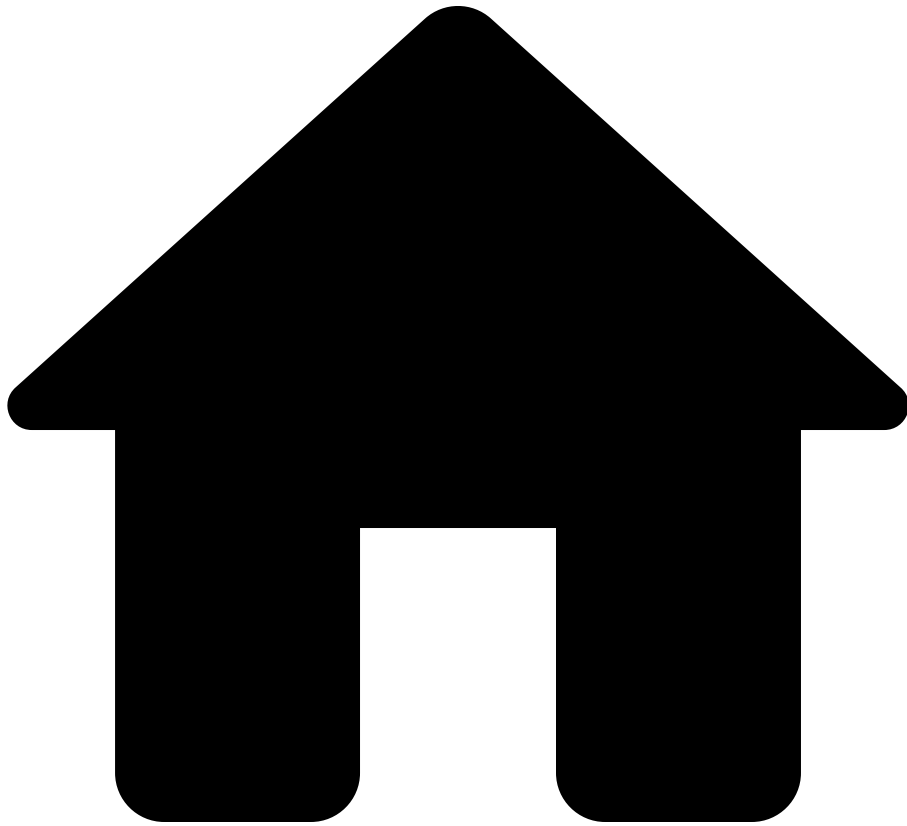
**network** **dataConnectors** **httpResults** **status**

See also

Data model path

- network: [Network](#)
  - dataConnectors: Bag< [DataConnector](#) >
    - httpResults: Bag< [HttpApiEndpointResult](#) >
      - status: [HttpApiEndpointStatus](#)

.



- Reference
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- [Data Model](#)
- HttpEndpoint

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# HttpEndpoint

## Fields

| name         | type    | nullable | description                                     |
|--------------|---------|----------|-------------------------------------------------|
| name         | String  | false    | --                                              |
| statusCode   | Integer | false    | --                                              |
| reasonPhrase | String  | false    | --                                              |
| responseBody | String  | true     | Absent if the response is not textual (binary). |

## Used by

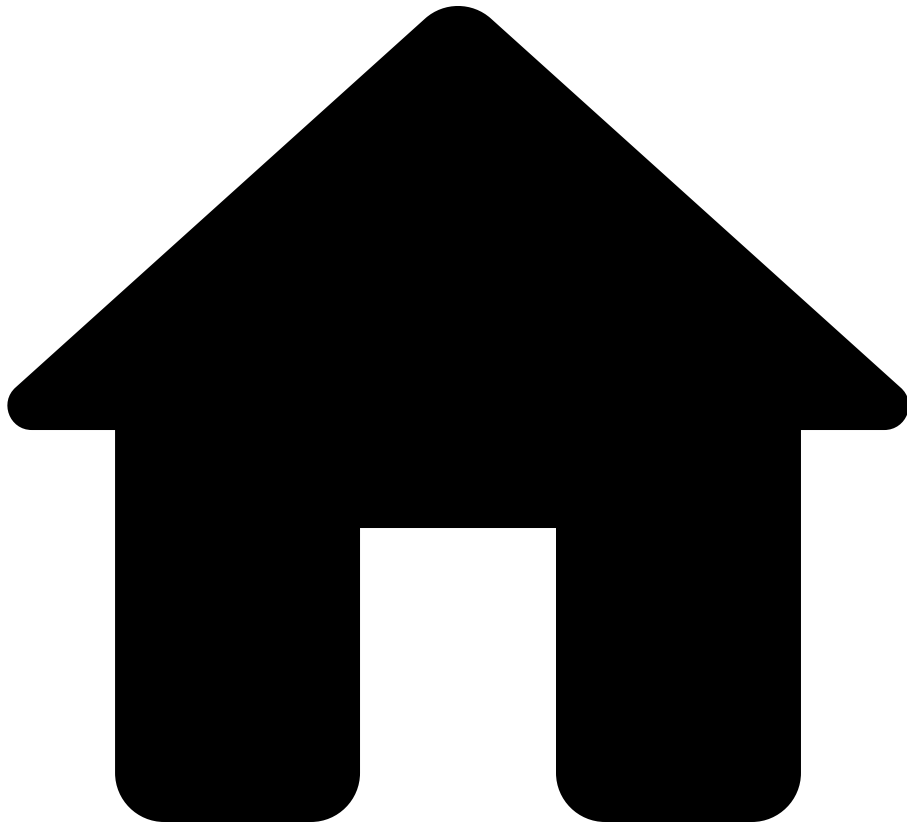
network externalSources httpSources endpoints

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - httpSources: Bag< [HttpSource](#) >
      - endpoints: Bag< [HttpEndpoint](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- HttpPage

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# HttpPage

## Fields

| name         | type    | nullable | description                                     |
|--------------|---------|----------|-------------------------------------------------|
| statusCode   | Integer | false    | --                                              |
| reasonPhrase | String  | false    | --                                              |
| responseBody | String  | true     | Absent if the response is not textual (binary). |

## Used by

**network** **endpoints** **httpResults** **pages**

See also

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - httpResults: Bag< [HttpApiEndpointResult](#) >
      - pages: Bag< [HttpPage](#) >

**network** **externalSources** **httpSources** **paginatedEndpoints** **pages**

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - httpSources: Bag< [HttpSource](#) >
      - paginatedEndpoints: Bag< [PaginatedHttpEndpoint](#) >
        - pages: Bag< [HttpPage](#) >

**network** [dataConnectors](#) [httpResults](#) [pages](#)

See also

Data model path

- network: [Network](#)
  - dataConnectors: Bag< [DataConnector](#) >
    - httpResults: Bag< [HttpApiEndpointResult](#) >
      - pages: Bag< [HttpPage](#) >

.



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- HttpSource

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# HttpSource

## Fields

| name               | type                                         | nullable | description |
|--------------------|----------------------------------------------|----------|-------------|
| name               | String                                       | false    | --          |
| endpoints          | Bag< <a href="#">HttpEndpoint</a> >          | false    | --          |
| paginatedEndpoints | Bag< <a href="#">PaginatedHttpEndpoint</a> > | false    | --          |

## Used by

network externalSources httpSources

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - httpSources: Bag< [HttpSource](#) >

.



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- Iface

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# Iface

## Fields

| name          | type                      | nullable | description                                                                                                                                                                                                                                                        |
|---------------|---------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name          | String                    | false    | The name of the interface.                                                                                                                                                                                                                                         |
| interfaceType | <a href="#">IfaceType</a> | false    | The type of the interface.                                                                                                                                                                                                                                         |
| layer         | <a href="#">Layer</a>     | false    | Denotes whether this is a Layer 2 (physical) or Layer 3 (logical) interface.                                                                                                                                                                                       |
| mtu           | Integer                   | true     | The max transmission unit size in octets for the physical interface. Note that this is the L2 MTU, which is normalized to include the length of the L2 payload, but not the L2 header and trailers (such as the MAC source and destination, Ethernet type, or VLAN |



| name         | type | nullable | description                                                                                                                                                                                                                                                                                                                                             |
|--------------|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |      |          | headers). Some vendors include the length of the L2 header when reporting the L2 MTU, and as a result the MTU value reported by the device will differ from the value reported here. For example, Cisco IOS-XR includes L2 headers. An interface that reports a 1514 byte MTU on IOS-XR is normalized to 1500 after subtracting 14 bytes of L2 headers. |
| loopbackMode | Bool | false    | When set to true, the interface is logically looped back, such that packets that are forwarded via the interface are received on                                                                                                                                                                                                                        |

| name        | type                        | nullable | description                                                                                                                                                                                              |
|-------------|-----------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |                             |          | the same interface.                                                                                                                                                                                      |
| aliases     | Bag< String >               | false    | The set of all names that refer to this interface. For example, on Cisco, an interface named "eth1" might also be referred to as "Ethernet1". This field would then include both "eth1" and "Ethernet1". |
| description | String                      | true     | A textual description of the interface.                                                                                                                                                                  |
| adminStatus | <a href="#">AdminStatus</a> | false    | The desired state of the interface. Here, it reflects the administrative state as set by enabling or disabling the interface.                                                                            |
| operStatus  | <a href="#">OperStatus</a>  | false    | The current operational state of the interface.                                                                                                                                                          |

| name          | type                                | nullable | description                                                                          |
|---------------|-------------------------------------|----------|--------------------------------------------------------------------------------------|
| subinterfaces | Bag< <a href="#">SubInterface</a> > | false    | The list of subinterfaces (logical interfaces) associated with a physical interface. |
| ethernet      | <a href="#">Ethernet</a>            | false    | Ethernet-specific configuration and state of the interface.                          |
| acls          | <a href="#">IfaceAcls</a>           | false    | Access control list details specific to this interface.                              |
| routedVlan    | <a href="#">RoutedVlan</a>          | true     | Non-null when the interface is a logical interface providing L3 routing for VLANs.   |
| bridge        | <a href="#">Bridge</a>              | true     | Non-null when the interface is a bridge (BVI, BDI, etc.)                             |
| aggregation   | <a href="#">Aggregation</a>         | true     | Non-null when the interface is set to type IF_AGGREGATE.                             |
| tunnel        | <a href="#">Tunnel</a>              | true     | In the case that the interface is                                                    |

| name     | type                                     | nullable | description                                                                                                                                                |
|----------|------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                          |          | logical tunnel interface, the parameters for the tunnel are specified within this subtree.                                                                 |
| teTunnel | <a href="#">TeTunnel</a>                 | true     | In the case that the interface is a logical Traffic Engineering (TE) tunnel interface, the parameters for the TE tunnel are specified within this subtree. |
| links    | Bag< <a href="#">PhysicalLink</a> >      | false    | The other interfaces that this interface connects to.                                                                                                      |
| cdp      | <a href="#">DeviceDiscoveryIfaceInfo</a> | false    | --                                                                                                                                                         |
| lldp     | <a href="#">DeviceDiscoveryIfaceInfo</a> | false    | --                                                                                                                                                         |

## Used by[↗](#)

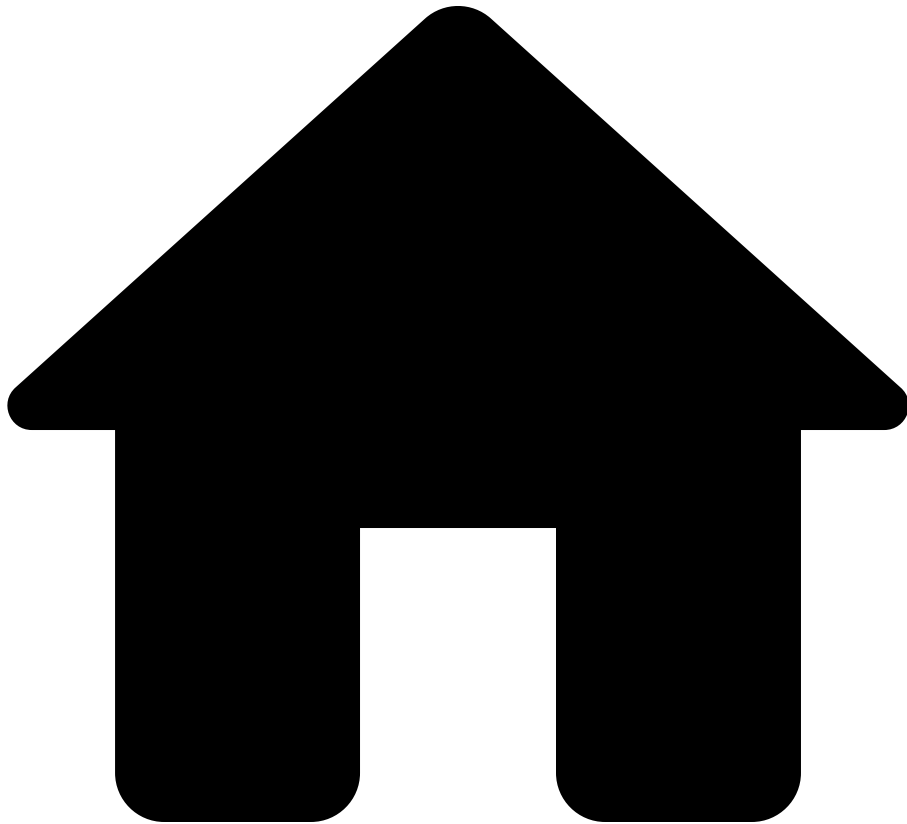
**network** [↗](#) **devices** [↗](#) **interfaces**[↗](#)

**See also**[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >

.



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- [NQE Language](#)
- [Data Model](#)
- IfaceAcls

On this page

## IfaceAcls

Access control list details for an interface.

## Fields

| name                          | type           | nullable | description                                                                                                                                     |
|-------------------------------|----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>inboundAclNames</code>  | List< String > | false    | Names of the ACL entries that secure inbound traffic to this interface. The ACLs are applied in the order in which they appear in this list.    |
| <code>outboundAclNames</code> | List< String > | false    | Names of the ACL entries that secure outbound traffic from this interface. The ACLs are applied in the order in which they appear in this list. |

## Used by

network devices interfaces subinterfaces acls

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - subinterfaces: Bag< SubInterface >
        - acls: IfaceAcls

**network** **devices** **interfaces** **acls**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - acls: [IfaceAcls](#)



.



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- IfaceFhrp

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# IfaceFhrp

## Fields

| name | type          | nullable | description |
|------|---------------|----------|-------------|
| hsrp | IfaceFhrpInfo | false    | --          |
| vrrp | IfaceFhrpInfo | false    | --          |

## Used by

network devices interfaces subinterfaces ipv4 fhrp

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - subinterfaces: Bag< SubInterface >
        - ipv4: IfaceIpv4Info
        - fhrp: IfaceFhrp

network devices interfaces subinterfaces ipv6 fhrp

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - subinterfaces: Bag< SubInterface >
        - ipv6: IfaceIpv6Info
        - fhrp: IfaceFhrp

**network** **devices** **interfaces** **routedVlan** **ipv4** **fhrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)

**network** **devices** **interfaces** **routedVlan** **ipv6** **fhrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)

**network** **devices** **interfaces** **bridge** **ipv4** **fhrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)

**network** **devices** **interfaces** **bridge** **ipv6** **fhrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)

**network** **devices** **interfaces** **tunnel** **ipv4** **fhrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)

**network** **devices** **interfaces** **tunnel** **ipv6** **fhrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)

.



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- IfaceFhrpGroup

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# IfaceFhrpGroup

## Fields

| name            | type                      | nullable | description                                              |
|-----------------|---------------------------|----------|----------------------------------------------------------|
| virtualRouterId | Integer                   | false    | The virtual router id for use by the VRRP or HSRP group. |
| state           | <a href="#">FhrpState</a> | false    | --                                                       |
| virtualAddress  | IpAddress                 | false    | --                                                       |

## Used by

network devices interfaces subinterfaces ipv4 fhrp hsrp fhrpGroups

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **subinterfaces** **ipv4** **fhrp** **vrrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **subinterfaces** **ipv6** **fhrp** **hsrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **subinterfaces** **ipv6** **fhrp** **vrrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
        - fhrp: [IfaceFhrp](#)
          - vrrp: [IfaceFhrpInfo](#)
            - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **routedVlan** **ipv4** **fhrp** **hsrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
      - ipv4: [IfaceIpv4Info](#)
      - fhrp: [IfaceFhrp](#)
        - hsrp: [IfaceFhrpInfo](#)
          - fhrpGroups: Bag< [IfaceFhrpGroup](#) >



**network** **devices** **interfaces** **routedVlan** **ipv4** **fhrp** **vrrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **routedVlan** **ipv6** **fhrp** **hsrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **routedVlan** **ipv6** **fhrp** **vrrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **bridge** **ipv4** **fhrp** **hsrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **bridge** **ipv4** **fhrp** **vrrp**  
**fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **bridge** **ipv6** **fhrp** **hsrp**  
**fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **bridge** **ipv6** **fhrp** **vrrp**  
**fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **tunnel** **ipv4** **fhrp** **hsrp**  
**fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **tunnel** **ipv4** **fhrp** **vrrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **tunnel** **ipv6** **fhrp** **hsrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

**network** **devices** **interfaces** **tunnel** **ipv6** **fhrp** **vrrp** **fhrpGroups**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)
              - fhrpGroups: Bag< [IfaceFhrpGroup](#) >

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- [Data Model](#)
- IfaceFhrpInfo

On this page

# IfaceFhrpInfo

## Fields

| name       | type                                  | nullable | description |
|------------|---------------------------------------|----------|-------------|
| fhrpGroups | Bag< <a href="#">IfaceFhrpGroup</a> > | false    | --          |

## Used by

network devices interfaces subinterfaces ipv4 fhrp hsrp

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)



**network** **devices** **interfaces** **subinterfaces** **ipv4** **fhrp** **vrrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **subinterfaces** **ipv6** **fhrp** **hsrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **subinterfaces** **ipv6** **fhrp** **vrrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **routedVlan** **ipv4** **fhrp** **hsrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **routedVlan** **ipv4** **fhrp** **vrrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **routedVlan** **ipv6** **fhrp** **hsrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **routedVlan** **ipv6** **fhrp** **vrrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
      - ipv6: [IfaceIpv6Info](#)
      - fhrp: [IfaceFhrp](#)
        - vrrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **bridge** **ipv4** **fhrp** **hsrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
      - ipv4: [IfaceIpv4Info](#)
      - fhrp: [IfaceFhrp](#)
        - hsrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **bridge** **ipv4** **fhrp** **vrrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **bridge** **ipv6** **fhrp** **hsrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **bridge** **ipv6** **fhrp** **vrrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **tunnel** **ipv4** **fhrp** **hsrp**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **tunnel** **ipv4** **fhrp** **vrrp**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **tunnel** **ipv6** **fhrp** **hsrp**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - hsrp: [IfaceFhrpInfo](#)

**network** **devices** **interfaces** **tunnel** **ipv6** **fhrp** **vrrp**

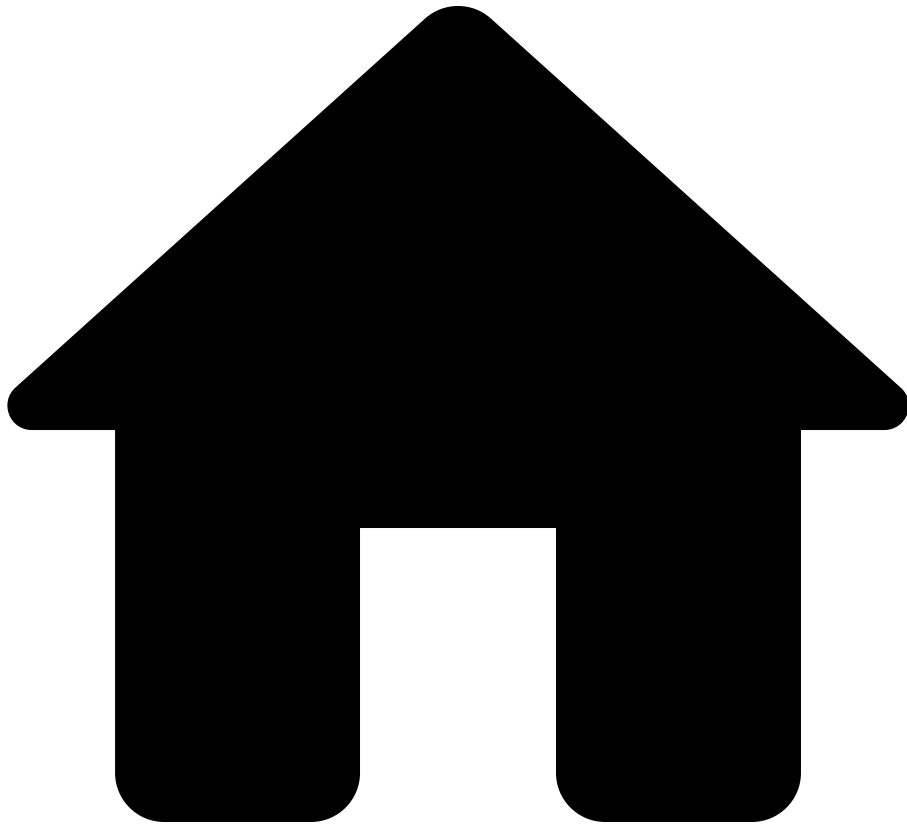
See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrp: [IfaceFhrp](#)
            - vrrp: [IfaceFhrpInfo](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Ifacelpv4Info

On this page

## Ifacelpv4Info

The IPv4 addresses configured on an interface.

## Fields

| name          | type                               | nullable | description                                                                                                                |
|---------------|------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------|
| addresses     | Bag< <a href="#">Ipv4Address</a> > | false    | --                                                                                                                         |
| fhrpAddresses | Bag< <a href="#">Ipv4Address</a> > | false    | --                                                                                                                         |
| neighbors     | Bag< <a href="#">ArpEntry</a> >    | false    | IPv4 address and link-layer address of neighbors discovered on this interface.                                             |
| fhrp          | <a href="#">IfaceFhrp</a>          | false    | --                                                                                                                         |
| multicastMode | <a href="#">MulticastMode</a>      | true     | The Protocol-Independent Multicast (PIM) mode enabled on this interface for IPv4 multicast traffic. if PIM is not enabled. |

## Used by

network devices interfaces subinterfaces ipv4

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv4: [IfaceIpv4Info](#)

**network** **devices** **interfaces** **routedVlan** **ipv4**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)

**network** **devices** **interfaces** **bridge** **ipv4**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)

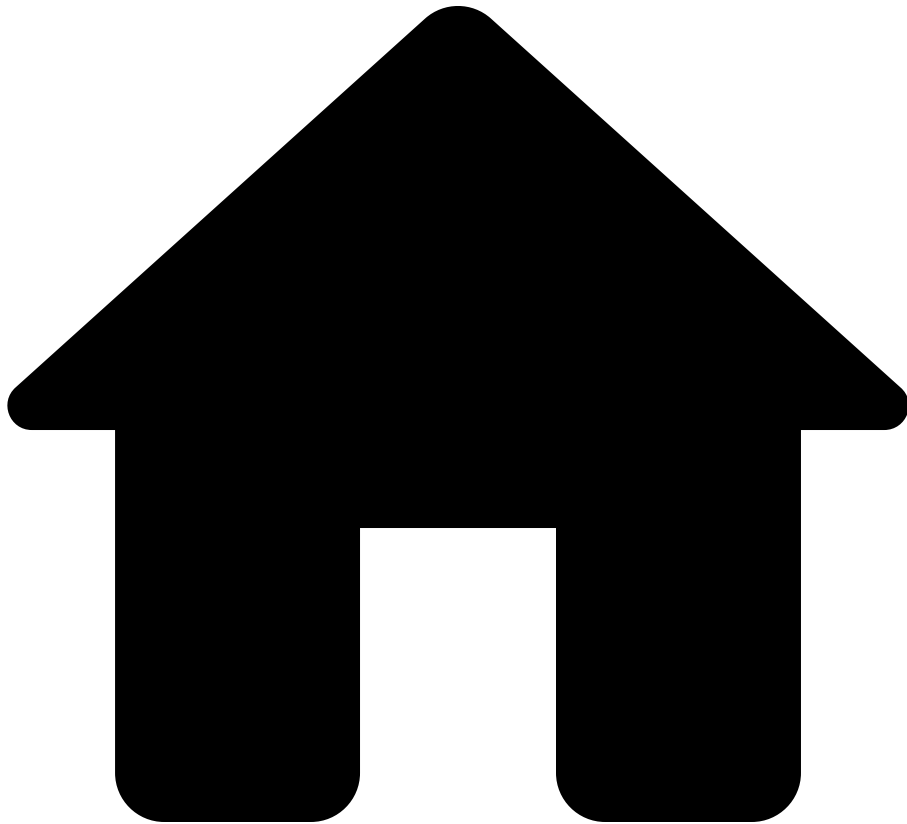
**network** **devices** **interfaces** **tunnel** **ipv4**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Ifacelpv6Info

On this page

## Ifacelpv6Info

The IPv6 addresses configured on an interface.

## Fields

| name          | type                               | nullable | description                                                                                                                       |
|---------------|------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| addresses     | Bag< <a href="#">Ipv6Address</a> > | false    | --                                                                                                                                |
| fhrpAddresses | Bag< <a href="#">Ipv6Address</a> > | false    | --                                                                                                                                |
| neighbors     | Bag< <a href="#">ArpEntry</a> >    | false    | IPv6 address and link-layer address of neighbors discovered on this interface.                                                    |
| fhrp          | <a href="#">IfaceFhrp</a>          | false    | --                                                                                                                                |
| multicastMode | <a href="#">MulticastMode</a>      | true     | The Protocol-Independent Multicast (PIM) mode enabled on this interface for IPv6 multicast traffic. Absent if PIM is not enabled. |

## Used by

network devices interfaces subinterfaces ipv6

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)

**network** **devices** **interfaces** **routedVlan** **ipv6**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)

**network** **devices** **interfaces** **bridge** **ipv6**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)

**network** **devices** **interfaces** **tunnel** **ipv6**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- IfaceSubIface

On this page

## IfaceSubIface

The names of the interface and subinterface that are associated with the network instance.

## Fields

| name         | type   | nullable | description |
|--------------|--------|----------|-------------|
| ifaceName    | String | false    | --          |
| subIfaceName | String | true     | --          |

## Used by

**network devices networkInstances interfaces**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - interfaces: Bag< [IfaceSubIface](#) >

**network devices securityZones interfaces**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - securityZones: Bag< [DeviceSecurityZone](#) >
      - interfaces: Bag< [IfaceSubIface](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- IfaceType

On this page

# IfaceType

## Options

| name               | description                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IF_AGGREGATE       | An aggregated, or bonded, interface forming a Link Aggregation Group (LAG), or bundle, most often based on the IEEE 802.1AX (or 802.3ad) standard.                               |
| IF_BRIDGE          | A bridge interface                                                                                                                                                               |
| IF_ETHERNET        | Ethernet interfaces based on IEEE 802.3 standards, as well as FlexEthernet.                                                                                                      |
| IF_LOOPBACK        | A virtual interface designated as a loopback used for various management and operations tasks.                                                                                   |
| IF_ROUTED_VLAN     | A logical interface used for routing services on a VLAN. Such interfaces are also known as switch virtual interfaces (SVI) or integrated routing and bridging interfaces (IRBs). |
| IF_SONET           | SONET/SDH interface.                                                                                                                                                             |
| IF_TE_TUNNEL       | A TE tunnel interface.                                                                                                                                                           |
| IF_TUNNEL_GRE4     | A GRE tunnel over IPv4 transport.                                                                                                                                                |
| IF_TUNNEL_GRE6     | A GRE tunnel over IPv6 transport.                                                                                                                                                |
| IF_TUNNEL_IPSEC    | An IPSEC tunnel.                                                                                                                                                                 |
| IF_TUNNEL_IP_IN_IP | An IP-in-IP tunnel.                                                                                                                                                              |
| IF_TUNNEL_L2_VPN   | An L2 VPN tunnel.                                                                                                                                                                |
| IF_TUNNEL_UNKNOWN  | An unknown tunnel interface.                                                                                                                                                     |
| IF_TUNNEL_VXLAN    | A VXLAN VTEP.                                                                                                                                                                    |

## Used by[↗](#)

**network** [↗](#) **devices** [↗](#) **interfaces** [↗](#) **interfaceType**[↗](#)

**See also**[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - interfaceType: [IfaceType](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- InetGateway

On this page

# InetGateway

## Fields

| name          | type          | nullable | description |
|---------------|---------------|----------|-------------|
| name          | String        | true     | --          |
| id            | String        | false    | --          |
| tags          | Bag< String > | false    | --          |
| region        | String        | true     | --          |
| routeTableIds | Bag< String > | false    | --          |

## Used by

network cloudAccounts vpcs inetGateways

## See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - vpcs: Bag< VpcData >
      - inetGateways: Bag< InetGateway >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Instancelface

On this page

# Instanceface

## Fields

| name     | type   | nullable | description      |
|----------|--------|----------|------------------|
| subnetId | String | false    | Ref to Subnet.id |
| ifaceId  | String | false    | Ref to Iface.id  |

## Used by

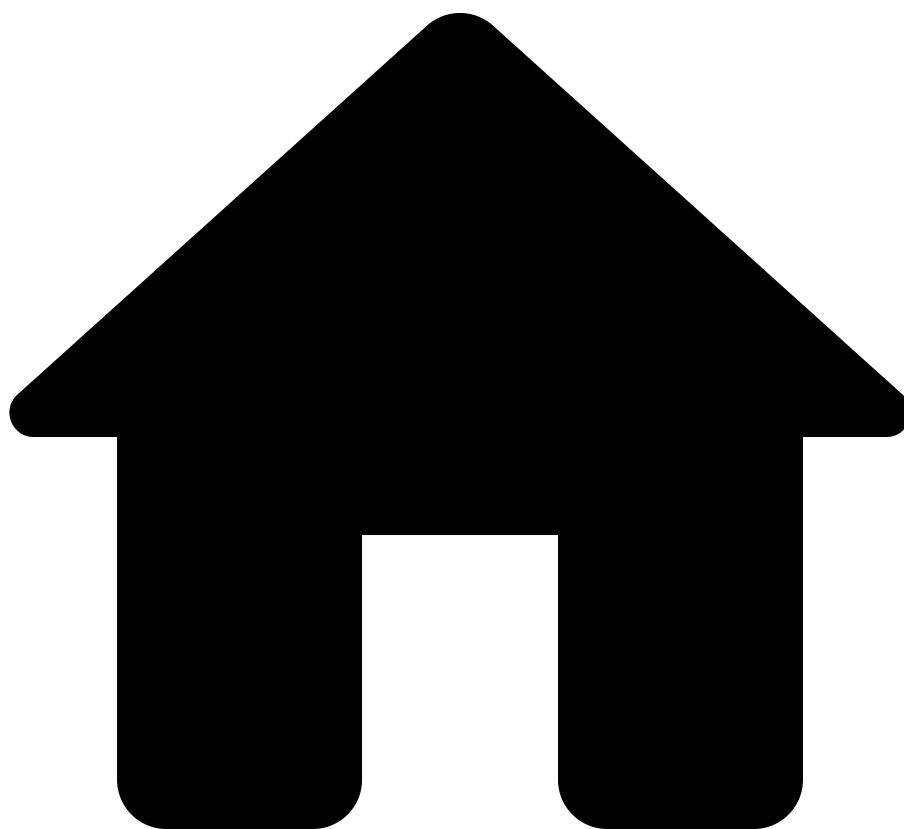
network cloudAccounts vpcs computeInstances instancefaces

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - vpcs: Bag< VpcData >
      - computeInstances: Bag< ComputeInstance >
        - instancefaces: Bag< InstanceIface >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- InterfaceConnectivityParameters

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# InterfaceConnectivityParameters

## Fields

| name                 | type | nullable | description                                                                                                                                                  |
|----------------------|------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| toExternalDevice     | Bool | true     | Whether the interface connects to the interface of a device that is in another network (a network in a different administrative domain).                     |
| toAccessLayerSwitch  | Bool | true     | Whether the interface is connected to an access layer switch.                                                                                                |
| toUntrustedDevice    | Bool | true     | Whether the interface connects to one or more untrusted devices, such as a user device or an untrusted gateway. Such interfaces should not be in trunk mode. |
| toCoreLayer          | Bool | true     | Whether the interface is connected to the core layer.                                                                                                        |
| viaFiberOptics       | Bool | true     | Whether the interface connects to its neighbor via fiber optics.                                                                                             |
| toCustomerEdgeDevice | Bool | true     | Whether the interface connects to a customer edge device (i.e. is CE-facing).                                                                                |
| toDodinBackbone      | Bool | true     | Whether the interface connects to a DODIN backbone device.                                                                                                   |

## Used by[â](#)

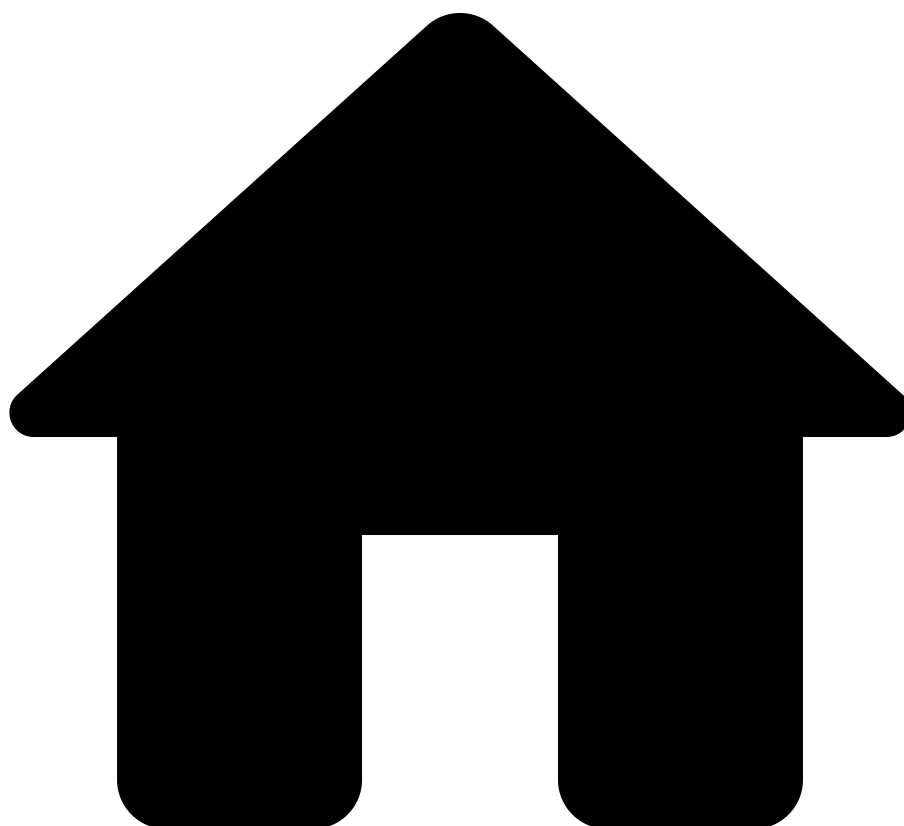
**network** [â](#) **stigDatabase** [â](#) **defaultParameters** [â](#) **ifaceParameters** [â](#) **connectivity**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - defaultParameters: [DefaultStigParameters](#)
      - ifaceParameters: [DefaultInterfaceParameters](#)
        - connectivity: [InterfaceConnectivityParameters](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- InterfaceConnectivityPolicy

On this page

# InterfaceConnectivityPolicy

## Fields

| name                 | type                                                               |
|----------------------|--------------------------------------------------------------------|
| toExternalDevice     | Bag< <a href="#">IsExternalInterfacePolicy</a> >                   |
| toAccessLayerSwitch  | Bag< <a href="#">ConnectsToAccessLayerSwitchInterfacePolicy</a> >  |
| toUntrustedDevice    | Bag< <a href="#">IsUntrustedInterfacePolicy</a> >                  |
| toCoreLayer          | Bag< <a href="#">ConnectsToCoreLayerInterfacePolicy</a> >          |
| viaFiberOptics       | Bag< <a href="#">ConnectsViaFiberOpticsInterfacePolicy</a> >       |
| toCustomerEdgeDevice | Bag< <a href="#">ConnectsToCustomerEdgeDeviceInterfacePolicy</a> > |
| toDodinBackbone      | Bag< <a href="#">ConnectsToDodinDeviceInterfacePolicy</a> >        |

## Used by

network [stigDatabase](#) [policy](#) [ifacePolicy](#) [connectivity](#)

### See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - connectivity: [InterfaceConnectivityPolicy](#)

.



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- [NQE Language](#)
- [Data Model](#)
- InterfacePolicy

On this page

# InterfacePolicy

## Fields

| name                       | type                                                      |
|----------------------------|-----------------------------------------------------------|
| requiresDefaultVlan        | Bag< <a href="#">RequiresDefaultVlanInterfacePolicy</a> > |
| requiresMulticast          | Bag< <a href="#">RequiresMulticastInterfacePolicy</a> >   |
| isHostGateway              | Bag< <a href="#">IsHostGatewayInterfacePolicy</a> >       |
| pseudowireVirtualCircuitId | Bag< <a href="#">VirtualCircuitIdInterfacePolicy</a> >    |
| connectivity               | <a href="#">InterfaceConnectivityPolicy</a>               |

## Used by

network [StigDatabase](#) [policy](#) [ifacePolicy](#)

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- InterfacesService

On this page

# InterfacesService

## Fields

| name           | type                                  | nullable | description |
|----------------|---------------------------------------|----------|-------------|
| configurations | Bag< InterfacesServiceConfiguration > | false    | --          |

## Used by

network devices outputs bluecat config items data  
services interfaces

## See also

Data model path

- network: Network
  - devices: Bag< Device >
    - outputs: Outputs
      - bluecat: BluecatOutputs
        - config: BluecatConfig
          - items: Bag< BluecatConfigItem >
            - data: BluecatConfigItemData
              - services: BluecatServices
                - interfaces: InterfacesService



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- InterfacesServiceConfiguration

On this page

# InterfacesServiceConfiguration

## Fields

| name                    | type                                                 | nullable |
|-------------------------|------------------------------------------------------|----------|
| interfacesConfiguration | <a href="#">InterfacesServiceConfigurationDetail</a> | false    |

## Used by

network devices outputs bluecat config items data  
services interfaces configurations

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - interfaces: [InterfacesService](#)
                  - configurations: Bag< [InterfacesServiceConfigurati](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- InterfacesServiceConfigurationDetail

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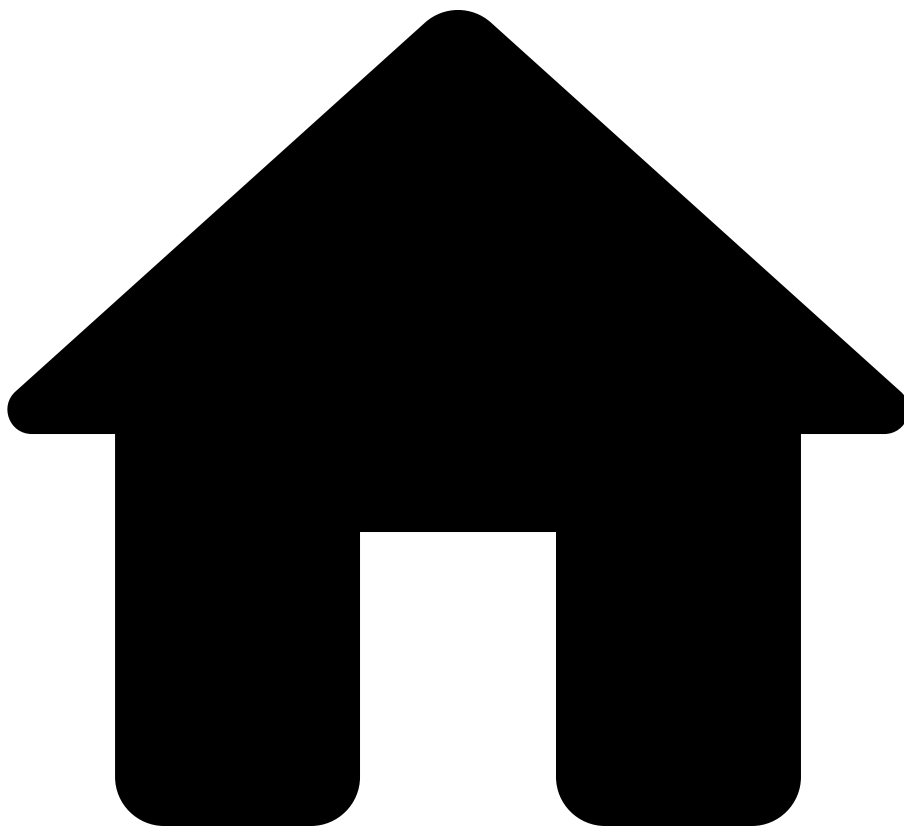
## Fields

Used by â

**See also**[ââ](#)

[illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- IntRange

On this page

# IntRange

## Fields

| name  | type    | nullable | description |
|-------|---------|----------|-------------|
| start | Integer | false    | --          |
| end   | Integer | false    | --          |

## Used by

**network** **devices** **aclEntries** **headerMatches** **ipProtocol**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - ipProtocol: Bag< [IntRange](#) >

**network** **devices** **aclEntries** **headerMatches** **tpSrc**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - tpSrc: Bag< [IntRange](#) >

**network** **devices** **aclEntries** **headerMatches** **tpDst**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - tpDst: Bag< [IntRange](#) >

**network** **devices** **aclEntries** **headerMatches** **icmpType**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - icmpType: Bag< [IntRange](#) >

**network** **devices** **aclEntries** **headerMatches** **icmpCode**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - icmpCode: Bag< [IntRange](#) >

**network** **devices** **natEntries** **headerMatches** **ipProtocol**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - ipProtocol: Bag< [IntRange](#) >

**network** **devices** **natEntries** **headerMatches** **tpSrc**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - tpSrc: Bag< [IntRange](#) >

**network** **devices** **natEntries** **headerMatches** **tpDst**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - tpDst: Bag< [IntRange](#) >



**network** **devices** **natEntries** **headerMatches** **icmpType**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - icmpType: Bag< [IntRange](#) >

**network** **devices** **natEntries** **headerMatches** **icmpCode**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - headerMatches: [HeaderRegion](#)
        - icmpCode: Bag< [IntRange](#) >

**network** **devices** **natEntries** **rewrites** **tpSrc**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - rewrites: Bag< [HeaderRewrite](#) >
        - tpSrc: Bag< [IntRange](#) >

**network** **devices** **natEntries** **rewrites** **tpDst**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - rewrites: Bag< [HeaderRewrite](#) >
        - tpDst: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **ingressRules** **match** **ipProtocol**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - ipProtocol: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **ingressRules**  
**match** **tpSrc**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - tpSrc: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **ingressRules**  
**match** **tpDst**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - tpDst: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **ingressRules**  
**match** **icmpType**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - icmpType: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **ingressRules**  
**match** **icmpCode**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - icmpCode: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **egressRules**  
**match** **ipProtocol**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - ipProtocol: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **egressRules**  
**match** **tpSrc**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - tpSrc: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **egressRules**  
**match** **tpDst**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - tpDst: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **egressRules**  
**match** **icmpType**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - icmpType: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **securityGroups** **egressRules**  
**match** **icmpCode**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - icmpCode: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **ingressRules**  
**match** **ipProtocol**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - ipProtocol: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **ingressRules**  
**match** **tpSrc**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - tpSrc: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **ingressRules**  
**match** **tpDst**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - tpDst: Bag< [IntRange](#) >



**network** **cloudAccounts** **vpcs** **networkAcls** **ingressRules**  
**match** **icmpType**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - icmpType: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **ingressRules**  
**match** **icmpCode**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - icmpCode: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **egressRules** **match** **ipProtocol**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - ipProtocol: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **egressRules** **match** **tpSrc**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - tpSrc: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **egressRules** **match** **tpDst**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - tpDst: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **egressRules** **match** **icmpType**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - icmpType: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **egressRules** **match** **icmpCode**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - match: [HeaderRegion](#)
            - icmpCode: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **loadBalancers** **loadBalancerRules** **frontendPorts**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - loadBalancers: Bag< [LoadBalancer](#) >
        - loadBalancerRules: Bag< [LoadBalancerRule](#) >
          - frontendPorts: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **definition**  
**sourcePorts**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - definition: [CloudFirewallRuleDefinition](#)
            - sourcePorts: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **definition**  
**destinationPorts**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - definition: [CloudFirewallRuleDefinition](#)
            - destinationPorts: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **match** **ipProtocol**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - match: [HeaderRegion](#)
            - ipProtocol: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **match** **tpSrc**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - match: [HeaderRegion](#)
            - tpSrc: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **match** **tpDst**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - match: [HeaderRegion](#)
            - tpDst: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **match** **icmpType**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - match: [HeaderRegion](#)
            - icmpType: Bag< [IntRange](#) >

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **match** **icmpCode**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - match: [HeaderRegion](#)
            - icmpCode: Bag< [IntRange](#) >

**network** **cloudAccounts** **externalLoadBalancers** **loadBalancerRules** **frontendPorts**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - externalLoadBalancers: Bag< [LoadBalancer](#) >
      - loadBalancerRules: Bag< [LoadBalancerRule](#) >
        - frontendPorts: Bag< [IntRange](#) >



**network** **cloudAccounts** **firewallRules** **securityRule** **match**  
**ipProtocol**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >
      - securityRule: [SecurityRule](#)
        - match: [HeaderRegion](#)
          - ipProtocol: Bag< [IntRange](#) >

**network** **cloudAccounts** **firewallRules** **securityRule** **match**  
**tpSrc**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >
      - securityRule: [SecurityRule](#)
        - match: [HeaderRegion](#)
          - tpSrc: Bag< [IntRange](#) >

**network** **cloudAccounts** **firewallRules** **securityRule** **match**  
**tpDst**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >
      - securityRule: [SecurityRule](#)
        - match: [HeaderRegion](#)
          - tpDst: Bag< [IntRange](#) >

**network** **cloudAccounts** **firewallRules** **securityRule** **match**  
**icmpType**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >
      - securityRule: [SecurityRule](#)
        - match: [HeaderRegion](#)
          - icmpType: Bag< [IntRange](#) >

**network** **cloudAccounts** **firewallRules** **securityRule** **match**  
**icmpCode**

**See also**

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >
      - securityRule: [SecurityRule](#)
        - match: [HeaderRegion](#)
          - icmpCode: Bag< [IntRange](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- IpEntry

On this page

# IpEntry

## Fields

| name     | type                           | nullable | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|--------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prefix   | IpSubnet                       | false    | Reference to the IP destination prefix which must be matched to utilise the AFT entry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| nextHops | Bag< <a href="#">NextHop</a> > | false    | <p>The list of next-hops that are to be used for entry within the AFT table. The structure of each next-hop is address family independent, such that it is possible to resolve fully how the next-hop is treated. For example:</p> <ul style="list-style-type: none"><li>- Where ingress IPv4 unicast packets are to be forwarded via an MPLS LSP, the next-hop list should indicate the MPLS label stack that is used to the next-hop.</li><li>- Where ingress MPLS labeled packets are to be forwarded to an IPv6 nexthop (for example, a CE within a VPN, then the popped label stack, and IPv6 next-hop address should be indicated).</li></ul> |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts** [â](#) **ipv4Unicast** [â](#) **ipEntries**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv4Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts** [â](#) **ipv6Unicast** [â](#) **ipEntries**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv6Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- IpUnicast

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# IpUnicast

## Fields

| name      | type                           | nullable | description                                                                                                                  |
|-----------|--------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------|
| ipEntries | Bag< <a href="#">IpEntry</a> > | false    | List of the IP unicast entries within the abstract forwarding table. This list is keyed by the destination IPv4/IPv6 prefix. |

## Used by

**network** **devices** **networkInstances** **afts** **ipv4Unicast**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv4Unicast: [IpUnicast](#)

**network** **devices** **networkInstances** **afts** **ipv6Unicast**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv6Unicast: [IpUnicast](#)



.



- Reference
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- [Data Model](#)
- Ipv4Address

On this page

# Ipv4Address

## Fields

| name         | type      | nullable | description                       |
|--------------|-----------|----------|-----------------------------------|
| ip           | IpAddress | false    | The IPv4 address on an interface. |
| prefixLength | Integer   | false    | The length of the subnet prefix.  |

## Used by

network devices interfaces subinterfaces ipv4 addresses

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - subinterfaces: Bag< SubInterface >
        - ipv4: IfaceIpv4Info
          - addresses: Bag< Ipv4Address >

**network** **devices** **interfaces** **subinterfaces** **ipv4** **fhrpAddresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv4: [IfaceIpv4Info](#)
          - fhrpAddresses: Bag< [Ipv4Address](#) >

**network** **devices** **interfaces** **routedVlan** **ipv4** **addresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - addresses: Bag< [Ipv4Address](#) >

**network** **devices** **interfaces** **routedVlan** **ipv4** **fhrpAddresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrpAddresses: Bag< [Ipv4Address](#) >

**network** **devices** **interfaces** **bridge** **ipv4** **addresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)
          - addresses: Bag< [Ipv4Address](#) >

**network** **devices** **interfaces** **bridge** **ipv4** **fhrpAddresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
      - ipv4: [IfaceIpv4Info](#)
        - fhrpAddresses: Bag< [Ipv4Address](#) >

**network** **devices** **interfaces** **tunnel** **ipv4** **addresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
      - ipv4: [IfaceIpv4Info](#)
        - addresses: Bag< [Ipv4Address](#) >

**network** **devices** **interfaces** **tunnel** **ipv4** **fhrpAddresses**

**See also**

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - fhrpAddresses: Bag< [Ipv4Address](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Ipv6Address

On this page

# Ipv6Address

## Fields

| name         | type      | nullable | description                       |
|--------------|-----------|----------|-----------------------------------|
| ip           | IpAddress | false    | The IPv6 address on an interface. |
| prefixLength | Integer   | false    | The length of the subnet prefix.  |

## Used by

network devices interfaces subinterfaces ipv6 addresses

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - subinterfaces: Bag< SubInterface >
        - ipv6: IfaceIpv6Info
          - addresses: Bag< Ipv6Address >



**network** **devices** **interfaces** **subinterfaces** **ipv6** **fhrpAddresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
          - fhrpAddresses: Bag< [Ipv6Address](#) >

**network** **devices** **interfaces** **routedVlan** **ipv6** **addresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - addresses: Bag< [Ipv6Address](#) >

**network** **devices** **interfaces** **routedVlan** **ipv6** **fhrpAddresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrpAddresses: Bag< [Ipv6Address](#) >

**network** **devices** **interfaces** **bridge** **ipv6** **addresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - addresses: Bag< [Ipv6Address](#) >

**network** **devices** **interfaces** **bridge** **ipv6** **fhrpAddresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrpAddresses: Bag< [Ipv6Address](#) >

**network** **devices** **interfaces** **tunnel** **ipv6** **addresses**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - addresses: Bag< [Ipv6Address](#) >

**network** **devices** **interfaces** **tunnel** **ipv6** **fhrpAddresses**

**See also**

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - fhrpAddresses: Bag< [Ipv6Address](#) >

.



- Reference
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- [Data Model](#)
- IsExternalInterfacePolicy

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# IsExternalInterfacePolicy

## Fields

| name      | type                                | nullable | description                                                                                                                              |
|-----------|-------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                                                                                                       |
| value     | Bool                                | false    | Whether the interface connects to the interface of a device that is in another network (a network in a different administrative domain). |

## Used by

network [stigDatabase](#) [policy](#) [ifacePolicy](#) [connectivity](#) [toExternalDevice](#)

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - connectivity: [InterfaceConnectivityPolicy](#)
          - toExternalDevice:  
Bag< [IsExternalInterfacePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- IsHostGatewayInterfacePolicy

On this page

# IsHostGatewayInterfacePolicy

## Fields

| name      | type                                | nullable | description                                              |
|-----------|-------------------------------------|----------|----------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                       |
| value     | Bool                                | false    | Whether the interface is the gateway for a set of hosts. |

## Used by

network [stigDatabase](#) [policy](#) [ifacePolicy](#) [isHostGateway](#)

## See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - isHostGateway:  
Bag< [IsHostGatewayInterfacePolicy](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- IsProviderEdgeDevicePolicy

On this page

# IsProviderEdgeDevicePolicy

## Fields

| name      | type                                | nullable | description                                   |
|-----------|-------------------------------------|----------|-----------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                            |
| value     | Bool                                | false    | Whether the device is a provider edge device. |

## Used by

network [stigDatabase](#) [policy](#) [devicePolicy](#) [isProviderEdge](#)

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - devicePolicy: [DevicePolicy](#)
        - isProviderEdge: Bag< [IsProviderEdgeDevicePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- IsUntrustedInterfacePolicy

On this page

# IsUntrustedInterfacePolicy

## Fields

| name      | type                                | nullable | description                                                                                                                                                  |
|-----------|-------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                                                                                                                           |
| value     | Bool                                | false    | Whether the interface connects to one or more untrusted devices, such as a user device or an untrusted gateway. Such interfaces should not be in trunk mode. |

## Used by

network [stigDatabase](#) [policy](#) [ifacePolicy](#) [connectivity](#) [toUntrustedDevice](#)

## See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - connectivity: [InterfaceConnectivityPolicy](#)
          - toUntrustedDevice:
            - Bag< [IsUntrustedInterfacePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- KeepAliveStatus

On this page

# KeepAliveStatus

## Options

| name          | description |
|---------------|-------------|
| ALIVE         | --          |
| DISABLED      | --          |
| NOT_REACHABLE | --          |
| SUSPENDED     | --          |

## Used by

network devices ha vpc vpcKeepAliveStatus

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - ha: Ha
      - vpc: VpcDomain
        - vpcKeepAliveStatus: KeepAliveStatus

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Layer

On this page

## Layer

Network layer in the OSI model.

## Options

| name | description                                          |
|------|------------------------------------------------------|
| L2   | Corresponds to the data link layer of the OSI model. |
| L3   | Corresponds to the network layer of the OSI model.   |

## Used by

**network** **devices** **interfaces** **layer**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - layer: [Layer](#)

**network** **devices** **interfaces** **subinterfaces** **layer**

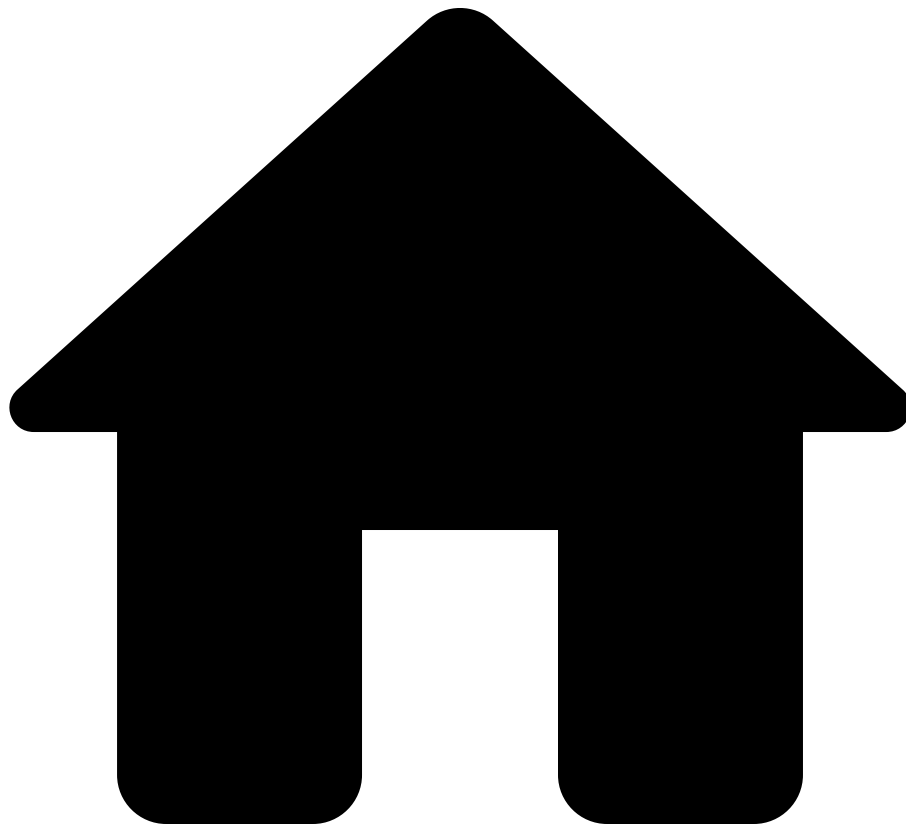
See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - layer: [Layer](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- LifecycleData

On this page

## LifecycleData

Firewall hit count and timestamps for their creation, last modification, and last used. Present for systems that were successfully collected and have one of these operating systems: SRX, PAN\_OS, CHECKPOINT, FORTINET and ASA, if reported by the operating system. At least one of the fields will be non-null.

## Fields

| name              | type      | nullable | description                                                                                                                                                                                                                                          |
|-------------------|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| createdAt         | Timestamp | true     | The time at which this rule was created.                                                                                                                                                                                                             |
| lastModified      | Timestamp | true     | The time at which this rule's configuration was last updated.                                                                                                                                                                                        |
| lastUsed          | Timestamp | true     | The time at which this rule last processed a packet.                                                                                                                                                                                                 |
| truncatedHitCount | Integer   | true     | Represents the number of packets processed by this rule since its creation. If the count exceeds the maximum integer limit, the value is set to -1 to indicate an overflow. In a future update, this field will be replaced with a larger data type. |

## Used by

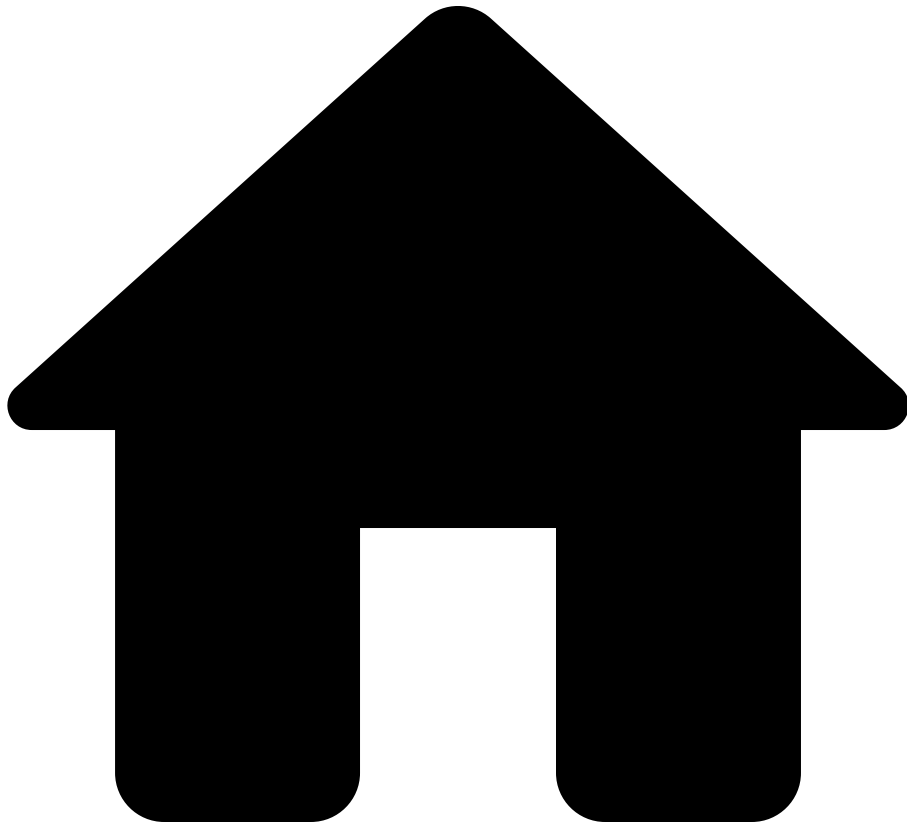
network devices aclEntries lifecycleData

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - lifecycleData: [LifecycleData](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- LoadBalancer

On this page

# LoadBalancer

## Fields

| name              | type                                     | nullable | description                            |
|-------------------|------------------------------------------|----------|----------------------------------------|
| name              | String                                   | true     | --                                     |
| id                | String                                   | false    | --                                     |
| tags              | Bag< String >                            | false    | --                                     |
| isUp              | Bool                                     | false    | --                                     |
| region            | String                                   | true     | --                                     |
| loadBalancerRules | List< <a href="#">LoadBalancerRule</a> > | false    | --                                     |
| outboundRules     | List< <a href="#">OutboundRule</a> >     | false    | The field outboundRules is Azure only. |

## Used by

network [cloudAccounts](#) [vpcs](#) [loadBalancers](#)

## See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - loadBalancers: Bag< [LoadBalancer](#) >

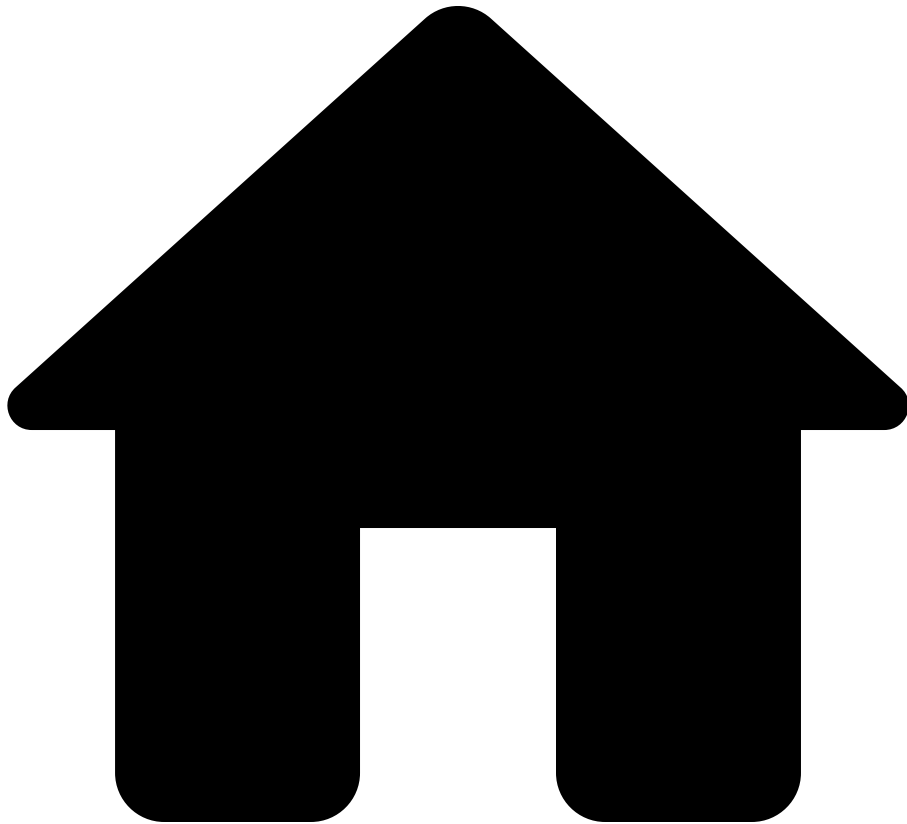
**network** **cloudAccounts** **externalLoadBalancers**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - externalLoadBalancers: Bag< [LoadBalancer](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- LoadBalancerRule

On this page

# LoadBalancerRule

## Fields

| name          | type             | nullable | description |
|---------------|------------------|----------|-------------|
| frontendIps   | Bag< IPAddress > | false    | --          |
| protocol      | Integer          | true     | --          |
| frontendPorts | Bag< IntRange >  | false    | --          |
| backends      | Bag< Backend >   | false    | --          |

## Used by

network cloudAccounts vpcs loadBalancers loadBalancerRules

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - vpcs: Bag< VpcData >
      - loadBalancers: Bag< LoadBalancer >
        - loadBalancerRules: Bag< LoadBalancerRule >

**network** **cloudAccounts** **externalLoadBalancers**  
**loadBalancerRules**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - externalLoadBalancers: Bag< [LoadBalancer](#) >
      - loadBalancerRules: Bag< [LoadBalancerRule](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Location

On this page

# Location

## Fields

| name          | type   | nullable | description                                                                                                                                                                                       |
|---------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name          | String | false    | The user-defined name of the location.                                                                                                                                                            |
| id            | String | false    | The ID of the location.                                                                                                                                                                           |
| city          | String | true     | The city can be null if the location has been defined in the platform (with latitude and longitude values), but no city was defined for that location. Present whenever adminDivision is present. |
| adminDivision | String | true     | Could be a state in the United States like California or a territory in Canada like Ontario.                                                                                                      |
| country       | String | true     | Present if and only if city is present.                                                                                                                                                           |

## Used by

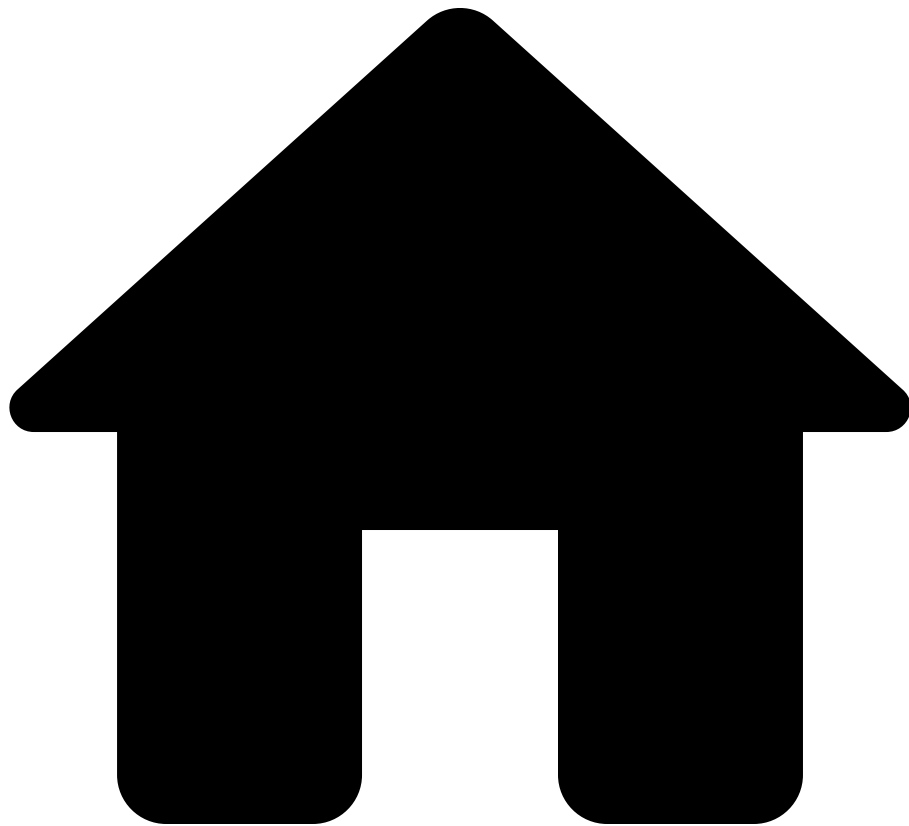
network locations

See also

Data model path

- network: Network
  - locations: Bag< Location >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- MacEntry

On this page

# MacEntry

## Fields

| name       | type                                         | nullable | description                                                                    |
|------------|----------------------------------------------|----------|--------------------------------------------------------------------------------|
| macAddress | MacAddress                                   | false    | MAC address for the dynamic or static MAC table entry.                         |
| vlan       | Integer                                      | true     | VLAN from which this MAC address was received.                                 |
| interfaces | Bag< <a href="#">MacEntryInterfaceInfo</a> > | false    | Information of interfaces/subinterfaces at which this MAC address was learned. |
| entryType  | <a href="#">MacEntryType</a>                 | false    | Type of MAC entry.                                                             |
| sourceType | <a href="#">MacEntrySourceType</a>           | false    | Type of source of the MAC entry.                                               |

## Used by[↗](#)

**network** [↗](#) **devices** [↗](#) **networkInstances** [↗](#) **fdb** [↗](#) **macEntries**[↗](#)

See also[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - fdb: [NetworkInstanceFdb](#)
        - macEntries: Bag< [MacEntry](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- MacEntryInterfaceInfo

On this page

# MacEntryInterfaceInfo

## Fields

| name             | type   | nullable | description                                     |
|------------------|--------|----------|-------------------------------------------------|
| interfaceName    | String | false    | Interface at which the MAC entry is learned.    |
| subinterfaceName | String | true     | Subinterface at which the MAC entry is learned. |

## Used by

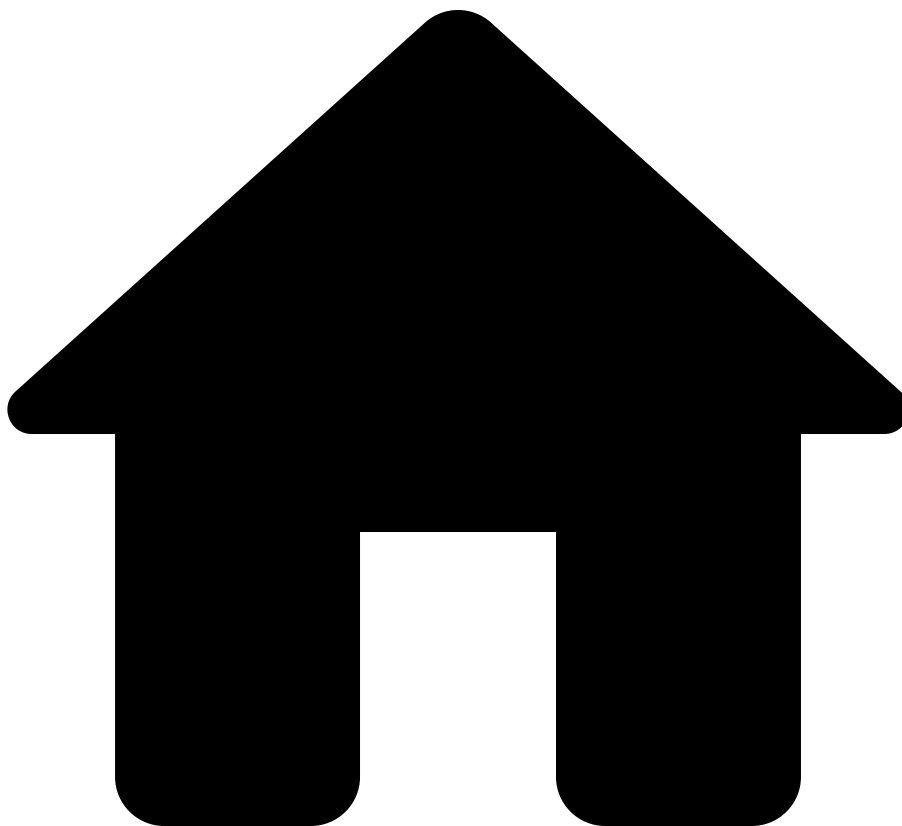
network devices networkInstances fdb macEntries interfaces

## See also

Data model path

- network: Network
  - devices: Bag< Device >
    - networkInstances: Bag< NetworkInstance >
      - fdb: NetworkInstanceFdb
        - macEntries: Bag< MacEntry >
          - interfaces: Bag< MacEntryInterfaceInfo >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- MacEntrySourceType

On this page



# MacEntrySourceType

## Options

| name        | description                               |
|-------------|-------------------------------------------|
| FABRIC_PATH | MAC learned from FabricPath.              |
| LOCAL       | Regular MAC learned from local L2 domain. |
| MPLS        | MAC learned over MPLS.                    |
| VXLAN       | MAC learned over VXLAN tunnel.            |

## Used by

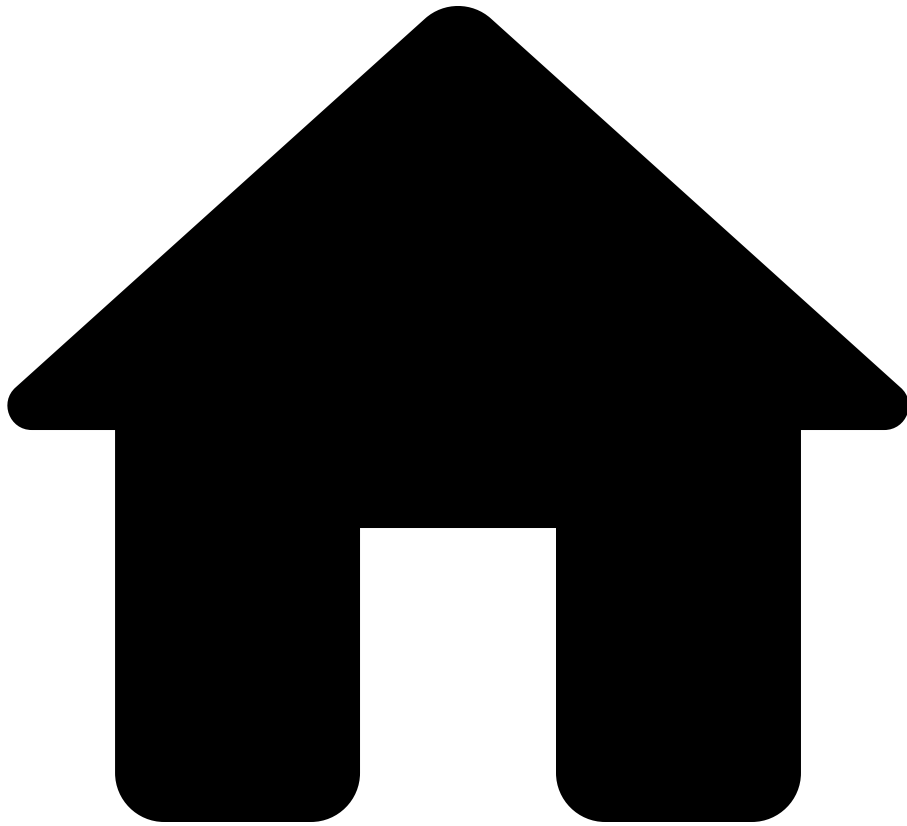
`network` `devices` `networkInstances` `fdb` `macEntries` `sourceType`

See also

Data model path

- network: `Network`
  - devices: Bag< `Device` >
    - networkInstances: Bag< `NetworkInstance` >
      - fdb: `NetworkInstanceFdb`
        - macEntries: Bag< `MacEntry` >
          - sourceType: `MacEntrySourceType`

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- MacEntryType

On this page

# MacEntryType

## Options

| name    | description                            |
|---------|----------------------------------------|
| DYNAMIC | Dynamically learned MAC table entry.   |
| STATIC  | Statically programmed MAC table entry. |

## Used by

network devices networkInstances fdb macEntries entryType

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - networkInstances: Bag< NetworkInstance >
      - fdb: NetworkInstanceFdb
        - macEntries: Bag< MacEntry >
          - entryType: MacEntryType

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- ManagementSubnetsDevicePolicy

On this page

# ManagementSubnetsDevicePolicy

## Fields

| name      | type                                | nullable | description                                                      |
|-----------|-------------------------------------|----------|------------------------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                               |
| value     | Bag< <a href="#">IpSubnet</a> >     | false    | The list of subnets in CIDR notation used for management access. |

## Used by

network [stigDatabase](#) [policy](#) [devicePolicy](#) [managementSubnets](#)

## See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - devicePolicy: [DevicePolicy](#)
        - managementSubnets:  
Bag< [ManagementSubnetsDevicePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- MetadataMatches

On this page

# MetadataMatches

## Fields

| name              | type          | nullable | description                                                                                                                                                                   |
|-------------------|---------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ingressInterfaces | Bag< String > | false    | The set of ingress interfaces declared in the ACL or NAT rule, either explicitly or via declared zones. Empty for ACL or NAT rules that are separately applied to interfaces. |
| egressInterfaces  | Bag< String > | false    | The set of egress interfaces declared in the ACL or NAT rule, either explicitly or via declared zones. Empty for ACL or NAT rules that are separately applied to interfaces.  |

## Used by

network devices aclEntries metadataMatches

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - aclEntries: Bag< AclEntry >
      - metadataMatches: MetadataMatches

**network** **devices** **natEntries** **metadataMatches**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >
      - metadataMatches: [MetadataMatches](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- MstInstance

On this page

# MstInstance

## Fields

| name                   | type                                 | nullable | description                                                                                                                                  |
|------------------------|--------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| id                     | Integer                              | false    | In an MSTP Bridge, an MSTID, i.e., a value used to identify a spanning tree (or MST) instance. Reference: IEEE8021-MIB IEEE8021MstIdentifier |
| vlan                   | Bag< <a href="#">VlanIdOrRange</a> > | false    | list of vlans mapped to the MST instance                                                                                                     |
| bridgePriority         | Integer                              | true     | The manageable component of the Bridge Identifier. References IEEE 802.1D 17.13.7 Bridge Identifier Priority                                 |
| bridgeAddress          | MacAddress                           | true     | A unique 48-bit Universally Administered MAC Address assigned to the bridge. Reference IEEE 802.1D 7.12.5 Unique identification of bridge.   |
| designatedRootPriority | Integer                              | true     | The bridge priority of root of the spanning tree as determined by the Spanning Tree Protocol as executed by this network. Reference: RFC4188 |

| name                  | type       | nullable | description                                                                                                                                                                     |
|-----------------------|------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |            |          | BRIDGE-MIB<br>dot1dStpDesignatedR                                                                                                                                               |
| designatedRootAddress | MacAddress | true     | The bridge address of the root of the spanning tree as determined by the Spanning Tree Protocol as executed by this network. References: RFC4188 BRIDGE-MIB dot1dStpDesignatedR |
| rootPortName          | String     | true     | The port name of the bridge which offers the lowest cost path from this bridge to the root bridge. References: RFC4188 BRIDGE-MIB dot1dStpRootPort.                             |
| rootCost              | Integer    | true     | The cost of the path to the root as seen from this bridge. References: RFC4188 BRIDGE-MIB dot1dStpRootCost.                                                                     |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **stp** [â](#) **mstp** [â](#) **instances**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - stp: [Stp](#)
      - mstp: [Mstp](#)
        - instances: Bag< [MstInstance](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Mstp

On this page

## Mstp

Multi Spanning-tree protocol configuration and operation data.

## Fields

| name      | type                               | nullable | description                                                                                                                  |
|-----------|------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------|
| name      | String                             | true     | The Configuration Name in the MST Configuration Identifier. Reference: IEEE 802.1Q 13.8 MST Configuration Identifier (MCID). |
| instances | Bag< <a href="#">MstInstance</a> > | false    | Configuration and operation data for MSTP instances.                                                                         |

## Used by

network devices stp mstp

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - stp: [Stp](#)
      - mstp: [Mstp](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- MulticastMode

On this page

# MulticastMode

## Options

| name                      | description |
|---------------------------|-------------|
| BIDIRECTIONAL             | --          |
| DENSE                     | --          |
| SOURCE_SPECIFIC_MULTICAST | --          |
| SPARSE                    | --          |
| SPARSE_DENSE              | --          |

## Used by

network devices interfaces subinterfaces ipv4 multicastMode

## See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - subinterfaces: Bag< SubInterface >
        - ipv4: IfaceIpv4Info
          - multicastMode: MulticastMode



**network** **devices** **interfaces** **subinterfaces** **ipv6** **multicastMode**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - ipv6: [IfaceIpv6Info](#)
          - multicastMode: [MulticastMode](#)

**network** **devices** **interfaces** **routedVlan** **ipv4** **multicastMode**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv4: [IfaceIpv4Info](#)
          - multicastMode: [MulticastMode](#)

**network** **devices** **interfaces** **routedVlan** **ipv6** **multicastMode**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)
        - ipv6: [IfaceIpv6Info](#)
          - multicastMode: [MulticastMode](#)

**network** **devices** **interfaces** **bridge** **ipv4** **multicastMode**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv4: [IfaceIpv4Info](#)
          - multicastMode: [MulticastMode](#)

**network** **devices** **interfaces** **bridge** **ipv6** **multicastMode**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - bridge: [Bridge](#)
        - ipv6: [IfaceIpv6Info](#)
          - multicastMode: [MulticastMode](#)

**network** **devices** **interfaces** **tunnel** **ipv4** **multicastMode**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv4: [IfaceIpv4Info](#)
          - multicastMode: [MulticastMode](#)

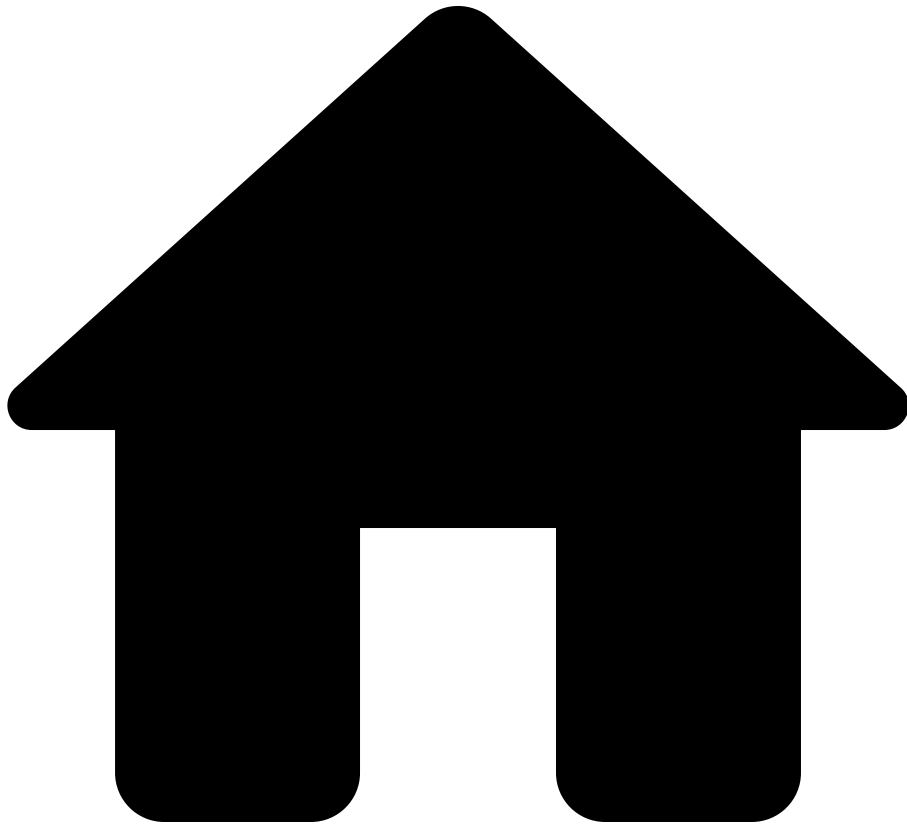
**network** **devices** **interfaces** **tunnel** **ipv6** **multicastMode**

**See also**

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - tunnel: [Tunnel](#)
        - ipv6: [IfaceIpv6Info](#)
          - multicastMode: [MulticastMode](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- NamedVlanCollection

On this page

# NamedVlanCollection

## Fields

| name   | type                             | nullable | description                                                                |
|--------|----------------------------------|----------|----------------------------------------------------------------------------|
| name   | String                           | true     | Name of the VLAN object. Absent if no name was configured for these VLANs. |
| ranges | Bag< <a href="#">VlanRange</a> > | false    | List of disjoint VLAN ranges associated with this name.                    |

## Used by

network devices vlans

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - vlans: Bag< [NamedVlanCollection](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- NatEntry

On this page

# NatEntry

## Fields

| name            | type                                 | nullable | description                                                                                                                                      |
|-----------------|--------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| headerMatches   | <a href="#">HeaderRegion</a>         | false    | --                                                                                                                                               |
| metadataMatches | <a href="#">MetadataMatches</a>      | false    | --                                                                                                                                               |
| name            | String                               | false    | --                                                                                                                                               |
| description     | String                               | true     | A user-defined description of the NAT entry. Only present on PanOS devices.                                                                      |
| rewrites        | Bag< <a href="#">HeaderRewrite</a> > | false    | --                                                                                                                                               |
| interfaceNat    | Bool                                 | false    | Indicates that ipv4Src is rewritten to the IP address of the egress interface, and in this case the ipv4Src field of any rewrites will be empty. |
| natType         | <a href="#">NatType</a>              | false    | --                                                                                                                                               |



## Used by[â](#)

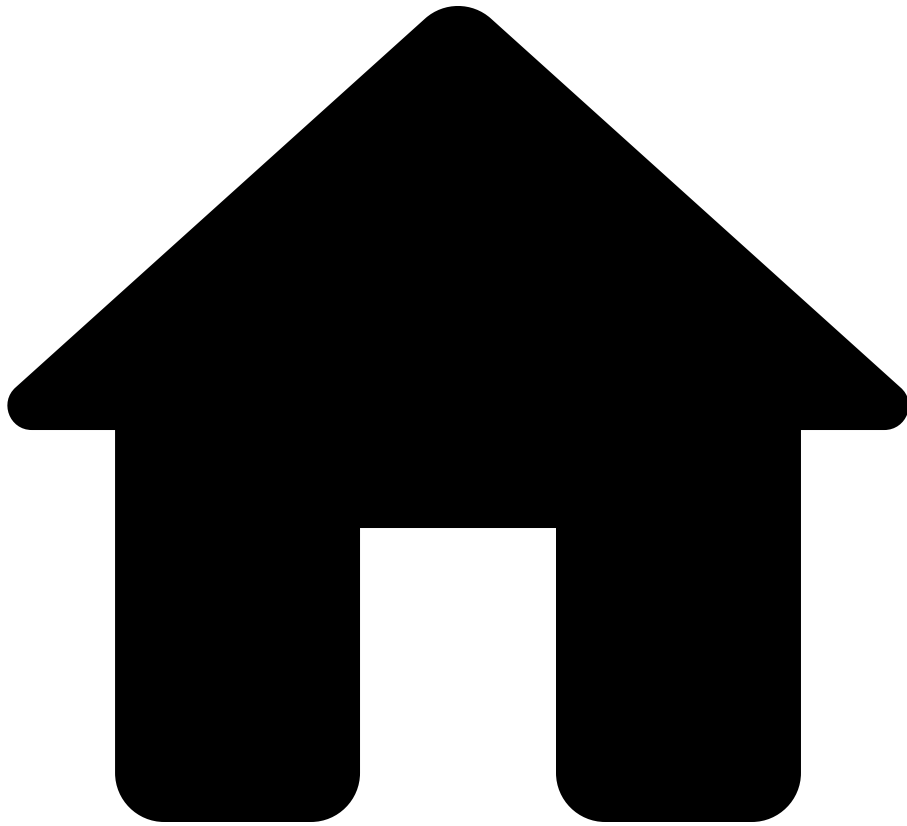
**network** [â](#) **devices** [â](#) **natEntries**[â](#)

**See also**[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - natEntries: Bag< [NatEntry](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- NatGateway

On this page

# NatGateway

## Fields

| name           | type            | nullable | description |
|----------------|-----------------|----------|-------------|
| name           | String          | true     | --          |
| id             | String          | false    | --          |
| tags           | Bag< String >   | false    | --          |
| region         | String          | true     | --          |
| privateSubnets | Bag< IpSubnet > | false    | --          |
| publicSubnets  | Bag< IpSubnet > | false    | --          |

## Used by

network cloudAccounts vpcs natGateways

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - vpcs: Bag< VpcData >
      - natGateways: Bag< NatGateway >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- NatType

On this page

# NatType

## Options

| name     | description                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNat     | --                                                                                                                                                     |
| Identity | A NAT entry that does no translation. Such entries are used to exclude specific traffic from NAT when there are broad NAT translation entries present. |
| LB       | --                                                                                                                                                     |
| SNat     | --                                                                                                                                                     |
| TwiceNat | --                                                                                                                                                     |

## Used by

`network` `devices` `natEntries` `natType`

See also

Data model path

- network: `Network`
  - devices: Bag< `Device` >
    - natEntries: Bag< `NatEntry` >
      - natType: `NatType`

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Network

On this page

# Network

## Fields

| name                     | type                                      | nullable | description                                                          |
|--------------------------|-------------------------------------------|----------|----------------------------------------------------------------------|
| devices                  | Bag< <a href="#">Device</a> >             | false    | Devices in the network, such as routers, switches, and firewalls.    |
| endpoints                | Bag< <a href="#">NetworkEndpoint</a> >    | false    | Devices collected via a custom operation, such as HTTP, CLI or SNMP. |
| locations                | Bag< <a href="#">Location</a> >           | false    | The set of all locations defined in the network.                     |
| cloudAccounts            | Bag< <a href="#">CloudAccount</a> >       | false    | --                                                                   |
| cloudOrganizationalUnits | Bag< <a href="#">OrganizationalUnit</a> > | false    | --                                                                   |
| externalSources          | <a href="#">ExternalSources</a>           | false    | Inferred data from structured data                                   |

| name           | type                                 | nullable | description                                                                                          |
|----------------|--------------------------------------|----------|------------------------------------------------------------------------------------------------------|
|                |                                      |          | sources, such as CSV and JSON sources served over HTTP.                                              |
| dataConnectors | Bag< <a href="#">DataConnector</a> > | false    | Data made available to the Forward platform by sources outside the network, such as an HTTP service. |
| cveDatabase    | <a href="#">CveDatabase</a>          | false    | --                                                                                                   |
| stigDatabase   | <a href="#">StigDatabase</a>         | false    | --                                                                                                   |
| extensions     | <a href="#">NetworkExtensions</a>    | false    | --                                                                                                   |

Used by

network

See also

Data model path

- network: [Network](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- NetworkAcl

On this page

# NetworkAcl

## Fields

| name         | type                                 | nullable | description |
|--------------|--------------------------------------|----------|-------------|
| name         | String                               | true     | --          |
| id           | String                               | false    | --          |
| tags         | Bag< String >                        | false    | --          |
| ingressRules | List< <a href="#">SecurityRule</a> > | false    | --          |
| egressRules  | List< <a href="#">SecurityRule</a> > | false    | --          |

## Used by

network cloudAccounts vpcs networkAcls

## See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >

.



- Reference
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- [Data Model](#)
- NetworkEndpoint

On this page

# NetworkEndpoint

## Fields

| name         | type                               | nullable | description                                                            |
|--------------|------------------------------------|----------|------------------------------------------------------------------------|
| name         | String                             | false    | The name of the device endpoint. Unique across all collection sources. |
| locationName | String                             | true     | The location name. Absent if no location is assigned.                  |
| tagNames     | Bag< String >                      | false    | Names of the device tags that contain this device.                     |
| snapshotInfo | <a href="#">DeviceSnapshotInfo</a> | false    | Metadata about the inclusion of this source in the snapshot.           |
| profileName  | String                             | false    | The name of the                                                        |

| name                | type                                         | nullable | description                                      |
|---------------------|----------------------------------------------|----------|--------------------------------------------------|
|                     |                                              |          | profile from which this source was derived.      |
| httpResults         | Bag< <a href="#">HttpApiEndpointResult</a> > | false    | Non-empty when this is an HTTP network endpoint. |
| cliCommandResponses | Bag< <a href="#">CliCommandResponse</a> >    | false    | Non-empty when this is a CLI network endpoint.   |
| snmpOutputs         | Bag< <a href="#">OidOutput</a> >             | false    | Non-empty when this is an SNMP network endpoint. |

## Used by[â](#)

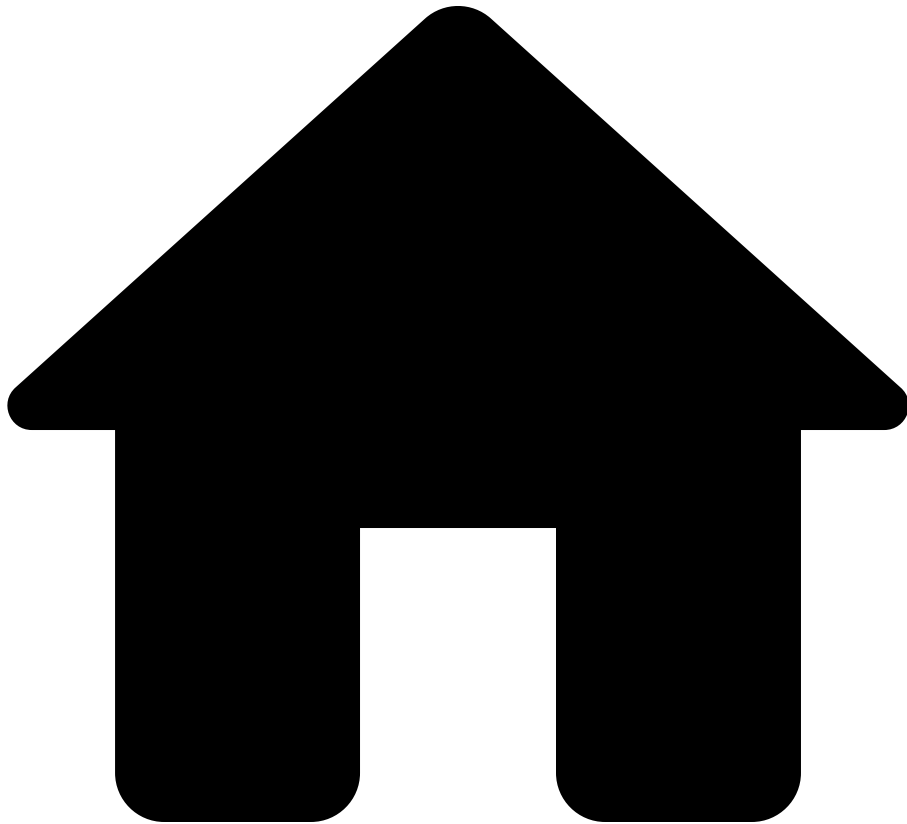
### network [â](#) endpoints[â](#)

### See also[â](#)

#### Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >

.



- Reference
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- NetworkExtensions

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## NetworkExtensions

Contains user-defined extensions to the standard Network data model. By default, this type has no fields. In any Forward Instance, this type will have one field for each Data File defined.

Fields[↗](#)

| name | type | nullable | description |
|------|------|----------|-------------|
|------|------|----------|-------------|

Used by[↗](#)

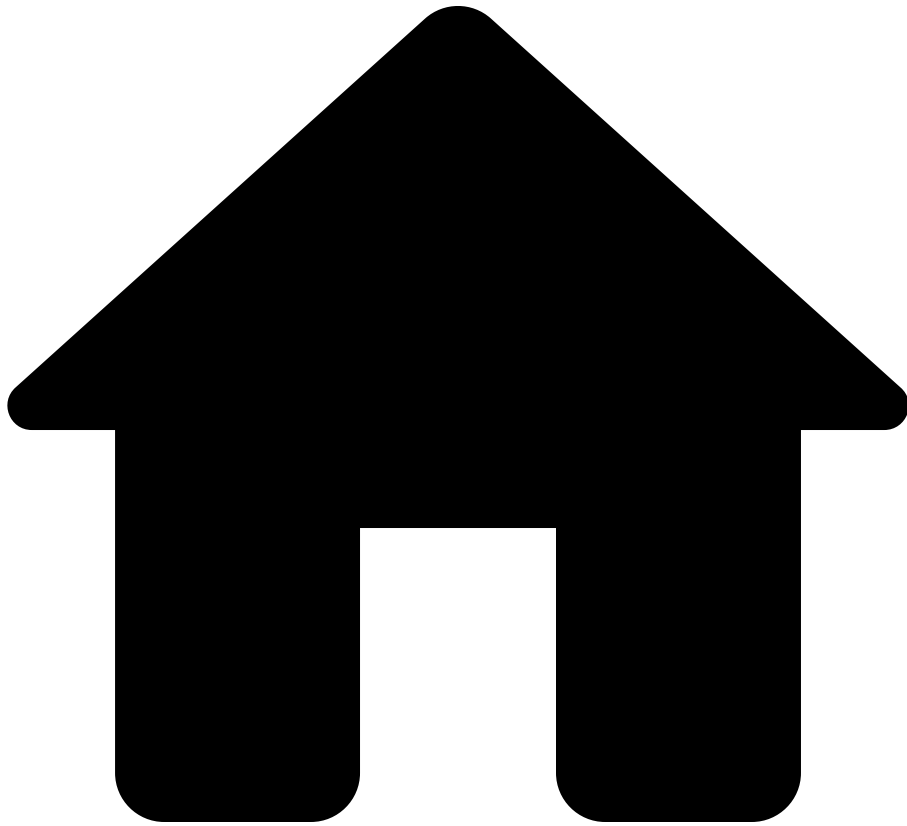
network [↗](#) extensions[↗](#)

See also[↗](#)

Data model path

- network: [Network](#)
  - extensions: [NetworkExtensions](#)

.



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- NetworkInstance

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# NetworkInstance

## Fields

| name         | type                                 | nullable | description                                                                                         |
|--------------|--------------------------------------|----------|-----------------------------------------------------------------------------------------------------|
| name         | String                               | false    | A unique name identifying the network instance. The value is either "default" or a unique VRF name. |
| instanceType | <a href="#">NetworkInstanceType</a>  | false    | Type of network instance.                                                                           |
| afts         | <a href="#">Afts</a>                 | false    | The abstract forwarding tables (AFTs) that are associated with the network instance.                |
| interfaces   | Bag< <a href="#">IfaceSubIface</a> > | false    | The interfaces that are associated with this                                                        |

| name      | type                                           | nullable | description                                                                                                                                                                                                                                                                                     |
|-----------|------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                                |          | network instance.                                                                                                                                                                                                                                                                               |
| v lans    | Bag< <a href="#">Vlan</a> >                    | false    | VLANs that are associated with this network instance. Only populated for the "default" network instance and only includes VLANs that are associated with some interfaces. In particular, VLANs that are defined globally on the device but not associated with any interface are excluded here. |
| protocols | Bag< <a href="#">NetworkInstanceProtocol</a> > | false    |                                                                                                                                                                                                                                                                                                 |

| name | type                               | nullable | description                                                       |
|------|------------------------------------|----------|-------------------------------------------------------------------|
|      |                                    |          | The routing protocols that are enabled for this network instance. |
| fdb  | <a href="#">NetworkInstanceFdb</a> | false    | Forwarding database for this network instance.                    |

## Used by<sup>â</sup>

network<sup>â</sup> devices<sup>â</sup> networkInstances<sup>â</sup>

See also<sup>â</sup>

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- NetworkInstanceFdb

On this page

# NetworkInstanceFdb

## Fields

| name       | type                            | nullable | description                                                         |
|------------|---------------------------------|----------|---------------------------------------------------------------------|
| macEntries | Bag< <a href="#">MacEntry</a> > | false    | Learned MAC addresses.<br>Present only in default network instance. |

## Used by

**network** **devices** **networkInstances** **fdb**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - fdb: [NetworkInstanceFdb](#)

.



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- [NQE Language](#)
- [Data Model](#)
- NetworkInstanceProtocol

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# NetworkInstanceProtocol

## Fields

| name       | type                                | nullable | description                                                                          |
|------------|-------------------------------------|----------|--------------------------------------------------------------------------------------|
| identifier | <a href="#">RoutingProtocolType</a> | false    | The protocol identifier for the instance. Only identifier currently possible is BGP. |
| bgp        | <a href="#">Bgp</a>                 | true     | Top-level configuration and state for the BGP router. Present if identifier is BGP.  |
| ospf       | <a href="#">Ospf</a>                | true     | Top-level state for the OSPF router. Present if identifier is OSPF.                  |

## Used by

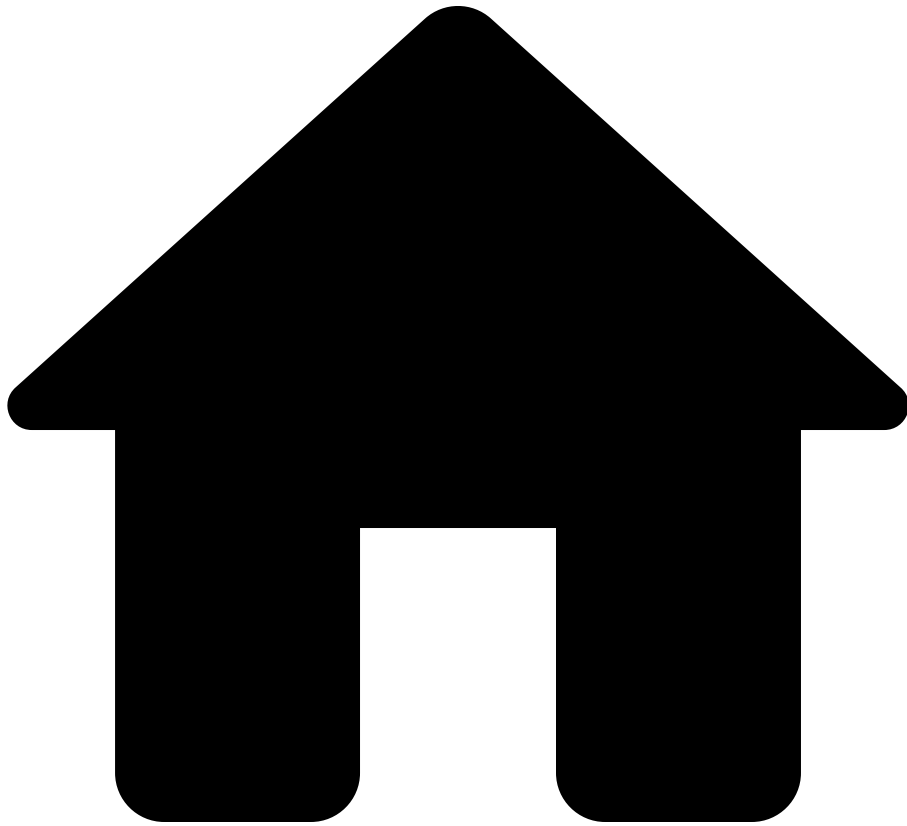
network devices networkInstances protocols

### See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >

.



- Reference
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- NetworkInstanceType

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# NetworkInstanceType

## Options

| name             | description                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------|
| DEFAULT_INSTANCE | A special routing instance which acts as the "default" or "global" routing instance for a network device. |
| L3VRF            | A private Layer 3 only routing instance which is formed of one or more RIBs.                              |

## Used by

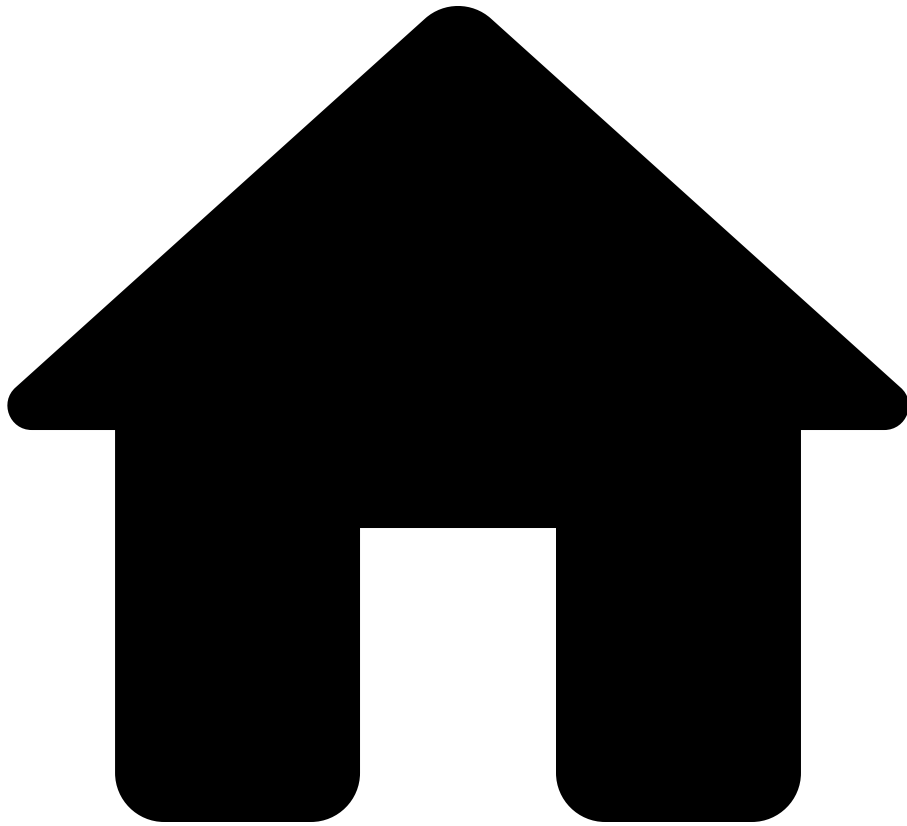
**network** **devices** **networkInstances** **instanceType**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - instanceType: [NetworkInstanceType](#)

.



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- NetworkObject

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# NetworkObject

## Fields

| name    | type            | nullable | description               |
|---------|-----------------|----------|---------------------------|
| name    | String          | false    | The name of the object.   |
| subnets | Bag< IpSubnet > | false    | IPv4 and/or IPv6 subnets. |

## Used by

network cloudAccounts firewallPolicy networkObjects

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - firewallPolicy: FirewallPolicy
      - networkObjects: Bag< NetworkObject >

.



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- [NQE Language](#)
- [Data Model](#)
- NextHop

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# NextHop

## Fields

| name      | type      | nullable | description                                                                                                                                                                                                                                                                                                                                   |
|-----------|-----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| weight    | Integer   | false    | The weight of the next-hop. Traffic is balanced according to the ratio described by the relative weights of the next hops that exist for the AFT entry. Note that all next-hops that are specified are assumed to be active next-hops and therefore eligible (and selected) to be installed in the FIB, and hence used for packet forwarding. |
| ipAddress | IpAddress | true     | The IP address of                                                                                                                                                                                                                                                                                                                             |

| name             | type           | nullable | description                                                                                                                                                                          |
|------------------|----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                |          | the next-hop system.                                                                                                                                                                 |
| macAddress       | MacAddress     | true     | The MAC address of the next-hop if resolved by the local network instance.                                                                                                           |
| vrf              | String         | true     | The next VRF to process the packet.                                                                                                                                                  |
| poppedMplsLabels | Bag< Integer > | false    | The MPLS labels to be popped from the packet when switched by the system. A swap operation is reflected by entries in the popped-mpls-label-stack and pushed-mpls-label-stack nodes. |
| pushedMplsLabels | Bag< Integer > | false    | The MPLS labels to be pushed from the packet when switched by                                                                                                                        |

| name              | type                                    | nullable | description                                                                                                                                     |
|-------------------|-----------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                                         |          | the system. A swap operation is reflected by entries in the popped-mpls-label-stack and pushed-mpls-label-stack nodes.                          |
| interfaceName     | String                                  | true     | The interface on this system to get to the next-hop.                                                                                            |
| subInterfaceName  | String                                  | true     | The sub-interface on this system to get to the next-hop.                                                                                        |
| nextHopType       | <a href="#">NextHopType</a>             | false    | Type of next-hop.                                                                                                                               |
| encapsulateHeader | <a href="#">EncapsulationHeaderType</a> | true     | When forwarding a packet to the specified next-hop the local system performs an encapsulation of the packet - adding the specified header-type. |

| name           | type                           | nullable | description                                        |
|----------------|--------------------------------|----------|----------------------------------------------------|
| originProtocol | <a href="#">OriginProtocol</a> | false    | The protocol from which the AFT entry was learned. |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts** [â](#) **ipv4Unicast** [â](#) **ipEntries** [â](#) **nextHops**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv4Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >
            - nextHops: Bag< [NextHop](#) >

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts** [â](#) **ipv6Unicast** [â](#) **ipEntries** [â](#) **nextHops**[â](#)

See also[â](#)

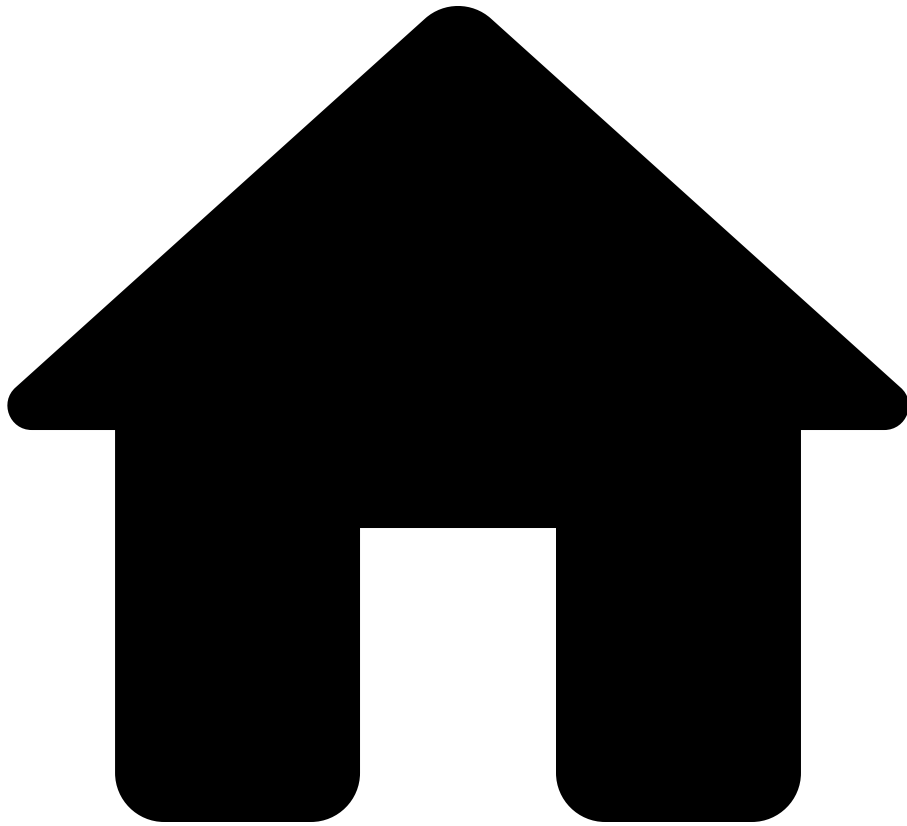
Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv6Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >
            - nextHops: Bag< [NextHop](#) >





.



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- NextHopType

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# NextHopType

## Options

| name          | description                                                            |
|---------------|------------------------------------------------------------------------|
| ATTACHED      | Packet will be L2 forwarded.                                           |
| BROADCAST     | Packet will be broadcasted to all ports, except the incoming port.     |
| DROP          | Packet will be dropped.                                                |
| RECEIVE       | Packet will be received and consumed by the device.                    |
| RECURSIVE     | Recursive lookup to determine next hop.                                |
| REDIRECT      | Packet will be redirected to another port.                             |
| REMOTE        | Packet will be L3 forwarded.                                           |
| UNDEFINED     | --                                                                     |
| VRF_FORWARDED | Packet will be forwarded based on VRF. Next hop will be decided there. |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts** [â](#) **ipv4Unicast** [â](#) **ipEntries** [â](#) **nextHops** [â](#) **nextHopType**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv4Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >
            - nextHops: Bag< [NextHop](#) >
              - nextHopType: [NextHopType](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts** [â](#) **ipv6Unicast** [â](#) **ipEntries** [â](#) **nextHops** [â](#) **nextHopType**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv6Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >
            - nextHops: Bag< [NextHop](#) >
              - nextHopType: [NextHopType](#)

.



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- [Data Model](#)
- NtpConfigurationDetail

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# NtpConfigurationDetail

## Fields

| name    | type                             | nullable | description |
|---------|----------------------------------|----------|-------------|
| servers | Bag< <a href="#">NtpServer</a> > | false    | --          |
| enable  | Bool                             | false    | --          |

## Used by

network devices outputs bluecat config items data  
services ntp configurations ntpConfiguration

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - ntp: [NtpService](#)
                  - configurations: Bag< [NtpServiceConfiguration](#) >
                    - ntpConfiguration: [NtpConfigurationDetail](#)

.



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- NtpServer

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# NtpServer

## Fields

| name    | type      | nullable | description |
|---------|-----------|----------|-------------|
| address | IpAddress | false    | --          |
| stratum | String    | false    | --          |

Used by an

network â devices â outputs â bluecat â config â items â data  
â services â ntp â configurations â ntpConfiguration â  
serversâ

**See also**[ââ](#)

## Data model path

- [illegible]



.



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- NtpService

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# NtpService

## Fields

| name           | type                                           | nullable | description |
|----------------|------------------------------------------------|----------|-------------|
| configurations | Bag< <a href="#">NtpServiceConfiguration</a> > | false    | --          |

## Used by

network devices outputs bluecat config items data  
services ntp

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - ntp: [NtpService](#)

.



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- [Data Model](#)
- NtpServiceConfiguration

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# NtpServiceConfiguration

## Fields

| name             | type                                   | nullable | description |
|------------------|----------------------------------------|----------|-------------|
| ntpConfiguration | <a href="#">NtpConfigurationDetail</a> | false    | --          |

## Used by

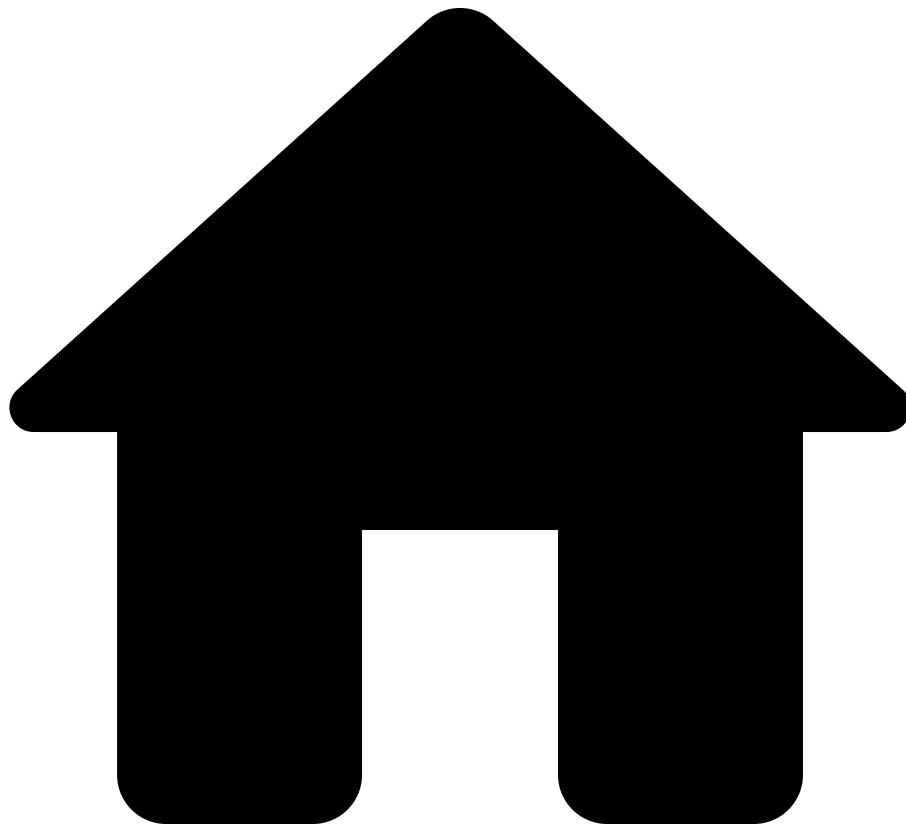
network devices outputs bluecat config items data  
services ntp configurations

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - bluecat: [BluecatOutputs](#)
        - config: [BluecatConfig](#)
          - items: Bag< [BluecatConfigItem](#) >
            - data: [BluecatConfigItemData](#)
              - services: [BluecatServices](#)
                - ntp: [NtpService](#)
                  - configurations: Bag< [NtpServiceConfiguration](#) >

.



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- [Data Model](#)
- OidCollectionStatus

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# OidCollectionStatus

## Options

| name  | type   | description |
|-------|--------|-------------|
| error | String | --          |
| ok    | --     | --          |

## Used by

**network** **devices** **outputs** **snmpOutputs** **status**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - snmpOutputs: Bag< [OidOutput](#) >
        - status: [OidCollectionStatus](#)

**network** **endpoints** **snmpOutputs** **status**

See also

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - snmpOutputs: Bag< [OidOutput](#) >
      - status: [OidCollectionStatus](#)

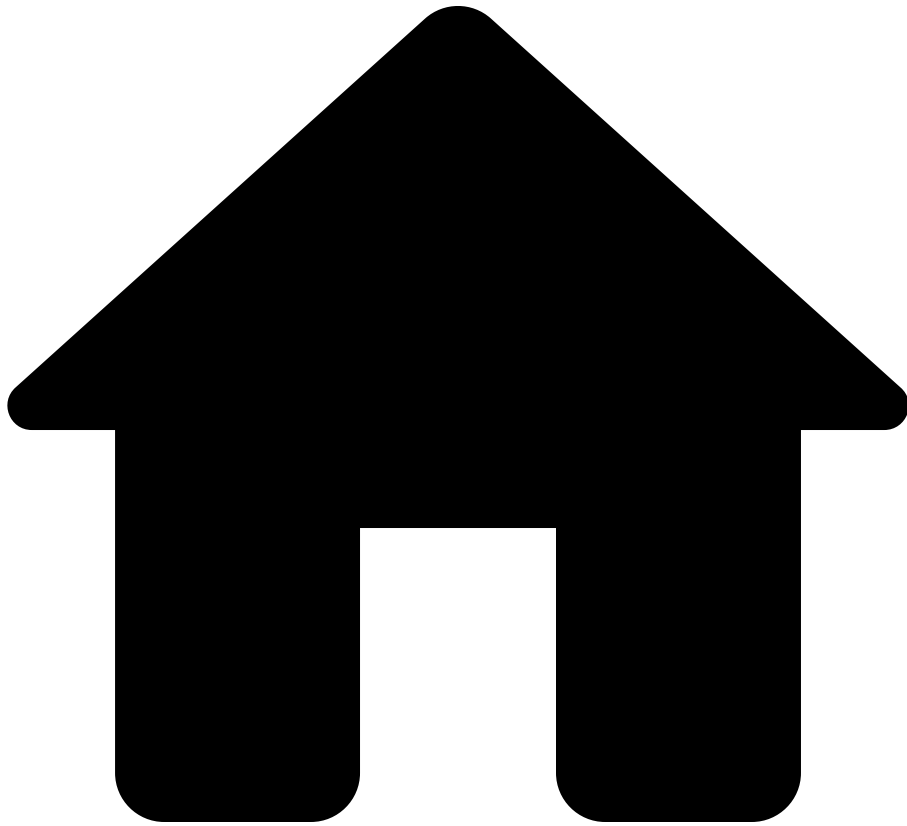
**network** [externalSources](#) [snmpSources](#) [outputs](#) [status](#)

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - snmpSources: Bag< [SnmpSource](#) >
      - outputs: Bag< [OidOutput](#) >
        - status: [OidCollectionStatus](#)

.



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- OidEntry

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# OidEntry

## Fields

| name       | type           | nullable | description                                                                                                                                                                                    |
|------------|----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| oid        | String         | false    | The full OID for the entry.                                                                                                                                                                    |
| oidNumbers | Bag< Integer > | false    | The full OID as a list of numbers.                                                                                                                                                             |
| rawValue   | String         | false    | The value stored in the OID entry, rendered as a string; e.g. "1" or "00:a0:bc:6a:e0:07". Consult the MIB documentation for the type of values that could be stored for the corresponding OID. |

## Used by

network devices outputs snmpOutputs rawOidEntries

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - outputs: Outputs
      - snmpOutputs: Bag< OidOutput >
        - rawOidEntries: Bag< OidEntry >

**network** **endpoints** **snmpOutputs** **rawOidEntries**

See also

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - snmpOutputs: Bag< [OidOutput](#) >
      - rawOidEntries: Bag< [OidEntry](#) >

**network** **externalSources** **snmpSources** **outputs** **rawOidEntries**

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - snmpSources: Bag< [SnmpSource](#) >
      - outputs: Bag< [OidOutput](#) >
        - rawOidEntries: Bag< [OidEntry](#) >

.



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- OidOutput

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# OidOutput

## Fields

| name          | type                                | nullable | description                                                                                                                                                      |
|---------------|-------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requestedOid  | String                              | false    | The prefix of the OID hierarchy to be returned in the SNMP response. For example, with a full OID of "1.3.6.1.2.1.2.2.1.1.1", this could be "1.3.6.1.2.1.2.2.1". |
| alias         | String                              | false    | --                                                                                                                                                               |
| status        | <a href="#">OidCollectionStatus</a> | false    | --                                                                                                                                                               |
| rawOidEntries | Bag< <a href="#">OidEntry</a> >     | false    | --                                                                                                                                                               |

## Used by

network devices outputs snmpOutputs

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)
      - snmpOutputs: Bag< [OidOutput](#) >

## network [endpoints](#) [snmpOutputs](#)

See also

Data model path

- network: [Network](#)
  - endpoints: Bag< [NetworkEndpoint](#) >
    - snmpOutputs: Bag< [OidOutput](#) >

## network [externalSources](#) [snmpSources](#) [outputs](#)

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - snmpSources: Bag< [SnmpSource](#) >
      - outputs: Bag< [OidOutput](#) >

.



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- [Data Model](#)
- OperStatus

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## OperStatus

The current operational state of the interface.

## Options

| name             | description                                              |
|------------------|----------------------------------------------------------|
| DORMANT          | Waiting for some external event.                         |
| DOWN             | The interface does not pass any packets.                 |
| LOWER_LAYER_DOWN | Down due to state of lower-layer interface(s).           |
| NOT_PRESENT      | Some component (typically hardware) is missing.          |
| TESTING          | In some test mode. No operational packets can be passed. |
| UNKNOWN          | Status cannot be determined for some reason.             |
| UP               | Ready to pass packets.                                   |

## Used by

**network** **devices** **interfaces** **operStatus**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - operStatus: [OperStatus](#)

**network** **devices** **interfaces** **subinterfaces** **operStatus**

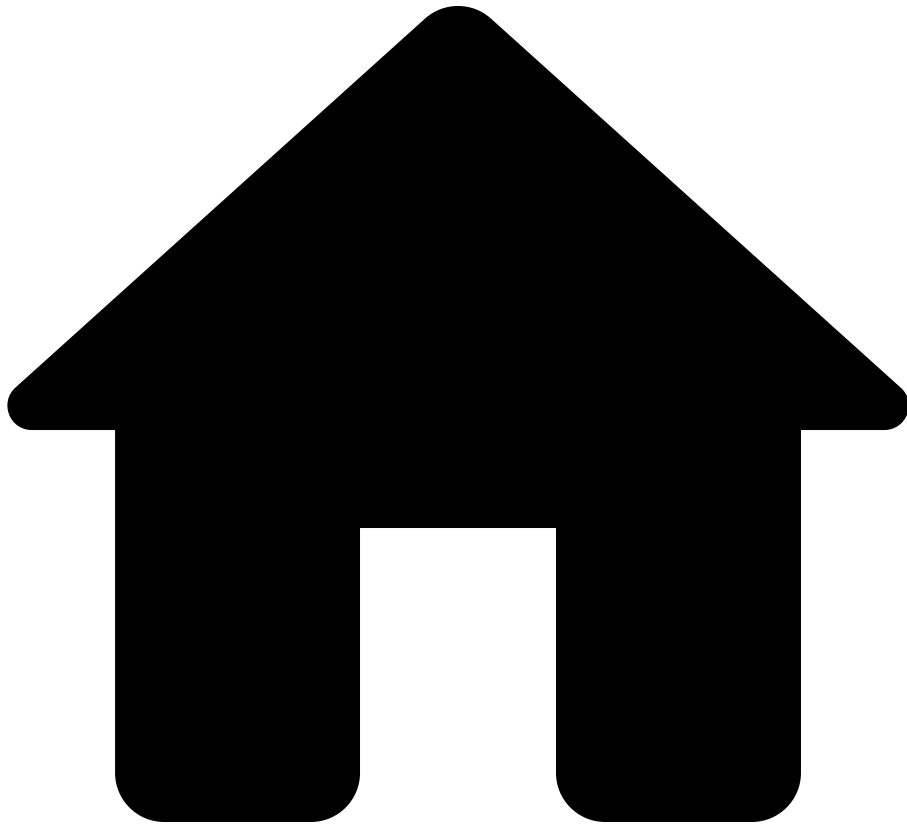
See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - subinterfaces: Bag< [SubInterface](#) >
        - operStatus: [OperStatus](#)



.



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- [Data Model](#)
- `OrganizationalUnit`

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## OrganizationalUnit

Defines a data model for representing NQE cloud account organizational unit.

## Fields

| name         | type   | nullable | description                                          |
|--------------|--------|----------|------------------------------------------------------|
| id           | String | false    | Organizational unit unique ID                        |
| name         | String | false    | Organizational unit name                             |
| parentUnitId | String | true     | Parent organizational unit ID or null for a root one |

## Used by

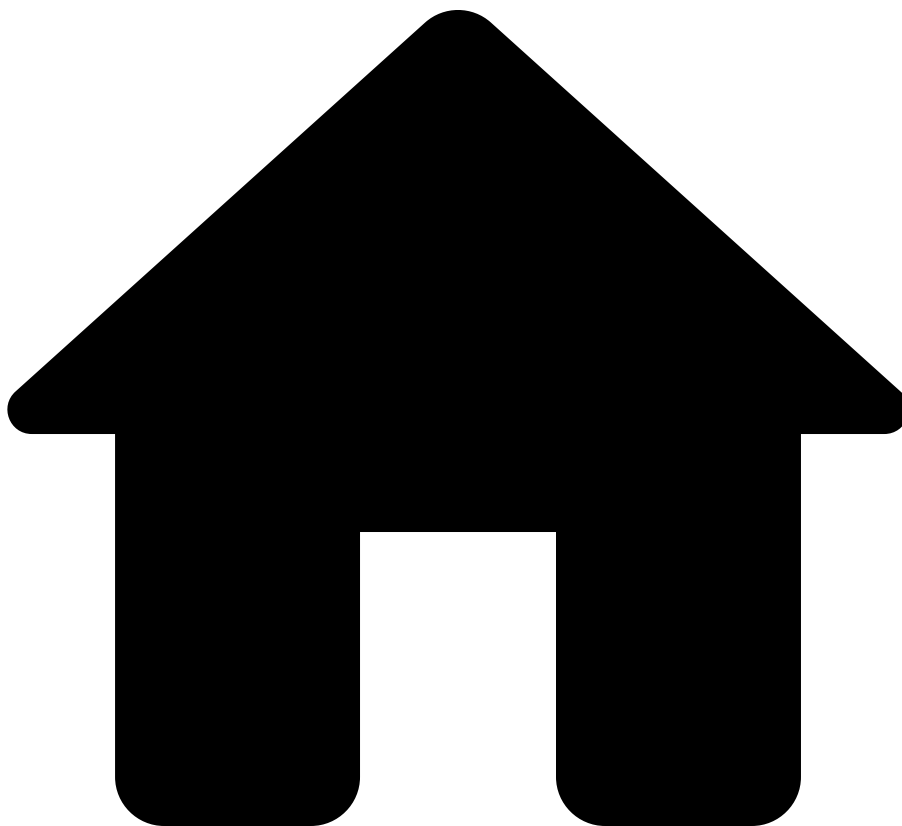
network cloudOrganizationalUnits

See also

Data model path

- network: Network
  - cloudOrganizationalUnits: Bag< OrganizationalUnit >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OriginProtocol

On this page

# OriginProtocol

## Optionsââ

| name             | description                                          |
|------------------|------------------------------------------------------|
| BGP              | Border Gateway Protocol                              |
| DIRECT_CONNECTED | A directly connected route.                          |
| EIGRP            | Enhanced Interior Gateway Routing Protocol.          |
| FRR              | Fast Re-Route protocol.                              |
| IGMP             | Internet Group Management Protocol.                  |
| ISIS             | Intermediate System to Intermediate System protocol. |
| LDP              | Label Distribution Protocol.                         |
| LOCAL_AGGREGATE  | Locally defined aggregate route.                     |
| ODR              | On Demand Routing protocol.                          |
| OSPF             | Open Shortest Path First protocol.                   |
| PIM              | Protocol Independent Multicast.                      |
| RIP              | Routing Information Protocol.                        |
| RSVP             | Resource Reservation Protocol.                       |
| STATIC           | Locally-installed static route.                      |
| UNKNOWN          | Origin protocol unknown.                             |
| VPLS             | Virtual Private LAN Service protocol.                |
| VPN              | Virtual Private Network protocol.                    |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts** [â](#) **ipv4Unicast** [â](#) **ipEntries** [â](#) **nextHops** [â](#) **originProtocol**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv4Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >
            - nextHops: Bag< [NextHop](#) >
              - originProtocol: [OriginProtocol](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **afts** [â](#) **ipv6Unicast** [â](#) **ipEntries** [â](#) **nextHops** [â](#) **originProtocol**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - afts: [Afts](#)
        - ipv6Unicast: [IpUnicast](#)
          - ipEntries: Bag< [IpEntry](#) >
            - nextHops: Bag< [NextHop](#) >
              - originProtocol: [OriginProtocol](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OS

On this page

# OS

## Options

| name           | description |
|----------------|-------------|
| ACOS           | --          |
| ALKIRA_CXP     | --          |
| APIC           | --          |
| ARISTA_EOS     | --          |
| ARUBA_AOS_CX   | --          |
| ARUBA_SWITCH   | --          |
| ARUBA_WIFI     | --          |
| ASA            | --          |
| AVAYA          | --          |
| AVAYA_ERS      | --          |
| AVI_VANTAGE    | --          |
| BLUECAT        | --          |
| BLUECOAT       | --          |
| BROCADE_SWITCH | --          |
| CHECKPOINT     | --          |
| CISCO_WIRELESS | --          |
| CLOUD_GENIX    | --          |
| CUMULUS        | --          |
| DELL_OS10      | --          |
| DELL_OS6       | --          |

| name                       | description |
|----------------------------|-------------|
| DELL_OS9                   | --          |
| DELL_SONIC                 | --          |
| EDGE_CORE_SONIC            | --          |
| ENCRYPTOR                  | --          |
| ENCS_NFV                   | --          |
| ESXI                       | --          |
| EXTREME_NOS                | --          |
| F5                         | --          |
| F5_OS_HYPERVISOR           | --          |
| FORCEPOINT                 | --          |
| FORTINET                   | --          |
| FXOS                       | --          |
| GENERAL_DYNAMICS_ENCRYPTOR | --          |
| HP_COMWARE                 | --          |
| HP_PROVISION               | --          |
| HUAWEI                     | --          |
| INTERNET_SERVICE           | --          |
| INTRANET_SERVICE           | --          |
| IOS                        | --          |
| IOS_XE                     | --          |
| IOS_XR                     | --          |
| JUNOS                      | --          |
| L2_VPN_SERVICE             | --          |
| LINUX                      | --          |



| name              | description |
|-------------------|-------------|
| LINUX_OVS_OFCTL   | --          |
| MERAKI_MR         | --          |
| MERAKI_MS         | --          |
| MERAKI_MX         | --          |
| MPLS_VPN_SERVICE  | --          |
| NETSCALER         | --          |
| NETSCREEN         | --          |
| NOKIA             | --          |
| NXOS              | --          |
| OTHER             | --          |
| PAN_OS            | --          |
| PENSANDO          | --          |
| PICA8_OVS_OFCTL   | --          |
| SG                | --          |
| SILVER_PEAK       | --          |
| SRX               | --          |
| T128              | --          |
| UNKNOWN           | --          |
| VELOCLOUD_EDGE    | --          |
| VELOCLOUD_GATEWAY | --          |
| VERSA             | --          |
| VIASAT_ENCRYPTOR  | --          |
| VIPTELA           | --          |

## Used by

**network** **devices** **platform** **os**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - platform: [Platform](#)
      - os: [OS](#)

**network** **cveDatabase** **cves** **criteria** **osCriteria** **os**

See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - criteria: Bag< [CveProductCriteria](#) >
        - osCriteria: Bag< [CveOsCriteria](#) >
          - os: [OS](#)

**network** **cveDatabase** **cves** **vendorInfos** **criteria** **osCriteria** **os**

See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - vendorInfos: Bag< [VendorCveInfo](#) >
        - criteria: Bag< [CveProductCriteria](#) >
          - osCriteria: Bag< [CveOsCriteria](#) >
            - os: [OS](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Ospf

On this page

# Ospf

## Fields

| name  | type                            | nullable | description |
|-------|---------------------------------|----------|-------------|
| areas | Bag< <a href="#">OspfArea</a> > | false    | --          |

## Used by

network devices networkInstances protocols ospf

## See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - ospf: [Ospf](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OspfArea

On this page

# OspfArea

## Fields

| name       | type                                | nullable | description                                                                                                                   |
|------------|-------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------|
| id         | Integer                             | false    | The area identifier.                                                                                                          |
| domain     | String                              | true     | The routing domain this area is part of. Absent if the device has no inferred neighbors and is not part of the OSPF topology. |
| processId  | String                              | true     | The ID of the process in which this area is configured. Absent if the device does not support multiple OSPF processes.        |
| lsaCount   | Integer                             | true     | The number of link state advertisements in this area. Absent if not reported by the device.                                   |
| areaType   | <a href="#">OspfAreaType</a>        | false    | --                                                                                                                            |
| routerType | <a href="#">OspfRouterType</a>      | true     | The router type. Absent if it is not a area border or AS boundary router.                                                     |
| neighbors  | Bag< <a href="#">OspfNeighbor</a> > | false    | OSPF neighbors in this area.                                                                                                  |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **protocols** [â](#) **ospf** [â](#) **areas**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - ospf: [Ospf](#)
          - areas: Bag< [OspfArea](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OspfAreaType

On this page

# OspfAreaType

## Options

| name           | description |
|----------------|-------------|
| BACKBONE       | --          |
| NSSA           | --          |
| STANDARD       | --          |
| STUB           | --          |
| TOTALLY_NSSA   | --          |
| TOTALLY_STUBBY | --          |

## Used by

network devices networkInstances protocols ospf areas areaType

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - ospf: [Ospf](#)
          - areas: Bag< [OspfArea](#) >
            - areaType: [OspfAreaType](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OspfNeighbor

On this page

# OspfNeighbor

## Fields

| name              | type                           | nullable | description                                                   |
|-------------------|--------------------------------|----------|---------------------------------------------------------------|
| localInterface    | String                         | false    | --                                                            |
| remoteInterfaceIp | IpSubnet                       | false    | The IP of the neighbor's interface.                           |
| remoteRouterId    | IpAddress                      | false    | The neighbor's router ID.                                     |
| cost              | Integer                        | false    | The configured cost of traffic egressing the local interface. |
| role              | <a href="#">OspfRole</a>       | true     | The local device's role, or absent if it has none.            |
| remotePeer        | <a href="#">OspfRemotePeer</a> | true     | The remote peer, if one was inferred. Otherwise, absent.      |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **protocols** [â](#) **ospf** [â](#) **areas** [â](#) **neighbors**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - ospf: [Ospf](#)
          - areas: Bag< [OspfArea](#) >
            - neighbors: Bag< [OspfNeighbor](#) >

.



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- [NQE Language](#)
- [Data Model](#)
- OspfRemotePeer

On this page

# OspfRemotePeer

## Fields

| name       | type    | nullable | description                                                                                                                          |
|------------|---------|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| deviceName | String  | false    | The device name of the remote peer.                                                                                                  |
| interface  | String  | false    | The interface of the remote peer to which a device connects.                                                                         |
| processId  | String  | true     | The process on the remote device under which this peer is configured. Absent if the device does not support multiple OSPF processes. |
| cost       | Integer | false    | The configured cost of traffic egressing the remote peer's interface.                                                                |

## Used by

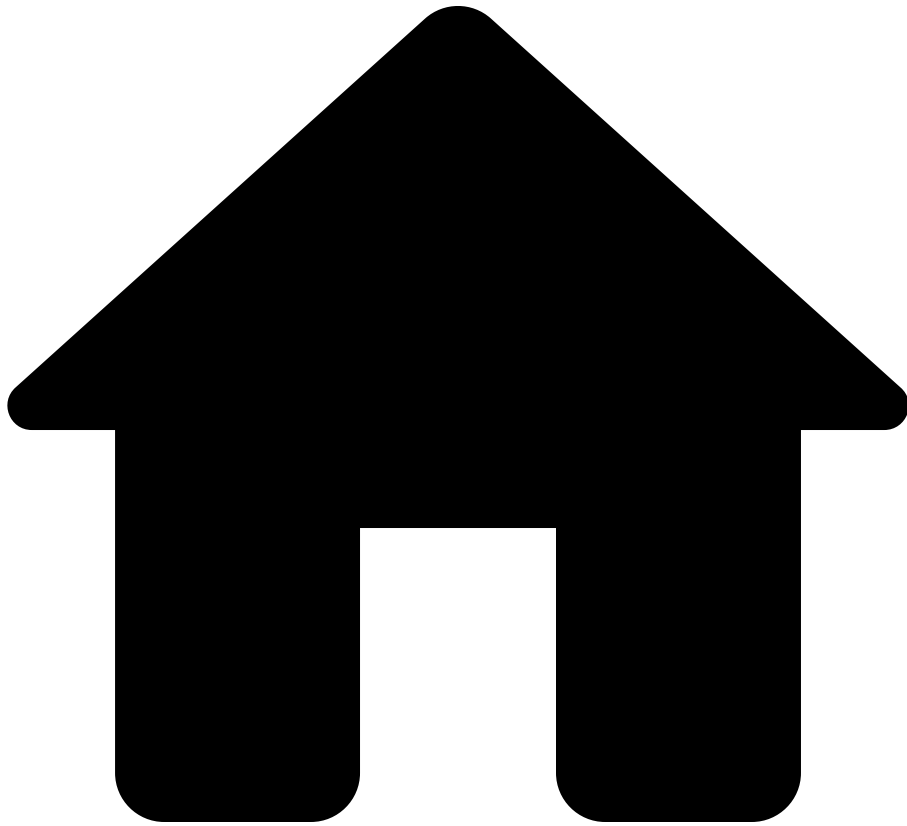
**network** **devices** **networkInstances** **protocols** **ospf** **areas** **neighbors** **remotePeer**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - ospf: [Ospf](#)
          - areas: Bag< [OspfArea](#) >
            - neighbors: Bag< [OspfNeighbor](#) >
              - remotePeer: [OspfRemotePeer](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OspfRole

On this page



# OspfRole

## Options[â](#)

| name                     | description |
|--------------------------|-------------|
| BACKUP_DESIGNATED_ROUTER | --          |
| DESIGNATED_ROUTER        | --          |

## Used by[â](#)

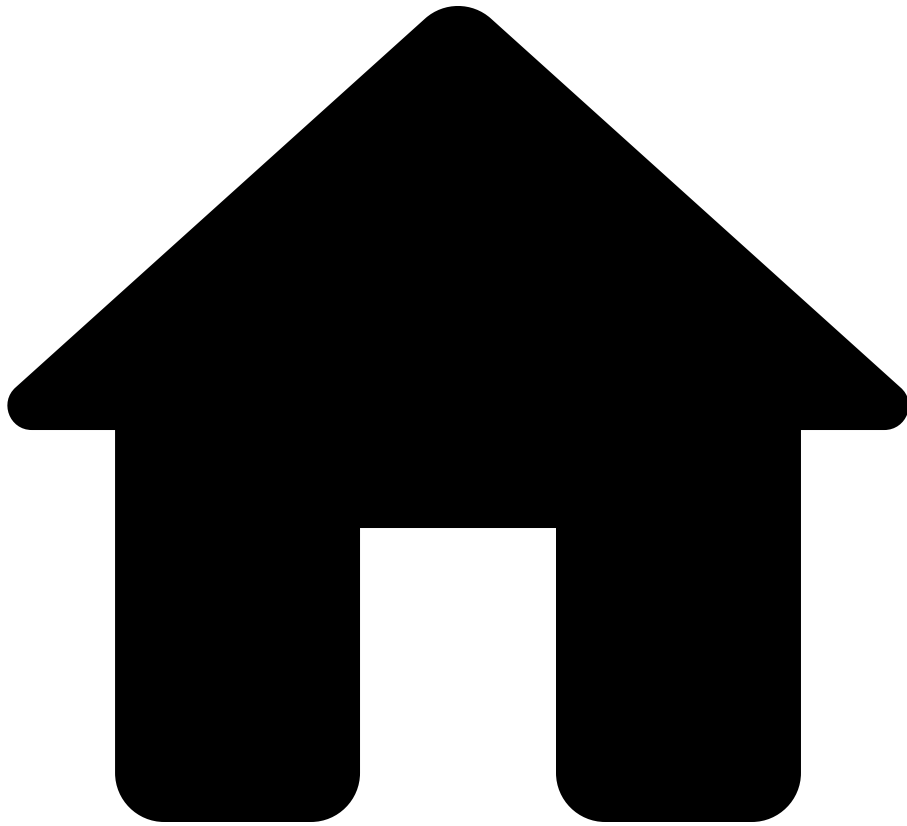
**network** [â](#) **devices** [â](#) **networkInstances** [â](#) **protocols** [â](#) **ospf** [â](#)  
**areas** [â](#) **neighbors** [â](#) **role**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - ospf: [Ospf](#)
          - areas: Bag< [OspfArea](#) >
            - neighbors: Bag< [OspfNeighbor](#) >
              - role: [OspfRole](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OspfRouterType

On this page

# OspfRouterType

## Options[↗](#)

| name     | description                                                  |
|----------|--------------------------------------------------------------|
| ABR      | An area border router.                                       |
| ABR_ASBR | An area border router and autonomous system boundary router. |
| ASBR     | An autonomous sytem boundary router.                         |

## Used by[↗](#)

**network** [↗](#) **devices** [↗](#) **networkInstances** [↗](#) **protocols** [↗](#) **ospf** [↗](#)  
**areas** [↗](#) **routerType**[↗](#)

See also[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - ospf: [Ospf](#)
          - areas: Bag< [OspfArea](#) >
            - routerType: [OspfRouterType](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OsSupport

On this page

## OsSupport

Contains dates for some of the OS support milestones of this device.

## Fields

| name                  | type   | nullable | description                                                                                                               |
|-----------------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------|
| announcementUrl       | String | false    | --                                                                                                                        |
| lastMaintenanceDate   | Date   | false    | The last date with maintenance support for the OS of this device.                                                         |
| lastVulnerabilityDate | Date   | true     | The last date with vulnerability support for the OS of this device. Absent if this date is not given in the announcement. |
| lastSupportDate       | Date   | false    | The last date with any support for the OS of this device.                                                                 |

## Used by

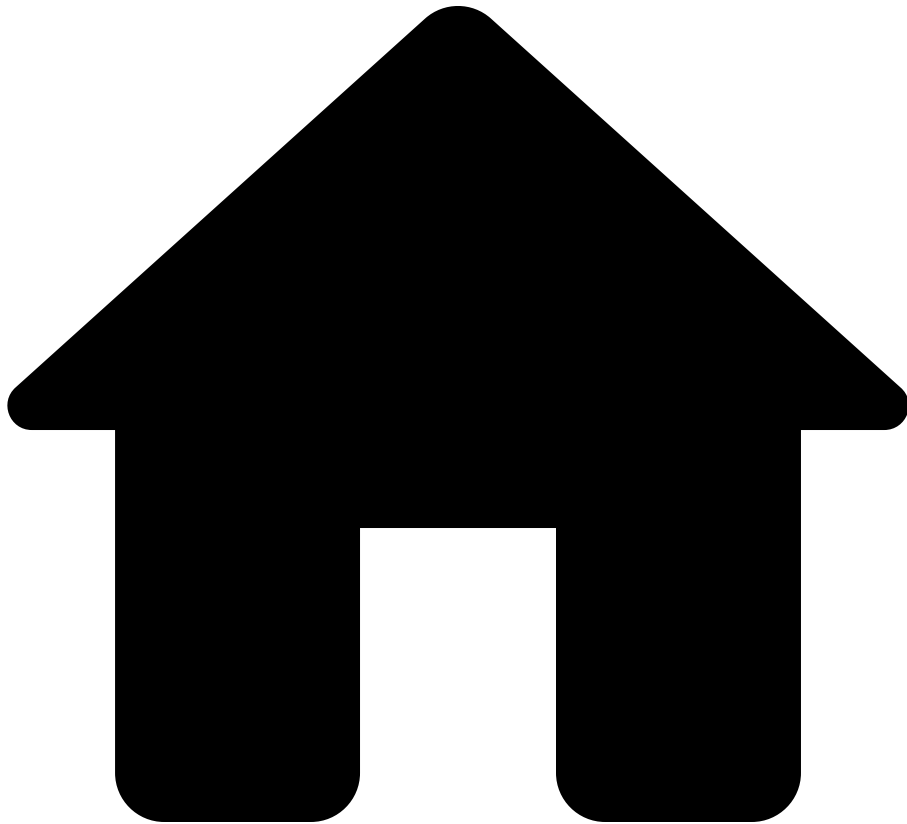
network devices platform osSupport

### See also

Data model path

- network: Network
  - devices: Bag< Device >
    - platform: Platform
      - osSupport: OsSupport

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- OutboundRule

On this page

# OutboundRule

## Fields

| name           | type            | nullable | description |
|----------------|-----------------|----------|-------------|
| privateSubnets | Bag< IpSubnet > | false    | --          |
| publicSubnets  | Bag< IpSubnet > | false    | --          |
| protocol       | Integer         | true     | --          |

## Used by

network [cloudAccounts](#) [vpcs](#) [loadBalancers](#) [outboundRules](#)

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - loadBalancers: Bag< [LoadBalancer](#) >
        - outboundRules: Bag< [OutboundRule](#) >

network [cloudAccounts](#) [externalLoadBalancers](#) [outboundRules](#)

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - externalLoadBalancers: Bag< [LoadBalancer](#) >
      - outboundRules: Bag< [OutboundRule](#) >





.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Outputs

On this page

# Outputs

## Fields

| name        | type                             | nullable | description                                       |
|-------------|----------------------------------|----------|---------------------------------------------------|
| commands    | Bag< <a href="#">Command</a> >   | false    | --                                                |
| snmpOutputs | Bag< <a href="#">OidOutput</a> > | false    | Collected custom SNMP OID outputs for the device. |
| bluecat     | <a href="#">BluecatOutputs</a>   | true     | --                                                |

## Used by

network devices outputs

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - outputs: [Outputs](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PaginatedHttpEndpoint

On this page

# PaginatedHttpEndpoint

## Fields

| name   | type                                        | nullable | description            |
|--------|---------------------------------------------|----------|------------------------|
| name   | String                                      | false    | --                     |
| status | <a href="#">PaginatedHttpEndpointStatus</a> | false    | --                     |
| error  | String                                      | true     | Absent if status is OK |
| pages  | Bag< <a href="#">HttpPage</a> >             | false    | --                     |

## Used by

network [externalSources](#) [httpSources](#) [paginatedEndpoints](#)

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - httpSources: Bag< [HttpSource](#) >
      - paginatedEndpoints: Bag< [PaginatedHttpEndpoint](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PaginatedHttpEndpointStatus

On this page

# PaginatedHttpEndpointStatus

## Options

| name              | description                                                                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ERROR             | The collection was terminated due to an error retrieving a page.                                                                                                                                   |
| MAX_PAGES_REACHED | The collection was terminated after the number of collected pages reached the max number of pages (defined for the endpoint) before the stopping condition (defined for the endpoint) was reached. |
| OK                | All pages of the endpoint were collected successfully.                                                                                                                                             |
| TIMED_OUT         | The collection was terminated after the maximum time allowed for collection was exceeded.                                                                                                          |

## Used by

**network** **externalSources** **httpSources** **paginatedEndpoints** **status**

See also

Data model path

- network: [Network](#)
  - externalSources: [ExternalSources](#)
    - httpSources: Bag< [HttpSource](#) >
      - paginatedEndpoints: Bag< [PaginatedHttpEndpoint](#) >
        - status: [PaginatedHttpEndpointStatus](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PanosFiles

On this page

# PanosFiles

## Fields

| name           | type                               | nullable | description                    |
|----------------|------------------------------------|----------|--------------------------------|
| configTemplate | List< <a href="#">ConfigLine</a> > | false    | Lines of template config.      |
| configPolicy   | List< <a href="#">ConfigLine</a> > | false    | Lines of shared policy config. |

## Used by

network devices files panos

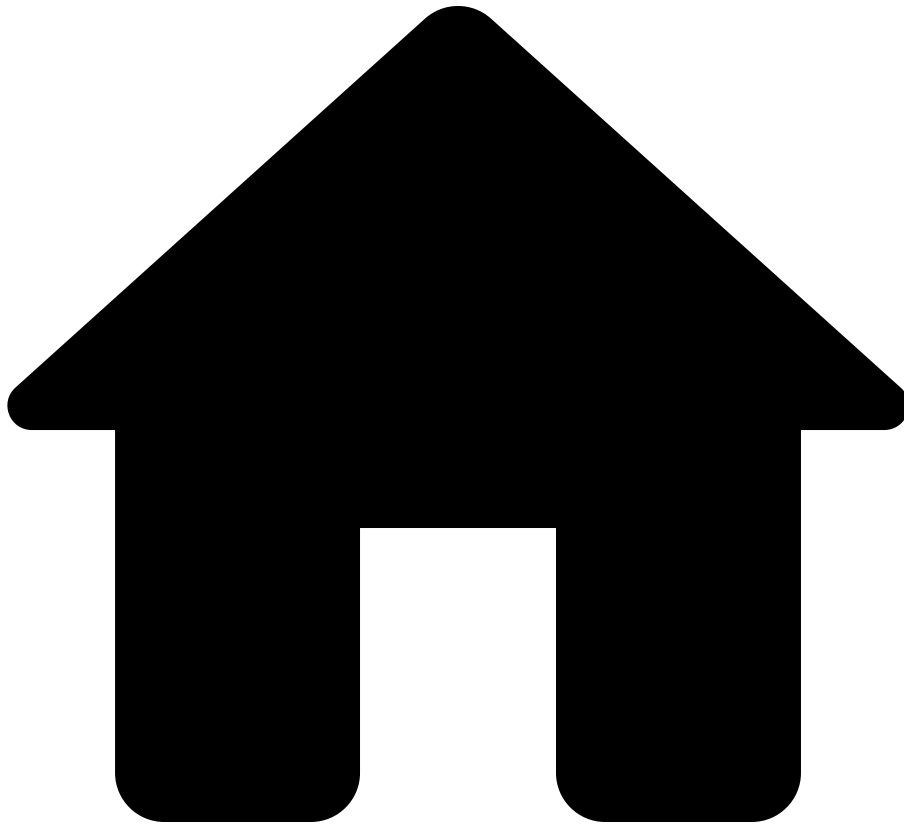
See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - files: [Files](#)
      - panos: [PanosFiles](#)



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Path

On this page

## Path

Represents a path (series of hops) from one device to another.

## Fields

| name | type        | nullable | description |
|------|-------------|----------|-------------|
| hops | List< Hop > | false    | --          |

## Used by

### path

### See also

Data model path

- path: Path

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PbrAction

On this page

## PbrAction

PBR action data will be added in upcoming releases

## Fields[↗](#)

| name | type | nullable | description |
|------|------|----------|-------------|
|------|------|----------|-------------|

## Used by[↗](#)

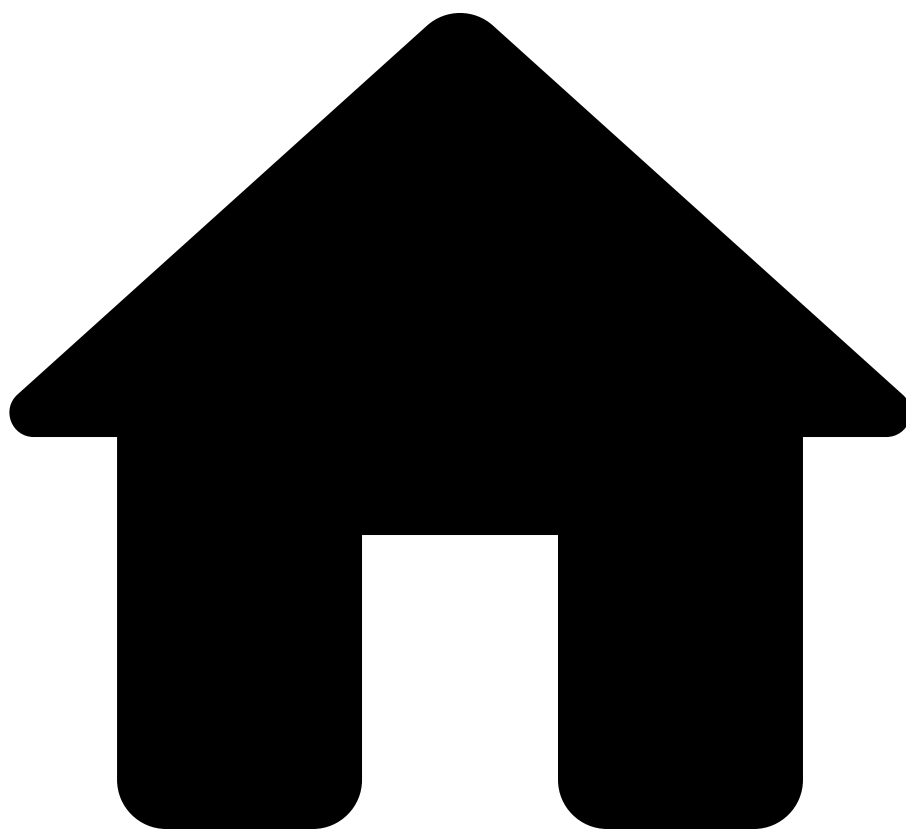
**network** [↗](#) **devices** [↗](#) **aclEntries** [↗](#) **action**[↗](#)

## See also[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - aclEntries: Bag< [AclEntry](#) >
      - action: [AclAction](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PeerLink

On this page

# PeerLink

## Fields

| name        | type           | nullable | description |
|-------------|----------------|----------|-------------|
| portName    | String         | false    | --          |
| isUp        | Bool           | false    | --          |
| activeVlans | Bag< Integer > | false    | --          |

## Used by

network devices ha vpc peerLinkStatus

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - ha: Ha
      - vpc: VpcDomain
        - peerLinkStatus: PeerLink

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PeerStatus

On this page

# PeerStatus

## Options

| name                | description |
|---------------------|-------------|
| ADJACENCY_OK        | --          |
| LINK_DOWN           | --          |
| LINK_NOT_CONFIGURED | --          |

## Used by

network devices ha vpc peerStatus

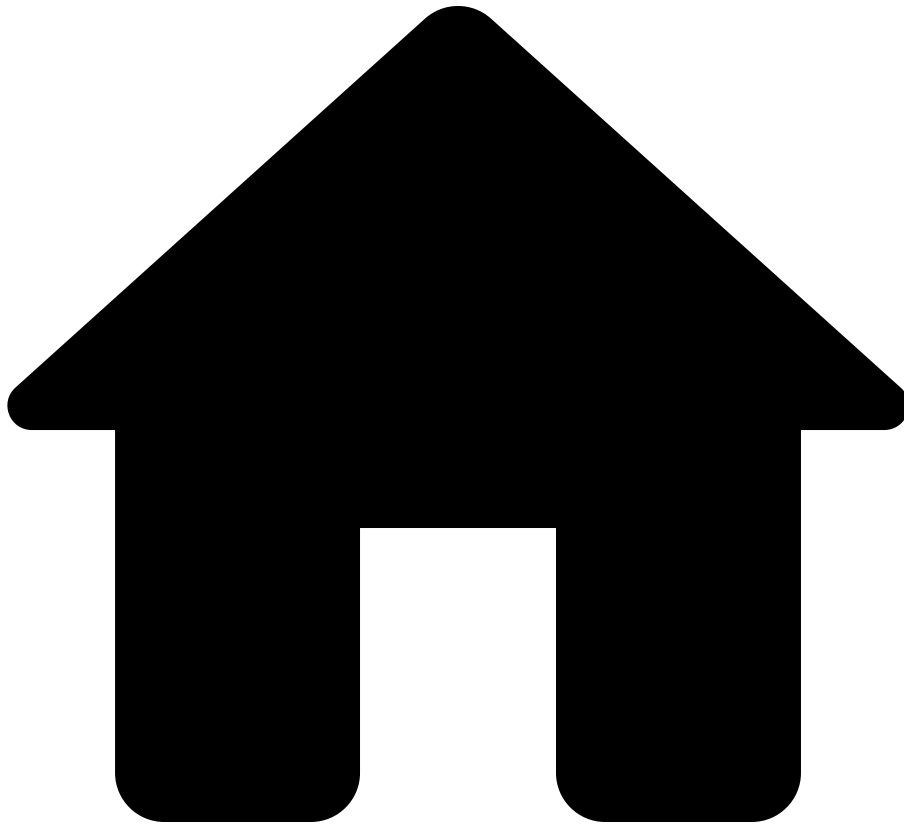
See also

Data model path

- network: Network
  - devices: Bag< Device >
    - ha: Ha
      - vpc: VpcDomain
        - peerStatus: PeerStatus



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PeerType

On this page

## PeerType

Labels a peer or peer group as explicitly internal or external.

# Options

| name     | description           |
|----------|-----------------------|
| EXTERNAL | External (eBGP) peer. |
| INTERNAL | Internal (iBGP) peer. |

## Used by

`network devices networkInstances protocols bgp neighbors peerType`

See also

Data model path

- network: `Network`
  - devices: Bag< `Device` >
    - networkInstances: Bag< `NetworkInstance` >
      - protocols: Bag< `NetworkInstanceProtocol` >
        - bgp: `Bgp`
          - neighbors: Bag< `BgpNeighbor` >
            - peerType: `PeerType`

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PhysicalLink

On this page

# PhysicalLink

## Fields

| name       | type   | nullable | description |
|------------|--------|----------|-------------|
| deviceName | String | false    | --          |
| ifaceName  | String | false    | --          |

## Used by

network devices interfaces links

### See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - links: Bag< PhysicalLink >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Platform

On this page

# Platform

## Fields

| name       | type                       | nullable | description                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|----------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| deviceType | <a href="#">DeviceType</a> | false    | --                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| vendor     | <a href="#">Vendor</a>     | false    | --                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| model      | String                     | true     | --                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| os         | <a href="#">OS</a>         | false    | --                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| osVersion  | String                     | true     | --                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| osSupport  | <a href="#">OsSupport</a>  | true     | End-of-life dates for the OS of this device.<br>Absent if the OS value is not ARISTA_EOS, CHECKPOINT, CLOUD_GENIX, F5, FORTINET, IOS, IOS_XE, IOS_XR, JUNOS, NXOS, or PAN_OS. Also absent if there is no end-of-life announcement from the vendor for this device or if such an announcement is unknown to Forward Networks. When the field lastVulnerabilityDate is present, the fields of OsSupport satisfy the following condition: lastMaintenanceDate |

| name          | type                              | nullable | description                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                                   |          | <= lastVulnerabilityDate<br><= lastSupportDate.                                                                                                                                                                                                                                                                                                     |
| components    | Bag< <a href="#">DevicePart</a> > | false    | A (non-comprehensive) list of components. On all platforms, the main device component and its serial number are available as the single component with CHASSIS partType. Other (Non-Chassis) components are provided on devices with the following OS values: ARISTA_EOS, ASA, DELL_SONIC, ENCS_NFV, F5, FXOS, IOS, IOS_XE, IOS_XR, JUNOS and NXOS. |
| managementIps | Bag< <a href="#">IpAddress</a> >  | false    | --                                                                                                                                                                                                                                                                                                                                                  |

## Used by [â](#)

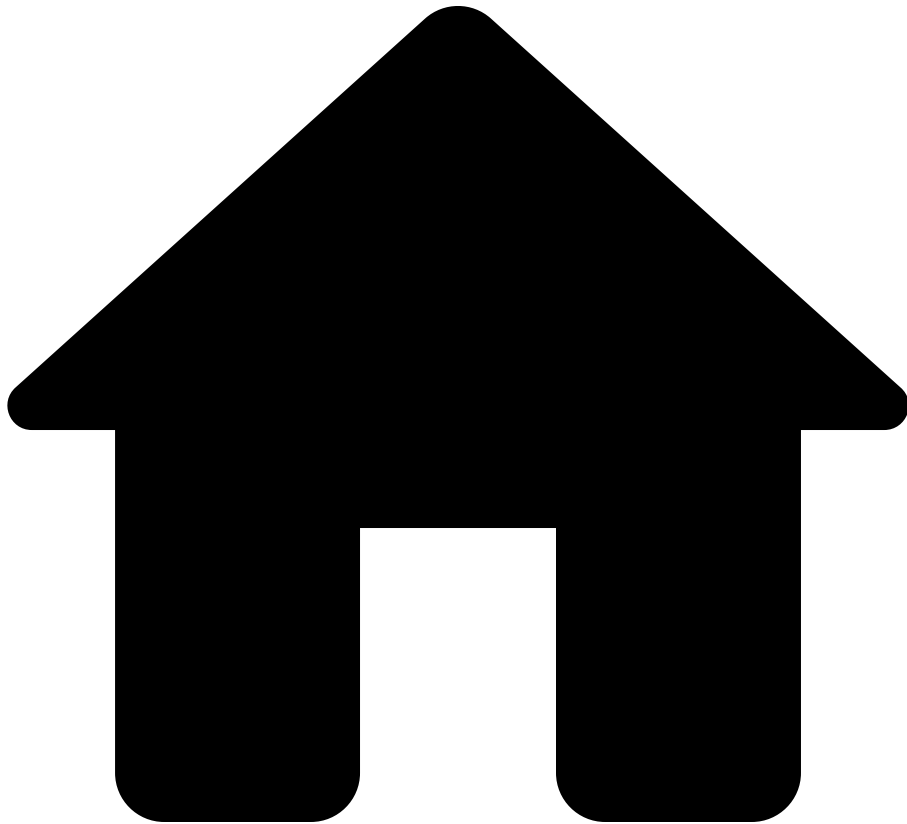
**network** [â](#) **devices** [â](#) **platform** [â](#)

**See also** [â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - platform: [Platform](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PortSpeed

On this page

## PortSpeed

Type to specify available Ethernet link speeds.



# Options

| name          | description                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------|
| SPEED_100GB   | --                                                                                                                                   |
| SPEED_100MB   | --                                                                                                                                   |
| SPEED_10GB    | --                                                                                                                                   |
| SPEED_10MB    | --                                                                                                                                   |
| SPEED_1GB     | --                                                                                                                                   |
| SPEED_2500MB  | --                                                                                                                                   |
| SPEED_25GB    | --                                                                                                                                   |
| SPEED_40GB    | --                                                                                                                                   |
| SPEED_50GB    | --                                                                                                                                   |
| SPEED_5GB     | --                                                                                                                                   |
| SPEED_UNKNOWN | Interface speed is unknown. Systems may report speed UNKNOWN when an interface is down or unpopulated (e.g., pluggable not present). |

# Used by

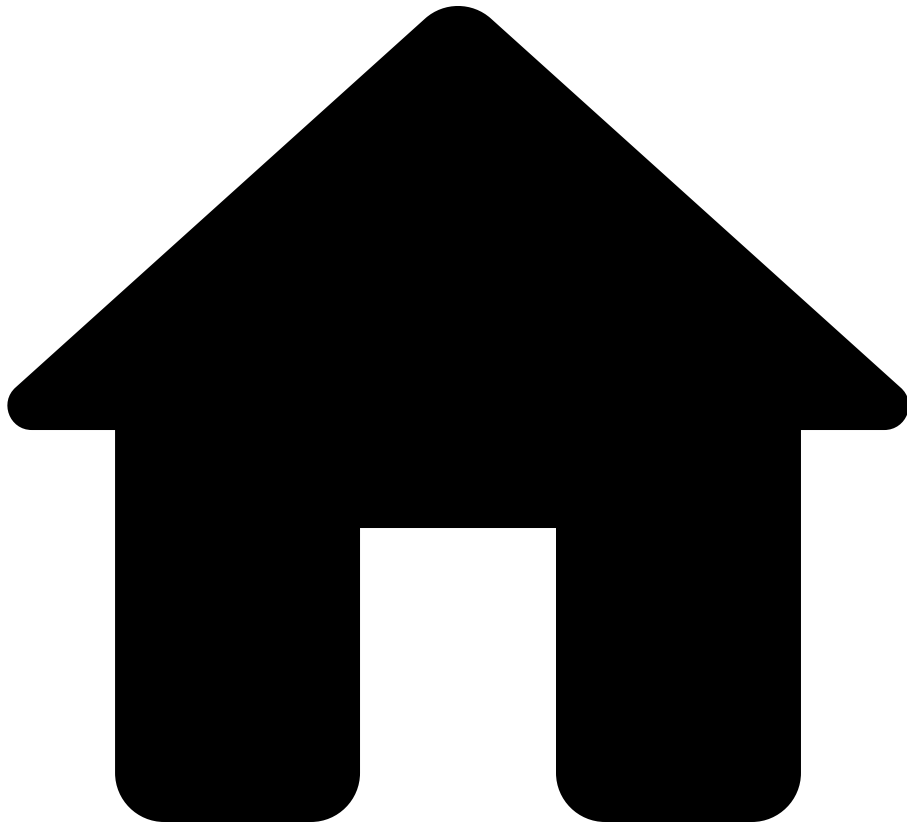
network devices interfaces ethernet negotiatedPortSpeed

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - interfaces: Bag< Iface >
      - ethernet: Ethernet
        - negotiatedPortSpeed: PortSpeed

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PublicUnallocatedIp

On this page

## PublicUnallocatedIp

A public unallocated IPv4 address and related metadata.

## Fields

| name               | type      | nullable | description                                                                                                              |
|--------------------|-----------|----------|--------------------------------------------------------------------------------------------------------------------------|
| ipAddress          | IpAddress | false    | Public unallocated IPv4 address                                                                                          |
| description        | String    | true     | The user-specified description of this IP address, or absent if not specified.                                           |
| networkBorderGroup | String    | false    | Describes a unique set of Availability Zones, Local Zones, or Wavelength Zones from which this IP address is advertised. |

## Used by

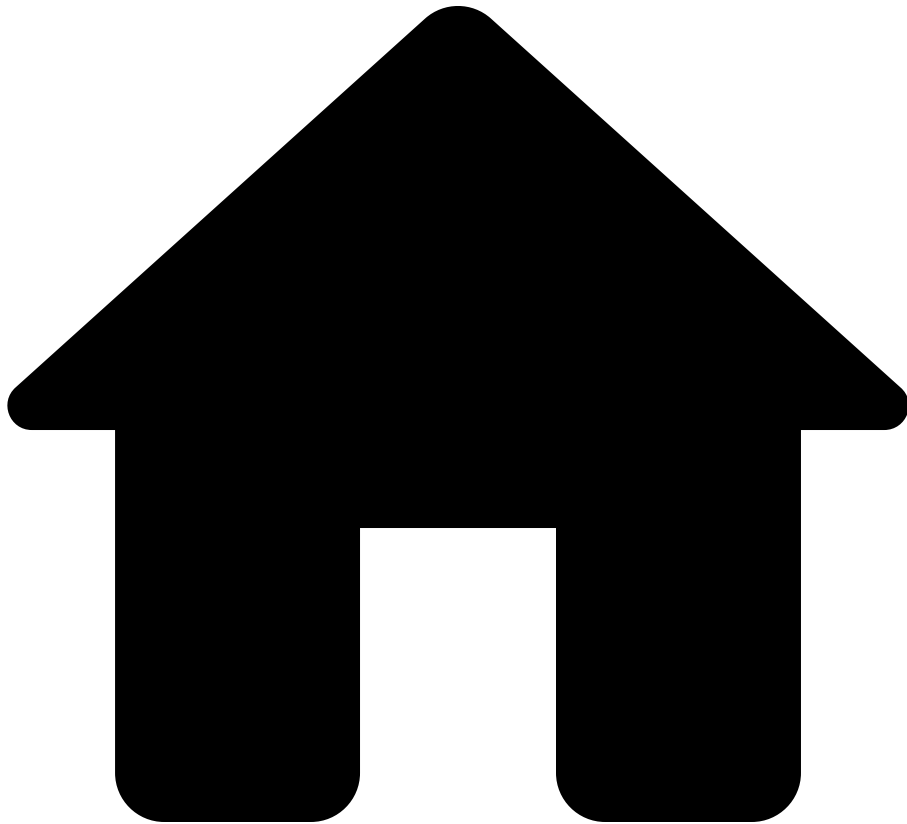
network cloudAccounts publicUnallocatedIps

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - publicUnallocatedIps: Bag< PublicUnallocatedIp >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- PublicUnallocatedIpSubnet

On this page

## PublicUnallocatedIpSubnet

A public unallocated IPv4 subnet and related metadata.

## Fields

| name               | type     | nullable | description                                                                                                             |
|--------------------|----------|----------|-------------------------------------------------------------------------------------------------------------------------|
| ipSubnet           | IpSubnet | false    | Public unallocated IPv4 subnet                                                                                          |
| description        | String   | true     | The user-specified description of this IP subnet, or absent if not specified.                                           |
| networkBorderGroup | String   | false    | Describes a unique set of Availability Zones, Local Zones, or Wavelength Zones from which this IP subnet is advertised. |

## Used by

network cloudAccounts publicUnallocatedSubnets

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - publicUnallocatedSubnets: Bag< PublicUnallocatedIpSubnet >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- QRadarConfig

On this page

# QRadarConfig

## Fields

| name   | type      | nullable | description |
|--------|-----------|----------|-------------|
| enable | Bool      | false    | --          |
| ip     | IpAddress | true     | --          |

Used by an

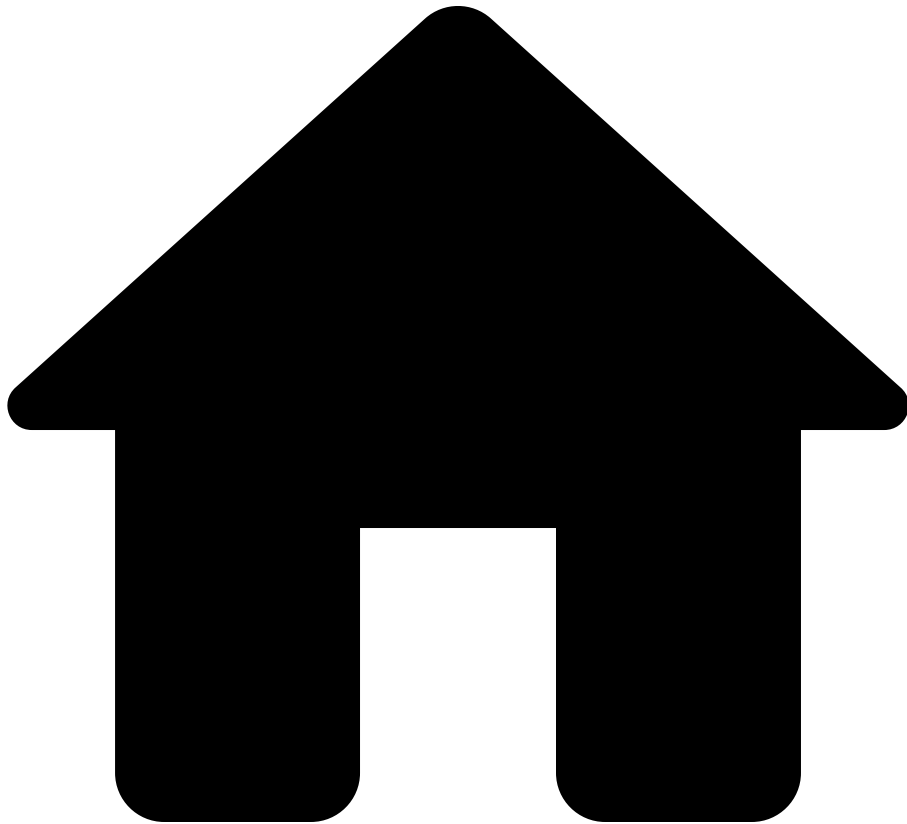
network & devices & outputs & bluecat & config & items & data  
& services & syslog & configurations & syslogConfiguration &  
qradar&

**See also**[ââ](#)

## Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RapidPvst

On this page

## RapidPvst

Rapid per vlan Spanning-tree protocol configuration and operation data.



## Fields

| name      | type                                 | nullable | description |
|-----------|--------------------------------------|----------|-------------|
| v l a n s | Bag< <a href="#">RapidPvstVlan</a> > | false    | --          |

## Used by

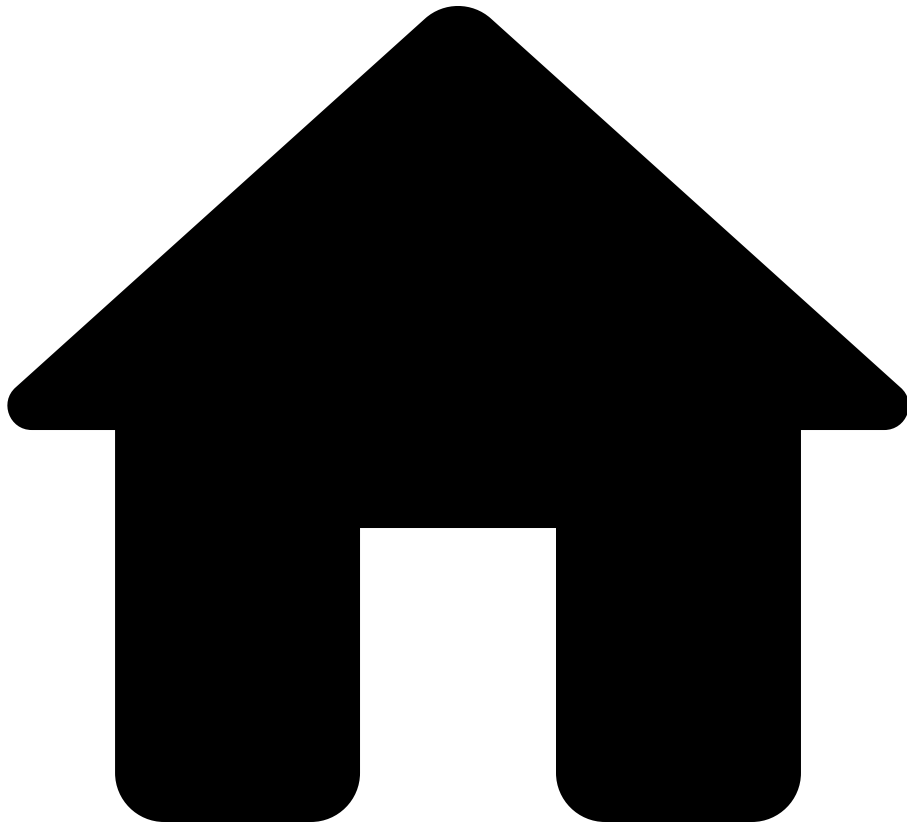
**network**  **devices**  **stp**  **rapidPvst**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - stp: [Stp](#)
      - rapidPvst: [RapidPvst](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RapidPvstVlan

On this page

## RapidPvstVlan

Rapid Spanning-tree protocol configuration and operation data.

# Fields

| name                   | type       | nullable | description                                                                                                                                                                        |
|------------------------|------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vlanId                 | Integer    | false    | Interface VLAN ID                                                                                                                                                                  |
| bridgePriority         | Integer    | true     | The manageable component of the Bridge Identifier. References: IEEE 802.1D 17.13.7 Bridge Identifier Priority.                                                                     |
| bridgeAddress          | MacAddress | true     | A unique 48-bit Universally Administered MAC Address assigned to the bridge. Reference: IEEE 802.1D 7.12.5 Unique identification of a bridge.                                      |
| designatedRootPriority | Integer    | true     | The bridge priority of the root of the spanning tree, as determined by the Spanning Tree Protocol, as executed by this node. Reference: RFC4188 BRIDGE-MIB dot1dStpDesignatedRoot. |
| designatedRootAddress  | MacAddress | true     | The bridge address of the root of the spanning tree, as determined by the Spanning Tree Protocol, as executed by this node. References: RFC4188 BRIDGE-MIB dot1dStpDesignatedRoot. |
| rootPortName           | String     | true     | The port name of the port which offers the lowest                                                                                                                                  |

| name     | type    | nullable | description                                                                                                 |
|----------|---------|----------|-------------------------------------------------------------------------------------------------------------|
|          |         |          | cost path from this bridge to the root bridge. References: RFC4188 BRIDGE-MIB dot1dStpRootPort.             |
| rootCost | Integer | true     | The cost of the path to the root as seen from this bridge. References: RFC4188 BRIDGE-MIB dot1dStpRootCost. |

## Used by

network devices stp rapidPvst vlans

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - stp: [Stp](#)
      - rapidPvst: [RapidPvst](#)
        - vlans: Bag< [RapidPvstVlan](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RefObj

On this page

# RefObj

## Fields

| name       | type       | nullable | description |
|------------|------------|----------|-------------|
| refObjType | RefObjType | false    | --          |
| refObjId   | String     | false    | --          |

## Used by

network cloudAccounts vpcs routeTables routes  
nextHop

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - vpcs: Bag< VpcData >
      - routeTables: Bag< RouteTable >
        - routes: Bag< CloudRoute >
          - nextHop: CloudRouteNextHop

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RefObjType

On this page

# RefObjType

## Options [↗](#)

| name              | description                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| ADDRESS_REFERENCE | --                                                                                                                        |
| AWS_OUTPOST_LGW   | For AWS only, a next hop with this type points to the AWS Outpost Local Gateway.                                          |
| COMP_INSTANCE     | For GCP, a next hop with this type points to a compute instance. For AWS and Azure, this next hop route type is not used. |
| DCGW              | --                                                                                                                        |
| IGW               | --                                                                                                                        |
| INTERFACE         | For AWS, a next hop with this type points to the interface of a compute instance.                                         |
| NATGW             | --                                                                                                                        |
| SUBNET            | --                                                                                                                        |
| TGW               | --                                                                                                                        |
| VPC               | --                                                                                                                        |
| VPC_ENDPOINT      | --                                                                                                                        |
| VPC_PEERING       | --                                                                                                                        |
| VPNGW             | --                                                                                                                        |
| VPN_CONNECTION    | --                                                                                                                        |



## Used by[â](#)

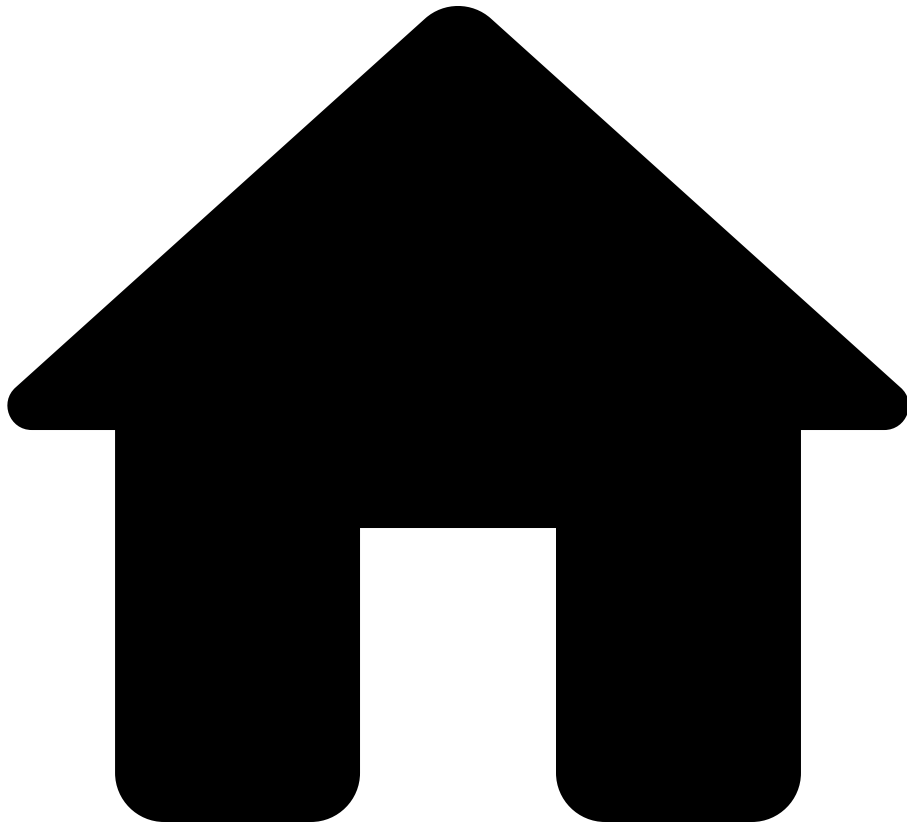
**network** [â](#) **cloudAccounts** [â](#) **vpcs** [â](#) **routeTables** [â](#) **routes** [â](#)  
**nextHop** [â](#) **refObjType**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - routeTables: Bag< [RouteTable](#) >
        - routes: Bag< [CloudRoute](#) >
          - nextHop: [CloudRouteNextHop](#)
            - refObjType: [RefObjType](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RequiresDefaultVlanInterfacePolicy

On this page

# RequiresDefaultVlanInterfacePolicy

## Fields

| name      | type                                | nullable | description                                      |
|-----------|-------------------------------------|----------|--------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                               |
| value     | Bool                                | false    | Whether the interface requires the default VLAN. |

## Used by

network [stigDatabase](#) [policy](#) [ifacePolicy](#) [requiresDefaultVlan](#)

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - requiresDefaultVlan: [Bag< RequiresDefaultVlanInterfacePolicy >](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RequiresMulticastInterfacePolicy

On this page

# RequiresMulticastInterfacePolicy

## Fields

| name      | type                                | nullable | description                                       |
|-----------|-------------------------------------|----------|---------------------------------------------------|
| condition | <a href="#">StigPolicyCondition</a> | false    | --                                                |
| value     | Bool                                | false    | Whether the interface requires multicast routing. |

## Used by

network [stigDatabase](#) [policy](#) [ifacePolicy](#) [requiresMulticast](#)

See also

Data model path

- network: [Network](#)
  - stigDatabase: [StigDatabase](#)
    - policy: [StigPolicy](#)
      - ifacePolicy: [InterfacePolicy](#)
        - requiresMulticast:  
Bag< [RequiresMulticastInterfacePolicy](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Route

On this page

# Route

## Fields

| name               | type                           | nullable | description                                                                                |
|--------------------|--------------------------------|----------|--------------------------------------------------------------------------------------------|
| vrf                | String                         | true     | Vrf for the route; may be undefined for route-reflector routes.                            |
| routeDistinguisher | String                         | true     | Route distinguisher for the route; may be undefined for some AfiSafis.                     |
| prefix             | IpSubnet                       | false    | Prefix for the route.                                                                      |
| pathAttributes     | Bag< <a href="#">AttrSet</a> > | false    | Attributes for paths for the route. More than one path may exist if multipath BGP is used. |

## Used by[â](#)

**network** [â](#) **devices** [â](#) **bgpRib** [â](#) **afiSafis** [â](#) **neighbors** [â](#) **adjRibInPost** [â](#) **routes**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibInPost: [AfiSafiNeighborAdjRib](#)
            - routes: Bag< [Route](#) >

**network** [â](#) **devices** [â](#) **bgpRib** [â](#) **afiSafis** [â](#) **neighbors** [â](#) **adjRibOutPost** [â](#) **routes**[â](#)

See also[â](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - bgpRib: [BgpRib](#)
      - afiSafis: Bag< [AfiSafi](#) >
        - neighbors: Bag< [AfiSafiNeighbor](#) >
          - adjRibOutPost: [AfiSafiNeighborAdjRib](#)
            - routes: Bag< [Route](#) >



.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RoutedVlan

On this page

# RoutedVlan

## Fields

| name                | type                          | nullable | description                                                                                                                                                                                           |
|---------------------|-------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vlan                | Integer                       | false    | References the VLAN for which this IP interface provides routing services -- similar to a switch virtual interface (SVI), or integrated routing and bridging interface (IRB) in some implementations. |
| ipv4                | <a href="#">IfaceIpv4Info</a> | false    | --                                                                                                                                                                                                    |
| ipv6                | <a href="#">IfaceIpv6Info</a> | false    | --                                                                                                                                                                                                    |
| networkInstanceName | String                        | false    | The network instance (i.e. VRF) that this interface is part of.                                                                                                                                       |

## Used by[↗](#)

**network** [↗](#) **devices** [↗](#) **interfaces** [↗](#) **routedVlan**[↗](#)

**See also**[↗](#)

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - interfaces: Bag< [Iface](#) >
      - routedVlan: [RoutedVlan](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RouteTable

On this page

# RouteTable

## Fields

| name   | type                              | nullable | description         |
|--------|-----------------------------------|----------|---------------------|
| name   | String                            | true     | Never null for GCP. |
| id     | String                            | false    | --                  |
| tags   | Bag< String >                     | false    | --                  |
| region | String                            | true     | --                  |
| routes | Bag< <a href="#">CloudRoute</a> > | false    | --                  |

## Used by

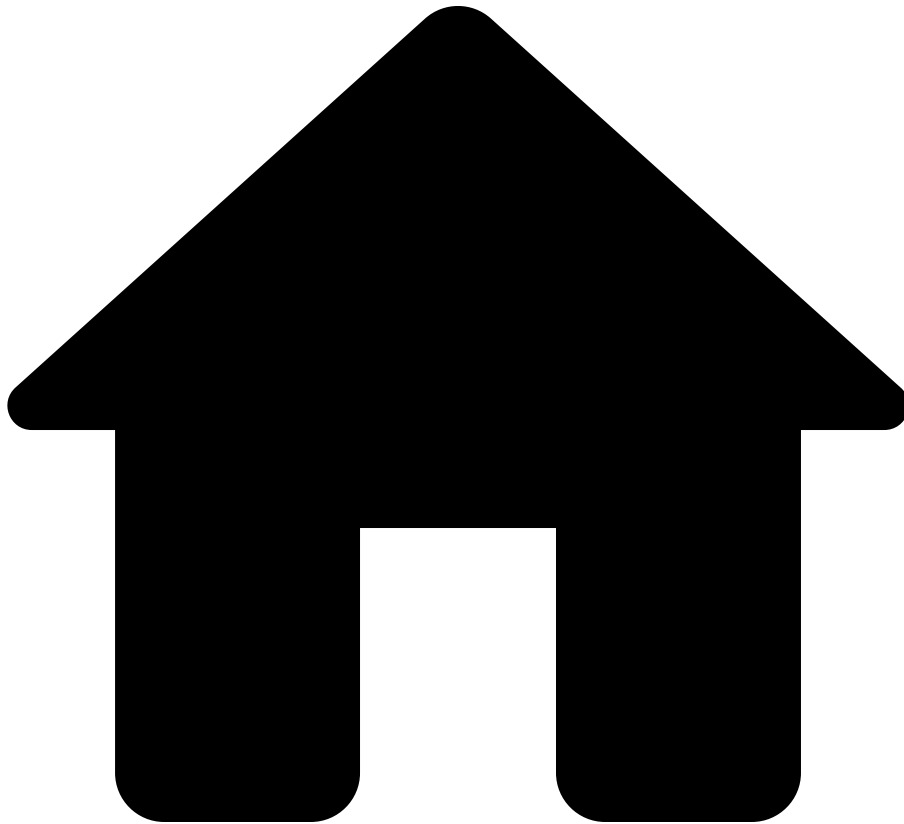
network [cloudAccounts](#) [vpcs](#) [routeTables](#)

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - routeTables: Bag< [RouteTable](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RoutingProtocolType

On this page

## RoutingProtocolType

Type for routing protocols, including those which may install prefixes into the RIB

## Options

| name               | description                        |
|--------------------|------------------------------------|
| BGP                | BGP                                |
| DIRECTLY_CONNECTED | A directly connected route         |
| IGMP               | Internet Group Management Protocol |
| ISIS               | IS-IS                              |
| LOCAL_AGGREGATE    | Locally defined aggregate route    |
| OSPF               | OSPFv2                             |
| OSPF3              | OSPFv3                             |
| PIM                | Protocol Independent Multicast     |
| STATIC             | Locally-installed static route     |

## Used by

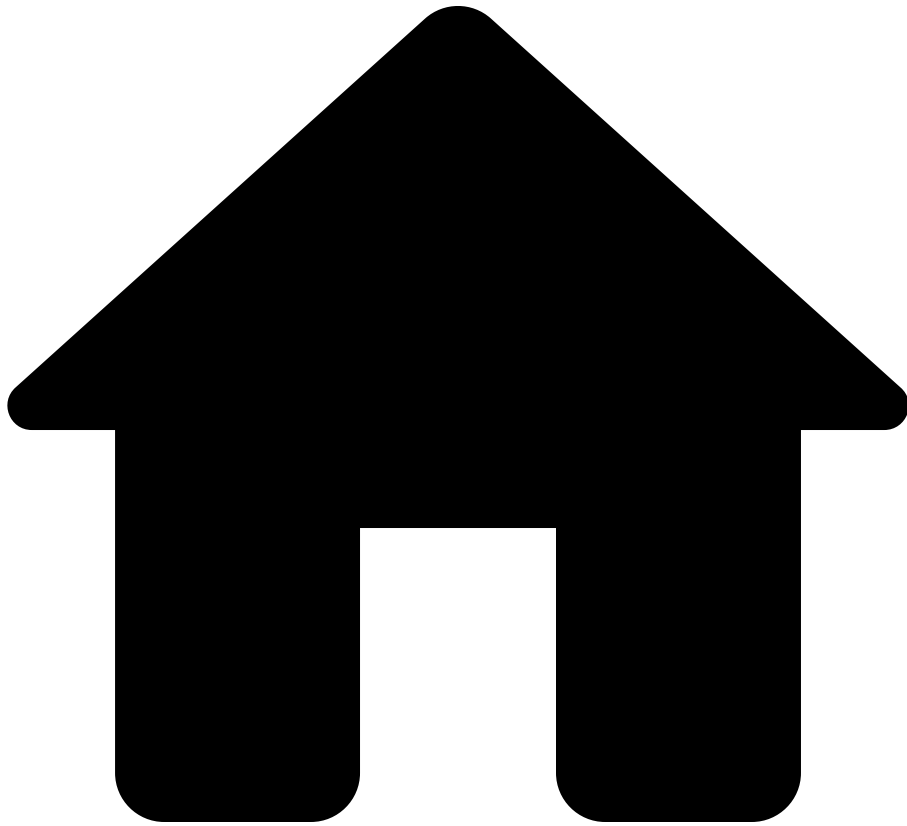
**network** **devices** **networkInstances** **protocols** **identifier**

See also

Data model path

- network: [Network](#)
  - devices: Bag< [Device](#) >
    - networkInstances: Bag< [NetworkInstance](#) >
      - protocols: Bag< [NetworkInstanceProtocol](#) >
        - identifier: [RoutingProtocolType](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- Rstp

On this page

## Rstp

Rapid Spanning-tree protocol configuration and operation data.



# Fields

| name                   | type       | nullable | description                                                                                                                                                                        |
|------------------------|------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bridgePriority         | Integer    | true     | The manageable component of the Bridge Identifier. References: IEEE 802.1D 17.13.7 Bridge Identifier Priority.                                                                     |
| bridgeAddress          | MacAddress | true     | A unique 48-bit Universally Administered MAC Address assigned to the bridge. Reference: IEEE 802.1D 7.12.5 Unique identification of a bridge.                                      |
| designatedRootPriority | Integer    | true     | The bridge priority of the root of the spanning tree, as determined by the Spanning Tree Protocol, as executed by this node. Reference: RFC4188 BRIDGE-MIB dot1dStpDesignatedRoot. |
| designatedRootAddress  | MacAddress | true     | The bridge address of the root of the spanning tree, as determined by the Spanning Tree Protocol, as executed by this node. References: RFC4188 BRIDGE-MIB dot1dStpDesignatedRoot. |
| rootPortName           | String     | true     | The port name of the port which offers the lowest cost path from this bridge to the root bridge.                                                                                   |

| name     | type    | nullable | description                                                                                                    |
|----------|---------|----------|----------------------------------------------------------------------------------------------------------------|
|          |         |          | References: RFC4188<br>BRIDGE-MIB<br>dot1dStpRootPort.                                                         |
| rootCost | Integer | true     | The cost of the path to the root as seen from this bridge. References: RFC4188 BRIDGE-MIB<br>dot1dStpRootCost. |

## Used by

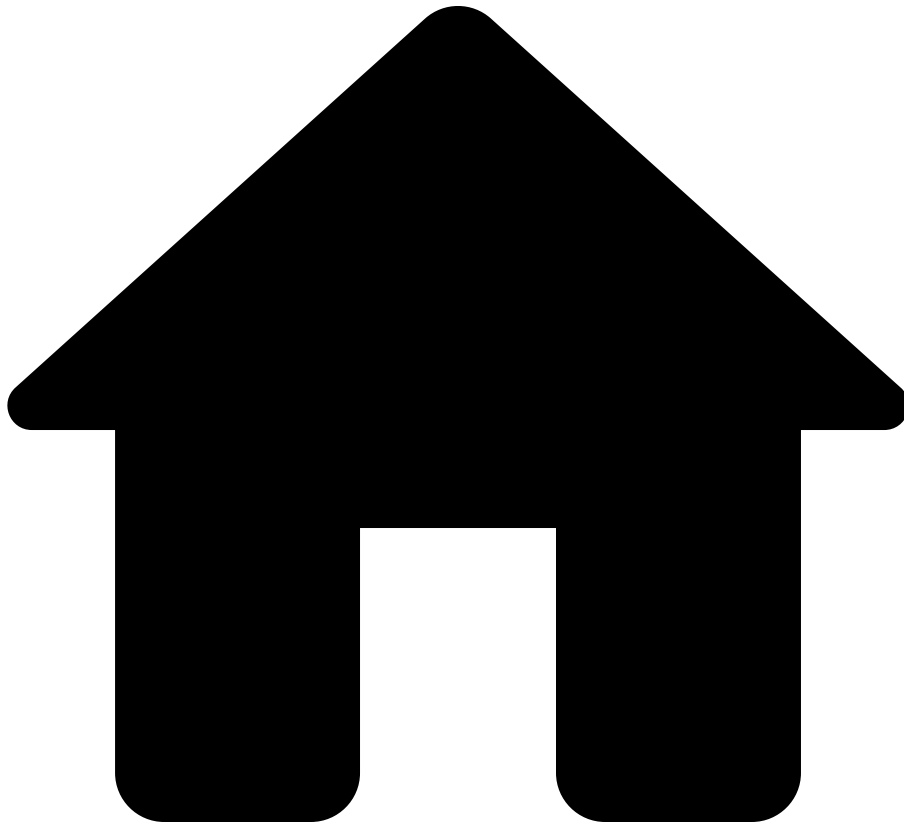
network devices stp rstp

See also

Data model path

- network: Network
  - devices: Bag< Device >
    - stp: Stp
      - rstp: Rstp

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- RuleContainer

On this page

# RuleContainer

The origin of a firewall rule.

## Fields

| name          | type          | nullable | description              |
|---------------|---------------|----------|--------------------------|
| name          | String        | false    | The name of rule origin. |
| containerType | ContainerType | false    | --                       |

## Used by

network cloudAccounts firewallPolicy ruleContainers

See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - firewallPolicy: FirewallPolicy
      - ruleContainers: Bag< RuleContainer >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- SecurityGroup

On this page

# SecurityGroup

## Fields

| name         | type                                 | nullable | description |
|--------------|--------------------------------------|----------|-------------|
| name         | String                               | true     | --          |
| id           | String                               | false    | --          |
| groupName    | String                               | true     | --          |
| description  | String                               | true     | --          |
| tags         | Bag< String >                        | false    | --          |
| ingressRules | List< <a href="#">SecurityRule</a> > | false    | --          |
| egressRules  | List< <a href="#">SecurityRule</a> > | false    | --          |

## Used by

network cloudAccounts vpcs securityGroups

### See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- SecurityRule

On this page

# SecurityRule

## Fields

| name   | type                               | nullable | description |
|--------|------------------------------------|----------|-------------|
| match  | <a href="#">HeaderRegion</a>       | false    | --          |
| action | <a href="#">SecurityRuleAction</a> | false    | --          |

## Used by

network [cloudAccounts](#) [vpcs](#) [securityGroups](#) [ingressRules](#)

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - ingressRules: Bag< [SecurityRule](#) >

network [cloudAccounts](#) [vpcs](#) [securityGroups](#) [egressRules](#)

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - egressRules: Bag< [SecurityRule](#) >



**network** **cloudAccounts** **vpcs** **networkAcls** **ingressRules**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - ingressRules: Bag< [SecurityRule](#) >

**network** **cloudAccounts** **vpcs** **networkAcls** **egressRules**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - egressRules: Bag< [SecurityRule](#) >

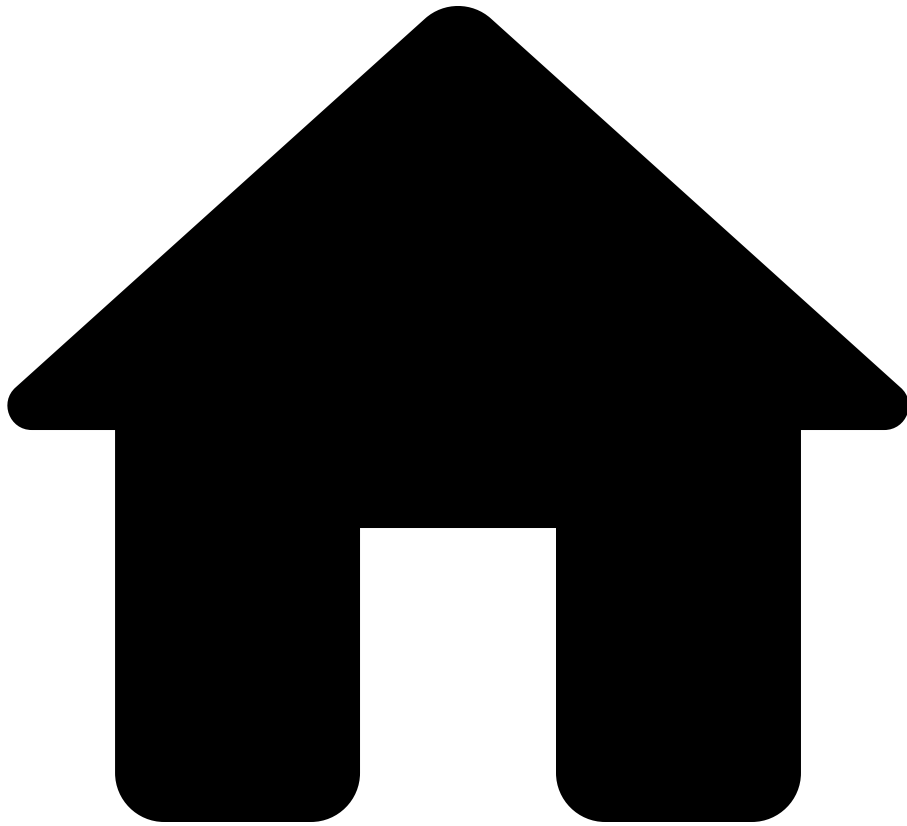
**network** **cloudAccounts** **firewallRules** **securityRule**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >
      - securityRule: [SecurityRule](#)

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- SecurityRuleAction

On this page

# SecurityRuleAction

## Options[âââ](#)

| name   | description |
|--------|-------------|
| DENY   | --          |
| PERMIT | --          |

## Used by[âââ](#)

**network** [âââ](#) **cloudAccounts** [âââ](#) **vpcs** [âââ](#) **securityGroups** [âââ](#) **ingressRules** [âââ](#) **action**[âââ](#)

See also[âââ](#)

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - action: [SecurityRuleAction](#)

**network** **cloudAccounts** **vpcs** **securityGroups** **egressRules**  
**action**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - securityGroups: Bag< [SecurityGroup](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - action: [SecurityRuleAction](#)

**network** **cloudAccounts** **vpcs** **networkAcls** **ingressRules**  
**action**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - ingressRules: Bag< [SecurityRule](#) >
          - action: [SecurityRuleAction](#)

**network** **cloudAccounts** **vpcs** **networkAcls** **egressRules** **action**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - networkAcls: Bag< [NetworkAcl](#) >
        - egressRules: Bag< [SecurityRule](#) >
          - action: [SecurityRuleAction](#)

**network** **cloudAccounts** **vpcs** **firewalls** **rules** **action**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - vpcs: Bag< [VpcData](#) >
      - firewalls: Bag< [CloudFirewall](#) >
        - rules: Bag< [CloudFirewallRule](#) >
          - action: [SecurityRuleAction](#)

**network** **cloudAccounts** **firewallRules** **securityRule** **action**

See also

Data model path

- network: [Network](#)
  - cloudAccounts: Bag< [CloudAccount](#) >
    - firewallRules: Bag< [FirewallRule](#) >
      - securityRule: [SecurityRule](#)
        - action: [SecurityRuleAction](#)



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- Reference
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- [Data Model](#)
- ServiceEndpoint

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# ServiceEndpoint

## Fields

| name                   | type            | nullable | description |
|------------------------|-----------------|----------|-------------|
| name                   | String          | true     | --          |
| id                     | String          | false    | --          |
| tags                   | Bag< String >   | false    | --          |
| region                 | String          | false    | --          |
| ipSubnets              | Bag< IpSubnet > | false    | --          |
| backendLoadBalancerIds | Bag< String >   | false    | --          |

## Used by

network cloudAccounts vpcs serviceEndpoints

## See also

Data model path

- network: Network
  - cloudAccounts: Bag< CloudAccount >
    - vpcs: Bag< VpcData >
      - serviceEndpoints: Bag< ServiceEndpoint >



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- Severity

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# Severity

## Options

| name     | description                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------|
| CRITICAL | When CVSS v3.0 Rating and CVSS v4.0 Rating is in the range 9.0 - 10.0. Not a CVSS v2.0 Rating.                                  |
| HIGH     | When the CVSS v3.0 Rating and CVSS v4.0 Rating is in the range 7.0 - 8.9. When the CVSS v2.0 Rating is in the range 7.0 - 10.0. |
| LOW      | When the CVSS v3.0 Rating and CVSS v4.0 Rating is in the range 0.1 - 3.9. When the CVSS v2.0 Rating is in the range 0.0 - 3.9.  |
| MEDIUM   | When the CVSS v3.0 Rating and CVSS v4.0 Rating is in the range 4.0 - 6.9. When the CVSS v2.0 Rating is in the range 4.0 - 6.9.  |
| NONE     | CVSS v3.0 Rating is 0.0. Not a CVSS v2.0 Rating.                                                                                |

## Used by

network cveDatabase cves severity

See also

Data model path

- network: Network
  - cveDatabase: CveDatabase
    - cves: Bag< Cve >
      - severity: Severity

**network** **cveDatabase** **cves** **vendorInfos** **severity**

See also

Data model path

- network: [Network](#)
  - cveDatabase: [CveDatabase](#)
    - cves: Bag< [Cve](#) >
      - vendorInfos: Bag< [VendorCveInfo](#) >
        - severity: [Severity](#)

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- Reference
- [NQE Language](#)
- [Data Model](#)
- SnmpAgentService

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# SnmpAgentService

## Fields

| name          | type                                   | nullable | description |
|---------------|----------------------------------------|----------|-------------|
| system        | <a href="#">SnmpAgentServiceSystem</a> | false    | --          |
| v2c           | <a href="#">SnmpAgentV2C</a>           | false    | --          |
| loglevel      | String                                 | false    | --          |
| pollingPeriod | Integer                                | false    | --          |
| v1            | <a href="#">SnmpAgentV1</a>            | false    | --          |
| v3            | <a href="#">SnmpAgentV3</a>            | false    | --          |

Used by [100](#)

network devices outputs bluecat config items data  
services snmp configurations snmpConfiguration  
agentService

## See also

## Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- SnmpAgentServiceSystem

On this page

# SnmpAgentServiceSystem

## Fields

| name        | type   | nullable | description |
|-------------|--------|----------|-------------|
| contact     | String | false    | --          |
| name        | String | false    | --          |
| description | String | false    | --          |
| location    | String | false    | --          |



Used by [1000](#)

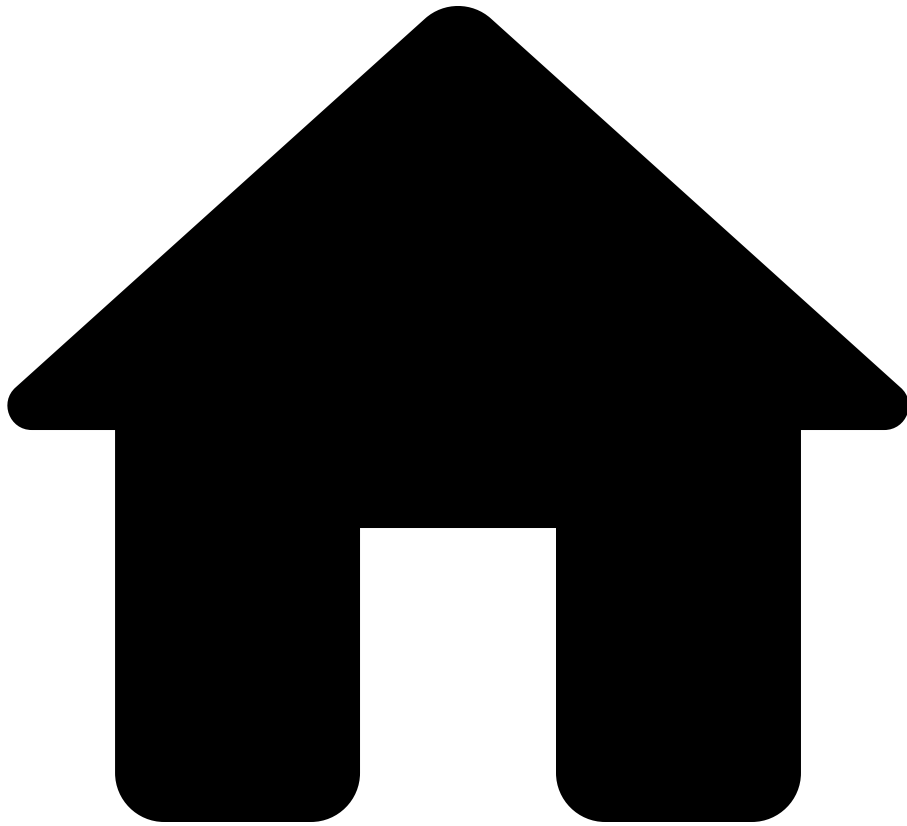
network â devices â outputs â bluecat â config â items â data  
â services â snmp â configurations â snmpConfiguration â  
agentService â system[â](#)

## See also

## Data model path

- [illegible]

.



- Reference
- [NQE Language](#)
- [Data Model](#)
- SnmpAgentV1

On this page

# SnmpAgentV1

## Fields

| name      | type   | nullable | description |
|-----------|--------|----------|-------------|
| enable    | Bool   | false    | --          |
| community | String | false    | --          |

Used by an

network devices outputs bluecat config items data  
services snmp configurations snmpConfiguration  
agentService v1

**See also**[ââ](#)

## Data model path

- [illegible]



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- Reference
- [NQE Language](#)
- [Data Model](#)
- SnmpAgentV2C

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